Class-11th + 12th

Kattar Advanced

EaJEE NOTES

for

INORGANIC CHEMISTRY

JEE Main & Advanced
Theory and PYQ's

By Om Pandey
(IIT Delhi)

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JEE Mains Syllabus

Classification of Elements and Periodicity in Properties

- Modern periodic law and present form of the periodic table, s, p, d and f block elements, periodic trends in properties of elements-atomic and ionic radii, ionization enthalpy, electron gain enthalpy, valence, oxidation states and chemical reactivity.

Chemical Bonding and Molecular Structure

- Kossel-Lewis approach to chemical bond formation, the concept of ionic and covalent bonds.
- **Ionic Bonding:** Formation of ionic bonds, factors affecting the formation of ionic bonds; calculation of lattice enthalpy.
- **Covalent Bonding:** Concept of electronegativity, Fajan's rule, dipole moment; Valence Shell Electron Pair Repulsion (VSEPR) theory and shapes of simple molecules.
- **Quantum mechanical approach to covalent bonding:** Valence bond theory- its important features, the concept of hybridization involving s, p and d orbitals, resonance.
- **Molecular Orbital Theory:** Its important features, LCAOs, types of molecular orbitals (bonding, antibonding), sigma and pi-bonds, molecular orbital electronic configurations of homonuclear diatomic molecules, the concept of bond order, bond length and bond energy.
- Elementary idea of metallic bonding, hydrogen bonding and its applications

Coordination Compounds

- Introduction to coordination compounds. Werner's theory, ligands, coordination number, denticity, chelation, IUPAC nomenclature of mononuclear coordination compounds, isomerism. Bonding: Valence bond approach and basic ideas of Crystal field theory, colour and magnetic properties, importance of coordination compounds (in qualitative analysis, extraction of metals and in biological systems).

p- Block Elements (Group-13 and Group 14 Elements)

- **General Introduction:** Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups, unique behaviour of the first element in each group.

p- Block Elements (Group-15 to Group 18 Elements)

- **General Introduction:** Electronic configuration and general trends in physical and chemica properties of elements across the periods and down the groups, unique behaviour of the firs element in each group.

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d- and f- Block Elements

- **Transition Elements:** General introduction, electronic configuration, occurrence and characteristics, general trends in properties of the first-row transition elements- physical properties, ionization enthalpy, oxidation states, atomic radii, colour, catalytic behaviour, magnetic properties, complex formation, interstitial compounds, alloy formation, preparation, properties and uses of $K_2Cr_2O_7$ and $KMnO_4$.
- **Inner Transition Elements:**

  Lanthanoids- Electronic configuration, oxidation states and Lanthanoid contraction.

  Actinoids- Electronic configuration and oxidation states

Salt Analysis

- **Chemical principles involved in the qualitative salt analysis:**

  Cations- Pb^{2+}, Cu^{2+}, Al^{3+}, Fe^{3+}, Zn^{2+}, Ni^{2+}, Ca^{2+}, Ba^{2+}, Mg^{2+}, NH_4^+

  Anions- CO_3^{2-}, S^{2-}, SO_4^{2-}, NO_3^-, NO_2^-, Cl^-, Br^-, I^- (Insoluble salts excluded).

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मैं ज्वाला हूँ
बुझी राख नहीं,
मैं कर्म के साथ हूँ
किस्मत के हाथ नहीं।

Scan the QR code to access detailed content of 3 chapters of JEE Advanced Syllabus (Hydrogen, s-Block, General Principles and Processes of isolation of Elements).

1 Classification of Elements and Periodicity in Properties

(JEE Advanced)

Syllabus Modern periodic law and the present form of periodic table; electronic configuration of elements; periodic trends in atomic radius, ionic radius, ionization enthalpy, electron gain enthalpy, valence, oxidation states, electronegativity, and chemical reactivity.

QUANTUM NUMBERS

The set of four numbers required to defined an electron completely in an atom are called quantum numbers. The first three have been derived from Schrodinger wave equation.

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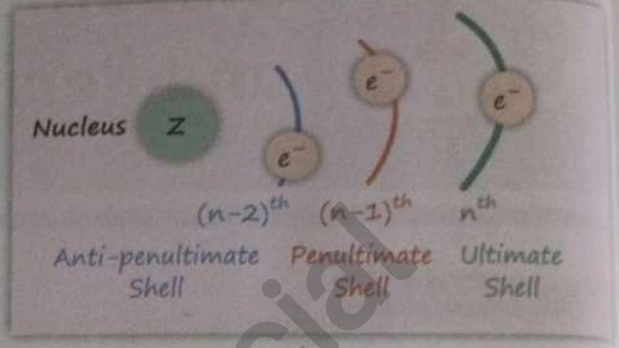
[I] Principal Quantum Number (n)

It describes the size of the electron wave and the total energy of the electron. It has integral values 1, 2, 3, 4... etc. and is denoted by K, L, M, N, ... etc.

Number of subshell present in n^{th} shell = n

n	subshell
1	s
2	s, p
3	s, p, d
4	s, p, d, f



- Number of orbitals present in n^{th} shell = n^2

[II] Azimuthal Quantum Number (ℓ)

It describes the shape of electron cloud and the number of subshells in a shell.

- It can have values from 0 to [n-1].
-

Value of l		Subshell
0	→	s
1	→	p
2	→	d
3	→	f
4	→	g

Note

The notations for the sub-energy levels come from the spectroscopic terms that were used to describe the atomic spectra and have the following full form:

s	→ sharp	f	→ fundamental
p	→ principal	g	→ generalized
d	→ diffused		

[III] Magnetic Quantum Number (m)

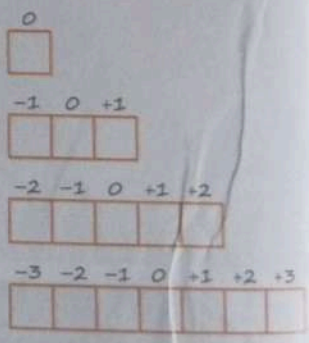
It describes the orientation of an orbital within a subshell. It can have values from $-\ell$ to $+\ell$ including zero, i.e., total $(2\ell + 1)$ values. Each value corresponds to an orbital, s-subshell has one orbital, p-subshell three orbitals (p_x, p_y and p_z), d-subshell five orbitals (d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$, d_{z^2}) and f-subshell has seven orbitals.

s → $\ell = 0, m = 0$

p → $\ell = 1, m = -1, 0, +1$

d → $\ell = 2, m = -2, -1, 0, +1, +2$

f → $\ell = 3, m = -3, -2, -1, 0, +1, +2, +3$



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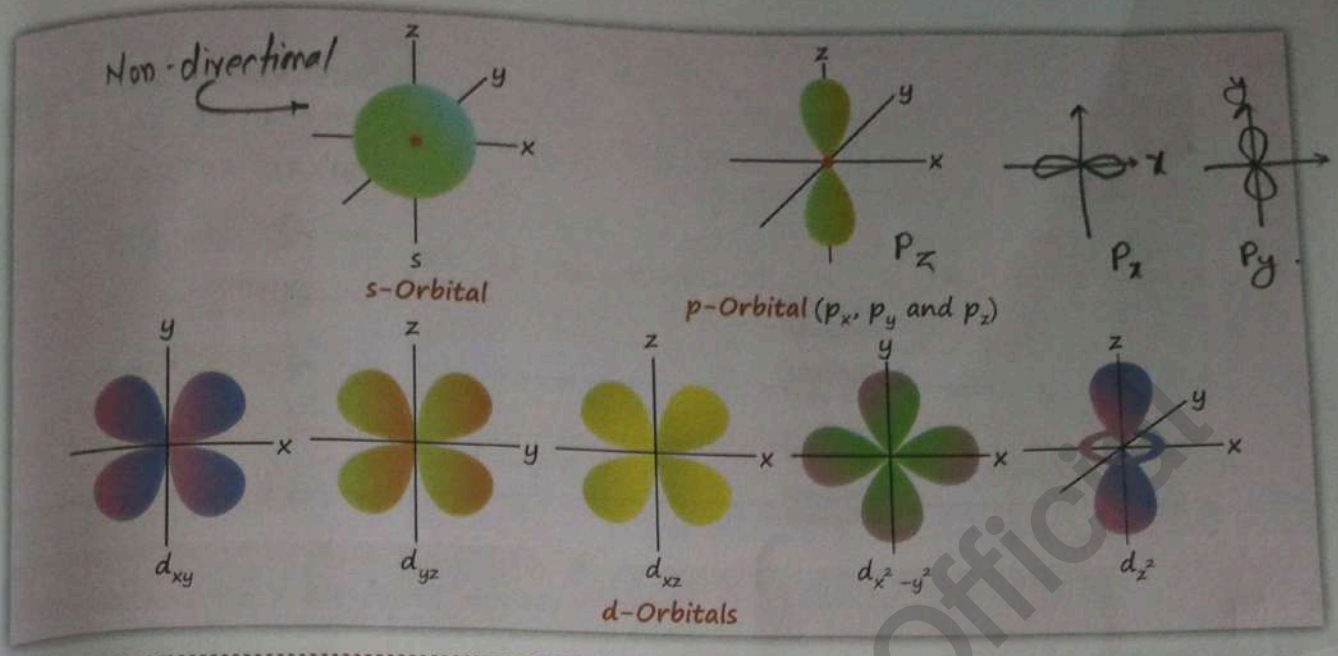
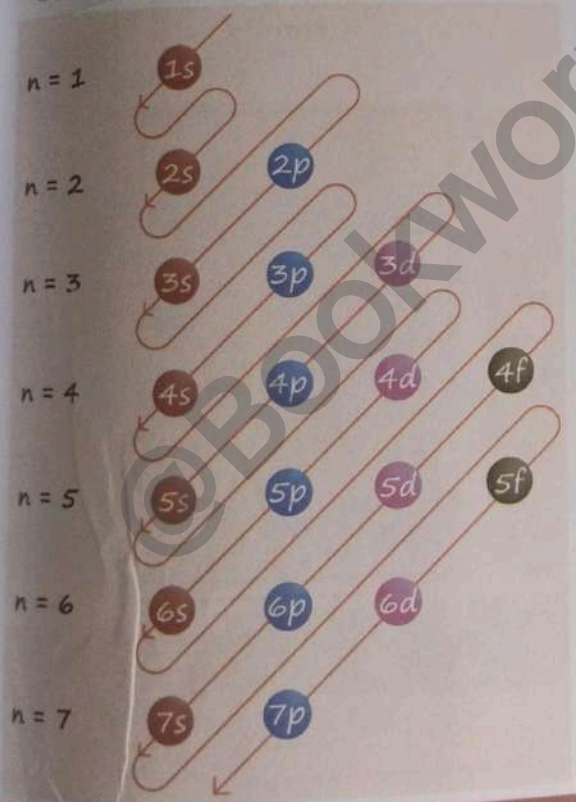


[IV] Spin Quantum Number (s)

It describes the spin of the electron. It has values +1/2 and −1/2. (+) signifies clockwise spinning and (−) signifies anticlockwise spinning.

n+l RULE AND MODERN PERIODIC TABLE



Max. n	Energy Sequence and Set of Subshells				No. of electrons in set of subshell
1	1s				2
2	2s			2p	2 + 6 = 8
3	3s			3p	2 + 6 = 8
4	4s		3d	4p	2 + 10 + 6 = 18
5	5s		4d	5p	2 + 10 + 6 = 18
6	6s	4f	5d	6p	2 + 14 + 10 + 6 = 32
7	7s	5f	6d	7p	2 + 14 + 10 + 6 = 32

Block: s, p, d, f

7 Periods: Horizontal Rows

18 Groups: Vertical Columns

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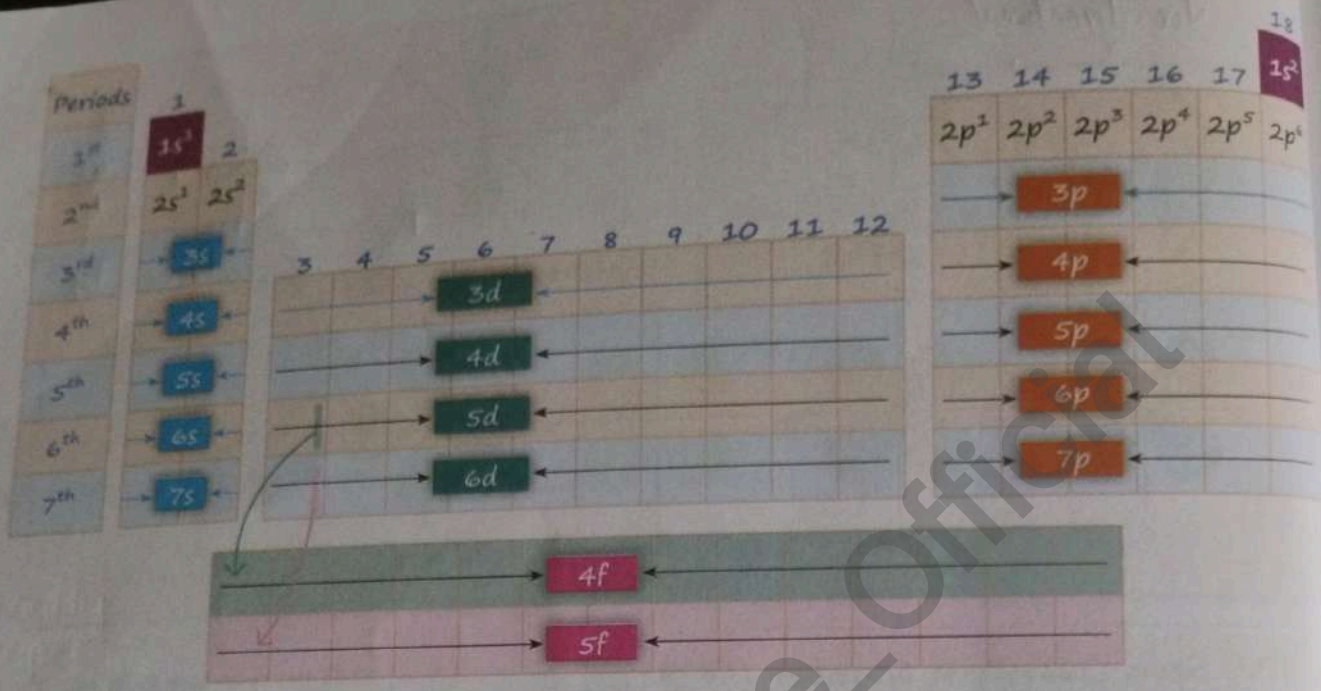
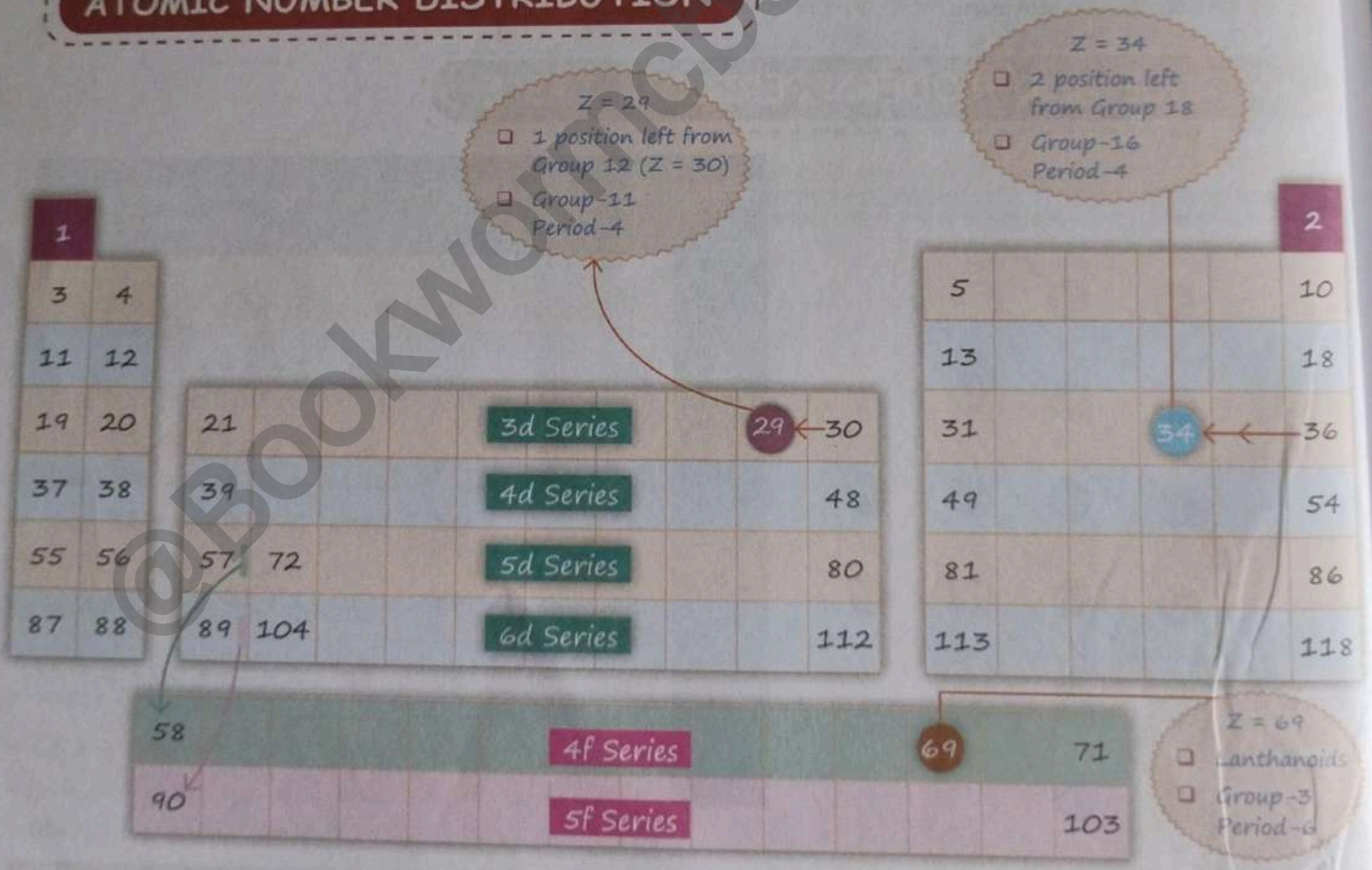


ATOMIC NUMBER DISTRIBUTION



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Period Number	Range
1	1-2
2	3-10
3	11-18
4	19-36
5	37-54
6	55-86
7	87-118

La_{57} $Ce_{58} \longrightarrow Lu_{71}$ Hf_{72}

Ac_{89} $Th_{90} \longrightarrow Lr_{103}$ Rf_{104}

Lanthanoid Series

Period No.	6
Group No.	3

Actinoid Series

Period No.	7
Group No.	3

MODERN PERIODIC TABLE

s-Block (ns) | d-Block (n − 1)d | p-Block (np)

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 (1s)	H																	He
2 (2s, 2p)	Li	Be											B	C	N	O	F	Ne
3 (3s, 3p)	Na	Mg											Al	Si	P	S	Cl	Ar
4 (4s, 3d, 4p)	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5 (5s, 4d, 5p)	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6 (6s, 4f, 5d, 6p)	Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7 (7s, 5f, 6d, 7p)	Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

Transition metals (d-Block)

The Rare Earths, (n − 2)f

f-Block														
Lanthanide series (4f) Period : 6	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Actinide series (5f) Period : 7	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

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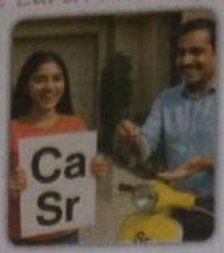
देशी जुगाड़

Alkali Metals (Group-1)

HLiNa ne Ki Rb Cs Fariyad

Alkaline Earth Metals (Group-2)



Beti Mage Car Scooter Baap Razi

Group-13: Boron Family

Bengan Alu Gajar In Thaila

Group-14: Carbon Family

Chemistry Sir gives Sabki Problems & flowers

Cahei Sita Ge Suno Prabhu

Group-15: Nitrogen Family

Nana Patekar Aishwarya Sb Bimar

Group-16: Oxygen Family

O S Se Te Po

Group-17: Halogen Family

Fir Cal* Bahar I Aunty

Group-18: Inert Gas Family

Heena Neena Aur Kareena Xerox Rangeen

1st Transition Series: 3d Series

Science Ticher Very Cruel Mange Fees Copy Niha Cukker Zindabad

2nd Transition Series: 4d Series

Yari Zra Nibhana Mout Tc Rukawat Rah Pde Age Cudo

3rd Transition Series: 5d Series

La HalTa Warna Re Osama Idher Pitayi Aur Hogi

4th Transition Series: 6d Series

Ac Rutherford Dube Sagar me Bohr Hs Mat Darse Royga Caun

Group-3

ScYLa Ac me rahti hai

Ek Raftar ko aage bhole hum mat dosti khatm karenge।

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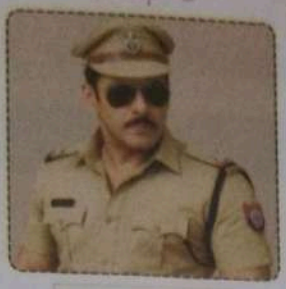
Group-4

Tina Zor Haaf Rafi* hai

Group-5

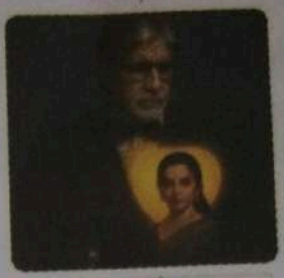


Vo Nabab Tha Dabang

Group-6

Crying Moti Wife

Group-7



Maan Tac* Rekha

Group-8 and 9

Fer* Rouya Osama Con* Rahega Iran me

Group-10, 11, 12

Nahi Padoge to Pitoge | Cyu Aage Aau | Zindagi Cadbury Hogyi

Lanthanoids → I[st] Inner Transition Series (Part of Group 3)

Cene Pr Nadiyan Prem ki Samayi Eu Gadgad Tab Dyl Ho gya Engineer Tum Yebhi Lu

Actinoids → II[nd] Inner Transition Series (Part of Group 3)

Thor Ke Papa ne U bola Nepal me Purane Am Cam Bikenge Cafe me jana Ease Farmana Madam Noodles Lare

थोड़े पापड़ यू नेपो पुराने आम कम बिकेंगे

Place of Hydrogen is not fixed in Periodic Table because it shares properties with both alkali metals and halogens.

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IUPAC NAME, COMMON NAME AND POSITION OF ELEMENTS HAVING ATOMIC NUMBER 101 TO 118

Digit	Name	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

Suffix – ium	
101	: Unnilunium (Unu)
102	: Unnilbium (Unb)
105	: Unnilpentium (Unp)
110	: Ununnilium (Uun)
118	: Ununoctium (Uuo)
119	: Ununennium (Uue)

Z = 143
IUPAC Name
↓
Unquadtrium
UQT

Atomic Number 101 to 118

1. Mendeleev [101] ko mila Nobel [102] in Lowrence [103]
2. Rutherford [104] Dube [105] Sea [106] me Bohr [107] Hass [108] Meit [109]
3. Dar [110] se Roe [111] Coper [112]
4. Niho [113] Fle [114] to Mosco [115] to Live [116] Ten [117] Ogan [118]

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Atomic Number	Name
101	Mendelevium (Md)
102	Nobelium (No)
103	Lawrencium (Lr)
104	Rutherfordium (Rf)
105	Dubnium (Db)
106	Seaborgium (Sg)
107	Bohrium (Bh)
108	Hassium (Hs)
109	Meitnerium (Mt)

Atomic Number	Name
110	Darmstadtium (Ds)
111	Roentgenium (Rg)
112	Copernicium (Cn)
113	Nihonium (Nh)
114	Flerovium (Fl)
115	Moscovium (Mc)
116	Livermorium (Lv)
117	Tennessine (Ts)
118	Oganesson (Og)

Position in Periodic Table

	101	102	103
Period No. →	7		
Group No. →	3	(Part of actinoids)	

	104	105	106	107	.....	118
Group No. →	4th	5th	6th	7th		18th
Period No. →	7					

1. The IUPAC symbol for the element with atomic number 119 would be- [8 April, 2019 (Shift-2)]

(a) Unh (b) Uue (c) Uun (d) Une

Sol. (b)

2. Lanthanoids and Actinoids are present in which of the following group-

(a) Group 1 (b) Group 3 (c) Group 17 (d) Group 18

Sol. (b)

3. If the atomic number of an element is 35, it will be placed in the periodic table in the-

(a) Group 1 (b) Group 3 (c) Group 17 (d) Group 18

Sol. (c)

Period-4, Group-17

							2
							10
							18
						35	36
							54
							86
							118

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[30 Jan, 2023 (Shift-1)]

4. Match List-I with List-II

List-I (Atomic Number)			List-II (Block of Periodic Table)	
(A)	37	S	(i)	p-block
(B)	78	D	(ii)	d-block
(C)	52	P	(iii)	f-block
(D)	65	f	(iv)	s-block

Choose the correct answer from the options given below:

(a) A-ii, B-iv, C-i, D-iii

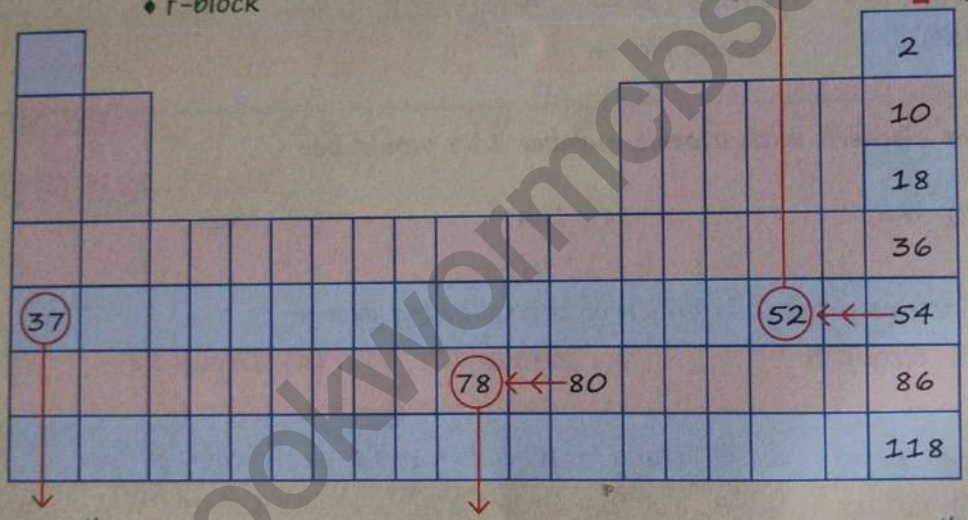
(b) A-i, B-iii, C-iv, D-ii

(c) A-iv, B-iii, C-ii, D-i

~~(d)~~ A-iv, B-ii, C-i, D-iii

Sol. (d) A-iv, B-ii, C-i, D-iii

- Lanthanoids (58 < 65 < 71)
  - 6th period
  - 3rd group
  - f-block
- 5th period
- 16th group (p-block)



- 5th period
- 1st group (s-block)
- 2 position left to Z = 80 (12th group)
- 10th group, 6th period

5. The element with atomic number 117, 119 , 120 , 91 will be- [30 Jan, 2023 (Shift-1)]

(a) Alkali's

(b) Halogen, Alkali, Alkaline earth metal & Lanthanide

~~(c)~~ Halogen, Alkali, Alkaline earth metal & Actinide

(d) Transition element, Halogen, Alkali, Alkaline earth metal

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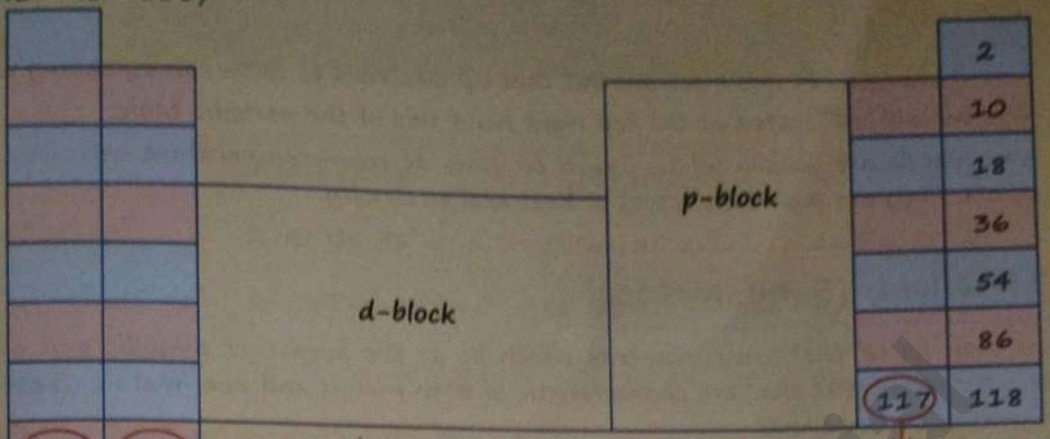
Sol. (c) Actinoids (90 < 91 < 103)



- 1st group
- Alkali metals
- 8th period

- 2nd group (Alkaline Earth metal)
- 8th period

- 17th group (Halogen family)
- 7th period

6. The elements with atomic number 101 and 104 belong to, respectively- [4 Sept. 2020 (Shift-1)]

(a) Actinoids and Group 4 (b) Group 11 and Group 4
(c) Group 6 and Actinoids (d) Actinoids and Group 6

Sol. (a)

7. Identify the incorrect match.

	Name		IUPAC Official Name
(A)	Unnilunium ✓	(i)	Mendelevium
(B)	Unniltrium ✓	(ii)	Lawrencium
(C)	Unnilhexium ✓	(iii)	Seaborgium
(D)	Unununium ✗	(iv)	Darmstadtium

(a) A-(i) (b) B-(ii) (c) C-(iii) (d) D-(iv)

Sol. (d) Unununium [111] → Roentgenium

8. The element with Z = 120 (not yet discovered) will be an/a [12 Jan, 2019 (Shift-2)]

(a) Transition metal (b) Inner transition metal
(c) Alkaline earth metal (d) Alkali metal

Sol. (c)

METAL, NON-METAL AND METALLOID

Metal

$M \rightarrow M^+ + e^-$

- The metals are characterised by their nature of readily giving up the electron(s) and form shining lustre.
- Metals comprises more than 78% of all known elements and appear on the left hand side of the periodic table.
- Metals are usually solids at room temperature (except Hg, Ga).
- They have high melting and boiling points and are good conductors of heat and electricity.

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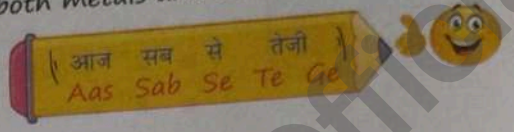
Non-metal

$N + e^- \rightarrow N^-$

- Non-metals do not lose electrons but take up electrons to form corresponding anions.
- Non-metals are located at the top right hand side of the periodic table.
- Non-metals are usually solids, liquids or gases at room temperature with low melting and boiling points. They are poor conductors of heat and electricity.

Metalloids (Semi-metals)

It has been found that some elements which lie at the border of metallic and nonmetallic behavior possess the properties that are characteristic of both metals and non-metals. These elements are called semi-metals or metalloids.
Examples - As, Sb, Se, Te, Ge



s-BLOCK

Alkali Metals (Group-1)		
General electronic configuration ⇒ [Inert gas] ns^1		
n = 2	Li_3	: [He] $2s^1$
n = 3	Na_{11}	: [Ne] $3s^1$
n = 4	K_{19}	: [Ar] $4s^1$
n = 5	Rb_{37}	: [Kr] $5s^1$
n = 6	Cs_{55}	: [Xe] $6s^1$

Alkaline Earth Metals (Group-2)	
General electronic configuration ⇒ [Inert gas] ns^2	
Be_4	: [He] $2s^2$
Mg_{12}	: [Ne] $3s^2$
Ca_{20}	: [Ar] $4s^2$
Sr_{38}	: [Kr] $5s^2$
Ba_{56}	: [Xe] $6s^2$

When shells upto (n – 1) are completely filled and the last electron enters the s-orbital of the outermost (n^{th}) shell, the elements of this class are called s-block elements.

- Group 1 & 2
- General electronic configuration : [inert gas] ns^{1-2}
- s-block elements lie on the extreme left of the periodic table.
- This block includes metals.

p-BLOCK

Group-13: [Inert Gas] ns^2np^1		Group-14: [Inert Gas] ns^2np^2	
B_5	: [He] $2s^2\ 2p^1$	C_6	: [He] $2s^2\ 2p^2$
Al_{13}	: [Ne] $3s^2\ 3p^1$	Si_{14}	: [Ne] $3s^2\ 3p^2$
Ga_{31}	: [Ar] $4s^2\ 3d^{10}\ 4p^1$	Ge_{32}	: [Ar] $4s^2\ 3d^{10}\ 4p^2$
In_{49}	: [Kr] $5s^2\ 4d^{10}\ 5p^1$	Sn_{50}	: [Kr] $5s^2\ 4d^{10}\ 5p^2$
Tl_{81}	: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^1$	Pb_{82}	: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^2$
Nh_{113}	: [Rh] $7s^2\ 5f^{14}\ 6d^{10}\ 7p^1$	Fl_{114}	: [Rn] $7s^2\ 5f^{14}\ 6d^{10}\ 7p^2$

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Period No. / Group No.	Group 15 (Pnictogens)	Group 16 (Chalcogens)	Group 17 (Halogens)	Group 18 (Inert gas)
1.				He: $1s^2$
2.	N: [He] $2s^2\ 2p^3$	O: [He] $2s^2\ 2p^4$	F: [He] $2s^2\ 2p^5$	Ne: [He] $2s^2\ 2p^6$
3.	P: [Ne] $3s^2\ 3p^3$	S: [Ne] $3s^2\ 3p^4$	Cl: [Ne] $3s^2\ 3p^5$	Ar: [Ne] $3s^2\ 3p^6$
4.	As: [Ar] $4s^2\ 3d^{10}\ 4p^3$	Se: [Ar] $4s^2\ 3d^{10}\ 4p^4$	Br: [Ar] $4s^2\ 3d^{10}\ 4p^5$	Kr: [Ar] $4s^2\ 3d^{10}\ 4p^6$
5.	Sb: [Kr] $5s^2\ 4d^{10}\ 5p^3$	Te: [Kr] $5s^2\ 4d^{10}\ 5p^4$	I: [Kr] $5s^2\ 4d^{10}\ 5p^5$	Xe: [Kr] $5s^2\ 4d^{10}\ 5p^6$
6.	Bi: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^3$	Po: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^4$	At: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^5$	Rn: [Xe] $6s^2\ 4f^{14}\ 5d^{10}\ 6p^6$

When shells upto $(n-1)$ are completely filled and differentiating electron enters the p-orbital of the n^{th} orbit, elements of this class are called p-block elements.

- Group 13 to 18
- General electronic configuration: [inert gas] $ns^2\ np^{1-6}$
- p-block elements lie on the extreme right of the periodic table.
- This block includes some metals, all non-metals and metalloids.
- Representative Elements: s + p block elements except inert gas elements. ★

d-BLOCK ELEMENTS

Series name ↓ | Group →	3		4	5	6	7	8	9	10	11	12
3d-series	Sc_{21}		Ti_{22}	V	Cr	Mn	Fe	Co	Ni	Cu	Zn_{30}
4d-series	Y_{39}		Zr_{40}	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd_{48}
5d-series	La_{57}	4f Series	Hf_{72}	Ta	W	Re	Os	Ir	Pt	Au	Hg_{80}
6d-series	Ac_{89}	5f Series	Rf_{104}	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn_{112}

Electronic Configuration

3d series: [Ar] 4s 3d

4d series: [Kr] 5s 4d

5d series: [Xe] 6s 5d : for La
[Xe] 6s 5d $4f^{14}$: from Hf → Hg

6d series: [Rn] 7s 6d : for Ac
[Rn] 7s 6d $5f^{14}$: from Rf → Cn

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देशी जुगाड़ For Electronic Configuration of 11 Elements

Rahgir(Rg) Kyu(Cu) Aage(Ag) Aau(Au) Pitayi(Pt) Croge(Cr), Nabab(Nb) Mout(Mo) Rukawat(Ru) Rah(Rh) Padegi(Pd)

Rg	Cu	Ag	Au	Pt	Cr	Nb	Mo	Ru	Rh
				ns^1					

Pd → ns^0

Sc $4s^2\ 3d^1$	Ti $4s^2\ 3d^2$	V $4s^2\ 3d^3$	Cr $4s^1\ 3d^5$	Mn $4s^2\ 3d^5$	Fe $4s^2\ 3d^6$	Co $4s^2\ 3d^7$	Ni $4s^2\ 3d^8$	Cu $4s^1\ 3d^{10}$	Zn $4s^2\ 3d^{10}$
Y $5s^2\ 4d^1$	Zr $5s^2\ 4d^2$	Nb $5s^1\ 4d^4$	Mo $5s^1\ 4d^5$	Tc $5s^2\ 4d^5$	Ru $5s^1\ 4d^7$	Rh $5s^1\ 4d^8$	Pd $5s^0\ 4d^{10}$	Ag $5s^1\ 4d^{10}$	Cd $5s^2\ 4d^{10}$
La $6s^2\ 5d^1$	Hf $6s^2\ 5d^2\ 4f^{14}$	Ta $6s^2\ 5d^3\ 4f^{14}$	W $6s^2\ 5d^4\ 4f^{14}$	Re $6s^2\ 5d^5\ 4f^{14}$	Os $6s^2\ 5d^6\ 4f^{14}$	Ir $6s^2\ 5d^7\ 4f^{14}$	Pt $6s^1\ 5d^9\ 4f^{14}$	Au $6s^1\ 5d^{10}\ 4f^{14}$	Hg $6s^2\ 5d^{10}\ 4f^{14}$
Ac $7s^2\ 6d^1$	Rf $7s^2\ 6d^2\ 5f^{14}$	Db $7s^2\ 6d^3\ 5f^{14}$	Sg $7s^2\ 6d^4\ 5f^{14}$	Bh $7s^2\ 6d^5\ 5f^{14}$	Hs $7s^2\ 6d^6\ 5f^{14}$	Mt $7s^2\ 6d^7\ 5f^{14}$	Ds $7s^2\ 6d^8\ 5f^{14}$	Rg $7s^1\ 6d^{10}\ 5f^{14}$	Cn $7s^2\ 6d^{10}\ 5f^{14}$

- Group : 3 to 12
- General electronic configuration : [inert gas] $ns^{0-2}\ (n-1)d^{1-10}$ (except, Pd : $4d^{10}5s^0$)

When outermost (n^{th}) and penultimate shells $(n-1)^{th}$ shells are incompletely filled and differentiating electron enters the $(n-1)$ d orbitals (d-orbital of penultimate shell) then elements of this class are called d-block elements.

- All the transition elements are metals.

1st Transition Series	2nd Transition Series
3d series	4d series
10 elements	10 elements
Starts from $_{21}Sc - _{30}Zn$.	Starts from $_{39}Y - _{48}Cd$
Filling of electrons takes place in 3d subshell.	Filling of electrons takes place in 4d subshell.

3rd Transition Series	4th Transition Series
5d series	6d series
10 elements	10 elements
Starts from $_{57}La$, $_{72}Hf - _{80}Hg$.	Starts from $_{89}Ac$, $_{104}Rf - _{112}Uub$.
Filling of electrons takes place in 5d subshell.	Filling of electrons takes place in 6d subshell.

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Transition Metals ★

The elements whose atoms or simple ions contain partially filled d-orbitals are called transition metals.

1. Sc is a transition metal because Sc has partially filled d-orbitals.

   $Sc : [Ar]\ 4s^2\ 3d^1$

2. Zn is not a transition metal because Zn and Zn^{2+} both have filled d-orbitals.

   $Zn : [Ar]\ 4s^2\ 3d^{10}$
   $Zn^{2+} : [Ar]\ 3d^{10}$

3. Cu is a transition metal because Cu^{2+} has partially filled d-orbitals.

   $Cu : [Ar]\ 4s^1\ 3d^{10}$
   $Cu^{2+} : [Ar]\ 3d^9$

Note

- Zn, Cd, Hg are not considered as transition elements because they have filled 'd^{10} configuration' in atomic (M) and ionic (M^{2+}) form.
- All transition elements are d-block elements but not all d-block elements are transition elements.
  - Elements in Liquid State at STP - Hg (metal), Br_2 (Non-metal)
  - Metals in Liquid State (at T < 40°C) - Rubi Mar Gayi Fir se
    Rb Hg Ga Fr Cs

f-BLOCK ELEMENTS

The f-block consists of the two series, lanthanoids (the fourteen elements following lanthanum) and actinoids (the fourteen elements following actinium).

Lanthanoids	Ce_{58}	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu_{71}
Actinoids	Th_{90}	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr_{103}

- They are metals
- All f-block elements belong to 3rd group.

1st Inner Transition Series - 4f series	2nd Inner Transition Series- 5f series
14 elements - $_{58}Ce$ to $_{71}Lu$	14 elements - $_{90}Th$ to $_{103}Lr$
Filling of electrons takes place in 4f subshell.	Filling of electrons takes place in 5f subshell.

Transuranium Elements $Z > 92$

- The elements coming after uranium are called transuranium elements.

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General electronic configuration : [Inert gas] $ns^2\ (n-2)f^{1-14}\ (n-1)d^{0-1}$

- When n, (n–1) and (n–2) shells are incompletely filled and last electron enters into f-orbital of anti-penultimate $(n-2)^{th}$ shell, elements of this class are called f-block elements.
- They are also called as inner transition elements as they contain three incomplete outermost shells and were also referred to as rare earth elements since their oxides were rare in earlier days.

9. The element Z = 114 has been discovered recently. It will belong to which of the following family/ group and electronic configuration?

(a) Carbon family, $[Rn]5f^{14}6d^{10}7s^27p^2$ (b) Oxygen family, $[Rn]5f^{14}6d^{10}7s^27p^4$

(c) Nitrogen family, $[Rn]5f^{14}6d^{10}7s^28p^6$ (d) Halogen family, $[Rn]5f^{14}6d^{10}7s^27p^5$

Sol. (a)

10. Which of the following is representative group of elements in the periodic table?

(a) Lanthanum (La) (b) Argon (Ar)

(c) Chromium (Cr) (d) Aluminium (Al)

Sol. (d)

11. If Z is given (Z = 32, 57, 71, 87), then what will be Period number, Group number, Block, Family, Electronic Configuration?

Sol. Z = 32
- p-Block
- Group Number – 14
- Period – 4 (Carbon family)
- EC = $[Ar]\ 4s^2\ 3d^{10}\ 4p^2$

Z = 57
- d-Block
- Group Number – 3
- Period Number – 6
- EC = $[Xe]\ 6s^2\ 5d^1$

Z = 71
- f-Block (Lanthanoids)
- Group Number – 3
- Period Number – 6
- EC = $[Xe]\ 6s^2\ 5d^1\ 4f^{14}$

Z = 87
- Alkali metal
- s-Block
- Group 1 [7^{th} Period]
- EC = $[Rn]\ 7s^1$

Note

If electronic configuration of an element is given and you have to find the position of element in periodic table then add all electrons and find atomic number (For neutral element → Total no. of electrons = atomic number).

- Through atomic number, you can find the position of element.

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12. The IUPAC nomenclature of an element with electronic configuration $[Rn]5f^{14}6d^{1}7s^{2}$ is :

[23 July, 2022 (Shift-1)]

(a) Unnilbium (b) Unnilunium (c) Unnilquadium (d) Unniltrium

Sol. (d) Total number of electrons = 86 +14 + 1 + 2 = 103 [IUPAC Name - Unniltrium]

13. An atom has electronic configuration $1s^{2}2s^{2}2p^{6}3s^{2}3p^{6}3d^{3}4s^{2}$, you will place it in-

(a) fifth group (b) fifteenth group (c) second group (d) third group

Sol. (a) Total number of electrons = 2 + 2 + 6 + 2 + 6 + 3 + 2 = 23

14. The characteristics of element X, Y and Z with atomic numbers, respectively, 33, 53 and 83 are:

[16 March, 2021 (Shift-2)]

(a) X and Y are metalloids and Z is a metal

(b) X is a metalloid, Y is a non-metal and Z is a metal

(c) X, Y and Z are metals.

(d) X and Z are non-metals and Y is metalloid

Sol. (b) Metalloid X → 33 [As] Non-metal Y → 53 [I] Metal Z → 83 [Bi]

EFFECTIVE NUCLEAR CHARGE

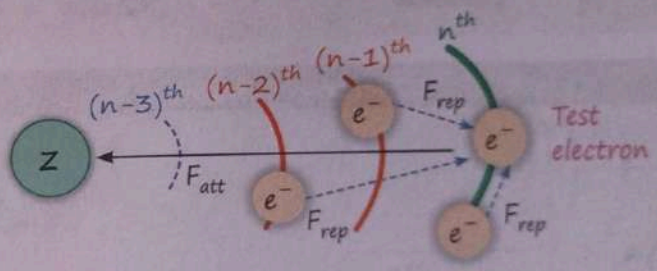
$F_{effective} = F_{att.} - F_{rep.}$

$Z_{eff} = Z - \sigma$

Z_{eff} = औकात of Nucleus

Z = Atomic number

σ = Shielding constant / Screening constant / Slater's constant



- Number of inner shell electrons ↑ ⇒ σ ↑ ⇒ Z_{eff} ↓
- There is a reduction in nuclear charge due to presence of screen b/w test electron and nucleus.
- **Screening/Shielding Effect:** The phenomenon where inner electrons in an atom reduce the attractive force of the nucleus on outer electrons. These inner electrons effectively "screen or shield" the outer electrons from the full nuclear charge.
- **Effective nuclear charge (Z_{eff}):** The net positive charge experienced by an electron in a multi-electron atom, which is less than the full nuclear charge due to shielding by inner electrons.

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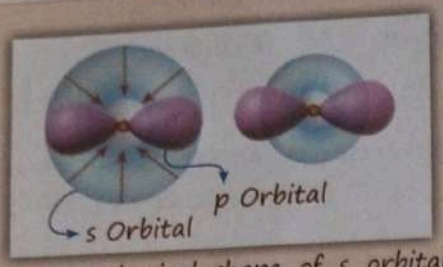
PENETRATION EFFECT

- Penetration Power means the ability of an electron in an orbital to get close to the nucleus.
- The Penetration effect of s-orbital is the maximum because it is closer to the nucleus than the p, and f-orbitals.



Due to spherical shape of s orbital, it is more attracted towards nucleus (centre of sphere) than p orbital having dumb bell shape.

Closeness to the nucleus : s > p > d > f

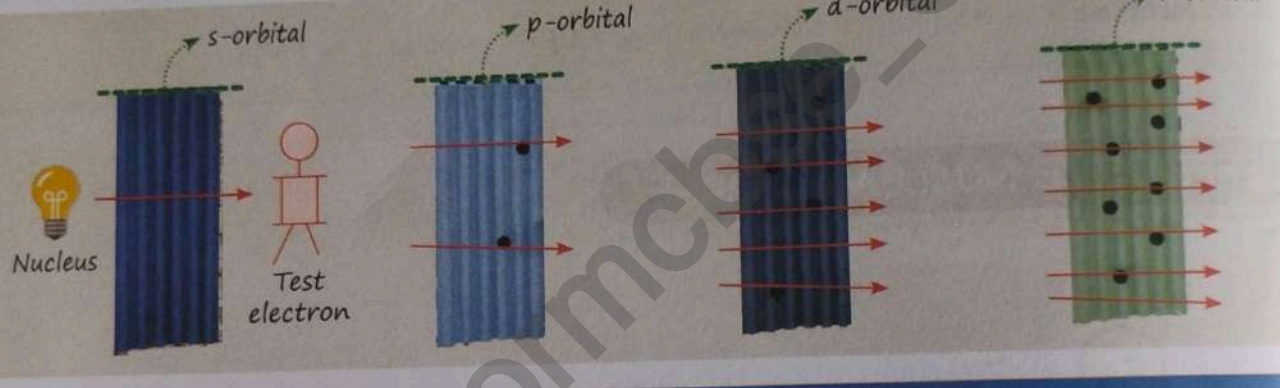
Penetration effect : s > p > d > f

Screening effect/Shielding effect : s > p > d > f

Poor shielding

Contribution of subshell in shielding constant (σ) :

ns > np > nd > nf



1 Electron System	More than 1 Electron System
H, He^+, Li^{2+}	
E 2s 2p 1s	2p 2s 1s
2s & 2p subshells are degenerate. Energy same	2s and 2p subshells are non-degenerate

15. The order of screening effect of electrons of s, p, d and f orbitals of a given shell of an atom on its outer shell electrons is :

(a) s > p > d > f (b) f > d > p > s (c) p > d > s > f (d) f > p > s > d

Sol. (a)

16. Screening effect is not observed in :

(a) He^+ (b) Li^{2+} (c) Be^{3+} (d) In all cases

Sol. (d) Screening effect is not observed in 1 electron system.

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Variation of Z_{eff} in Periodic Table

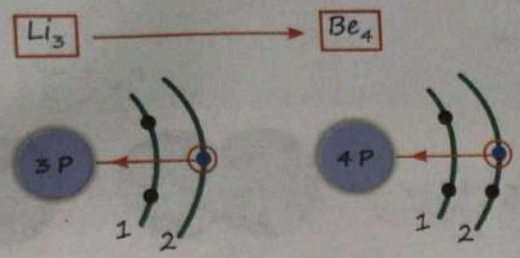
$Z_{eff} = Z - \sigma$

1. $Z_{eff} \propto Z$
2. $Z_{eff} \propto \frac{1}{\sigma} \propto \frac{1}{n_e}$

$Z_{eff} \propto \frac{Z}{n_e}$

$n_e \rightarrow$ Total number of electrons in atom

- **Variation in a Period:** Left → Right ⇒ Z_{eff} ↑ [Generally it is valid in s, p, d and f block]

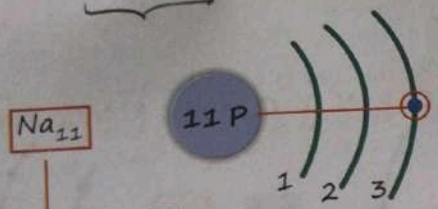


Z_{eff} : Li < Be

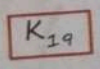
- Same shell number
- No. of protons ↑ ; Z_{eff} ↑

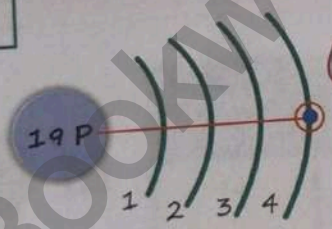
- **Variation in a Group:** When we move from top to bottom

1. Zeff ⇒ Constant [Generally valid in s and p blocks but not always]



Zeff is constant as no. of proton↑ but distance between nucleus and test electron↑.





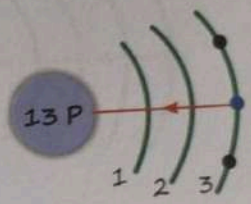
For Example

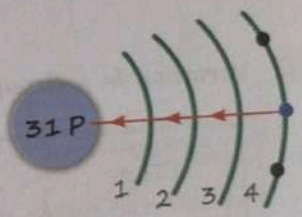
Al_{13} → [Ne] $3s^2\ 3p^1$

Ga_{31} → [Ar] $4s^2\ 3d^{10}\ 4p^1$

Due to poor shielding of $3d^{10}$ electrons, shielding constant σ decreases and Z_{eff}↑

Z_{eff} : Al < Ga





ATOMIC RADIUS

Atomic radius is the average or typical distance from the center of an atom's nucleus to the outermost shell.

$$\text{Atomic Radius} = \frac{\text{Inter-nuclear Distance}}{2}$$

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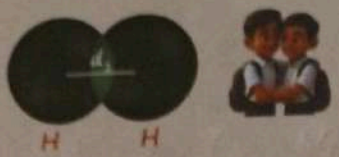
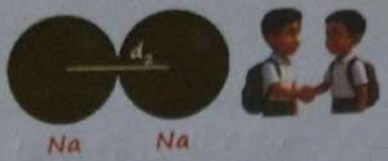
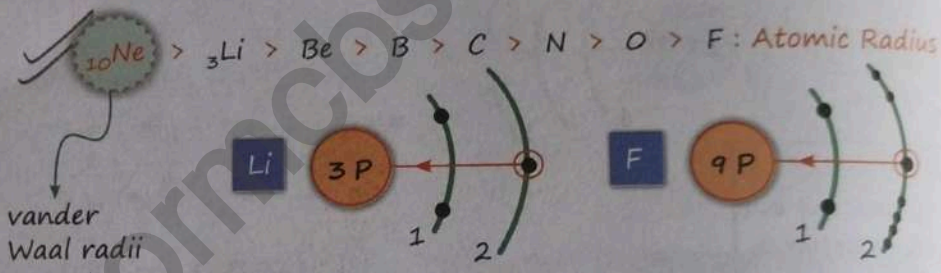
Covalent Radius	Metallic Radius	vander Waal Radius
♦ Covalent Bond ♦ Non metal – non metal (H)	♦ Metallic bond ♦ Metal – metal (Na-Na)	♦ VW – Force of attraction ♦ Inert gas / molecules (He, Ne, Ar) (H_2, Cl_2)
		
$r_c = \frac{d_1}{2}$	$r_M = \frac{d_2}{2}$	$r_{vw} = \frac{d_3}{2}$

- $d_1 < d_2 < d_3$
- $r_C < r_M < r_{VW}$

$(r_{VW})_{H_2} = \frac{d_4}{2}$

ATOMIC SIZE

- **Variation in a Period:** Left → Right ⇒ Z_{eff} ↑ ⇒ Atomic Radius ↓

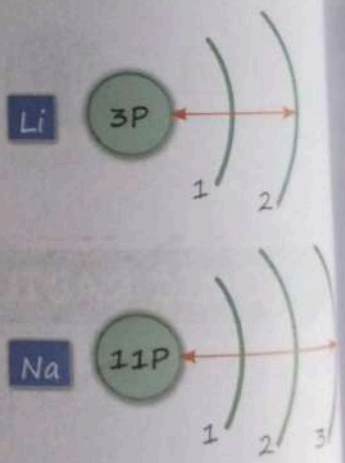


- **Variation in a Group:**

Top → Bottom → Z_{eff} ≈ constant

→ Due to addition of new shell ⇒ Atomic Radius ↑

Period No.	Atomic Size
2	Li
	^
3	Na
	^
4	K
	^
5	Rb
	^
6	Cs



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Same Period Elements	Different Period Elements
❑ L → R : Atomic Radius ↓	★ ❑ Period Number ↑ : Atomic Radius ↑

♦ Br > C > N > F
n = 4 n = 2

♦ C < S < Al < Ca
n = 2 n = 3 n = 4

B P > S > Cl Br
n = 2 n = 3 n = 3 n = 3 n = 4

B < Cl < S < P < Br → Atomic Size

Boron Family

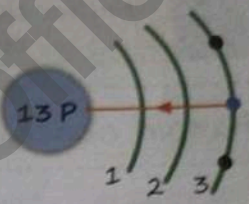
नौटंकी Family ⇐ Manoj yogi

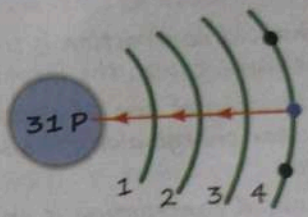
B
^
Al$_{13}$ → [Ne] $3s^2$ $3p^1$
v
Ga$_{31}$ → [Ar] $4s^2$ $3d^{10}$ $4p^1$
^
In
^
Tl

Due to Poor shielding of 3d orbital electrons, Z_{eff} is more in Ga w.r.t Al

Z_{eff}↑ : Size↓

MANOJ yogi





Atomic Size Order in s and p Block

H
^
Li
^
Na
^
K
^
Rb
^
Cs

Be
^
Mg
^
Ca
^
Sr
^
Ba

B
^
Al
v
Ga
^
In
^
Tl

C
^
Si
^
Ge
^
Sn
^
Pb

N
^
P
^
As
^
Sb
^
Bi

O
^
S
^
Se
^
Te
^
Po

F
^
Cl
^
Br
^
I
^
At

He
^
Ne
^
Ar
^
Kr
^
Xe
^
Rn

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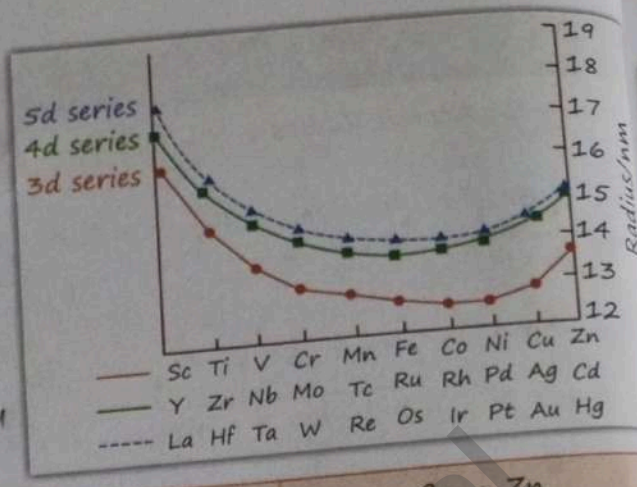
Atomic Size of 3d Series

There are two factors in order to decide atomic radius of an element.

1. Nuclear charge [Nuclear charge↑ : size↓]
2. Interelectronic repulsion (IER) [IER↑ : size↑]

→ Sc > Ti > V > Cr > Mn ≃ Fe ≃ Co ≃ Ni < Cu < Zn



Elements	Sc → Ti → V → Cr	Mn → Fe → Co → Ni	Cu → Zn
Size	Decreases	Nearly constant	Slightly increases
Reason	Nuclear charge dominates over IER	Nuclear charge ≈ IER	IER dominates over nuclear charge

Lanthanoid Contraction

- The decrease in atomic or ionic radii from Ce to Lu due to imperfect shielding of 4f e⁻ from nuclear charge is known as lanthanoid contraction.
- Lanthanoid contraction is similar to that observed in an ordinary transition series and is attributed to the same cause, the imperfect shielding of one electron by another in the same subshell.
- The shielding of one 4f electron by another is less than one d electron by another with the increase in nuclear charge along the series. There is fairly regular decrease in the sizes with increasing atomic number.
- Lanthanoid contraction is applicable on both metal (M) and metal ion (M^{+3}) size orders.

Ce → Lu
Ce^{3+} → Lu^{3+}
→ Size decreases

Effect of Lanthanide Contraction on Size of d-Block Elements

3d →	$_{21}Sc$	$_{22}Ti$	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
4d →	$_{39}Y$	$_{40}Zr$	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
5d →	$_{57}La$ ($_{58}Ce$ ⟶ $_{71}Lu$)	$_{72}Hf$	Ta	W	Re	Os	Ir	Pt	Au	Hg

→ Free from effect of Lanthanide contraction

- Due to lanthanoid contraction, radii of the members of 5d series are found to be very similar to those of corresponding members of the 4d series.
- For d-block elements, the trend in atomic radii is: $r_{3d\ series} < r_{4d\ series} \approx r_{5d\ series}$

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17. The lanthanide contraction is responsible for the fact that :
(a) Zr and Y have about the same radius (b) Zr and Nb have similar oxidation state
(c) Zr and Hf have about the same radius. (d) Zr and Zn have same oxidation state.

Sol. (c)

18. Which of the following has similar atomic radius with respect to Ag :
(a) La (b) Zn (c) Au (d) Rf

Sol. (c)

IONIC RADIUS

Cation → ⊕ vely charged ion	Anion → ⊖ vely charged ion
$Be \xrightarrow{-e^-} Be^+ \xrightarrow{-e^-} Be^{2+}$	$O \xrightarrow{+e^-} O^- \xrightarrow{+e^-} O^{2-}$
$Z_{eff} \propto \frac{Z}{e} = \frac{4}{4}$ $Z_{eff} \propto \frac{4}{3}$ $Z_{eff} \propto \frac{4}{2}$	$Z_{eff} \propto \frac{8}{8}$ $Z_{eff} \propto \frac{8}{9}$ $Z_{eff} \propto \frac{8}{10}$
Z_{eff} : $Be < Be^+ < Be^{2+}$ Size : $Be > Be^+ > Be^{2+}$: Mother atom > Cation	Z_{eff} : $O > O^- > O^{2-}$ Size : $O < O^- < O^{2-}$: Mother atom < Anion
♦ $Mg > Mg^+ > Mg^{2+}$: Size	♦ $I < I^-$: Size

anion > Mother atom > Cation.

ISOELECTRONIC SPECIES

- Species having same number of electrons are known as isoelectronic species.
- All ions– C^{4-}, N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+} are isoelectronic because they all have 10 electrons.

	C^{4-}	N^{3-}	O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
Number of Electrons :	(6 + 4)	(7+3)	(8+2)	(9+1)	(11-1)	(12-2)	(13-3)
Number of Protons :	6	7	8	9	11	12	13
$Z_{eff} \propto \frac{Z}{e}$:	$\frac{6}{10}$ <	$\frac{7}{10}$ <	$\frac{8}{10}$ <	$\frac{9}{10}$ <	$\frac{11}{10}$ <	$\frac{12}{10}$ <	$\frac{13}{10}$
Size :	C^{4-} >	N^{3-} >	O^{2-} >	F^-	> Na^+ >	Mg^{2+} >	Al^{3+}
			Anions			Cations	

- Si^{4-}, P^{3-}, S^{2-}, Cl^-, K^+, Ca^{2+}, Sc^{3+} → isoelectronic because they all have 18 electrons.

Size → $Si^{4-} > P^{3-} > S^{2-} > Cl^- > K^+ > Ca^{2+} > Sc^{3+}$

Size ∝ $\frac{1}{Z}$

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Note

- H_2S, HCl, Ar, SH^- → Isoelectronic with 18 electrons
- NH_2^-, NH_3, CH_4, H_2O, OH^-, NH_4^+, NH_2^- → Isoelectronic with 10 electrons
- CO_3^{2-}, NO_3^-, BO_3^{3-} → Isoelectronic with 32 electrons
- SiO_4^{4-}, PO_4^{3-}, SO_4^{2-}, ClO_4^- → Isoelectronic with 50 electrons
- $[Ni(CO)_4]$, $[Co(CO)_4]^-$, $[Fe(CO)_4]^{2-}$, $[Fe(CO)_2(NO)_2]$ → Isoelectronic with 84 electrons

$[Ni(CO)_4]$: $28 + 4(6 + 8)$ = 84

$[Co(CO)_4]^-$: $27 + 4(6 + 8) + 1$ = 84

$[Fe(CO)_4]^{2-}$: $26 + 4(6 + 8) + 2$ = 84

$[Fe(CO)_2(NO)_2]$: $26 + 2(6 + 8) + 2(7 + 8)$ = 84

Find the order of ionic radius?

- $S^{2-} > Cl^- > K^+ > Ca^{2+}$: Size
  [16+2] [17+1] [19−1] [20−2] : Number of electrons
- $Te^{2-} > I^-$: Size
  [52+2] [53+1] : Number of electrons
- Period Number → 2 / Period Number → 3
  $O^{2-} < N^{3-} < S^{2-} < P^{3-}$: Size
- $K^+ > Na^+ > Li^+$: Size [K > Na > Li : Size]
- $F^- < Cl^- < Br^- < I^-$: Size [I > Br > Cl > F : Size]
- $O^{2-} < S^{2-} < Se^{2-} < Te^{2-}$: Size

- Ar > $S^{2-} > Cl^- > K^+ > Ca^{2+}$: Size
  Ar → vander Waal radii

Note

The smallest anion is F^- and not H^- (ionic radius 208 pm). The radius order of anions is $F^- < Cl^- < Br^- < H^- < I^-$. The exception in the size of H^- is because this is the only anion with z/e ratio = 0.5.

19. Which of the following order is incorrect for size :

(a) Al > Ga (b) $Te^{2-} > I^- > Cs^+ > Ba^{2+}$ (c) $Cr^{3+} < Cr^{6+}$ (d) Pd ≈ Pt

Sol. (c)

20. The ionic radii (in Å) of N^{3-}, O^{2-} and F^- are respectively :

(a) 1.36, 1.40 and 1.71 (b) 1.36, 1.40 and 1.71

(c) 1.71, 1.40 and 1.36 (d) 1.71, 1.36 and 1.40

Sol. (c) $N^{3-} > O^{2-} > F^-$: size

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21. Ionic radii of (IIT JEE 1999)
(a) $Ti^{4+} < Mn^{7+}$ (b) $^{35}Cl^- < ^{37}Cl^-$ (c) $K^+ > Cl^-$ (d) $P^{3+} > P^{5+}$

Sol. (d)

22. Among $V(CO)_6$, $Cr(CO)_5$, $Cu(CO)_3$, $Mn(CO)_5$, $Fe(CO)_5$, $[Co(CO)_3]^{3-}$, $[Cr(CO)_4]^{4-}$, and $Ir(CO)_3$, the total number of species isoelectronic with $Ni(CO)_4$ is _____. (JEE Adv. 2024)

[Given, atomic number: V = 23, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Ir = 77]

Sol. [1] Total number of electron in $Ni(CO)_4$ = 84

Species	Total electron
$V(CO)_6$	107
$Cr(CO)_5$	94
$Cu(CO)_3$	71
$Mn(CO)_5$	95

Species	Total electron
$Fe(CO)_5$	96
$[Co(CO)_3]^{3-}$	72
$[Cr(CO)_4]^{4-}$	84
$Ir(CO)_3$	119

Property	Paramagnetic	Diamagnetic
Unpaired Electrons	Have one or more unpaired electrons	All electrons are paired
Magnetic Behavior	Attracted to an external magnetic field	Repelled by an external magnetic field
Magnetic Moment	Has a net magnetic moment	No net magnetic moment
Examples	Fe^{3+}, Mn^{2+}, NO 23 electrons 15 electrons Entities having odd number of electrons are always paramagnetic.	He, Ne, Zn^{2+}, Mg no unpaired electrons.

$\mu = \sqrt{n(n+2)}$ B.M. n = Unpaired e^-

IONISATION ENERGY

Minimum energy required to remove an electron from an isolated gaseous atom is known as ionisation energy (IE).

$$A_{(g)} \xrightarrow{\text{Min. Energy}} A^+_{(g)} + e^-$$

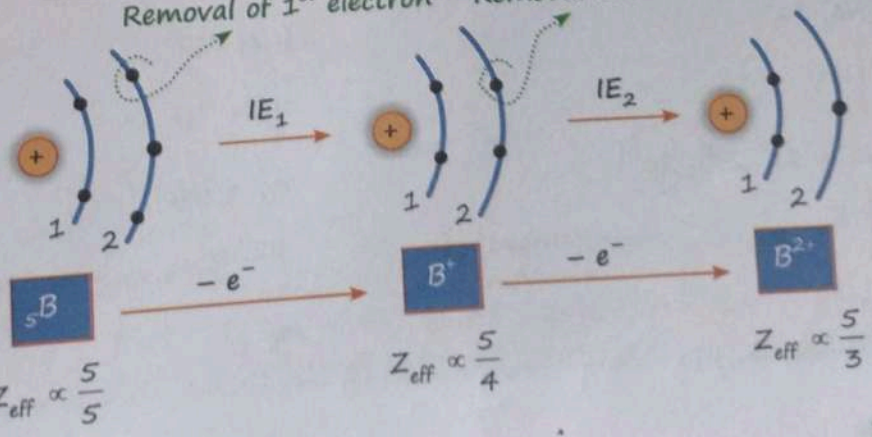
Successive IE :

Z_{eff} : B < B^+

IE : B < B^+

IE_1 of B < IE_2 of B

- IE_2 of B = IE_1 of B^+
- IE_3 of B = IE_1 of B^{2+} = IE_2 of B^+



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Factors Affecting I.E.

(1) Z_{eff} — Left → Right ⇒ Z_{eff} ↑ : IE ↑

- $_3Li < _4Be$: IE
- B < C : IE
- O < F : IE

(2) Size — Top → Bottom ⇒ Size ↑ : IE ↓

Li > Na > K > Rb > Cs

As we move from top to bottom ⇒ Size ↑ : IE ↓

Metallic Character

$M \rightarrow M^+ + e^-$

- Tendency of an element to lose electrons and form positive ions or cations.
- IE ↓ : Metallic character ↑
- Metallic character : Cs > Rb > K > Na > Li

(3) Electronic Configuration — Half filled / Fully filled subshell → Stable due to high exchange energy.
→ Removal of electron is tough so high IE.

IE : N ($2s^2$ $2p^3$) > O ($2s^2$ $2p^4$)

Removal of electron from half filled subshell is tough

IE :	P ($3p^3$)	>	S ($3p^4$)
IE :	As ($4p^3$)	>	Se ($4p^4$)

In the oxygen atom, two of the four 2p electrons must occupy the same 2p-orbital, resulting in an increased $e^- - e^-$ repulsion. So, it is easier to remove the fourth 2p electron from oxygen than it is to remove one of the three 2p-electrons from nitrogen.

Comparison of 2nd Ionisation Energy

Group-1

Na

$[Ne]\ 3s^1$ —IE_1→ $Na^+ \equiv 1s^2\ 2s^2\ 2p^6$ —IE_2→ Na^{2+}

Removal of electron from 2nd shell is tough.

Group-2

Mg

$[Ne]\ 3s^2$ —IE_1→ $Mg^+ \equiv [Ne]\ 3s^1$ —IE_2→ Mg^{2+}

Removal of electron from 3rd shell is easy.

IE_1 : group 1 < group 2
IE_2 : group 1 > group 2

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❑ N

$2s^2 (2p^3)$ → Half filled

$-e^-$

N^+

$2s^2 2p^2$

O

$2s^2 2p^4$

$-e^-$

O^+

$2s^2 (2p^3)$ → Half filled

$IE_1 : N > O$

$IE_2 : N < O$

$IE : N^+ < O^+$

❑ Cr

$[Ar]\ 4s^1 3d^5$

$-e^-$ → Cr^+ $[Ar]$ $(3d^5)$

For 2nd IE, removal of e^- from 3rd shell is tough.

Mn

$[Ar]\ 4s^2 3d^5$

$-e^-$ → $Mn^+ \equiv [Ar]$ $(4s^1)$ $3d^5$

For 2nd IE, removal of electron from 4th shell is easy.

IE_1 Mn > Cr.

IE_2 : Cr > Mn

4 Penetration Effect

❑ Penetration Effect for different subshell: s > p > d > f

♦ Be > B : IE

$2s^2$ $2s^2$ $(2p^1)$

$-e^-$

Removal of e^- from 2p subshell is easy with respect to 2s (Be).

The penetration of a 2s -electron to the nucleus is more than that of a 2p electron; hence the 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.

♦ Mg > Al: IE

♦ Ca > Ga: IE

23. The correct order of decreasing second ionisation enthalpy of Ti (22), V (23), Cr (24) and Mn (25) is:

(a) Cr > Mn > V > Ti

(b) V > Mn > Cr > Ti

(c) Mn > Cr > Ti > V

(d) Ti > V > Cr > Mn

Sol. (a)

24. Assertion: The first ionization energy of Be is greater than that of B.

Reason: 2p orbital is lower in energy than 2s.

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(b) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

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(c) Assertion is true but Reason is false.

(d) Assertion is false but Reason is true.

Sol. (c)

25. Four successive members of the first row transition elements are listed below with their atom numbers. Which one of them is expected to have the highest third ionisation enthalpy?

(a) Vanadium (Z = 23) (b) Chromium (Z = 24)

(c) Iron (Z = 26) (d) Manganese (Z = 25)

Sol. (d) $Mn\ [4s^2 3d^5] \xrightarrow[IE_1]{-e^-} Mn^+\ [4s^1 3d^5] \xrightarrow[IE_2]{-e^-} Mn^{2+}\ [3d^5]$ → Removal of 3rd electron from half filled subshell is tough, so, IE_3 will be highest among all.

26. The set representing the correct order of first ionisation potential is:

(a) K > Na > Li (b) Be > Mg > Ca (c) B > C > N (d) Ge > Si > C

Sol. (b)

27. Reason of lanthanoid contraction is:

(a) Negligible screening effect of f orbitals (b) Increasing nuclear charge

(c) Decreasing nuclear charge (d) Decreasing screening effect

Sol. (a)

❑ Ionisation Energy: $ns^1 < ns^2 > np^1 < np^2 < np^3 > np^4 < np^5 < np^6$

→ due to penetration effect

→ due to stable electronic configuration [np^3]

$ns^1 < np^1 < ns^2 < np^2 < np^4 < np^3 < np^5 < np^6$: IE

$2s^1$	$2s^2$	$2p^1$	$2p^2$	$2p^3$	$2p^4$	$2p^5$	$2p^6$
♦ Li	Be	B	C	N	O	F	Ne

Li < B < Be < C < O < N < F < Ne : IE

♦ Na	Mg	Al	Si	P	S	Cl	Ar

Na < Al < Mg < Si < S < P < Cl < Ar : IE

♦ K	Ca	Ga	Ge	As	Se	Br	Kr

K < Ga < Ca < Ge < Se < As < Br < Kr : IE

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Boron Family

B_5 : $[He]\ 2s^2\ 2p^1$

Al_{13} : $[Ne]\ 3s^2\ 3p^1$

Ga_{31} : $[Ar]\ 4s^2\ \boxed{3d^{10}}\ 4p^1$

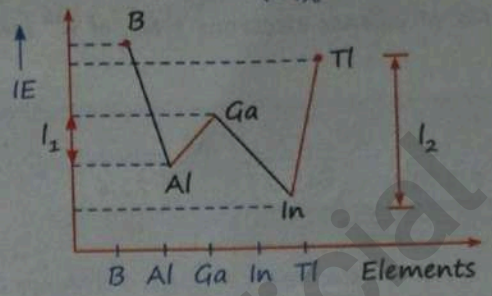
poor shielding of 3d electrons

In_{49} : $[Kr]\ 5s^2\ 4d^{10}\ 5p^1$

Tl_{81} : $[Xe]\ 6s^2\ \boxed{4f^{14}}\ 5d^{10}\ 6p^1$

poor shielding of 4f electrons

- B > Tl > Ga ≥ Al > In : IE
  - $(IE)_{Ga}$ = 5.99 eV and $(IE)_{Al}$ = 5.98 eV



- $(IE_{Ga} - IE_{Al} = I_1) < (IE_{Tl} - IE_{In} = I_2)$
- $\Delta IE_{(Ga, Al)} < \Delta IE_{(Tl, In)}$

s Block

Li, Be
Na, Mg
K, Ca
Rb, Sr
Cs, Ba

Highest I.E. : He
Lowest I.E. : Cs

Due to poor shielding of 10 d electrons in Ga

Due to poor shielding of 14 f electrons in Tl & Pb

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p Block

					He
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po		

Size → 4d series ≈ 5d series

4d	$_{39}Y$	$_{40}Zr$ ≈	Nb ≈	Mo ≈	Tc ≈	Ru ≈	Rh ≈	Pd ≈	Ag ≈	Cd ≈
5d	$_{57}La$	($_{58}Ce \rightarrow {}_{71}Lu$)	$_{72}Hf$	Ta	W	Re	Os	Ir	Pt	Au	Hg

Free from the effect of Lanthanoid contraction

Ionisation Energy → 4d series < 5d series

4d	$_{39}Y$	$_{40}Zr$ ^	Nb ^	Mo ^	Tc ^	Ru ^	Rh ^	Pd ^	Ag ^	Cd ^
5d	$_{57}La$	($_{58}Ce \rightarrow {}_{71}Lu$)	$_{72}Hf$	Ta	W	Re	Os	Ir	Pt	Au	Hg

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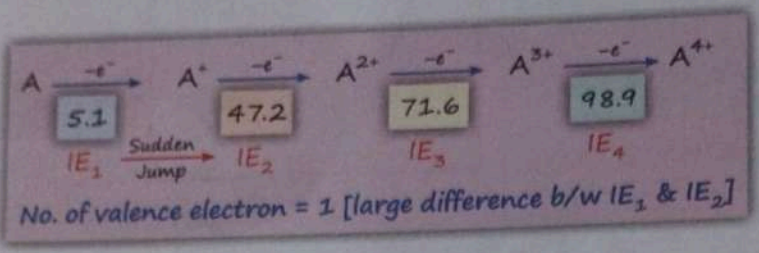
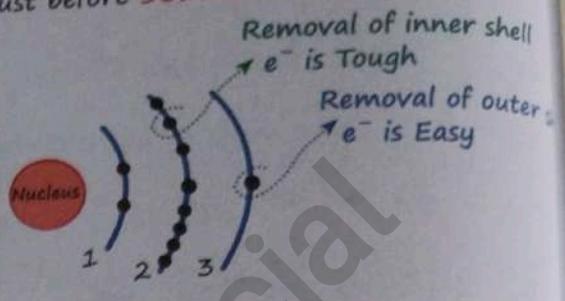
The first ionisation energy of 5d elements is higher than that of 4d elements due to a greater Z_{eff} ac on the outer valence electrons, combined with weak shielding by the 4f electrons.

How to find no. of Valence Shell Electrons?

- Removal of electron from valence shell is easy and from inner shell is tough. So there is a la difference in energies required to remove those electrons.
- No. of valence electrons = No. of n^{th} ionisation energy just before SUDDEN JUMP





Element	IE_1	IE_2	IE_3	IE_4	IE_5	Number of valence electro
X	5.98	18.22	28.44 $\xrightarrow{\text{Sudden Jump}}$	119.4	–	3
Y	11.26	26.38	47.44	77.41 $\xrightarrow{\text{Sudden Jump}}$	392.07	4

- If the difference in ionisation energy $\Delta(IE_2 - IE_1)$ is greater than 16 eV $atom^{-1}$, then the oxidation state is stable.

  For Na, $\Delta(IE_2 - IE_1) \approx 41$ eV $atom^{-1} \Rightarrow Na^+$ is more stable.
- If value of $\Delta(IE_2 - IE_1)$ is less than 11 eV $atom^{-1}$ then the (+2) oxidation state is stable.

  For Mg, value of $\Delta(IE_2 - IE_1) \approx 7$ eV $atom^{-1} \Rightarrow Mg^{2+}$ is more stable.

28. Consider the following ionisation enthalpies of two elements 'A' and 'B'. Which of the follow statements is correct? [8 April, 2017 (Shift-

Element	Ionization enthalpy (kJ/mol)		
	1^{st}	2^{nd}	3^{rd}
A	899	1757	14847
B	737	1450	7731

(a) Both 'A' and 'B' belong to group-1 where 'B' comes below 'A'

(b) Both 'A' and 'B' belong to group-2 where 'A' comes below 'B'

(c) Both 'A' and 'B' belong to group-2 where 'B' comes below 'A'

(d) Both 'A' and 'B' belong to group-1 where 'A' comes below 'B'

Sol. (c) Both 'A' and 'B' belong to group-2. On moving down the group, the ionisation energy decrea Since first ionisation energy of B is lower than that of A, 'B' comes below 'A'.

29. Which of the following represents the correct order of metallic character of the given elements [24 Jan, 2023 (Shift-

(a) Si < Be < Mg < K

(b) Be < Si < Mg < K

(c) K < Mg < Be < Si

(d) Be < Si < K < Mg

Sol. (a)

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30. Outermost electronic configurations of four element A, B, C, D are given below:

(1) $3s^2$ (2) $3s^23p^1$ (3) $3s^23p^3$ (4) $3s^23p^4$

The correct order of first ionization enthalpy for them is : [27 July, 2022 (Shift-2)]

(a) (1) < (2) < (3) < (4) (b) (2) < (1) < (4) < (3)

(c) (2) < (4) < (1) < (3) (d) (2) < (1) < (3) < (4)

Sol. (b) IE → Al < Mg < S < P

(1) $3s^2$ → Mg (2) $3s^23p^1$ → Al (3) $3s^23p^3$ → P (4) $3s^23p^4$ → S

31. Statement-I: The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement-II: The d orbitals in Ga are completely filled. [29 Jan, 2023 (Shift-II)]

(a) Statement-I is incorrect but Statement-II is correct

(b) Both the Statements-I and II are correct

(c) Statement-I is correct but Statement-II is incorrect

(d) Both the Statements-I and II are incorrect

Sol. (b)

32. Match List-I with List-II. [26 Feb, 2021 (Shift-I)]

List-I Electronic configuration of elements		List-II $\Delta_i H$ in kJ mol^{-1}	
A.	$1s^2 2s^2$	I.	801
B.	$1s^2 2s^2 2p^4$	II.	899
C.	$1s^2 2s^2 2p^3$	III.	1314
D.	$1s^2 2s^2 2p^1$	IV.	1402

Choose the most appropriate answer from the options given below:

(a) (A)-(II), (B)-(III), (C)-(IV), (D)-(I) (b) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(c) (A)-(I), (B)-(IV), (C)-(III), (D)-(II) (d) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Sol. (a) N > O > Be > B

33. The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol^{-1}. The number of valence electrons in the element is: [3 Sept, 2020 (Shift-II)]

(a) 4 (b) 2 (c) 5 (d) 3

Sol. (d) Due to large difference in 3rd and 4th ionisation energies, the number of valence electrons is 3.

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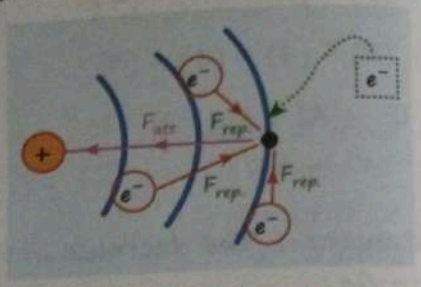
ELECTRON GAIN ENTHALPY

Enthalpy change in addition of an electron in an isolated gaseous atom is known as electron gain enthalpy.

- Electron gain enthalpy provides a measure of the ease with which an atom adds an electron to form anion.
  - $F_{att.} > F_{rep.}$: Release of Energy

    $A \xrightarrow{e^-} A^-$ + Energy
  - $F_{att.} < F_{rep.}$: Absorption of Energy

    $B \xrightarrow[\text{Energy}]{e^-} B^-$



Electron Affinity

Tendency of an isolated gaseous atom to add an electron is known as electron affinity.

- Electron affinity is measured in terms of EGE (ΔH_{eg}).
- More –ve $\Delta_{eg}H \Rightarrow$ more electron affinity
- Less –ve $\Delta_{eg}H \Rightarrow$ less electron affinity

- $H \xrightarrow{+e^-} H^-$ $\quad \Delta_{eg}H = -73$ KJ/mol
- $Li \xrightarrow{+e^-} Li^-$ $\quad \Delta_{eg}H = -60$ KJ/mol

→ Electron affinity : H > Li

- **Formation of multiple ⊖ve anion:** Overall energy change during formation of multiple ⊖ve anion is always ⊕ve.

1. $O \xrightarrow{e^-} O^-$ $\quad \Delta H_1 = -141$ KJ/mol (Exo)

   $O^- \xrightarrow[\text{Energy}]{e^-} O^{2-}$ $\quad \Delta H_2 = +844$ KJ/mol (Endo)

   $O \xrightarrow{2e^-} O^{2-}$ $\quad \Delta H_1 + \Delta H_2 = +703$ KJ/mol (Endo)

2. $N \xrightarrow{3e^-} N^{3-}$ $\quad \Delta H_{eg} =$ ⊕ve (Endo)
3. $P \xrightarrow{3e^-} P^{3-}$ $\quad \Delta H_{eg} =$ ⊕ve (Endo)

Factor Affecting Electron Affinity

1 Z_{eff} — Left → Right : Z_{eff} ↑ : EA ↑

$O_8 < F_9$: EA

2 Size — Top ↓ Bottom → Size ↑ → EA ↓

EA : Cl > Br > I > At

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3 Electronic Configuration

- Generally for stable electronic configuration (half filled and fully filled) $\Rightarrow \Delta H = \oplus ve$.

♦ $N \xrightarrow{e^-} N^-$ $(2s^2\ 2p^3)$ $\Delta_{eg}H = \oplus ve$

♦ $Ne \xrightarrow{e^-} Ne^-$ $(2s^2\ 2p^6)$ $\Delta_{eg}H = \oplus ve$

4 Penetration Effect

Addition of electron in s-orbital is more favorable than p-orbital because s-orbital is near to the nucleus and upcoming electron get more attraction by nucleus.

- EA : K > Ga
  : Rb > In
  : Cs > Tl

Li $(2s^1)$ > B $(2s^2\ 2p^1)$

$Li \xrightarrow{+e^-} Li^-\ (2s^2)$; $B \xrightarrow{+e^-} B^-\ (2s^2\ 2p^2)$

Na $(3s^1)$ > Al $(3s^2\ 3p^1)$

Important Points

- Generally For neutral atom : $\Delta_{eg}H \equiv (-)ve$

$O \xrightarrow{e^-} O^-$ (Exo)

$C \xrightarrow{e^-} C^-$ (Exo)

- But For (Be, Mg, N) (Inert Gas) : $\Delta_{eg}H \equiv (+)ve$

$He \xrightarrow{e^-} He^-$ (Endo)

$Ne \xrightarrow{e^-} Ne^-$ (Endo)

$N \xrightarrow{e^-} N^-$ (Endo)

$Be \xrightarrow{e^-} Be^-$ (Endo)

$Mg \xrightarrow{e^-} Mg^-$ (Endo)

- For all cations : $\Delta_{eg}H \equiv (-)ve$

$C^+ \xrightarrow{e^-} C$ (Exo)

$Na^+ \xrightarrow{e^-} Na$ (Exo)

- For all anions : $\Delta_{eg}H \equiv (+)ve$

$O^- \xrightarrow{e^-} O^{2-}$ (Endo)

$N^- \xrightarrow{e^-} N^{2-}$ (Endo)

↓

Adding an electron to a negatively charged ion is more difficult because it requires energy to overcome the repulsion between the negative charge on atom and electron. This makes the second electron gain enthalpy positive (endothermic).

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- 1st EGE [ΔH_{eg_1}] may be +ve/–ve. 2nd EGE [ΔH_{eg_2}] is always ⊕ ve.

$O \xrightarrow{e^-} O^-$ $\quad \Delta H_{eg_1} = \ominus$ ve	$O^- \xrightarrow{e^-} O^{2-}$ $\quad \Delta H_{eg_2} = \oplus$ ve
$N \xrightarrow{e^-} N^-$ $\quad \Delta H_{eg_1} = \oplus$ ve	$N^- \xrightarrow{e^-} N^{2-}$ $\quad \Delta H_{eg_2} = \oplus$ ve

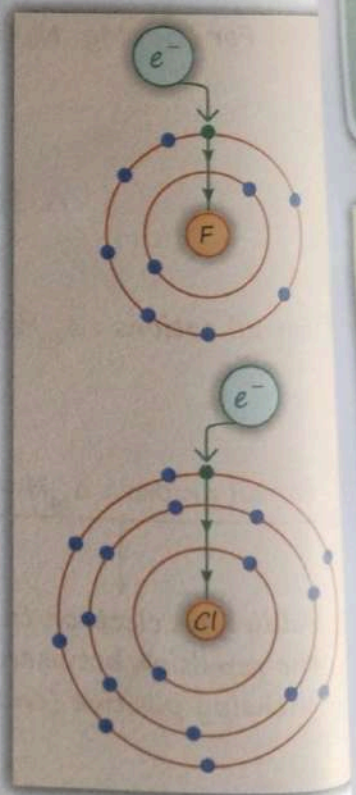
- EA : Ne < Be < N < B < Li < C < O < F
  - Ne, Be, N : $\Delta_{eg}H = \oplus$ve
  - B, Li, C, O, F : $\Delta_{eg}H = \ominus$ve
- EA : Ar < Mg < Al < Na < P < Si < S < Cl
  - Ar, Mg : $\Delta_{eg}H = \oplus$ve
  - Al, Na, P, Si, S, Cl : $\Delta_{eg}H = \ominus$ve
- EA : alkali metals > alkaline earth metals

alkali metals	alkaline earth metals
• $ns^1 \xrightarrow{+e} ns^2$	• $ns^2 \xrightarrow{+e} ns^2 np^1$
• Fully-filled electronic configuration is achieved	• Fully-filled electronic configuration is disturbed
• $\Delta_{eg}H$ is (–)ve	• $\Delta_{eg}H$ is (+)ve

Comparison b/w Electron Affinity of F & Cl

- $F \xrightarrow{e^-} F^-$; $2s^2\,2p^5 \rightarrow 2s^2\,2p^6$; $\Delta_{eg}H = -328$ KJ/mol
- $Cl \xrightarrow{e^-} Cl^-$; $3s^2\,3p^5 \rightarrow 3s^2\,3p^6$; $\Delta_{eg}H = -349$ KJ/mol

→ EA ≡ Cl > F



- Adding an electron to the 2p-orbital (F) leads to greater repulsion than adding an electron to the larger 3p-orbital (Cl) . Hence electron gain enthalpy of the Cl is more negative than F.

Note

- B C N O F (Period 2)
- Al Si P S Cl (Period 3)

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Group 1	$\Delta_{eg}H$	Group 16	$\Delta_{eg}H$	Group 17	$\Delta_{eg}H$	Group 0	$\Delta_{eg}H$
H	−73					He	+48
Li	−60	O	−141	F	−328	Ne	+116
Na	−53	S	−200	Cl	−349	Ar	+96
K	−48	Se	−195	Br	−325	Kr	+96
Rb	−47	Te	−190	I	−295	Xe	+77
Cs	−46	Po	−174	At	−270	Rn	+68

- Order of (−)ve electron gain enthapy : H > Li > Na > K > Rb > Cs (K, Rb, Cs: approx. same value)
- Order of (−)ve electron gain enthalpy: S > Se > Te > Po > O
  Cl > F > Br > I > At
- Order of (+)ve electron gain enthalpy: Ne > Ar = Kr > Xe > Rn > He

Noble gases have (+)ve electron gain enthalpy as they have completely filled orbitals of valence shell.

Note

- Cl has most (−)ve ΔH_{eg} and Ne has most (+)ve ΔH_{eg} among all elements.
- Oxygen has min. EGE in family. EA : S > Se > Te > Po > O
- Halogen family has max. EGE in periodic table.
- Any element from G-17 has higher EGE wrt other element in periodic table. EA : I > S

MANOJ NATH

34. Find the order of electron affinity in following:

(a) Cl, Br, S, O (b) O, S, Se, I (c) Li, B (d) F, Cl, O, N (e) P, S, Cl, F

Sol. (a) Cl > Br > S > O : ΔH_{eg} (more negative) or electron affinity

(b) I > S > Se > O

(c) Li > B

(d) Cl > F > O > N

(e) Cl > F > S > P

35. In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical? [28 July, 2022 (Shift-1)]

1. Rb and Cs 2. Na and K 3. Ar and Kr 4. I and At

Choose the correct answer from the options given below :

(a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 3 and 4 only

Sol. (c)

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36. Inert gases have positive electron gain enthalpy. Its correct order is : [24 Jan, 2023 (Shift-1)]

(a) Xe < Kr < Ne < He (b) He < Ne < Kr < Xe

(c) He < Xe < Kr < Ne (d) He < Kr < Xe < Ne

Sol. (c)

37. The first electron gain enthalpy (ΔH_{eg}) of oxygen is −141 kJ/mol, its second electron gain enthalpy is :

(a) Almost the same as that of the first (b) a more negative value than the first

(c) negative, but less negative than the first (d) a positive value

Sol. (d)

38. The formation of the oxide in O^{2-} requires first an exothermic and then an endothermic step as shown below :

$O \longrightarrow O^-$ $\quad \Delta H_1 = -141$ kJ / mol

$O^- \longrightarrow O^{2-}$ $\quad \Delta H_2 = +844$ kJ / mol

This is because:

(a) Oxygen is more electronegative

(b) Oxygen has high electron affinity.

(c) O^- ion will tend to resist the addition of another electron.

(d) O^- ion has comparatively larger size than oxygen atom.

Sol. (c)

39. For electron gain enthalpies of the elements denoted as $\Delta_{eg}H$, the incorrect option is: [1 Feb, 2023 (Shift-I)]

(a) $\Delta_{eg}H(Cl) < \Delta_{eg}H(F)$ (b) $\Delta_{eg}H(Se) < \Delta_{eg}H(S)$

(c) $\Delta_{eg}H(I) < \Delta_{eg}H(At)$ (d) $\Delta_{eg}H(Te) < \Delta_{eg}H(Po)$

Sol. (b) $\Delta_{eg}H$ (S) is more (−)ve value than $\Delta_{eg}H$ (Se).

40. The process that is NOT endothermic in nature is: [4 Sept, 2020 (Shift-II)]

(a) $H_{(g)} + e^- \rightarrow H^-_{(g)}$ (b) $O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)}$

(c) $Na_{(g)} \rightarrow Na^+_{(g)} + e^-$ (d) $Ar_{(g)} + e^- \rightarrow Ar^-_{(g)}$

Sol. (a)

[IIT JEE 2002]

41. Identify the least stable ion amongst the following:

(a) Li^- (b) Be^- (c) B^- (d) C^-

Sol. (b) Be^- ion is formed by absorption of energy, so they have high energy → means least stable

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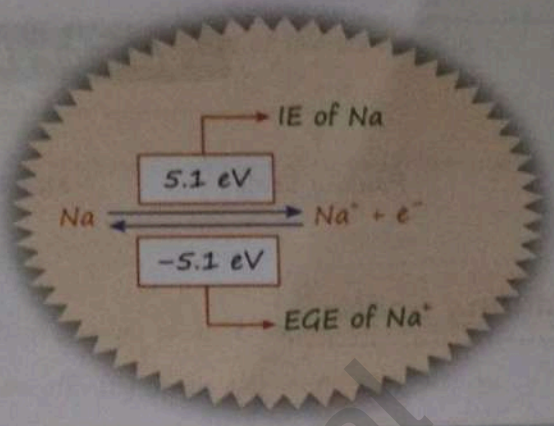
Relation Between I.E. & E.G.E.

- $A \underset{EGE}{\overset{IE}{\rightleftharpoons}} A^+ + e^-$



IE of A ≡ | EGE of $A^{\oplus}$ |

Order of IE : F > Cl

Order of EA : F^- > Cl^-

- $A^- \underset{\Delta H_{eg} \text{ of } A}{\overset{\text{IE of } A^-}{\rightleftharpoons}} A + e^-$

IE of A^- = $|\Delta_{eg}H$ of A|

- $F^- \underset{\text{EA of F}}{\overset{\text{IE of } F^-}{\rightleftharpoons}} F + e^-$
- $Cl^- \underset{\text{EA of Cl}}{\overset{\text{IE of } Cl^-}{\rightleftharpoons}} Cl + e^-$

EA = Cl > F

IE = Cl^- > F^-

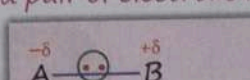
Note

If energy is released when an electron is added to an atom, the electron affinity is taken as positive, contrary to thermodynamic convention. If energy has to be supplied to add an electron to an atom, then the electron affinity of the atom is assigned a negative sign. However, electron affinity is defined as absolute zero and, therefore at any other temperature (T) heat capacities of the reactants and the products have to be taken into account in $\Delta_{eg}H = -A_e - 5/2\ RT$.

ELECTRONEGATIVITY

- Tendency of an atom to attract shared pair of electrons in a covalent bond is known as electronegativity.

$-\delta$: partial $\ominus$ve charge

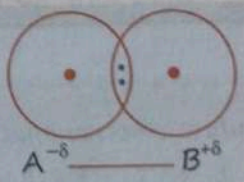


Z_{eff} : A > B ⇒ EN : A > B

1. Z_{eff} ↑ : EN ↑ $(Z_{eff})_A > (Z_{eff})_B$, $(EN)_A > (EN)_B$

Left → Right ⇒ EN ↑

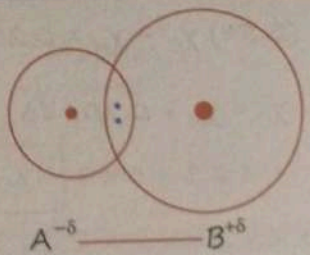
B < C < N < O < F : EN



2. Size ↑ : EN ↓ $(Size)_A < (Size)_B$

Top ↓ Bottom → Size ↑, → EN ↓

F > Cl > Br > I



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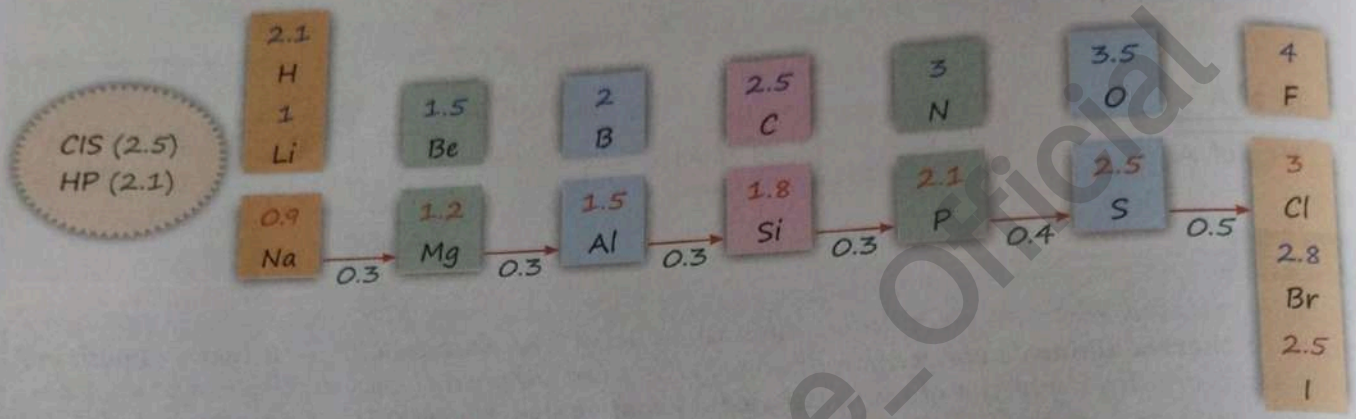
Scale for Electronegativity

- Pauling Scale
- Mulliken Scale
- Allred Rochow Scale

Pauling Scale



- Most electronegative element : F
- Second most electronegative element : O
- It is based on an empirical relation between energy of a bond and electronegativity.

$$\chi_A - \chi_B = 0.208\sqrt{\Delta} \text{ kcal mol}^{-1} \qquad \Delta = E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}$$

Δ = Extra bond Energy

= Actual bond Energy (A – B) – Calculated bond energy for 100% covalent bond (A – B)

$$\chi_A - \chi_B = 0.208\sqrt{E_{A-B} - \sqrt{E_{A-A} \cdot E_{B-B}}}$$

$$\chi_A - \chi_B = 0.101\sqrt{\Delta} \text{ kJ mol}^{-1}$$

42. Calculate electronegativity of carbon on Pauling scale given that E_{H-H}, E_{C-C} and E_{C-H} are 104.2, 83.1, 98.8 kcal / mol respectively, also electronegativity of Hydrogen atom is 2.1.

Sol. (2.59) $\chi_C - \chi_H = 0.208\sqrt{\Delta}$, $\left(\Delta = E_{C-H} - \sqrt{E_{C-C} \cdot E_{H-H}} = 98.8 - \sqrt{104.2 \times 83.1}\right)$

$\chi_C = \chi_H + 0.208\sqrt{\Delta}$

$= 2.1 + 0.49$ $\boxed{\chi_C = 2.59}$

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43. The correct option with respect to the Pauling electronegativity values of the elements is :
(a) Te > Se (b) Ga < Ge (c) Si < Al (d) P > S

Sol. (b)

Mulliken Scale

$$\chi_M = \frac{\text{I.E.} + \text{E.A. (eV/atom)}}{2}$$

$$\chi_M = 2.8\,\chi_P$$

I.E. ⟶ Ionisation energy of an element in eV/atom.
E.A ⟶ Electron affinity of an element in eV/atom.
χ_M ⟶ Electronegativity of an element on Mulliken scale
χ_P ⟶ Electronegativity of an element on Pauling scale

Allred Rochow Scale

Electronegativity as the electrostatic force of attraction exerted by the nucleus of an atom on the valence electrons.

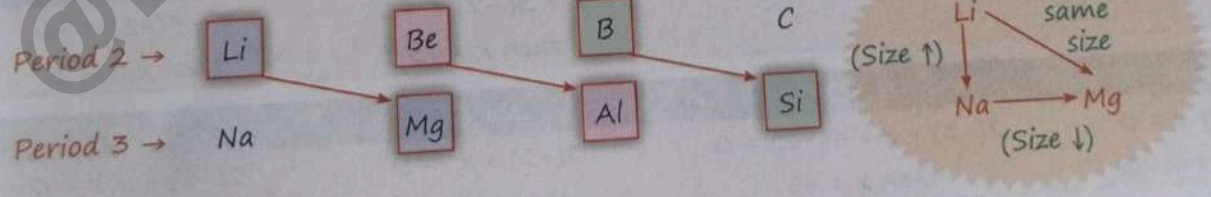
$$\chi_{AR} = \left(\frac{0.359 \times Z^*}{r^2}\right) + 0.744$$

Where Z^* = Effective nuclear charge
r = Covalent radius of atom in Å.

DIAGONAL RELATIONSHIP

- Some elements of certain groups of second period resemble much in properties with the elements of third period, which are diagonally related in properties.
- Diagonal relationship arises because of similar size of atom and ions.

For Example: Atomic size : Li ≈ Mg



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44. The set of elements that differ in mutual relationship from those of the other sets is : [17 March, 2021 (Shift-2)]

(a) Li – Mg (b) B – Si (c) Be – Al (d) Li – Na

Sol. (d)

45. Match List-I with List-II

List-I (Elements)		List-II (Properties in their respective groups)	
A.	Cl, S	I.	Elements with highest electronegativity
B.	Ge, As	II.	Elements with largest atomic size
C.	Fr, Ra	III.	Elements which show properties of both metal and non metal
D.	F, O	IV.	Elements with highest negative electron gain enthalpy

Choose the correct answer from the options given below: [08 April, 2024 (Shift-

(a) A-II, B-III, C-IV, D-I (b) A-III, B-II, C-I, D-IV

(c) A-IV, B-III, C-II, D-I (d) A-II, B-I, C-IV, D-III

Sol. (c)

OXIDATION STATE

The oxidation state of an atom in a molecule or in an ion is the charge the atom would have if the electr in each bond were located on the more electronegative atom.

Rule :

1.	F : −1	(Li → Cs) Group 1 : +1	(Be → Ra) Group 2: +2	Zn : +2 Al, Sc, Ga : +3
2.	H : +1			
3.	O : −2			
4.	N : −3			
5.	Cl, Br, I : −1			

- Elements in rule −1 ALWAYS show fix oxidation state in all molecules.
- Elements in rules 2, 3, 4, and 5 generally show the listed oxidation state but can also exhibit variable oxidation states in different compounds.
- **Priority of rule: 1 > 2 > 3 > 4 > 5**

Na^+	N^{3-}	NO_3^-	NH_4^+
$x = +1$	$x = -3$	$x + 3(-2) = -1$ then $x = +5$	$x + 4(+1) = +1$ then $x = -3$

LiH	H_2SO_4	Na_2O	Na_2O_2
$(+1) + x = 0$ then $x = -1$	$2(+1) + x + 4(-2) = 0$ then $x = +6$	$2(+1) + x = 0$ then $x = -2$	$2(+1) + 2x = 0$ then $x = -1$

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OXIDES & HYDROXIDES

Basic Oxides: $Na_2O + 2HCl \longrightarrow 2NaCl + H_2O$
Basic Oxide — Acid — Salt

Acidic Oxides: $CO_2 + 2NaOH \longrightarrow Na_2CO_3 + H_2O$
Acidic Oxide — Base — Salt

Amphoteric Oxides:

$ZnO \xrightarrow{2HCl} ZnCl_2 + H_2O$

$ZnO \xrightarrow{2NaOH} Zn(OH)_2 \xrightarrow[NaOH]{Excess} Na_2[Zn(OH)_4]$

Neutral Oxides: CO / N_2O / NO
Carbon monoxide / Nitrous oxide / Nitric oxide

Amphoteric Oxides

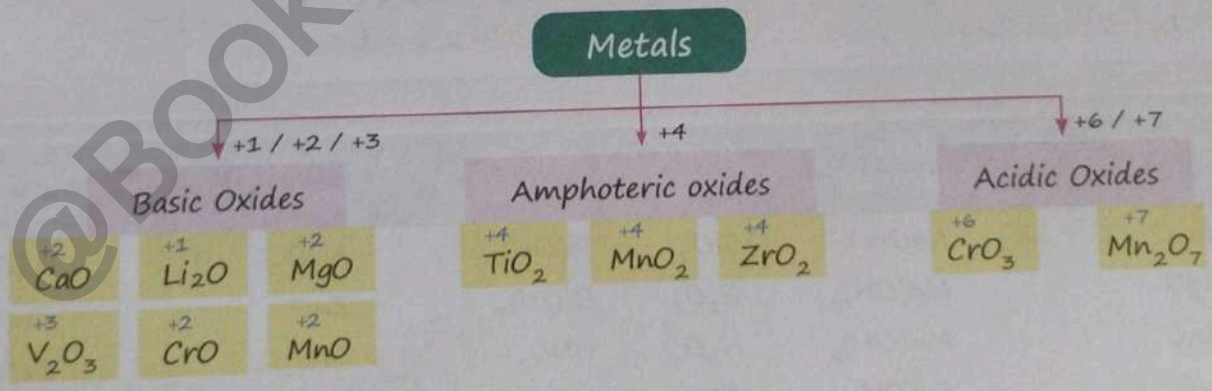
- As(+3) Sb(+3) Cr(+3) V(+5) Te(+4)
  Aas Sab Car Vento Tera
- Zn Sn Be Al Pb Ga
  Zaan Suno Be Aaliya Pub Gayi

Amphoteric but predominantly acidic

Example →		$^{+3}As_2O_3$		$^{+3}Sb_2O_3$	$^{+3}Cr_2O_3$	$^{+5}V_2O_5$	$^{+4}TeO_2$	
ZnO	BeO	Al_2O_3	Ga_2O_3	SnO	SnO_2	PbO	PbO_2	Pb_3O_4	Pb_3O_2

Basic Oxides

- Most common oxidation states for metals are +1, +2 and +3.
- Generally oxide of metals are basic in nature but some oxides are amphoteric and acidic also.



Acidic Oxides

- Generally non-metal oxides are acidic except some neutral oxide [CO, NO, N_2O].
- Metalloid oxides are acidic except those which are considered already in amphoteric.

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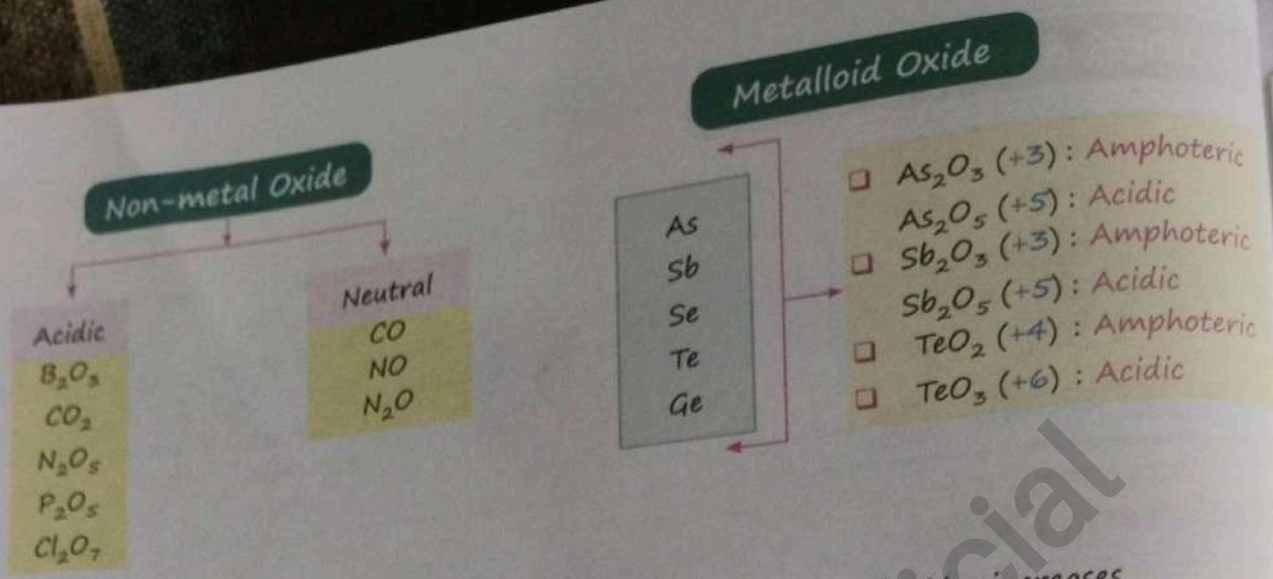

Acidic Nature of Oxides

- In an oxide, as oxidation state of element increases, acidic nature of oxides increases.
  - $\overset{+1}{Li_2O} < \overset{+2}{BeO} < \overset{+3}{B_2O_3} < \overset{+4}{CO_2} < \overset{+5}{N_2O_5}$
  - $\overset{+1}{Na_2O} < \overset{+2}{MgO} < \overset{+3}{Al_2O_3} < \overset{+4}{SiO_2} < \overset{+5}{P_2O_5} < \overset{+6}{SO_3} < \overset{+7}{Cl_2O_7}$
  - $\overset{+1}{N_2O} < \overset{+2}{NO} < \overset{+3}{N_2O_3} < \overset{+4}{NO_2} < \overset{+5}{N_2O_5}$
  - $\overset{+2}{CO} < \overset{+4}{CO_2}$
  - $\overset{+4}{SO_2} < \overset{+6}{SO_3}$

Note

M—O—H like compound will act as acid or as base can be predicted very easily.

a. Acts as a base when

$M-O-H \rightleftharpoons M^+ + OH^-$

$|\chi_O - \chi_M| > |\chi_O - \chi_H|$

- Cs—O—H is a base because

$\chi_O - \chi_{Cs} = 3.5 - 0.7 = 2.8$

$\chi_O - \chi_H = 3.5 - 2.1 = 1.4$

b. Acts as an acid when

$M-O-H \rightleftharpoons MO^- + H^+$

$|\chi_O - \chi_H| > |\chi_O - \chi_M|$

- Cl—O—H is an acid because

$\chi_O - \chi_{Cl} = 3.5 - 3.1 = 0.34$

$\chi_O - \chi_H = 3.5 - 2.1 = 1.4$

Basic Oxide	Basic hydroxide	Acidic oxide	Acidic hydroxide	Amphoteric oxide	Amphoteric hydroxide
Na_2O	NaOH	SiO_2	$Si(OH)_4$	Al_2O_3	$Al(OH)_3$
MgO	$Mg(OH)_2$	B_2O_3	$B(OH)_3$	ZnO	$Zn(OH)_2$
MnO	$Mn(OH)_2$	N_2O_5	HNO_3	BeO	$Be(OH)_2$
CuO	$Cu(OH)_2$	CO_2	H_2CO_3	PbO	$Pb(OH)_2$

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46. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is: [2 Sept, 2020 (Shift-II)]

(a) X < Z < Y (b) Z < Y < X (c) X < Y < Z (d) Y < X < Z

Sol. (c)

GENESIS OF PERIODIC CLASSIFICATION

1. Dobereiner's Traids
2. Newland's Law of Octaves
3. Lothar Meyer Classification
4. Mendeleev's Periodic Table
5. Modern Periodic Table

Dobereiner's Traids

3 elements → A, B, C

$$M_B = \frac{M_A + M_C}{2}$$

M_A = atomic mass of A
M_B = atomic mass of B
M_C = atomic mass of C

Dobereiner arranged certain elements with similar properties in groups of three (Triad) in such a way that the atomic mass of the middle element was nearly the same the average atomic masses of the first and the third elements.

Elements	Atomic weight	Element	Atomic weight	Element	Atomic weight
Li	7	Ca	40	Cl	35.5
Na	23	Sr	88	Br	80
K	39	Ba	137	I	127

Be	8	S	32	P	31	H	1	Sc : x
Mg →	24	Se →	79	As →	75	F →	19	Y → y = (x + z)/2
Ca	40	Te	127	Sb	120	Cl	35.5	La : z

Newland's Law of Octaves

- When elements are arranged in order of increasing atomic masses, every 8th element has properties similar to the first. Newlands called it law of octaves because similar relationship exists in the musical notes also.

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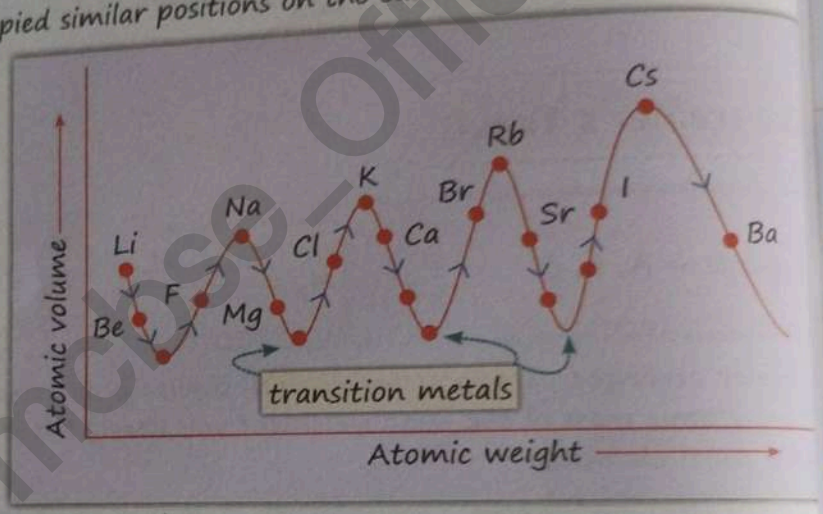
sa	re	ga	ma	pa	da	ni
H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe
Co and Ni	Cu	Zn	Y	In	As	Se
Br	Rb	Sr	Ce and La	Zr	–	–

Lothar Meyer Classification

- He calculated the atomic volumes by dividing atomic masses with their densities in solid states.
- He plotted a graph between atomic masses against their respective atomic volumes for a number of elements.
- Elements with similar properties occupied similar positions on the curve.
- Atomic volumes (a physical property) of the elements are the periodic functions of their atomic masses.
- Alkali metals (Li, Na, K, Rb, Cs) having larger atomic volumes occupied the crests.
- Alkaline earth metals (Be, Mg, Ca, Sr, Ba) occupied the positions at about the mid points of the descending portions of the curve.
- The halogens (F, Cl, Br, I) occupied the ascending portions of the curve before the inert gases.
- Transitions elements occupied the troughs.



Mendeleev's Periodic Table

Characteristic of Mendeleev's Periodic Table :

(i) It is based on atomic weight.

(ii) 63 elements were known, noble gases were not discovered.

(iii) He was the first scientist to classify the elements in a systematic manner (in horizontal rows and in vertical columns).

(iv) Horizontal rows are called periods and there were 7 periods in Mendeleev's Periodic table.

(v) Vertical columns are called groups and there were 8 groups in Mendeleev's Periodic table.

(vi) Each group upto VIIth is divided into A & B subgroups. 'A' sub group element are called normal element and 'B' sub groups elements are called transition elements.

(vii) The VIIIth group was consist of 9 elements in three rows (Transition metals group).

(viii) The elements belonging to same group exhibit similar properties.

Fe	Co	Ni
Ru	Rh	Pd
Os	Ir	Pt

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Series	Groups of Elements								
	0	I	II	III	IV	V	VI	VII	VIII
1	–	H 1.008	–	–					
2	He 4.0	Li 7.03	Be 9.1	B 11.0	C 12.0	N 14.04	O 16.00	F 19.0	
3	Ne 19.9	Na 23.5	Mg 24.3	Al 27.0	Si 28.4	P 31.0	S 32.06	Cl 35.45	
4	Ar 38	K 39.1	Ca 40.1	Sc 44.1	Ti 48.1	V 51.4	Cr 52.1	Mn 55.0	Fe Co Ni (Cu) 55.9 59 59
5		Cu 63.6	Zn 65.4	Ga 70.0	Ge 72.3	As 75	Se 79	Br 79.95	
6	Kr 81.8	Rb 85.4	Sr 87.6	Y 89.0	Zr 90.6	Nb 94.0	Mo 96.0	–	Ru Rh Pd (Ag) 101.7 103.0 106.5
7		Ag 107.9	Cd 112.4	In 114.0	Sn 119.0	Sb 120.0	Te 127.6	I 126.9	
8	Xe 128	Cs 132.9	Ba 137.4	La 139	Ce 140	–	–	–	
9		–	–	–	–	–	–	–	
10	–	–	–	Yb 173	–	Ta 183	W 184	–	Os Ir Pt (Au) 191 193 194.9
11		Au 197.2	Hg 200.0	Tl 204.1	Pb 206.9	Bi 208	–	–	
12	–	–	Ra 224	–	Th 232	–	U 239		
	R	R_2O	RO	R_2O_3	Higher Saline Oxides: RO_2 R_2O_5 RO_3 R_2O_7 Higher Gaseous Hydrogen Compounds: RH_4 RH_3 RH_2 RH				RO_4

Merits of Mendeleev Periodic Table :

- It has simplified and systematized the study of elements and their compounds.
- It has helped in predicting the discovery of new elements on the basis of the blank spaces given in its periodic table.
- Mendeleev predicted the properties of those missing elements from the known properties of the other elements in the same group.

  Eka-aluminium – Ga
  Eka-silicon – Ge

- Later, it was found that properties predicted by Mendeleev for these elements and those found experimentally were almost similar.

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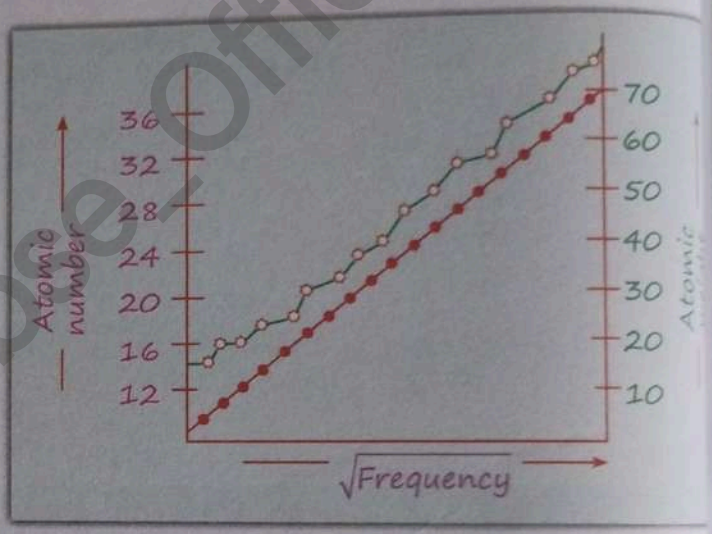
Demerits in Mendeleev's Periodic Table :

- Position of hydrogen is uncertain. It has been placed in IA and VIIA groups because of its resemblan with both the groups.
- No separate positions were given to isotopes.
- Anomalous positions of lanthanides and actinides in periodic table.
- Order of increasing atomic weights is not strictly followed.
  Te (127.6) is placed before I (126.9)
- Similar elements were placed in different groups (Cu in IB and Hg in IIB) and similarly the eleme with different properties were placed in same groups (alkali metals in IA and coinage metals in I

Modern Periodic Table

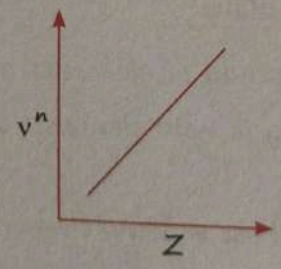
- Henry Moseley observed regularities in the characteristic X-ray spectra of the elements. A plot of $\sqrt{\nu}$ (where ν is frequency of X-rays emitted) against atomic number (Z) gave a straight line and not the plot of $\sqrt{\nu}$ vs atomic mass.



- $\sqrt{\nu} \propto Z$

$$\sqrt{\nu} = a(Z - b)$$ (a, b : constant)

47. It is observed that characteristic X-ray spectra of elements show regularity. When frequency to t power 'n' i.e. ν^n of X-rays emitted is plotted against atomic number 'Z', following graph is obtain [24 Jan, 2023 (Shift-



The value of 'n' is

(a) 1 (b) 2 (c) 1/2 (d) 3

Sol. (c)

48. Henry Moseley studied characteristic X-ray spectra of elements. The graph which represents h observation correctly is:

(Given: ν = frequency of X-ray emitted; Z = atomic number) [8 April, 2023 (Shift-I

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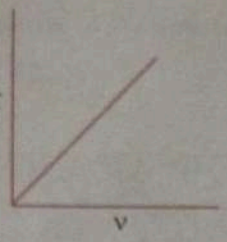
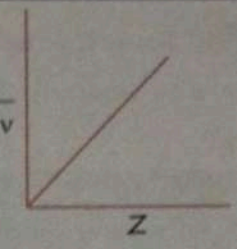
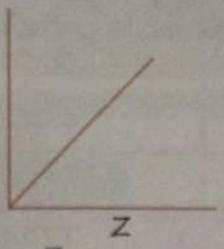
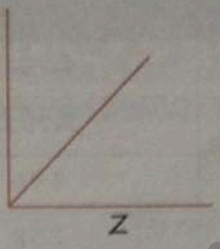
(a) 

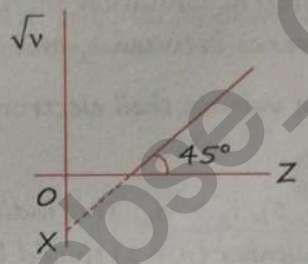
(b) 

(c) 

(d) 

Sol. (b) $\sqrt{\nu} \propto Z$

49. In the graph between $\sqrt{\nu}$ and Z for the Moseley's equation, the intercept OX is −1 on $\sqrt{\nu}$ axis. What will be in Frequency when atomic number (Z) is 51?



(a) 50 Hz (b) 100 Hz (c) 2500 Hz (d) None

Sol. (c)

$\sqrt{\nu} = a(z-b)$ $\sqrt{\nu} = az - ab$ $y = mx + c$	Slope (m) = a = tan 45° = 1 $-ab = -1$ $-1.b = -1$ then $b = 1$	$a = 1$ & $b = 1$ $\sqrt{\nu} = a(z-b)$ $\sqrt{\nu} = 1(z-1)$ $= 51 - 1 = 50$	$\nu = (50)^2 = 2500$ Hz

50. The increasing order of atomic radii of the following Group 13 elements is: (JEE Adv. 2016)

(a) Al < Ga < In < Tl (b) Ga < Al < In < Tl

(c) Al < In < Ga < Tl (d) Al < Ga < Tl < In

Sol. (b)

51. The option(s) with only amphoteric oxides is (are) (JEE Adv. 2020)

(a) NO, B_2O_3, PbO, SnO_2 (b) Cr_2O_3, CrO, SnO, PbO

(c) Cr_2O_3, BeO, SnO, SnO_2 (d) ZnO, Al_2O_3, PbO, PbO_2

Sol. (c) and (d)

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52. The 1st, 2nd and the 3rd ionisation enthalpies I_1, I_2 and I_3, of four atoms with atomic numbers n, n + 1, n + 2 and n + 3, where n < 10, are tabulated below. (JEE Adv. 2020)

What is the value of n?

Atomic number	Ionisation Enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1681	3374	6050
n + 1	2081	3952	6122
n + 2	496	4562	6910
n + 3	738	1451	7733

Sol. [9]

By observing the values of different ionisation energies for atomic number (n + 2), it is observed that there is very large difference between I_2 and I_1.

This indicates that number of valence shell electrons is 1 and atomic number (n + 2) should be an alkali metal.

Also for atomic number (n + 3), $I_3 >> I_2$. This indicates that it will be an alkaline earth metal which suggests that atomic number (n + 1) should be a noble gas and atomic number (n) should belong to halogen family. Since, n < 10; hence, n = 9 (F atom)

"Chalo, NEET ke PYQs practice karte hain, QR code scan karo aur apni preparation ko boost karo!"

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Syllabus

2 Chemical Bonding and Molecular Structure

(JEE Advanced)

Syllabus Orbital overlap and covalent bond; Hybridisation involving s, p and d orbitals only; Molecular orbital energy diagrams for homonuclear diatomic species (up to Ne_2); Hydrogen bond; Polarity in molecules, dipole moment; VSEPR model and shapes of molecules (linear, angular, triangular, square planar, pyramidal, square pyramidal, trigonal bipyramidal, tetrahedral and octahedral).

WHAT IS BOND?

- Bond ≡ Force of attraction
- The attractive force which holds various constituents (atoms, ions, etc.) together in different chemical species is called a chemical bond.
- Bond Formation → Exothermic Process (energy release)

Bond

- Strong Bond (≈ 200 kJ/mole) (Inter-atomic)
  - (a) Covalent Bond
  - (b) Coordinate Bond
  - (c) Ionic Bond
  - (d) Metallic Bond
- Weak Bond (4-20 kJ/mole) (Inter-molecular)
  - (a) Hydrogen Bond
  - (b) vander Waal force of attraction

Reason for Bond Formation

- Tendency to gain stability
- Tendency to acquire minimum energy
- Tendency to attain Inert gas configuration

VALENCE ELECTRON & LEWIS SYMBOL

Lewis Symbol → No of dots = no of valence electrons

Group No.	1	2	13	14	15	16	17	18
Valence Electron	1	2	3	4	5	6	7	8
Lewis Symbol	·Li	·Be·	·B·	·C·	:N·	:O:	:F:	:Ne:

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Lewis Symbol of Cl → :Cl:
Lewis Symbol of S → :S:
Lewis Symbol of P → :P:
Lewis Symbol of Xe → :Xe:

OCTET RULE

Atoms can combine either by transfer of electron from one atom to another atom or by sharing electrons in order to have an octet in their valence shell.

Na· ·Cl: ⇒ Na⊕ :Cl:⊖ (Ionic bond)

Na : $1s^2\ 2s^2\ 2p^6\ 3s^1$

Na^+ : $1s^2$ $\boxed{2s^2\ 2p^6}$ ⇒ Set of 8 electrons [Octet]

Cl : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

Cl^- : $1s^2\ 2s^2\ 2p^6$ $\boxed{3s^2\ 3p^6}$ ⇒ Set of 8 electrons [Octet]

Ionic Bond : The bond formed, as a result of the electrostatic attraction between the positive and negative ions was termed as electrovalent bond.

:F—F: (Covalent bond)

F : $1s^2\ 2s^2\ 2p^5$

After bonding F attain 8 electrons in its valence shell.

Covalent Bond : When two atoms share an electron pair they are said to be joined by a single covalent bond.

1. Non-bonding Electrons [NBE]
2. Lone Pair [LP]
3. Bond Pair [BP]

H_2 H—H

1. NBE = Zero
2. LP = Zero
3. BP = 1

O_2 :O::O:

1. NBE = 8
2. LP = 4
3. BP = 2

N_2 ⊙N≡N⊙

1. LP = 2
2. BP = 3
3. NBE = 4

Selection of Central Atom in a Molecule

Central Atom

- Less in number
- Least electronegative
- Tendency to form maximum number of bonds

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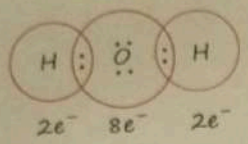
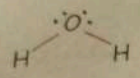
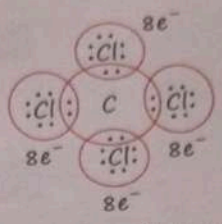
H_2O



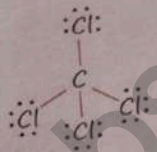
H atoms attain a duplet of electrons and O attain the octet



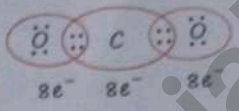
CCl_4



Each of the four Cl atoms along with the C atom attains octet of electrons

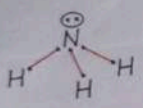


CO_2

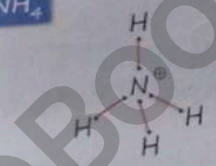


If two atoms share two pairs of electrons, the covalent bond between them is called a double bond.

NH_3

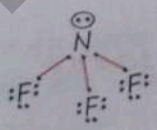


NH_4^+

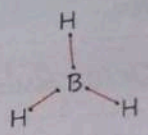


We remove $1e^-$ from nitrogen atom to get N^+, So N^+ has $4e^-$ in valence shell.

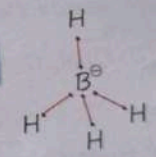
NF_3



BH_3

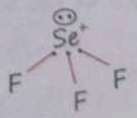


BH_4^-



We add $1e^-$ to boron atom to get B^-.

SeF_3^+



Note

F & H can never act as central atom.

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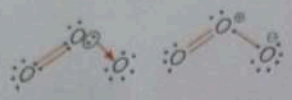
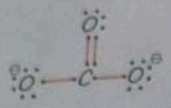
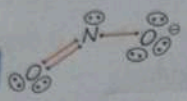
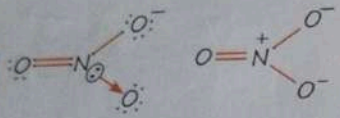
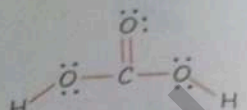
O_3 

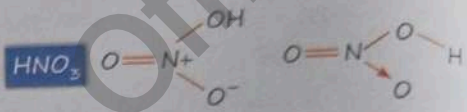
CO_3^{2-} 

NO_2^- 

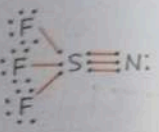
NO_3^- 

H_2CO_3 

HNO_2

HNO_3 

I_3^-

SNF_3 

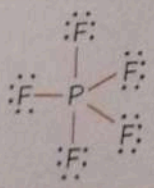
Limitations of Octet Rule

1. It applies mainly to 2nd period elements.

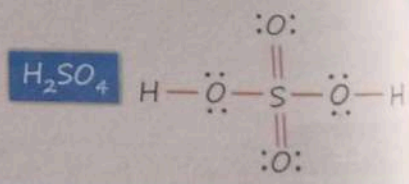
- 2nd Period elements [B, C, N, O, F] → s + p subshell in valence shell
  → maximum no. of electrons = 2 + 6 = 8

2. Expanded or Super or Hypervalent Octet: Elements in and beyond the 3rd period of periodic t… have d orbitals for bonding.

- 3rd Period elements [Al, Si, P, S, Cl] → s + p + d subshell in valence shell
  → maximum no. of electrons = 2 + 6 + 10 = 18

PF_5 

SF_6

H_2SO_4 

- In a number of compounds of these elements there are more than eight valence electrons around central atom. This is termed as the expanded octet.

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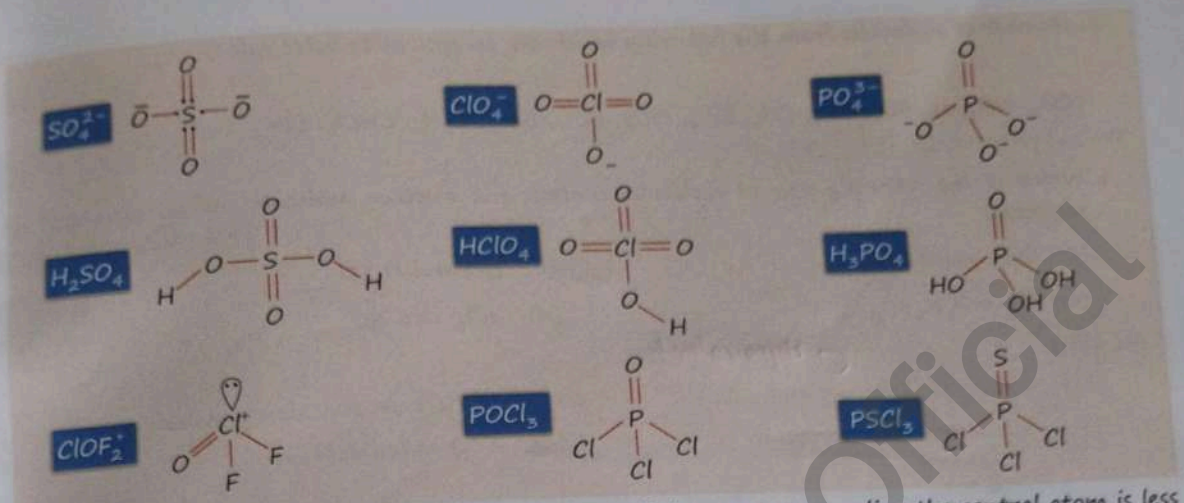
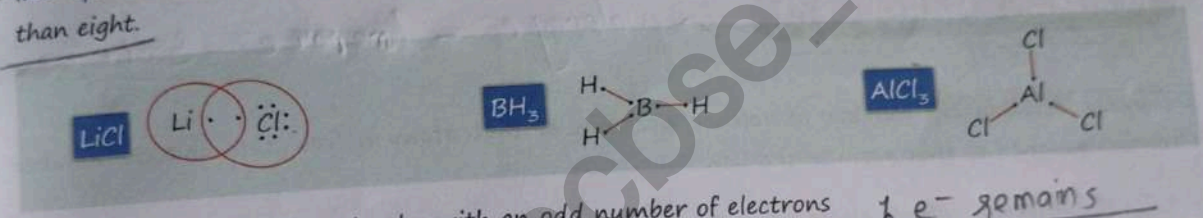
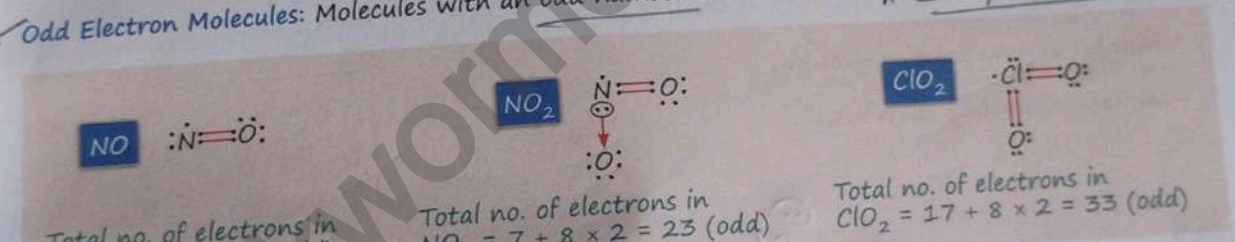


3. Incomplete Octet: In some compounds, the number of electrons surrounding the central atom is less than eight.



4. Odd Electron Molecules: Molecules with an odd number of electrons 1 e⁻ remains



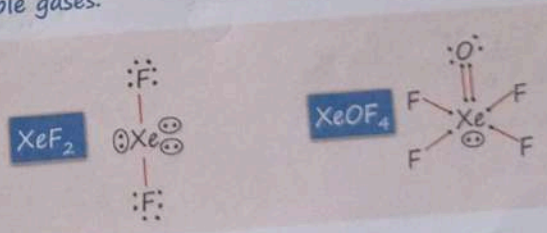
Total no. of electrons in NO = 7 + 8 = 15 (odd)

Total no. of electrons in NO_2 = 7 + 8 × 2 = 23 (odd)

Total no. of electrons in ClO_2 = 17 + 8 × 2 = 33 (odd)

5. Octet rule is based upon the chemical inertness of noble gases.

Some noble gases (for example xenon and krypton) also combine with oxygen and fluorine to form a number of compounds like XeF_2, KrF_2, $XeOF_4$ etc.



1. The number of molecules or ions from the following, which do not have odd number of electrons are _____. [29 Jan, 2023 (Shift-I)]

(A) NO_2 (B) ICl_4^- (C) BrF_3 (D) ClO_2 (E) NO_2^+ (F) NO

Sol. [3] ICl_4^-, BrF_3 and NO_2^+ do not have odd number of e^-

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2. Number of molecules from the following which are exceptions to octet rule is _____.
[08 April, 2024 (Shift

CO_2, NO_2, H_2SO_4, BF_3, CH_4, SiF_4, ClO_2, PCl_5, BeF_2, C_2H_6, $CHCl_3$, CBr_4

Sol. [6] NO_2, H_2SO_4, BF_3, ClO_2, PCl_5, BeF_2

3. Which of the following pair of molecules contain odd electron molecule and an expanded oc
[29 July, 2022 (Shift

molecule?

(a) BCl_3 and SF_6
(b) NO and H_2SO_4
(c) SF_6 and H_2SO_4
(d) BCl_3 and NO

Sol. (b)

BCl_3 – electron deficient molecule (Hypovalent)
SF_6 – expanded octet molecule
NO – odd electron containing molecule
H_2SO_4 – expanded octet molecule

FORMAL CHARGE (FC)

→ Valency e⁻ − lone pair e⁻ − $\frac{\text{Bond pair e}^-}{2}$

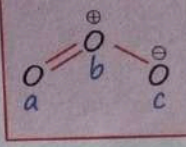
Difference between the number of valence electrons of that atom in free state and the number electrons assigned to that atom in the Lewis structure.

V = Total number of valence electrons in the free atom

N = Number of non-bonding electrons

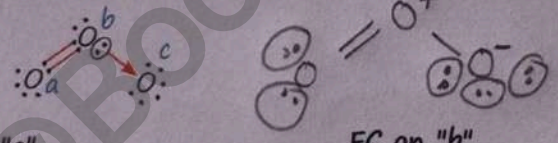
B = Number of bonding electrons

$$FC = V - N - \frac{B}{2}$$



	FC
a	zero
b	+1
c	−1

O_3



FC on "a"

V = 6, N = 4, B = 4 → $FC = 6 - 4 - \frac{4}{2} = 0$

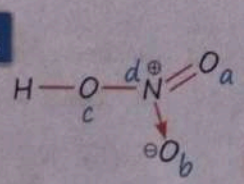
FC on "b"

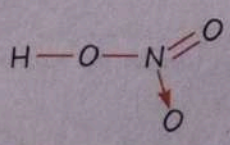
V = 6, N = 2, B = 6 → $FC = 6 - 2 - \frac{6}{2} = +1$

FC on "c"

V = 6, N = 6, B = 2 → $FC = 6 - 6 - \frac{2}{2} = -1$

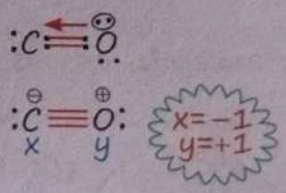
HNO_3





a = 0, b = −1, c = 0, d = +1

CO



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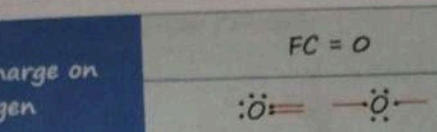
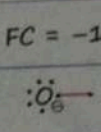
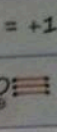
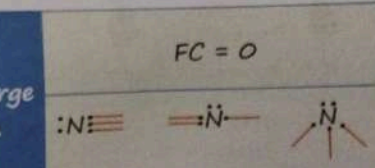
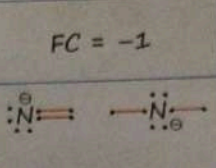
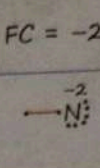
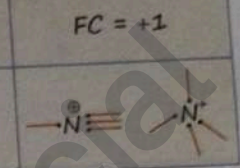
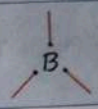
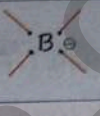
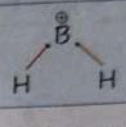
OP* Points

Formal charge on Oxygen	FC = 0	FC = −1	FC = +1
			

Formal charge on Nitrogen	FC = 0	FC = −1	FC = −2	FC = +1
				

Formal charge on Boron	FC = 0	FC = −1	FC = +1
			

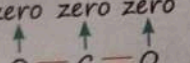
सोचो B के पास 3 ही थे 4 कैसे एक उसने लिया दिया तभी तो 4 है। लिया Hoon (-1)

- The structure of least energy (most stable) is usually the one with minimal formal charge and most distributed real charge.

* Stability ↑ → Formal charge ↓ → Energy ↓

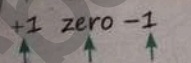
CO_2

zero zero zero



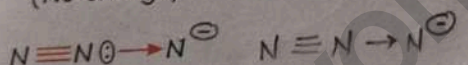
Neutral molecule (No charge) → Stability ↑

+1 zero −1



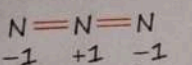
Molecule with formal charge → Stability ↓ → This structure is not acceptable.

$N_3^{\ominus}$



0 +1 −2

Less distributed charge → Less stability



−1 +1 −1

More distributed charge → More stability

C_3^{4-}

−1 zero −3 | −2 zero −2 : Stability

NATURE, SHAPE & PHASE OF ATOMIC ORBITAL

- s orbital (1 lobe)
- Non-directional nature
- + or − sign

- p orbital (2 lobes)
- Directional nature
- p_x, p_y, p_z

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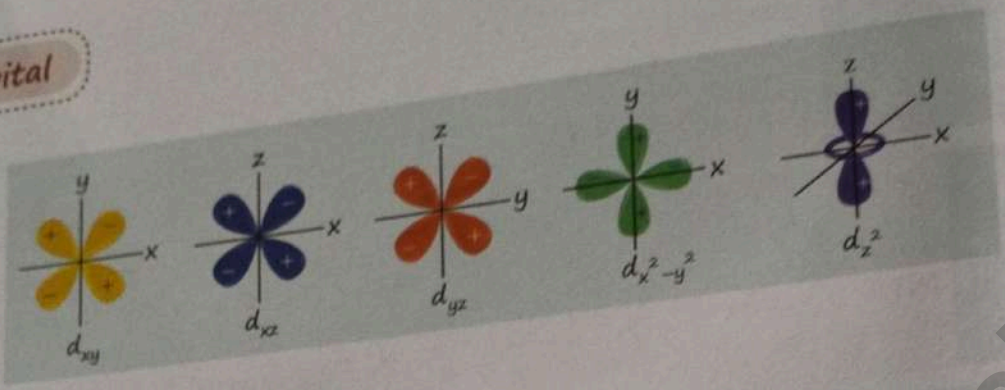
d-Orbital



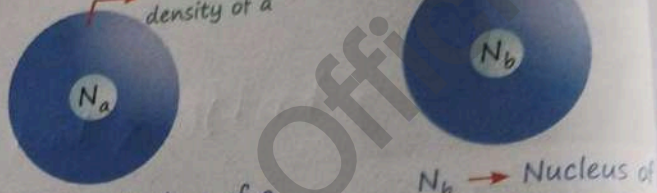
VALENCE BOND THEORY

$H_a + H_b \rightarrow H—H$

partial merging (overlap)

+ 435.8 KJ/mol

$(1S)_a$ $(1S)_b$ H_2 molecule — Bond Energy

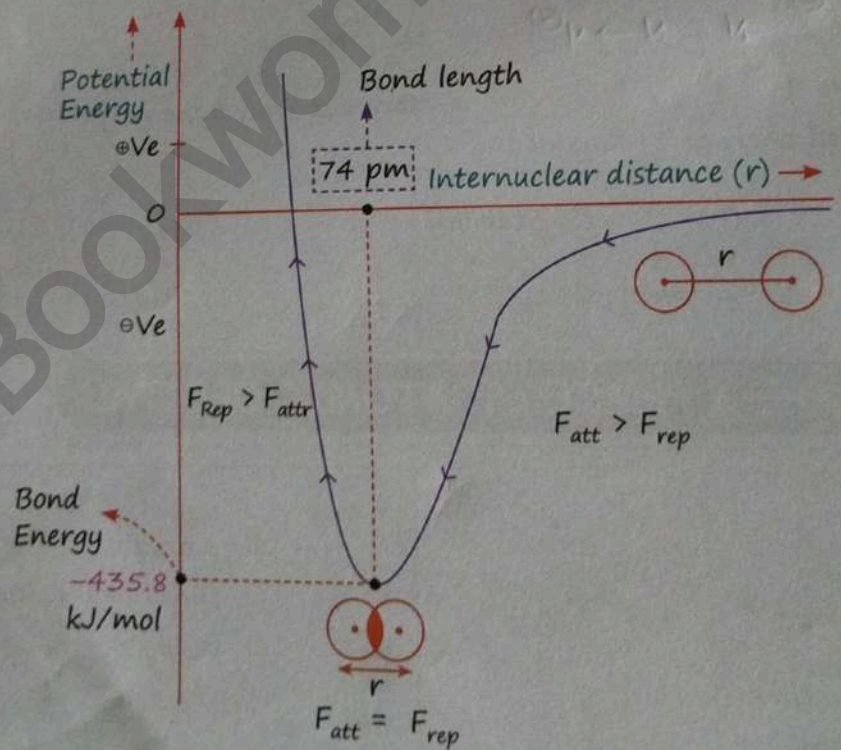


N_a → Nucleus of a
e_a^- → Electron density of a

N_b → Nucleus of
e_b^- → Electron density of

Force of attraction: (i) $(e^-)_a - (N)_b$
(ii) $(e^-)_b - (N)_a$

Force of repulsion: (i) $(e^-)_a - (e^-)_b$
(ii) $(N)_a - (N)_b$



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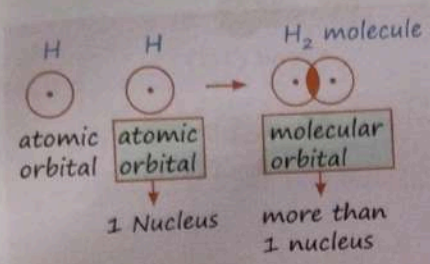
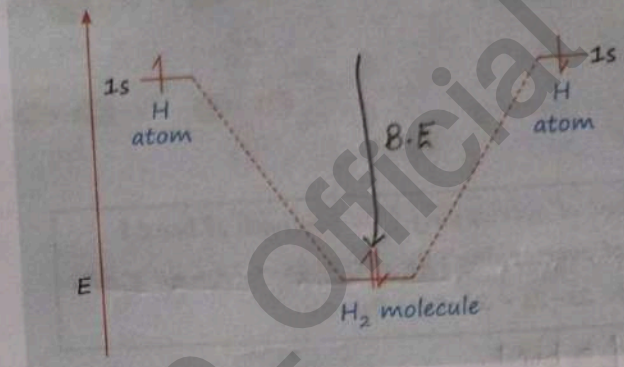
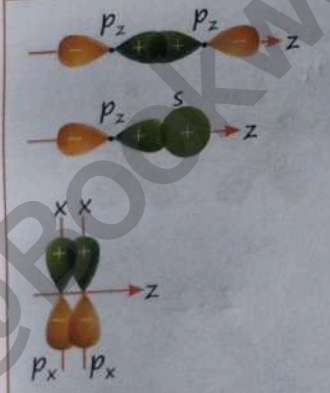
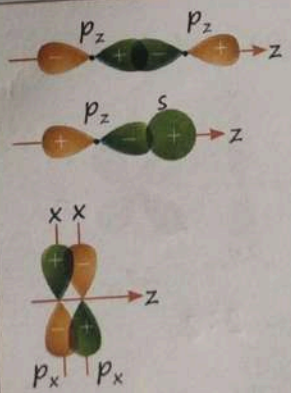
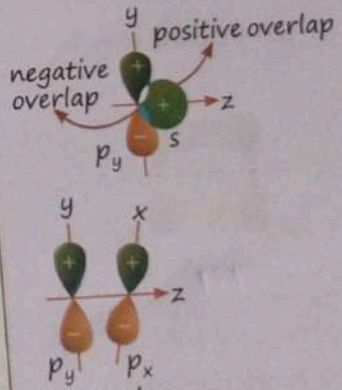
- Energy gets released when the bond is formed between two hydrogen atoms, the hydrogen molecule is more stable than that of isolated hydrogen atoms. The energy so released is called as bond enthalpy.
- VBT is an Orbital Overlap Concept.





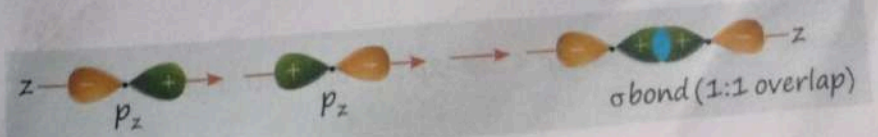
- Orbitals which are undergoing overlapping must have one electron with opposite spin for formation of covalent bond.

→ Bond formation

Positive Overlap	Negative Overlap	Zero Overlap
Same phase (sign) Overlap (+ +) (− −)	Opposite sign Overlap (+ −)	i. No Overlap between 2 orbitals ii. Both positive & negative overlap
		

Sigma Bond

Overlapping along the molecular axis/Head on overlapping



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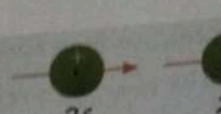
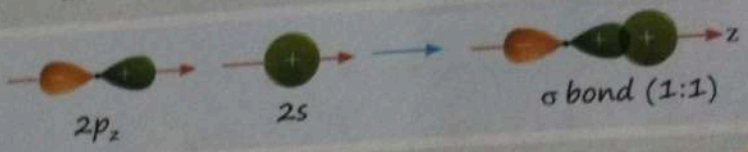
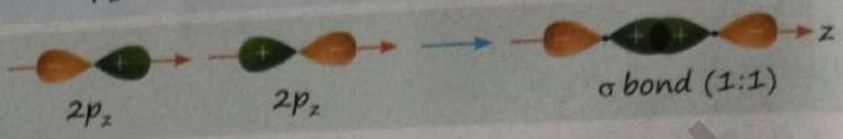
i. s–s overlapping



ii. s–p overlapping



iii. p–p overlapping



❑ Extent of overlapping ↑ → Strength of bond ↑

Extent of overlapping : 2s – 2s < 2s – 2p < 2p – 2p

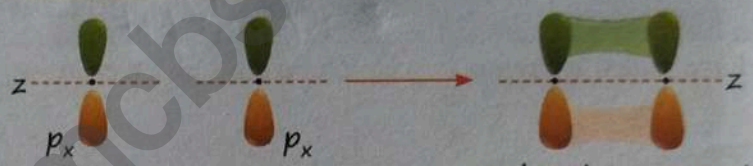
★ Strength: 2s–2s < 2s–2p < 2p–2p

↳ of σ bond.

Pi Bond

≡ Strength of σ bond > π bond

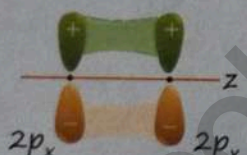
Overlapping ⊥ʳ to molecular axis (Side ways overlapping)



π bond (2:2 overlap)

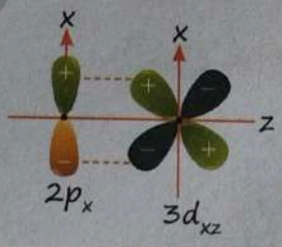
Strength : σ bond > π bond

(i) pπ–pπ bond



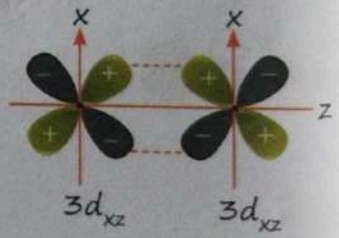
1:1

(ii) pπ–dπ bond



2:2

(iii) dπ–dπ bond



2:2

❑ Extent of overlapping ↑ : $(n_1 + n_2)$ ↓

Extent of overlapping : 2p–2p > 2p – 3d > 3d – 3d

$n_1 + n_2$: (2 + 2) < (2 + 3) < (3 + 3)

Strength of π bond : 2p–2p > 2p–3d > 3d–3d ★

Strength of Bond

(i) When $n_1 + n_2$ is same : 2s–2s < 2s–2p < 2p–2p : σ bond strength

$(n_1 + n_2)$→ (2 + 2) (2 + 2) (2 + 2)

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(ii) When $n_1 + n_2$ is different:

$1s - 1s > 2s - 2s > 3s-3s$: σ bond strength

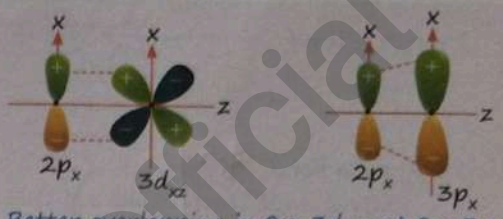
$(n_1 + n_2) \rightarrow$ (1 + 1) (2 + 2) (3 + 3)

$2p - 2p > 2p-3d$: π bond strength

$(n_1 + n_2) \rightarrow$ (2 + 2) (2 + 3)

★ $2p - 3p < 2p - 3d$: π bond strength

$(n_1 + n_2) \rightarrow$ (2 + 3) (2 + 3)



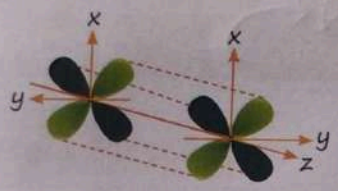
Better overlapping in 2p-3d w.r.t 2p-3p

Bonding in d-Orbitals

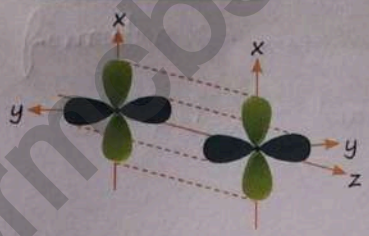
MANOJ YOGI

Delta Bond → 4 : 4 overlap

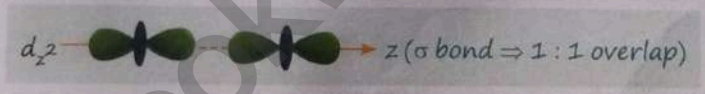
1. d_{xy} and d_{xy}

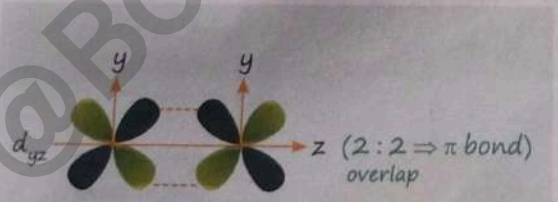


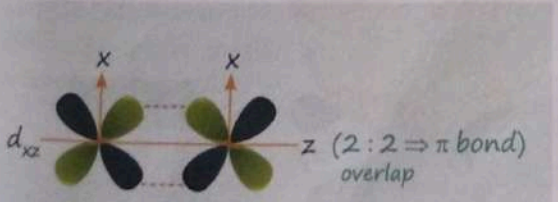
2. $d_{x^2-y^2}$ and $d_{x^2-y^2}$



❑ Overlapping of 2 $d_{x^2-y^2}$ or 2 d_{xy} orbitals gives a δ bond, when z axis is internuclear axis.







Orbital \ Bond	σ	π	δ
s	✓	×	×
p	✓	✓	×
d	✓	✓	✓

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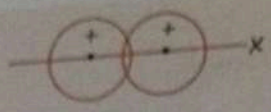
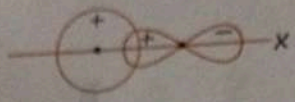
$s + s = \sigma$ bond



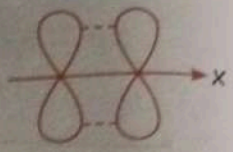
$p_x + p_x = \sigma$ bond

$s + p_x = \sigma$ bond



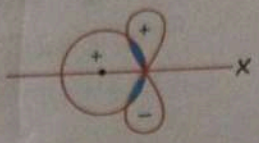
$p_y + p_y = \pi$ bond

$p_z + p_z = \pi$ bond



$s + p_y$ = No bond

$s + p_z$ = No bond

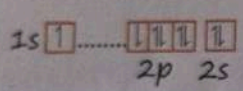


- Overlapping of orbitals when x axis is internuclear axis.

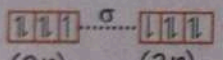
HF Molecule

$H\cdot + \cdot\ddot{F}: \longrightarrow H-\ddot{F}:$

σ bond ($(1s)_H - (2p)_F$)



F_2 Molecule

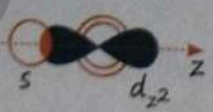


$(2p)_F$ σ $(2p)_F$ $\longrightarrow$ F—F

σ bond

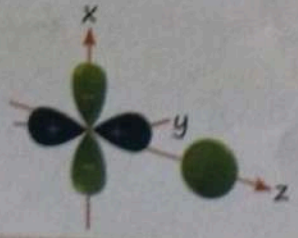
4. Draw the type of overlaps between given orbitals, Z is internuclear axis.
   (a) s and d_{z^2}
   (b) s and $d_{x^2-y^2}$
   (c) s and d_{yz}
   (d) p_z and d_{z^2}

Sol. (a)



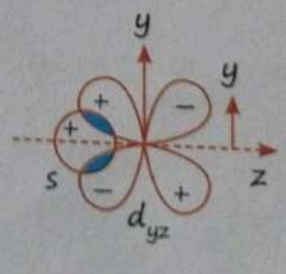
σ bond

(b)



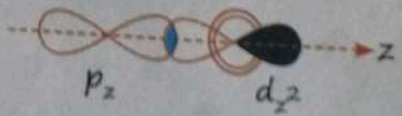
No bond

(c)



No bond

(d)



σ bond

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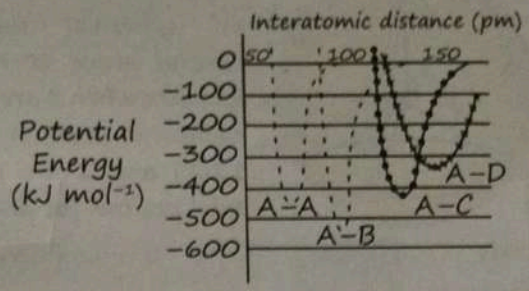
5. The intermolecular potential energy for the molecules A, B, C and D given below suggest that:
[4 Sept, 2020 (Shift-I)]



(a) A – A has the largest bond enthalpy (b) A – D has the shortest bond length

(c) D is more electronegative than other atoms (d) A – B has the stiffest bond.

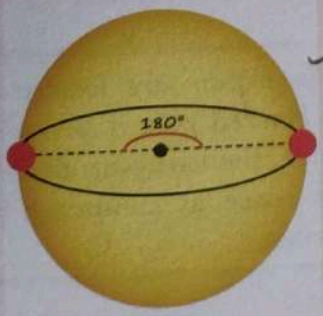
Sol. (d) Lower the potential energy, stronger will be the bond. A – B has lowest potential energy which means it has stronger bond.

VSEPR THEORY
VALENCE SHELL ELECTRON PAIR REPULSION THEORY

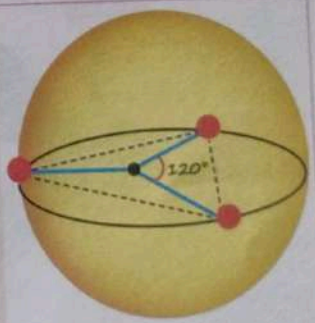
Main Postulates

- The shape of a molecule depends upon the number of valence shell electron pairs [bonded or non-bonded] around the central atom.
- The valence shell is taken as a sphere with the electron pairs localising on the spherical surface at maximum distance from one another.
- Pairs of electrons in the valence shell repel one another since their electron clouds are negatively charged.
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.

2 electron pairs are localised on spherical surface at maximum distance, when angle between them is 180°.



3 electron pairs are localised on spherical surface at maximum distance, when they are placed at corners at triangle.



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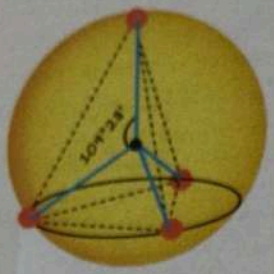
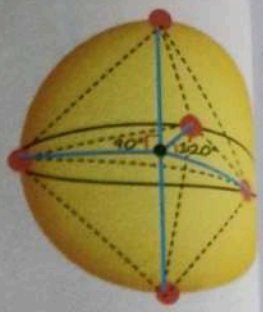
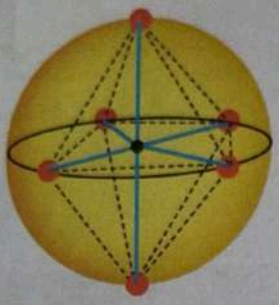
4 electron pairs are localised on spherical surface at maximum distance, when they are placed at corners of tetrahedron.



5 electron pairs are localised on spherical surface of central atom at maximum distance when 3 are placed at equatorial position [at angle 120°] and 2 are placed at axial positon [at angle 90°]

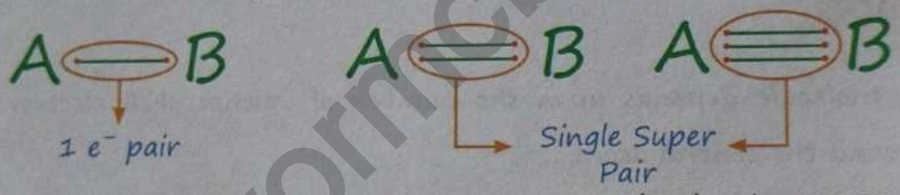


6 electron pairs are localised on spherical surface at maximum distance, when they are placed at corners of octahedron.



7 electron pairs at maximum distance is shown in figure

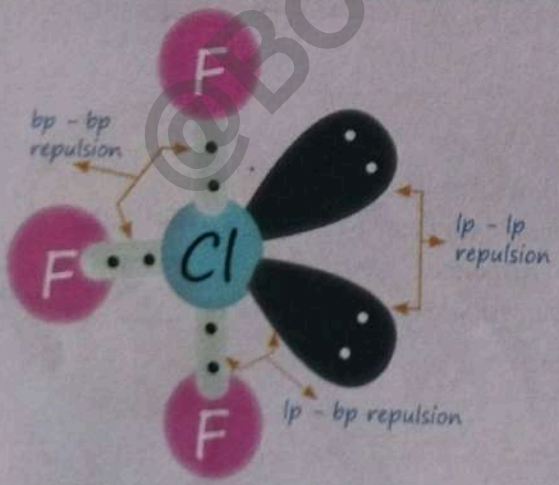
- A multiple bond is treated as if it is a single electron pair and the two or three electron pairs multiple bond are treated as a single super pair.



- Where two or more resonance structures can represent a molecule, the VSEPR model is applic to any such structure.

The repulsive interaction of electron pairs decrease in the order:

Lone pair (lp) - Lone pair (lp) > Lone pair (lp) - Bond pair (bp) > Bond pair (bp) - Bond pair (bp)



- The lone pair are localized on the central atom each bonded pair is shared between two atoms. A a result, the lone pair electrons in a molecule occup more space as compared to the bonding pairs electrons.

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- We can represent any molecule in AB_xL_y form where, A → Central atom, B → Side atom, L → Lone pair
- Number of valence shell electron pair decides the geometry of a molecule.

Valence shell electron pair [VSEP] = x + y

Side atom + l.p. = Steric no.

AB_2 Type Molecule

CO_2

- Valence shell electron pair = 2
- Geometry → Linear
- Bond angle → 180°

AB_2L_0

$x = 2$

$y = 0$

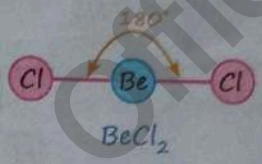
$BeCl_2$

- Valence shell electron pair = 2
- Geometry → Linear
- Bond angle → 180°

AB_2L_0

$x = 2$

$y = 0$



AB_3 Type Molecule

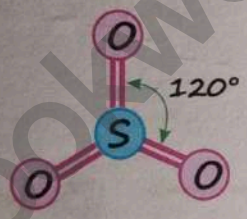
SO_3

- Valence shell electron pair = 3 + 0 = 3
- Geometry → Trigonal planar
- Bond angle → 120°

AB_3L_0

$x = 3$

$y = 0$



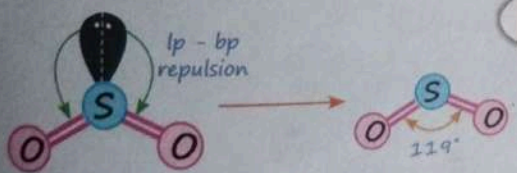
SO_2

- Valence shell electron pair = 2 + 1 = 3
- Due to lp - bp repulsion, bond angle reduces from 120° to 119°

AB_2L_1

$x = 2$

$y = 1$



Geometry → Trigonal planar Shape → Bent

★ Structure with lone pair → Geometry
Structure without lone pair → Shape

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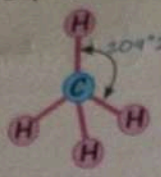
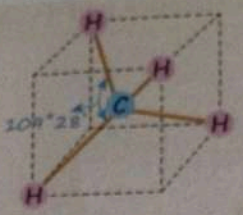
AB_4 Type Molecule

CH_4 → AB_4L_0 → $x = 4$, $y = 0$

- Valence shell electron pair = 4 + 0 = 4
- Geometry → Tetrahedral
- Bond angle → 109° 28'



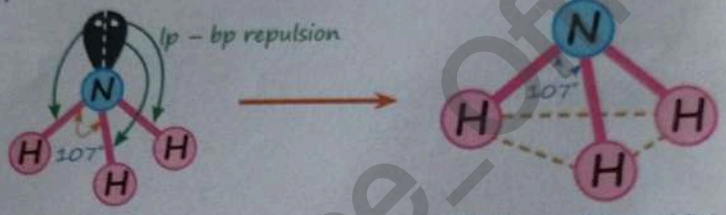


C → At body centre of cube

H → At alternate corners of cube

NH_3 → AB_3L_1 → $x = 3$, $y = 1$

- Valence shell electron pair = 3 + 1 = 4
- Due to lp – bp repulsion, bond angle reduces from ideal tetrahedral angle 109° 28' to

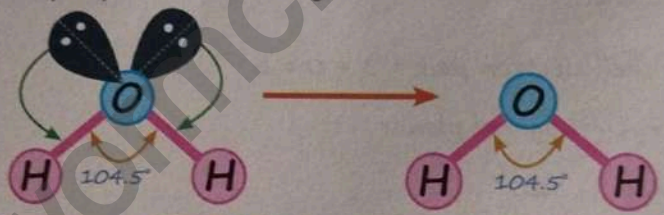


Geometry — Tetrahedral

Shape — Pyramidal

H_2O → AB_2L_2 → $x = 2$, $y = 2$

- Valence shell electron pair = 2 + 2 = 4
- Due to more lp – bp repulsion, bond angle reduces from 109° 28' to 104.5°

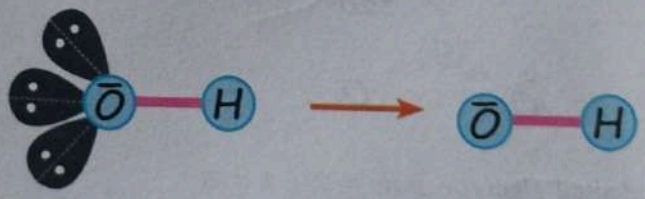


Geometry — Tetrahedral

Shape — Bent, V-shaped

OH^- → ABL_3 → $x = 1$, $y = 3$

- Valence shell electron pair = 1 + 3 = 4



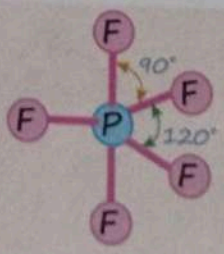
Geometry → Tetrahedral

Shape → linear

AB_5 Type Molecule

PF_5 → AB_5L_0

- Valence shell electron pair = 5 + 0 = 5
- Geometry → Trigonal bipyramidal
- θ → 90° and 120°



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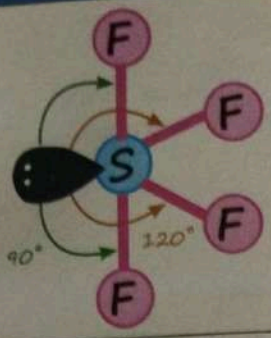
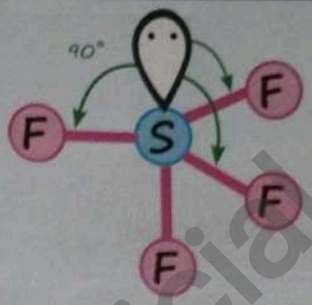
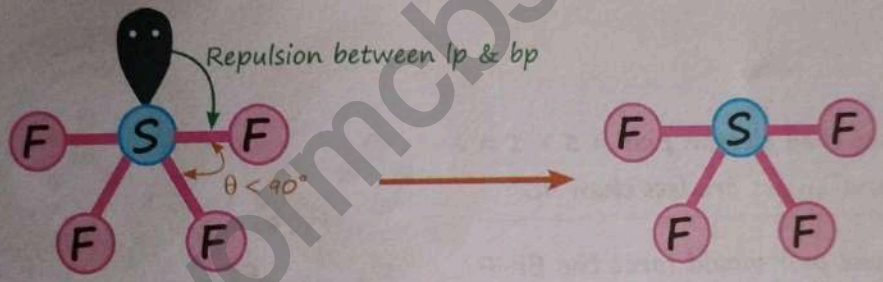
SF_4 → AB_4L_1

- Valence shell electron pair = 4 + 1 = 5

Case–1 If lp is present at equatorial position	Case–2 If lp is present at axial position
	
❑ There are 2 lp – bp repulsions at 90° and 2 lp – bp repulsions at 120°.	❑ There are 3 lp – bp repulsions at 90°.

- Repulsions at 90° are more effective than 120°. So we are going to compare on the basis of repulsions at 90°.
- Case – 1 structure is more favorable than case-2 due to less repulsion.
- So, case – 1 structure is Acceptable on the basis of postulates of VSEPR theory.



Geometry – TBP Shape – See Saw

Note

In TBP geometry due to less repulsion, more favorable position for lone pair is equatorial position.

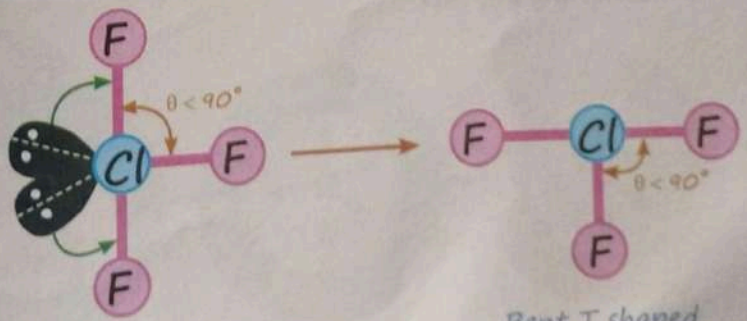
ClF_3

- Valence shell electron pair = 3 + 2 = 5

AB_3L_2

$x = 3$

$y = 2$



TBP Geometry Bent T shaped

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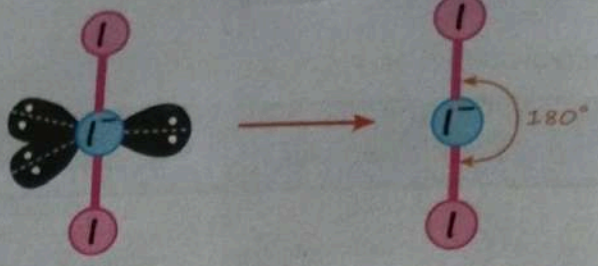
I_3^-

- Valence shell electron pair = 2 + 3 = 5

AB_2L_3

$x = 2$

$y = 3$



TBP Geometry → Linear shape

AB_6 Type Molecule

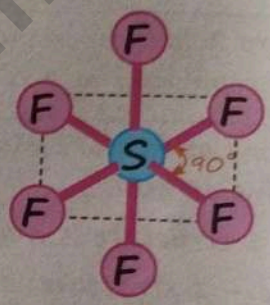
SF_6

- Valence shell electron pair = 6 + 0 = 6
- Geometry → Octahedral
- Bond angle → 90°

AB_6L_0

$x = 6$

$y = 0$



BrF_5

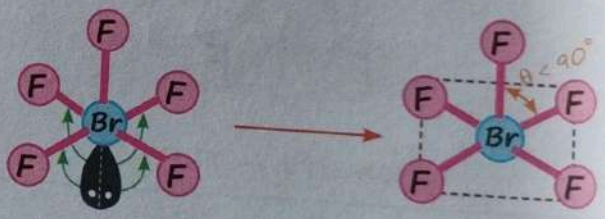
- Valence shell elctron pair = 5 + 1 = 6
- All bond angles are less than 90°

The lone pair would force the Br–F bond pairs upward, and all Br–F bond angles would contract.

AB_5L_1

$x = 5$

$y = 1$



Geometry – Octahedral Shape → Square Pyram

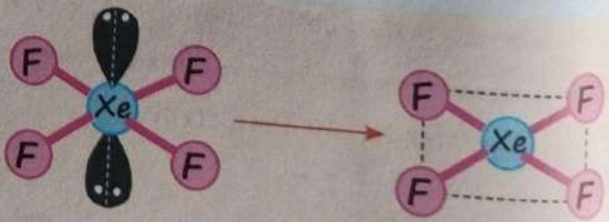
XeF_4

- Valence shell electron pair = 4 + 2 = 6
- Bond angle → 90°

AB_4L_2

$x = 4$

$y = 2$



Geometry – octahedral

shape – Squar planar

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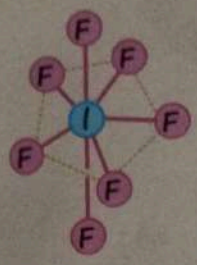
AB_7 Type Molecule

IF_7

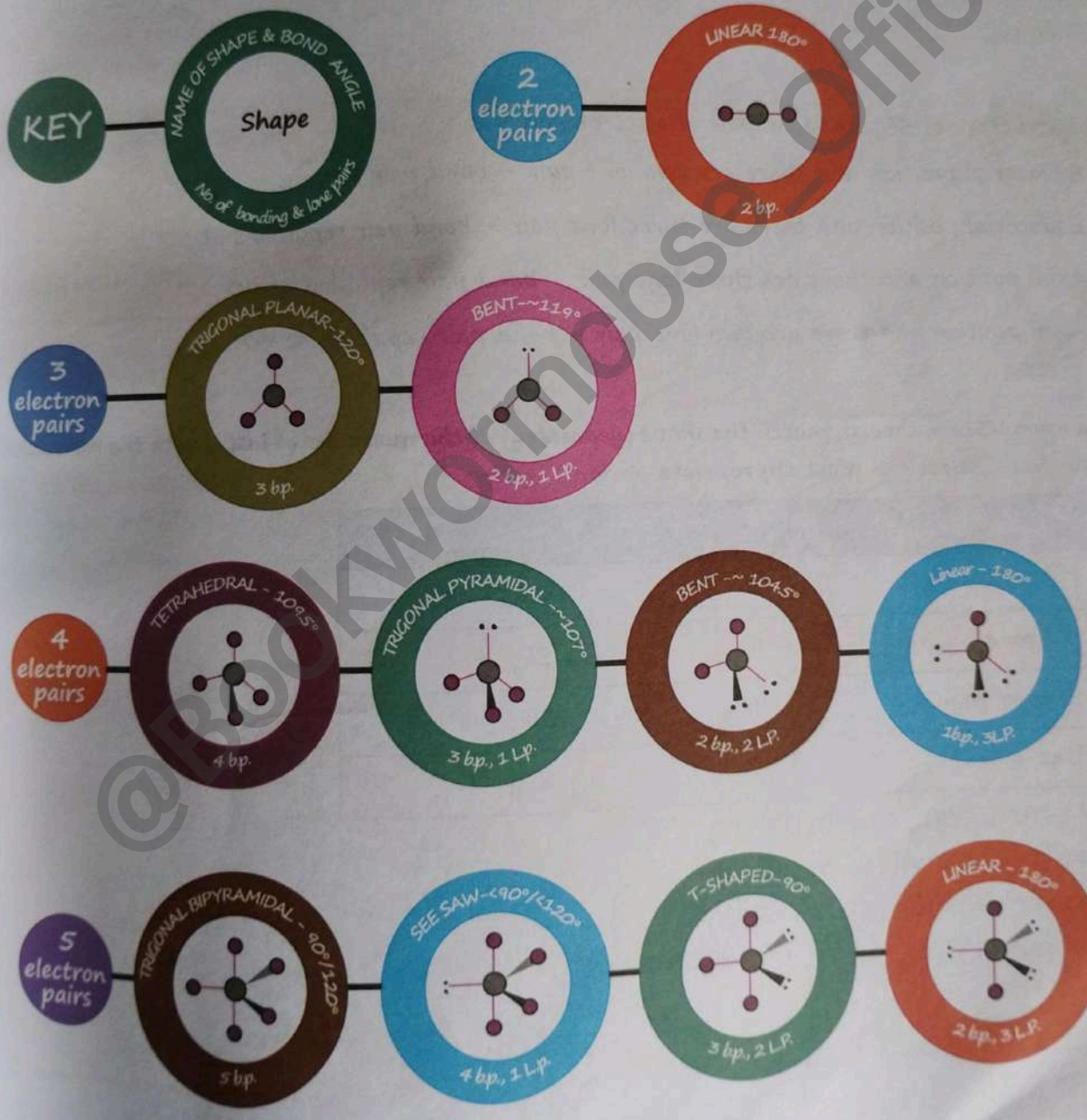
AB_7L_0

$x = 7$

$y = 0$



IF_7 [$\theta = 72°, 90'$]
Pentagonal bipyramidal Geometry



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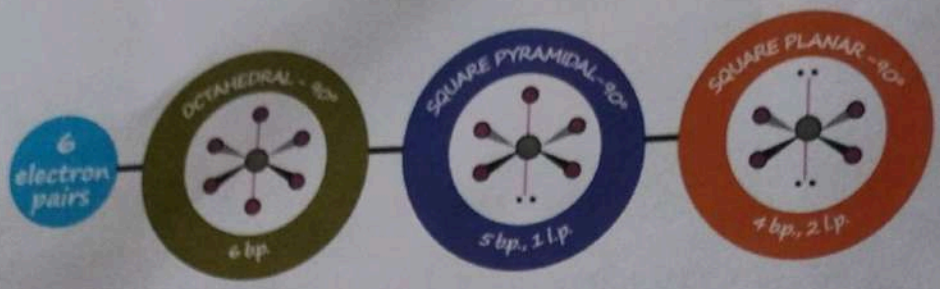

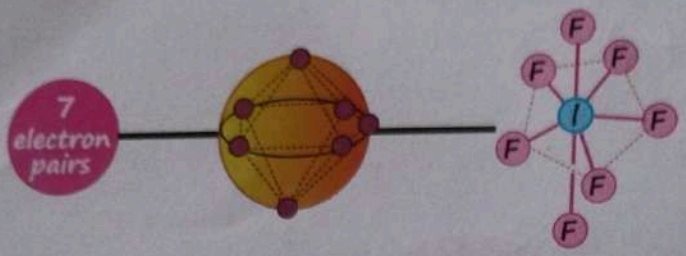

IF_7 [θ = 72°, 90']
Pentagonal bipyramidal Geometry

6. In the structure of SF_4, the lone pair of electrons on S is [27 June, 2022 (Shift

(a) Equatorial position and there are two lone pair – bond pair repulsions at 90°.

(b) Equatorial position and there are three lone pair – bond pair repulsions at 90°.

(c) Axial position and there are three lone pair – bond pair repulsion at 90°.

(d) Axial position and there are two lone pair – bond pair repulsion at 90°.

Sol. (a)

7. Based upon VSEPR theory, match the shape (geometry) of the molecules in List-I with the molec in List-II and select the most appropriate option. [27 June, 2022 (Shift

List-I (Shape)			List-II (Molecules)
A.	T-shaped	I.	XeF_4
B.	Trigonal planar	II.	SF_4
C.	Square planar	III.	ClF_3
D.	See-saw	IV.	BF_3

(a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(b) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(c) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(d) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Sol. (b)

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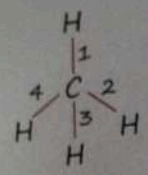
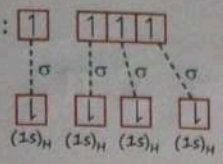
NEED OF NEW CONCEPT → (HYBRIDISATION)

Bonding in CH_4 according to VBT → CH_4 में 4 Bond length C-H same होती है But VBT नहीं बता पाया

$C_{ground\ state}$: 2s ↿⇂ 2p ↑ ↑

$C_{excited\ state}$: 2s ↑ 2p ↑ ↑ ↑





- 2 different types of overlapping
  - $(2s)_C - (1s)_H$
  - $(2p)_C - (1s)_H$
- 2 Different bond lengths.

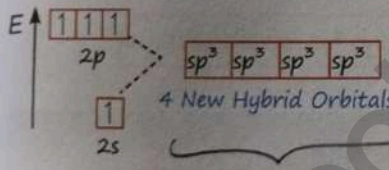
VBT predicts that the 3 bonds formed through s-p overlap would be the same and differ from the 4th bond formed through s-s overlap. This difference in bond types would result in unequal bond lengths which is not what is observed in methane.

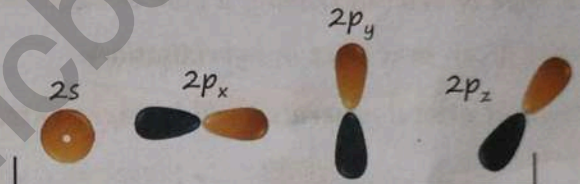
Experimental Fact All C — H bonds are same in CH_4. → This fact can be explained by new theory ⇒ Hybridisation

VBT नहीं कर पाया

Hybridisation in CH_4

$C_{excited\ state}$: 2s ↑ 2p ↑ ↑ ↑



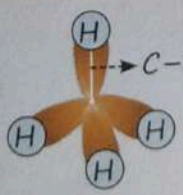


4 hybrid Orbitals (sp^3)

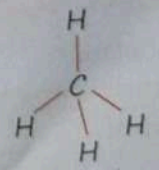
4 sp^3 hybrid orbitals of carbon at 109° 28'

(Min. Repulsion ⇒ Max. distance)

4H →



Overlapping between sp^3 hybrid orbital of carbon and 1s atomic orbital of hydrogen



All C—H bonds are equal in length.

Successful

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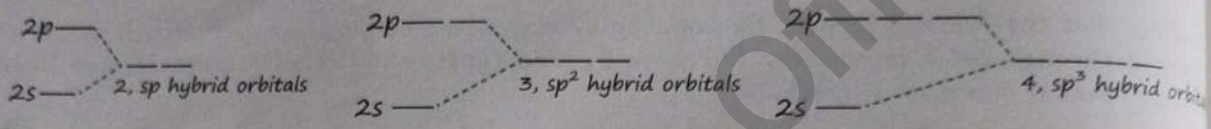
HYBRIDISATION

Inter Mixing of pure atomic orbitals before bonding to produce new hybrid orbitals, specially for bonding purpose.

Postulates

(i) It is a hypothetical concept.

(ii) Only those orbitals can take part in hybridisation which have comparable (almost equal) energ



(iii) The number of hybrid orbitals generated will be equal to the number of pure atomic orbitals part in hybridisation.

(iv) All 3 type of orbitals (having a pair of electrons ⇅ or having a unpaired electron 1 or comp empty ☐) can take part in hybridisation.

(v) The hybrid orbital generated will be represented by

Bigger lobe will be used for bonding.

Steric Number Rule

l.p + Side atom.

Steric number = No. of lone pairs + No. of side atoms

SN	Geometry	Hybridisation	Bond Angle
1	Linear	sp	θ = 180°
2	Trigonal planar	sp^2	θ = 120°
3	Tetrahedral	sp^3	θ = 109° 28'
4	Trigonal bi-pyramidal	sp^3d	θ = 90°, 120°
5	Octahedral	sp^3d^2	θ = 90°
6	Pentagonal bi-pyramidal	sp^3d^3	θ = 90°, 72°

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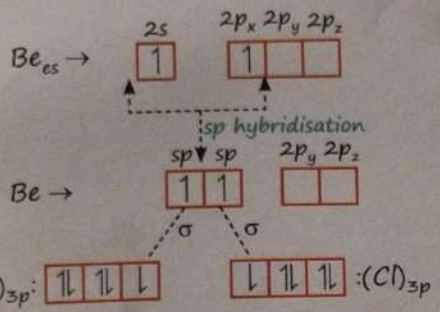
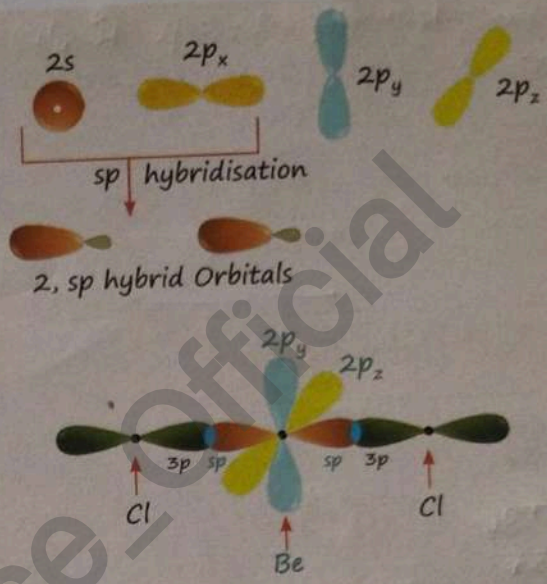
sp Hybridisation

1. steric no.
2. elec. confi.

$BeCl_2$ Cl — Be — Cl

- SN = LP + SA = 0 + 2
  = (2) → sp hybridisation

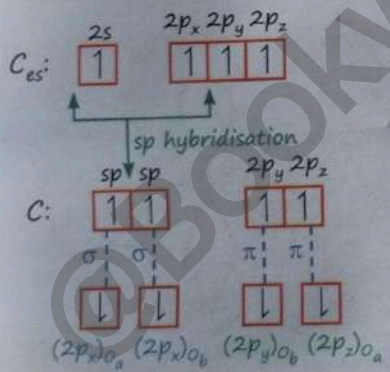


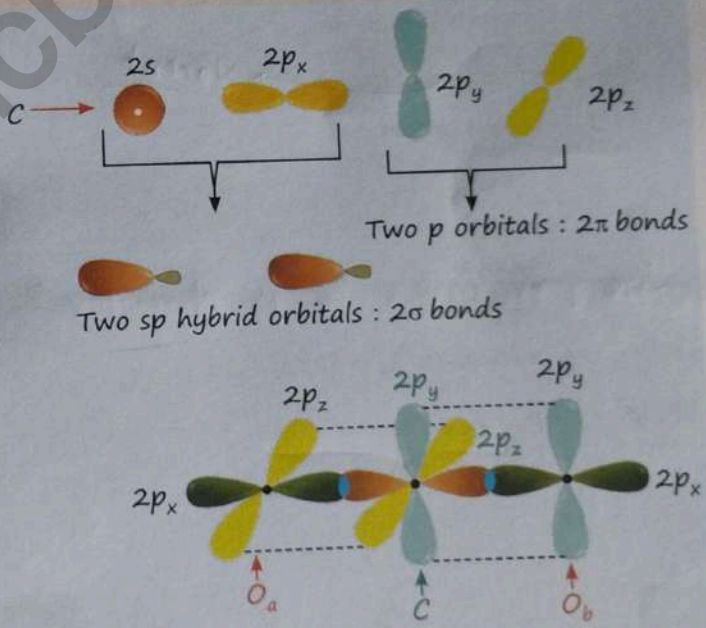


- 2 Be — Cl σ bonds : $(sp)_{Be} - (3p)_{Cl}$
- 2 vacant orbitals ($2p_y$ and $2p_z$) at Be.

CO_2 O = C = O

- SN = LP + SA
  = 0 + 2
  = 2 → sp hybridisation





$[(2p_z)_c - (2p_z)_O]$ π bond

π bond $[(2p_y)_c - (2p_y)_O]$

$_aO = C = O_b$

$[(sp)_c - (2p_x)_O]$ σ bond

σ bond $[(sp)_c - (2p_x)_O]$

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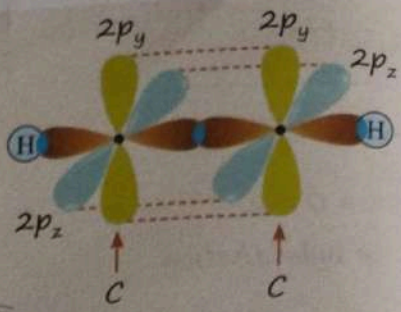
C_2H_2

$H-C\equiv C-H$

- 2 C–H σ bonds : $(sp)_C - (1s)_H$
- 1 C–C σ bond : $(sp)_C - (sp)_C$
- 2 C–C π bonds : $(2p_y)_C - (2p_y)_C$ and $(2p_z)_C - (2p_z)_C$

Molecular plane → $H-C\equiv C-H$

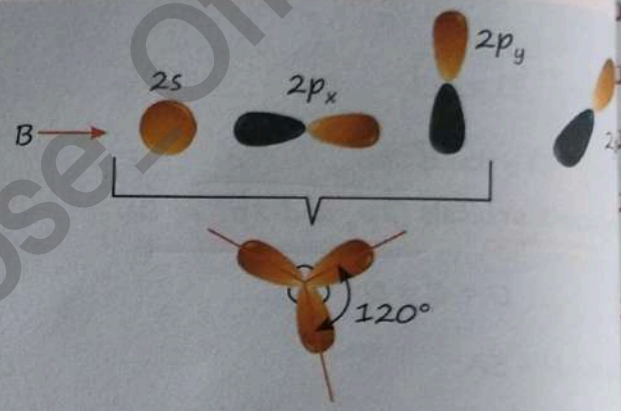


$\pi_1 \perp \pi_2$

sp^2 Hybridisation

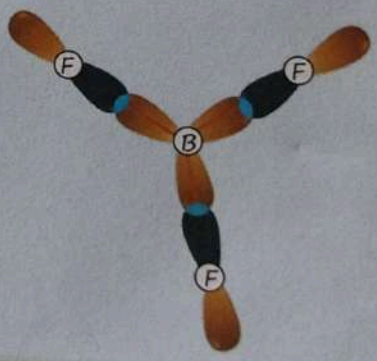
BF_3

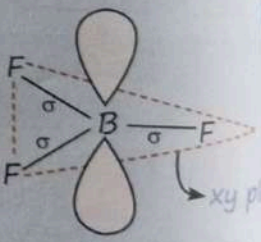
- B_{gs}: 2s | $2p_x\ 2p_y\ 2p_z$ — ↿⇂ | 1 |
- B_{es}: 2s | $2p_x\ 2p_y\ 2p_z$ — 1 | 1 | 1
- sp^2 Hybridisation
- B_{es}: 1 | 1 | 1 | □ $2p_z$ → Vacant
- 3, sp^2 hybrid orbitals



- 3 sp^2 hybrid orbitals are arranged at 120° b… bonding.

- F: 2s — ↿⇂ | $2p_z\ 2p_y\ 2p_x$ — ↿⇂ | ↿⇂ | 1
- 3 σ B – F bonds: $(sp^2)_B - (2p_x)_F$
- $sp^2 \equiv s + p_x + p_y$

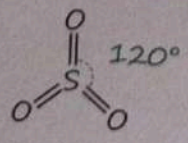




$2p_z$ (Vacant)

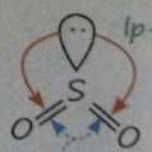
SO_3

- SN = 0 + 3 = 3
- hyb^n = sp^2
- Shape = Trigonal planar



SO_2

- SN = 1 + 2 = 3
- sp^2
- V shaped / Bent



θ < 120°

≈ 119°

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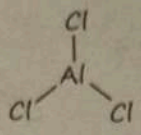
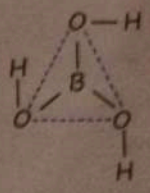
$AlCl_3$

- SN = 0 + 3 = 3
- sp^2
- Trigonal planar

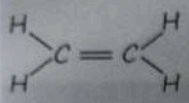


H_3BO_3

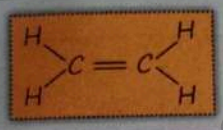
- SN = 0 + 3
- sp^2
- Trigonal planar

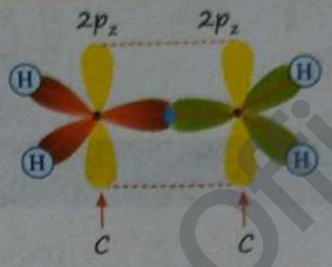


C_2H_4



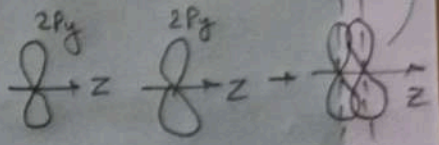
- $sp^2 \equiv s + p_x + p_y$
- 4 C—H σ bonds : $(sp^2)_c - (1s)_H$
- 1 C—C σ bond : $(sp^2)_c - (sp^2)_c$
- 1 C—C π bond : $(2p)_c - (2p)_c$
- Molecular plane →





- Max. no of atoms in a plane = 6

π bond

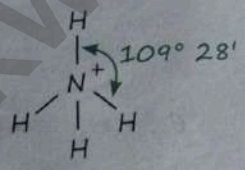


* Nodal plane of $2p_y$ → xz plane.
* Nodal plane of π bond → Molecular plane. (xz plane)

sp^3 Hybridisation → Geometry : Tetrahedral

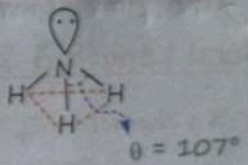
$NH_4^{\oplus}$

- SN = 0 + 4
- sp^3
- Tetrahedral



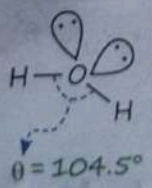
NH_3

- SN = 1 + 3 = 4
- sp^3
- Pyramidal



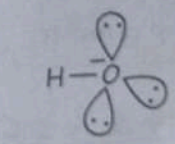
H_2O

- SN = LP + SA
  = 2 + 2 = 4
- sp^3
- V shape



HO^-

- SN = 3 + 1 = 4
- sp^3
- Linear



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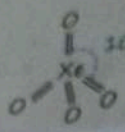
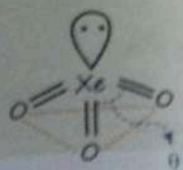
XeO_4

- SN = 0 + 4 = 4
- sp^3
- Tetrahedral



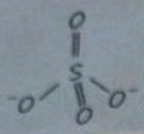
XeO_3

- SN = 1 + 3 = 4
- sp^3
- Pyramidal



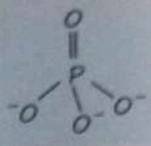
SO_4^{2-}

- SN = 0 + 4 = 4
- sp^3
- Tetrahedral



PO_4^{3-}

- SN = 0 + 4 = 4
- sp^3
- Tetrahedral

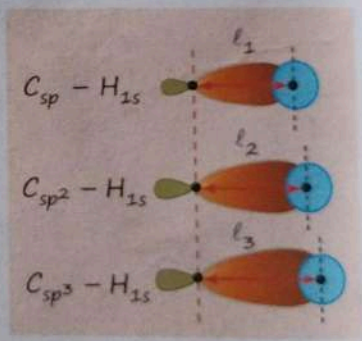


Orbital	% s Character	% p character
s orbital	100%	0%
p orbital	0%	100%
sp hybrid orbital	50%	50%
sp^2 hybrid orbital	33.3%	66.6%
sp^3 hybrid orbital	25%	75%

* Lone pair loves % S character

Bond Length & % s Character

- % s character: $C_{sp} > C_{sp^2} > C_{sp^3}$
- Bond length: $\ell_1 < \ell_2 < \ell_3$
- Bond length $\propto \frac{1}{\%\ s\ character}$ or $\%\ p\ character$



Electronegativity & % s Character

- $EN \propto \%\ s\ character$
- % s character: $C_{sp} > C_{sp^2} > C_{sp^3}$
- Electronegativity: $C_{sp} > C_{sp^2} > C_{sp^3}$

* electronegative element hates % S character
* जैसे E.N element S के पास जाना है S को चोच लेता है EN element को % S character ज्यादा EN है तो EN element S के पास नहीं आना चाहिए।

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sp^3d Hybridisation → Geometry: Trigonal bipyramidal (TBP)

PCl_5

$sp^3d \equiv s + p_x + p_y + p_z + d_{z^2}$

$3(P-Cl_e)$ Bond ← $[sp^2]_{xy}$ ← $[pd]_z$ → $2(P-Cl_a)$

$[sp^2]_{xy}$: approx 100 % s character

$[pd]_z$: approx 0 % s character

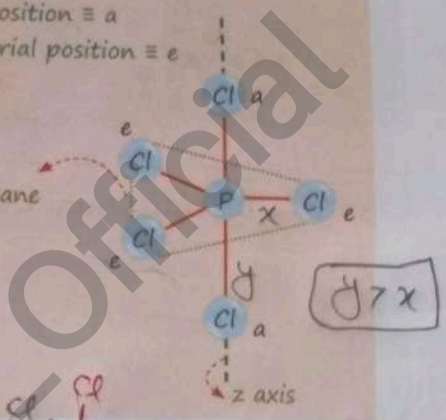
Bond length : $P-Cl_e$ < $P-Cl_a$

Stregth → % S charac. = $P-Cl_e > P-Cl_a$

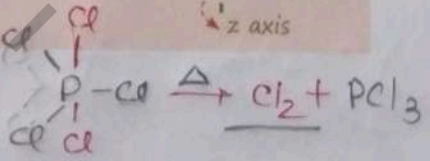
Bond length = $P-Cl_e < P-Cl_a$

axial position ≡ a

equatorial position ≡ e



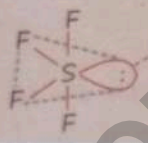
y > x



Bent's Rule

(1) Lone pair of electron prefers to occupy those hybrid orbitals which has greater % of s character.

SF_4



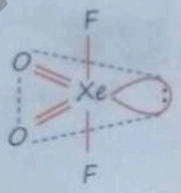
lone pair at equatorial position because it has more % s character than axial position.

(2) A more electronegative atom prefers to occupy that hybrid orbital which has smaller % of s character (axial position)

Bond length → P-Cl > P-Fa > P-Fe

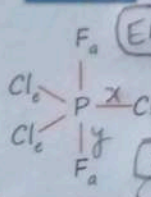
XeO_2F_2

- SN = LP + SA = 1 + 4 = 5
- Geometry - TBP
- Shape - see saw



(2) वाले elec. neg. को axial पर रखते हैं।

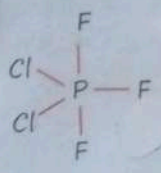
PCl_3F_2



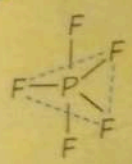
x > y

always.

PCl_2F_3

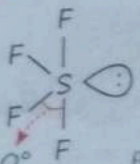


PF_5



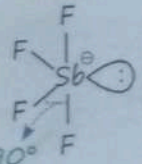
- SN = 0 + 5
- sp^3d
- TBP Shape
- BL : $P-F_a > P-F_e$

SF_4

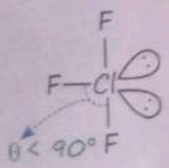


- SN = 1 + 4 = 5
- sp^3d
- See Saw Shape
- BL : $S-F_a > S-F_e$

SbF_4^-

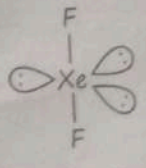


ClF_3



- SN = 2 + 3 = 5
- sp^3d
- Bent T shaped
- BL : $Cl-F_a > Cl-F_e$

XeF_2



- SN = 3 + 2 = 5
- sp^3d
- Linear Shape

* First priority रखती है कि EN element को equatorial पर ना रखे axial पर रखे।

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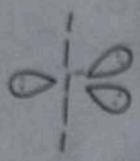
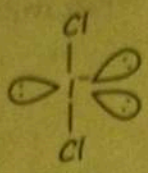
I_3^-



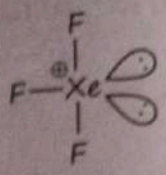
- SN = 3 + 2 = 5
- sp^3d
- Linear Shape

ICl_2^-



- SN = 3 + 2 = 5
- sp^3d
- Linear Shape

XeF_3^+



- SN = LP + SA = 2 + 3 = 5
- Bent T shaped

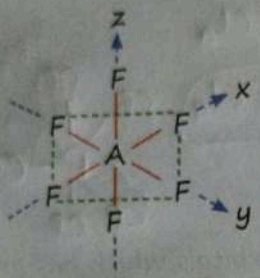
sp^3d^2 Hybridisation: Geometry → Octahedral

$sp^3d^2 = s + p_x + p_y + d_{x^2-y^2} + p_z + d_{z^2}$

($s + p_x + p_y + d_{x^2-y^2}$: xy plane; $p_z + d_{z^2}$: z axis)

AF_6

- SF_6
- PF_6^-
- SiF_6^{2-}
- AlF_6^{3-}



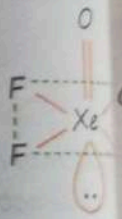
A	S	Si^{2-}	P^-	Al^{3-}
Valence shell electrons	6	4+2	5+1	3+3

BrF_5, XeF_5^+, ClF_5, IF_5, SF_5^-

- AB_5 → A = Br, Xe^+, I, Cl, S^-
- SN = 1 + 5 = 6
- sp^3d^2
- Square pyramidal → shape
- No of bond angles at 90° → zero

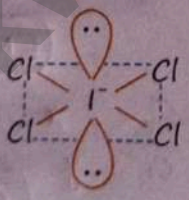
$XeOF_4$

- SN = 1 + 5 = 6
- Square pyramidal → shape



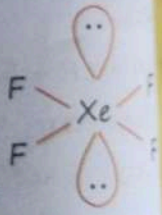
ICl_4^-

- SN = 2 + 4 = 6 (sp^3d^2)
- Square planar → shape



XeF_4

- Square planar → shape



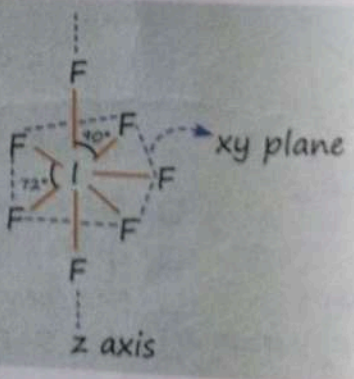
sp^3d^3 Hybridisation

IF_7

$s + p_x + p_y + d_{x^2-y^2} + d_{xy} + p_z + d_{z^2}$

($s + p_x + p_y + d_{x^2-y^2} + d_{xy}$: xy plane; $p_z + d_{z^2}$: z axis)

- SN = LP + SA = 0 + 7 = 7
- Pentagonal bipyramidal → shape
- θ → 90°/72°



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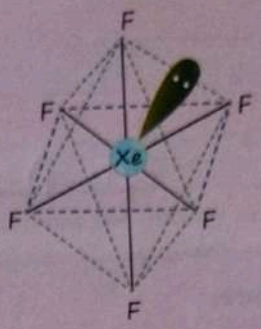
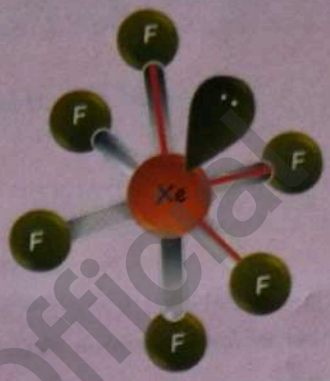
XeF_6 (g), IF_6^-, SeF_6^{2-}

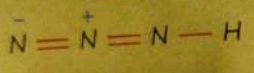
- SN = LP + SA = 1 + 6 = 7
- Shape → Distorted octahedral
  Capped octahedral



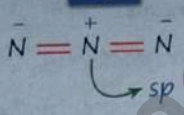


Some Important Molecules-

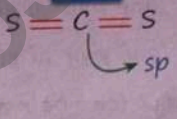
N_3H



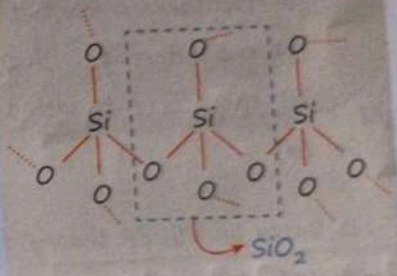
N_3^-



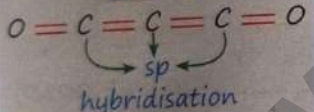
CS_2



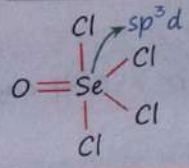
SiO_2



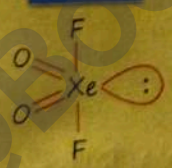
C_3O_2



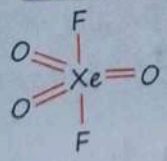
$SeOCl_4$



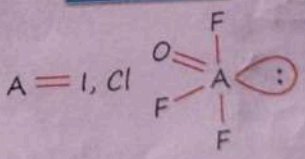
XeO_2F_2



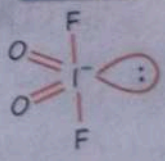
XeO_3F_2



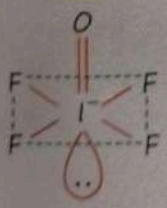
IOF_3/$ClOF_3$



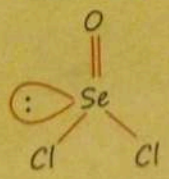
$IO_2F_2^-$



IOF_4^-



$SeOCl_2$



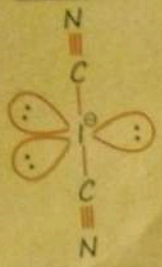
HCP

H — C ≡ P

- Structure is like HCN

H — C ≡ N

$I(CN)_2^-$



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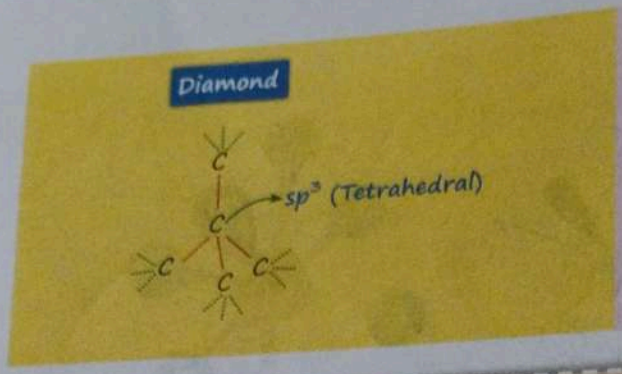
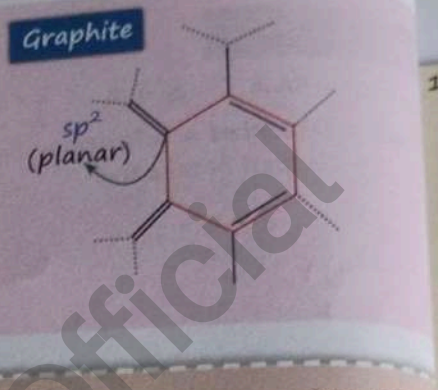




8. Number of lone pair(s) in $XeOF_4$ is/are (IIT JEE 20

(a) 0 (b) 1 (c) 2 (d) 3

Sol. (b)

9. Molecular shape of SF_4, CF_4 and XeF_4 are (IIT JEE 20

(a) the same, with 2, 0 and 1 lone pair of electrons respectively

(b) the same, with 1, 1 and 1 lone pair of electrons respectively

(c) the different, with 0, 1 and 2 lone pair of electrons respectively

(d) the different, with 1, 0 and 2 lone pair of electrons respectively

Sol. (d)

10. Based on VSEPR theory, the number of 90° F–Br–F angles in BrF_5 is _____. (IIT JEE 20

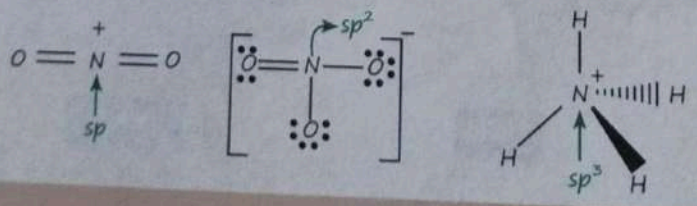
Sol. [0]

11. The hybridisation of atomic orbitals of nitrogen in NO_2^+, NO_3^- and NH_4^+ are (IIT JEE 20

(a) sp, sp^3 and sp^2 respectively (b) sp, sp^2 and sp^3 respectively

(c) sp^2, sp and sp^3 respectively (d) sp^3, sp^3 and sp respectively

Sol. (b)



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12. Match List–I with List–II: [27 Aug, 2021 (Shift-I)]

	List-I (Species)		List-II (Number of lone pairs of electrons on the central atom)
A.	XeF_2	I.	0
B.	XeO_2F_2	II.	1
C.	XeO_3F_2	III.	2
D.	XeF_4	IV.	3

Choose the most appropriate answer from the options given below:

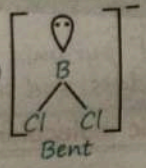
(a) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (b) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(c) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) (d) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

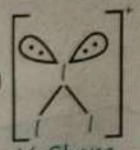
Sol. (d) XeF_2 Lone Pair = 3; XeO_2F_2 Lone Pair = 1; XeO_3F_2 Lone Pair = 0; XeF_4 Lone Pair = 2

13. Amongst the following, the number of species having the linear shape is ___.

XeF_2, I_3^+, C_3O_2, I_3^-, CO_2, SO_2, $BeCl_2$ and BCl_2^- [30 Jan, 2023 (Shift-II)]

Sol. [5] (i) 

Bent (ii) Cl—Be—Cl Linear (iii)
Linear (iv) XeF_2 – Linear

(v) 
V-Shape (vi) CO_2 – Linear (vii) C_3O_2 (O = C = C = C = O) Linear (viii) SO_2 – Bent

Iso-structural Species and Iso-electronic Species

Iso-structural Species

- Species having same shape

$CH_4 \rightarrow SN = 0 + 4$ (Tetrahedral)

$CCl_4 \rightarrow SN = 0 + 4$ (Tetrahedral)

XeF_6 BrF_6^- $SeCl_6^{2-}$ $TeCl_6^{2-}$

- Shape → Distorted octahedral
- $Xe \equiv Br^- \equiv Se^{2-} \equiv Te^{2-}$

No. of e^- in valence shell → 8

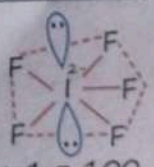
Iso-electronic Species

- Species having same no. of electrons

$CH_4 \rightarrow$ no. of $e^- \equiv 6 + 4 \times 1 = 10\ e^-$

$NH_4^+ \rightarrow$ no. of $e^- \equiv 7 + 4 \times 1 - 1 = 10\ e^-$

XeF_5^- IF_5^{2-}



- $Xe \equiv I^{2-} \rightarrow 8e^-$ in valence shell
- Shape → Pentagonal planar
- no. of e^- in $XeF_5^- \equiv 54 + 5 \times 9 + 1 = 100$
- no. of e^- in $IF_5^{2-} \equiv 53 + 5 \times 9 + 2 = 100$

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Hybridisation in Solid State

(1) $2PCl_{5(s)} \longrightarrow PCl_4^+ PCl_6^-$

- hyb^n of cationic part of $PCl_5 = PCl_4^+ = sp^3$ [Tetrahedral]
- hyb^n of anionic part of $PCl_5 = PCl_6^- = sp^3d^2$ [Octahedral]

(2) $PBr_{5(s)} \longrightarrow PBr_4^+ Br^-$

sp^3 hyb^n [Tetrahedral]

(3) $XeF_{6(s)} \longrightarrow XeF_5^+ F^-$

sp^3d^2 [Sq. Pyramidal]

(4) $N_2O_{5(s)} \longrightarrow NO_2^+ \quad NO_3^-$

Cation	Anion
$O=\overset{\oplus}{N}=O$	$\bar{O}-N(=O)-O$
sp hyb^n	sp^2 hyb^n
Linear	Trigonal planar

(5) $N_2O_{3(s)} \longrightarrow NO^+ NO_2^-$

(6) $Cl_2O_{6(s)} \longrightarrow ClO_2^+ ClO_4^-$

$XeOF_5^-$, IOF_5^{2-}

- Pentagonal Pyramidal shape

- (−)ve charge at oxygen
- (−)ve charge at xenon

Both structures are acceptable

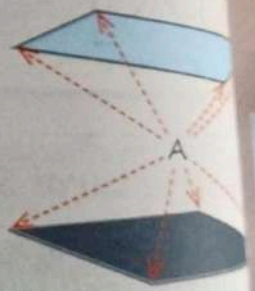
Pentagonal Pyramid

- A pentagonal pyramid has vertices, ten edges, and six fa

Square antiprismatic molecular ge

- TaF_8^{3-}
- IF_8^-
- XeF_8^{2-}

→ AB_8



Ideal square anti

14. Which of the following contains a maximum number of lone pairs on the central atom? (IIT JEE 2

(a) ClO_3^- (b) XeF_4 (c) SF_4 (d) I_3^-

Sol. (d)

15. Which of the following are isoelectronic and isostructural?
NO_3^-, CO_3^{2-}, ClO_3^-, SO_3

(a) NO_3^-, CO_3^{2-} (b) SO_3, NO_3^- (c) ClO_3^-, CO_3^{2-} (d) CO_3^{2-}, SO_3 (IIT JEE 2

Sol. (a)

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16. Among the following species, identify the isostructural pairs.

NF_3, NO_3^-, BF_3, H_3O^+, N_3H (IIT JEE 1996)

(a) [NF_3, NO_3^-] and [BF_3, H_3O^+] (b) [NF_3, HN_3] and [NO_3^-, BF_3]

(c) [NF_3, H_3O^+] and [NO_3^-, BF_3] (d) [NF_3, H_3O^+] and [HN_3, BF_3]

Sol. (c)

17. The sum of the number of lone pairs of electrons on each central atom in the following species is

$[TeBr_6]^{2-}$, $[BrF_2]^+$, SNF_3 and $[XeF_3]^-$

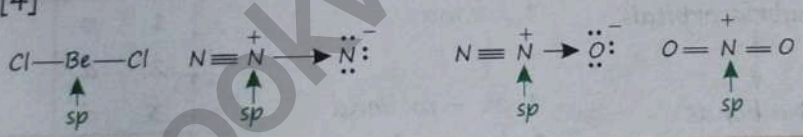
(Atomic numbers : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54) (JEE Adv. 2017)

Sol. [6]

S.N.	Species	No. of σ - bonds with central atom	No. of L.P at central atom
(i)	In $[TeBr_6]^{2-}$	6	1
(ii)	In $[BrF_2]^+$	2	2
(iii)	In SNF_3	4	0
(iv)	In $[XeF_3]^-$	3	3

18. Among the triatomic molecules/ions $BeCl_2$, N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^- and XeF_2, the total number of linear molecules(s)/ion(s) where the hybridisation of the central atom does not have contribution from the d-orbitals(s) is [atomic number of S = 16, Cl = 17, I = 53 and Xe = 54] (JEE Adv. 2015)

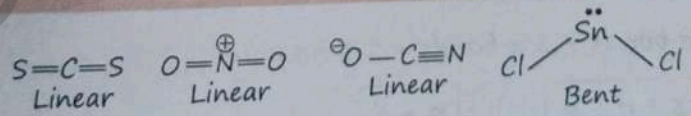
Sol. [4]



19. The linear structure assumed by (IIT JEE 1991)

(a) $SnCl_2$ (b) CS_2 (c) NO_2^+ (d) NCO^-

Sol. (b, c, d)



20. CO_2 is isostructural with (IIT JEE 1986)

(a) $HgCl_2$ (b) C_2H_2 (c) $SnCl_2$ (d) NO_2

Sol. (a, b) CO_2, $HgCl_2$, C_2H_2 are all linear.

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NO. OF pπ - pπ BONDS/NO. OF pπ - dπ BONDS

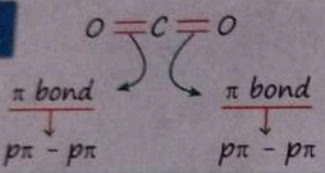
Orbital	π Bond
s	Can not form π bond
p	Can form π bond
d	Can form π bond

Period no = 2	π Bond
s	✗
p	✓
d	not available in 2nd period elements.

Period no > 2	π Bond
s	✗
p	✓
d	✓

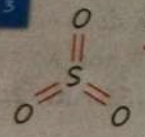
- If central atom or side atom is from period no 2 (B,C,N,O) then only p orbital can make π b
- If central atom or side atom is from period no > 2 (Al, Si, P, S, Cl, Xe) then p and d both can π bond.

CO_2



- overlapping between p orbital of carbon and p orbi of oxygen makes π bond.
- No. of pπ - pπ bonds = 2

SO_3



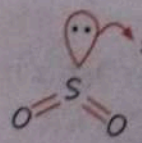
Sulphur (S) → 6 bonds → 3 σ bonds + 3 π bonds

s + p + p + p + d + d

- s + p + p → 3 sp^2 hybrid orbitals → 3 σ bonds
- p + d + d → 3 π bonds → 1 pπ – pπ bond, 2 dπ – pπ bonds

3 π bonds		
	Sulphur	Oxyg
1.	p	p
2.	d	p
3.	d	p

SO_2



Sulphur (S) → 2 σ bonds + 2 π bonds + 1 lone pair

s + p + p + p + d

- s + p + p → 3 sp^2 hybrid orbitals → 2 σ bonds + 1 lone pair
- p + d → 2 π bonds → 1 pπ – pπ bond, 1 dπ – pπ bond

2 π bonds		
	Sulphur	Oxygen
1.	p	p
2.	d	p

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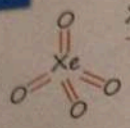
XeO_4



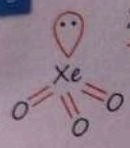
Xenon (Xe) ⟶ 4 σ bonds + 4 π bonds

s + p + p + p + d + d + d + d

- s + p + p + p → 4 sp^3 hybrid orbitals → 4 σ bonds
- d + d + d + d → 4 π bonds → 4 dπ – pπ bonds

4 π bonds	
Xenon	Oxygen
1. d	p
2. d	p
3. d	p
4. d	p

XeO_3



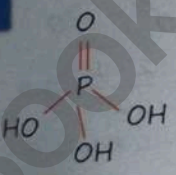
Xenon (Xe) ⟶ 3 σ bonds + 3 π bonds + 1 lone pair

s + p + p + p + d + d + d

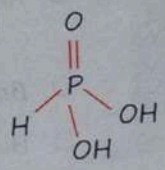
- s + p + p + p → 4 sp^3 hybrid orbitals → 3 σ bonds + 1 lone pair
- d + d + d → 3 π bonds → 3 dπ – pπ bonds

4 π bonds	
Xenon	Oxygen
1. d	p
2. d	p
3. d	p

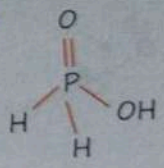
H_3PO_4



H_3PO_3

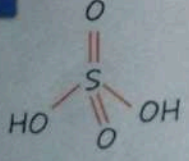


H_3PO_2



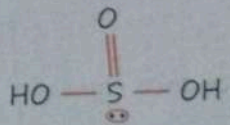
- Hybridisation of P : sp^3
- 1 pπ – dπ bond

H_2SO_4



- Hybridisation of S : sp^3
- 2 pπ – dπ bond

H_2SO_3



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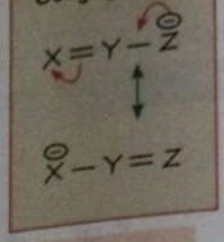
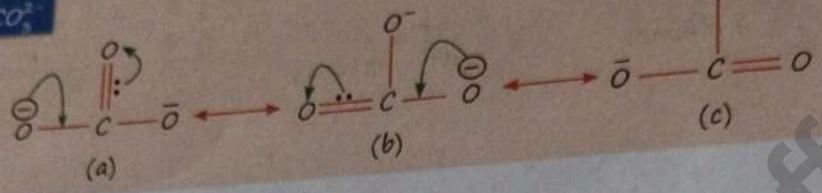
RESONANCE

- Delocalisation of pi electrons
- Conjugation – Necessary condition for resonance

Conjugation

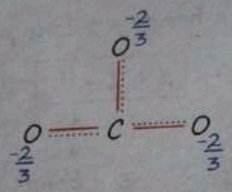


CO_3^{2-}



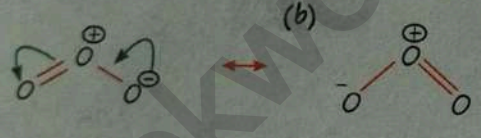
- Resonating structures ≡ a, b, c

♦ Resonance Hybrid ⟶

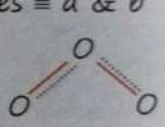


- No. of bonds b/w 2 atoms [Bond order] $= \frac{\sigma + \pi}{\sigma} = \frac{3+1}{3} = 1.33 \rightarrow 1\sigma$ bond $+ \frac{1}{3}\pi$ bonds.
- Formal charge on oxygen ≡ $\frac{-2}{3}$
- Bond length ≡ All C – O bonds are same due to resonance

O_3



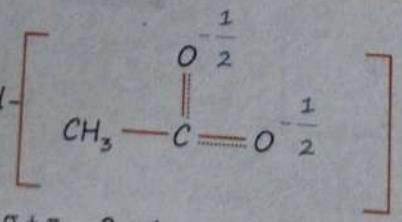
- Resonating structures ≡ a & b
- Resonance Hybrid ≡



- Bond order = $\frac{\sigma + \pi}{\sigma} = \frac{2+1}{2} = 1.5$
- Bond length ≡ All O – O bonds are same

CH_3COO^-

- Resonance Hybrid –



- CO bond order = $\frac{\sigma + \pi}{\sigma} = \frac{2+1}{2} = 1.5$
- Bond length ≡ All C – O bonds are same

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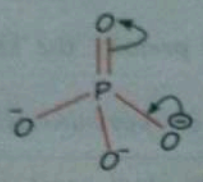
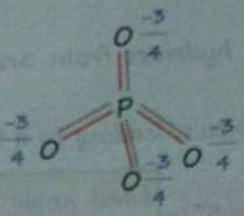
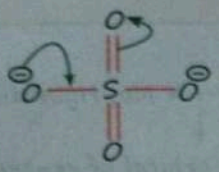
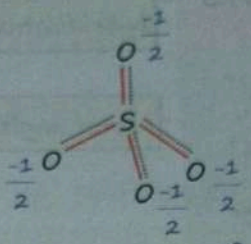
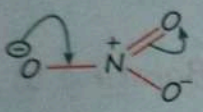
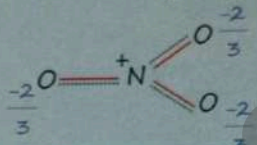
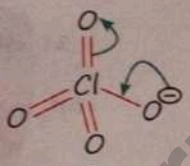
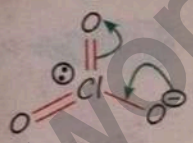
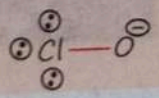
Sachin Jak

	Resonance Hybrid		Bond Order
PO_4^{3-}			$BO = \frac{\sigma+\pi}{\sigma} = \frac{4+1}{4} = 1.25$
SO_4^{2-}			$BO = \frac{4+2}{4} = 1.5$
NO_3^-			$BO = \frac{\sigma+\pi}{\sigma} = \frac{3+1}{3} = 1.33$

	Formal Charge	Bond Order
$HClO_4 \xrightleftharrow{-H^+} ClO_4^-$ 	$\frac{-1}{4}$	$BO = \frac{\sigma+\pi}{\sigma} = \frac{4+3}{4} = 1.75$
$HClO_3 \xrightleftharrow{-H^+} ClO_3^-$ 	$\frac{-1}{3}$	$BO = \frac{3+2}{3} = 1.66$
$HClO_2 \xrightleftharrow{-H^+} ClO_2^-$	$\frac{-1}{2}$	$BO = \frac{2+1}{2} = 1.5$
$HClO \xrightleftharrow{-H^+} ClO^-$ 	-1	$BO = 1$

- Acidity $\propto$ Stability of conjugate base $\propto \frac{1}{\text{formal charge}}$
- Acidic Strength : a > b > c > d

(JEE. Adv 2015)

21. The correct statement(s) regarding,
(i) $HClO$, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$ is (are)
(a) The number of Cl = O bonds in (ii) and (iii) together is two
(b) the number of lone pair of electrons on Cl in (ii) and (iii) together is three
(c) the hybridisation of Cl in (iv) is sp^3
(d) amongst (i) to (iv), the strongest acid is (i)

Sol. (b, c)

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DRAGO COMPOUNDS

- In some cases (like group 15, 16 hydrides from 3rd or higher period), the central atom participate in hybridisation.
- Bonding occurs through pure p orbitals, leading to a bond angle of approximately 90°.

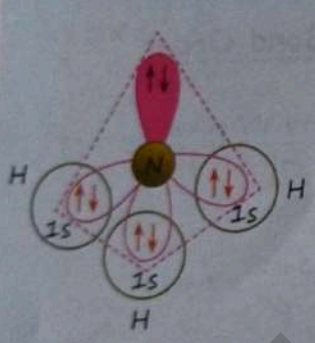
NH_3	107°	H_2O	104.5°
PH_3	93.6°	H_2S	92°
AsH_3	91.8°	H_2Se	91°
SbH_3	91.3°	H_2Te	90°

bond angle θ is near to ideal angle 109°28' of sp^3 hybridisation.

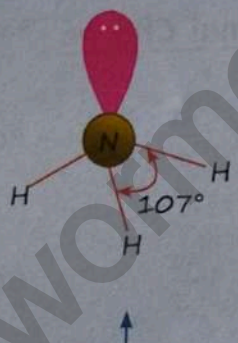
bond angle θ ≈ 90° → No. Hybridisation

Pure p orbital of central atom overlap with 1s atomic orbital of side atom H.

Bonding in NH_3

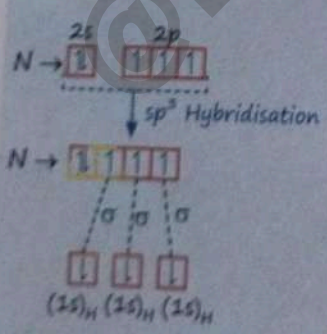


Orbital overlap in NH_3



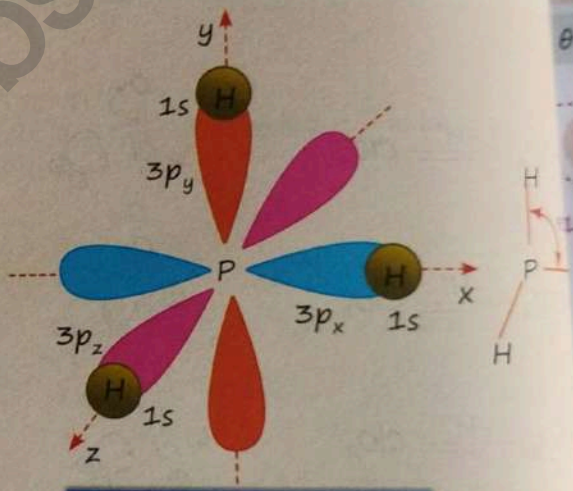
Pyramidal shape

In order to make N – H bonds sp^3 hybrid orbitals of nitrogen overlap with 1s atomic orbital of hydrogens.



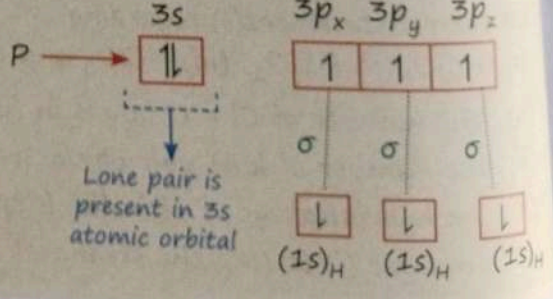
- 3 N – H σ bonds : $(sp^3)_N - (1s)_H$
- 1 lone pair in sp^3 hybrid orbital of nitrogen.

Bonding in PH_3



Orbital overlap in PH_3

In order to make P – H bonds p -atomic orbitals (p_x, p_y, p_z) of phosphorus overlap with 1s atomic orbital of hydrogens.



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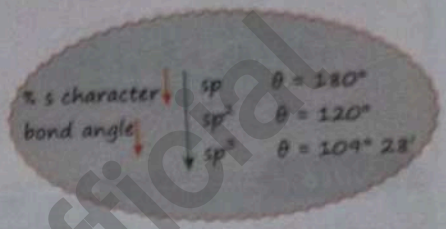
BOND ANGLE

- Angle between two bonds.

Molecules without Lone Pair

Hybridisation → θ : sp > sp^2 > sp^3



θ : CO_2 > CO_3^{2-} > CH_4

sp, sp^2, sp^3

θ : CF_4 = CH_4 = NH_4^+ = XeO_4 = 109°28' [sp^3 hybridisation]

θ : NO_2^+ = CO_2 > SO_3 > XeO_4

sp, sp, sp^2, sp^3

θ : BF_3 = BCl_3 = BBr_3 = 120° [sp^2 hybridisation]

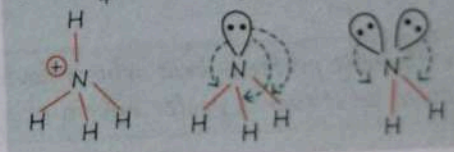
Molecules with Lone Pair

1. If same Hybridisation → No of lone pair ↑ : θ ↓

θ = CH_4 > $\ddot{N}H_3$ > $H_2\ddot{O}$

sp^3 hyb^n with zero lone pair; sp^3 hyb^n with 1 lone pair; sp^3 hyb^n with 2 lone pairs

θ : NH_4^+ > NH_3 > NH_2^-



2. No. of lone pair : same ⇒ Size of CA ↑ : Bond angle ↓

θ : $\ddot{N}H_3$ > $\ddot{P}H_3$ > $\ddot{A}sH_3$ > $\ddot{S}bH_3$

θ : $\ddot{N}Cl_3$ > $\ddot{P}Cl_3$ > $\ddot{A}sCl_3$ > $\ddot{S}bCl_3$

θ : OF_2 > SF_2

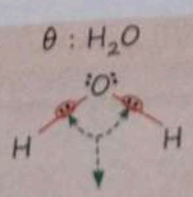
θ : OCl_2 > SCl_2 > $SeCl_2$ > $TeCl_2$

θ : NF_3 > PF_3 > AsF_3 > SbF_3

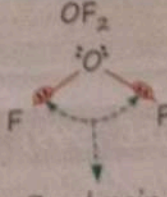
3. No of lone pair : same → E.N. of S.A. ↑ : Bond angle ↓

θ : NH_3 > NF_3

θ : H_2O > OF_2



- Bond pair - bond pair repulsion ↑
- Bond angle ↑
- θ = 104.5°



- Bond pair - bond pair repulsion ↓
- Bond angle ↓
- θ < 104.5°

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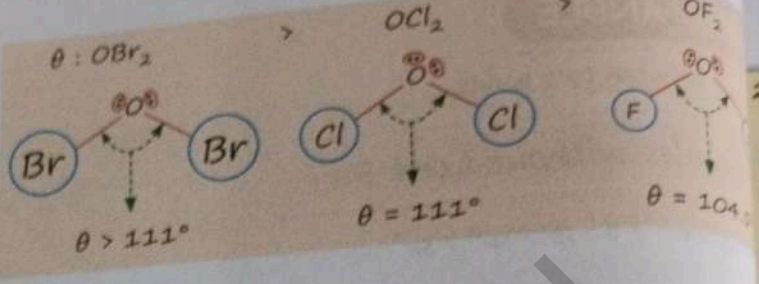
4. No of lone pair : same
↓
Size of S.A.↑ : Bond angle↑

$\theta : \ddot{P}Br_3 > \ddot{P}Cl_3 > \ddot{P}F_3$

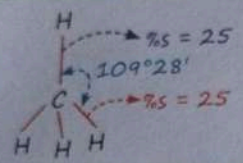
$\theta : OBr_2 > OCl_2 > OF_2$



Note

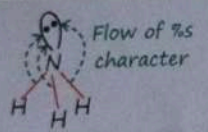
Bent's Rule — % s character ↑ : Bond Angle ↑

1. Lone pair orbitals must adopt a higher s character to stabilise the unshared, tightly b nonbonding electrons.



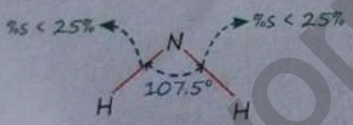
CH_4

- Each C-H bond has 25% s character and bond angle is 109°28'.



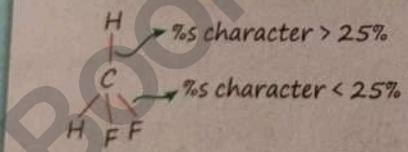
NH_3

- Flow of %s character from N-H bond to lone pair of nitrogen.



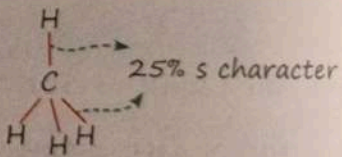
- %s character decreases in each N-H bond. So, bond angle decreases.

2. Electronegative element prefer hybrid orbital having less s character and more p character less electronegative substituents prefer hybrid orbital having more s character.



% s character : C – H bond > C – F bond

Bond angle: ∠HCH > ∠HCF > ∠FCF



∠HCH = 109°28'

$[\angle HCH]_{CH_2F_2} > [\angle HCH]_{CH_4}$

(IIT JEE 1999)

22. In compounds of type ECl_3, where E = B, P, As or Bi, the angles Cl—E—Cl is in order
(a) B > P = As = Bi (b) B > P > As > Bi (c) B < P = As = Bi (d) B < P < As < Bi

Sol. (b)

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23. For OF_2 molecule, consider the following: [30 Jan, 2023 (Shift-I)]

(A) Number of lone pairs on oxygen is 2.
(B) FOF angle is less than 104.5°.
(C) Oxidation state of O is -2.
(D) Molecule is bent 'V' shaped.
(E) Molecular geometry is linear.

Correct options are:

(a) C, D, E only (b) B, E, A only (c) A, C, D only (d) A, B, D only

Sol. (d)

Bond Length

Size ↑ : Bond length ↑

$H-F < H-Cl < H-Br < H-I$: Bond length

$C-F < C-Cl < C-Br < C-I$: Bond length

Bond order ↓ : Bond length ↑

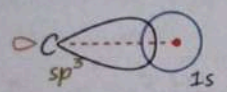
$C \equiv C < C = C < C - C$

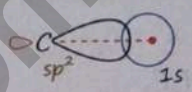
$C \equiv N < C = N < C - N$ → Bond length

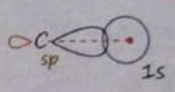
$N \equiv N < N = N < N - N$

%s character↑ : Bond length↓

C-H Bond length : $H-C(sp^3,\ 25\%s)-H$ > $H-C(sp^2,\ 33.3\%s)=C(sp^2,\ 33.3\%s)-H$ > $H-C(sp,\ 50\%s)\equiv C(sp,\ 50\%s)-H$







(a) $H_3C(25)-C(25)H_3$

(b) $H_3C(25)-C(33.3\%)H=CH_2$

(c) $H_3C(25)-C(50\%)\equiv CH$

- % s character : c > b > a
- C–C Bond length : c < b < a

(i) $H_2C=HC(33.3\%)-CH(33.3\%)=CH_2$

(ii) $H_2C=HC(33.3\%)-C(50\%)\equiv CH$

(iii) $HC\equiv C(50\%)-C(50\%)\equiv CH$

- %s character : iii > ii > i
- Bond length : iii < ii < i

Bond length : a > b > c > i > ii > iii

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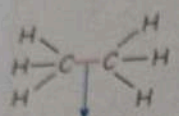
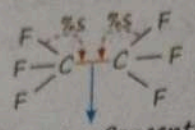
C_2H_6 > C_2F_6 : C–C bond length

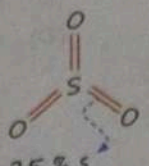


less % s character

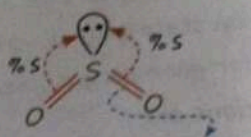


% s character Concentrates in C – C bond

SO_3 < SO_2 : S–O bond length

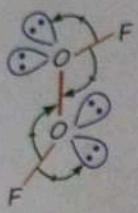


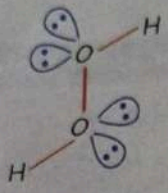
25 % s character



- % s character < 25%
- Less % s character → more bond

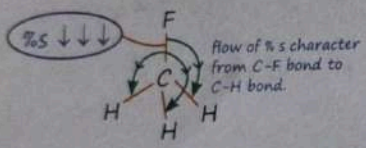
O_2F_2 < H_2O_2 : O–O bond length

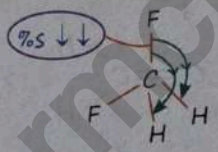


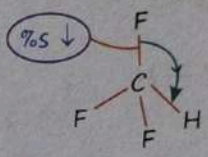


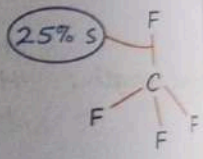
- Order of % s character in O – O bond : $O_2F_2 > H_2O_2$
- O – O bond length : $O_2F_2 < H_2O_2$
- % s character : O – F < O – H
  Bond length : O – F > O – H

C–F Bond Length: CH_3F > CH_2F_2 > CHF_3 > CF_4

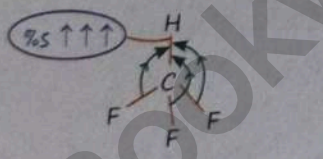


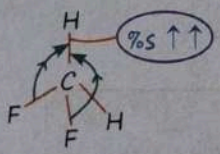


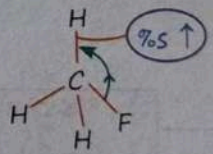


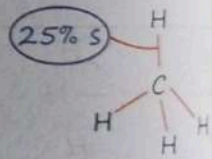


C–H Bond Length: CHF_3 < CH_2F_2 < CH_3F < CH_4

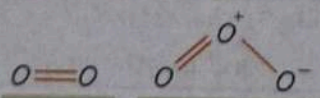






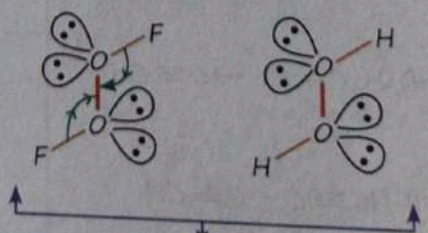


O–O Bond Length: O_2 < O_3 < O_2F_2 < H_2O_2 < Na_2O_2 $[2Na^+ O_2^{2-}]$

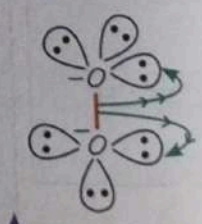


BO = 2

$BO = \frac{\sigma + \pi}{\sigma} = \frac{2+1}{2} = 1.5$



Order of % s character in O–O bond: $O_2F_2 > H_2O_2$
O–O bond length : $O_2F_2 < H_2O_2$



Less % s character in O–O bond in O_2^{2-} than O_2F_2 and H_2O_2

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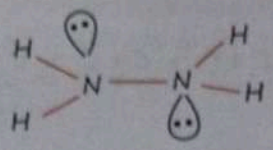
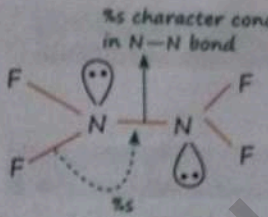
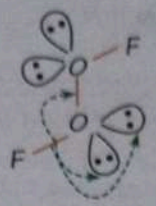
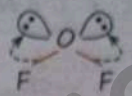
N-N Bond length: $N_2H_4 > N_2F_4$



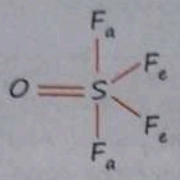


O-F Bond length: $O_2F_2 > OF_2$

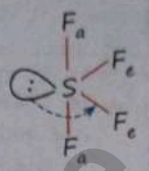


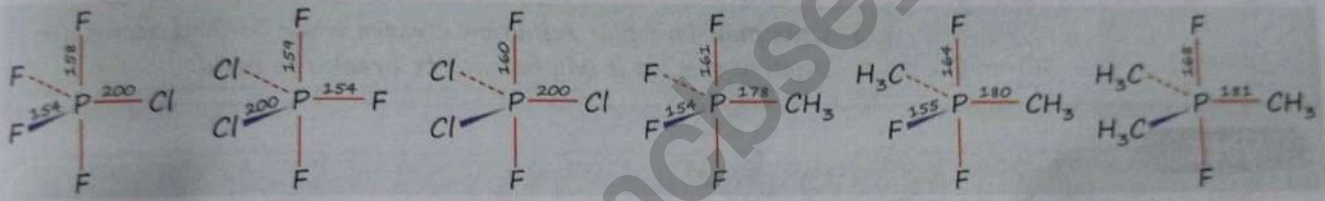


S-F(eq) Bond length:



$SOF_4 > SF_4$

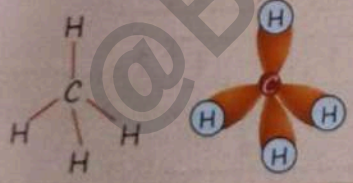
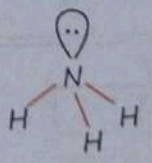
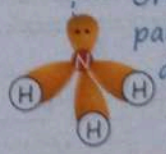




24. O–O bond length in H_2O_2 is X than the O–O bond length in F_2O_2. The O–H bond length in H_2O_2 is Y than that of the O–F bond in F_2O_2.
Choose the correct option for X and Y from the given below. [1 Feb, 2023 (Shift-II)]

(a) X - shorter, Y - shorter
(b) X - shorter, Y - longer
(c) X - longer, Y - longer
(d) X - longer, Y - shorter

Sol. (d)

Equivalent Hybrid Orbitals	Non-equivalent Hybrid Orbitals
 ❑ All 4 hybrid orbitals are equivalent in energy and size.	  Orbital having lone pair is small in size due to more %s character.
❑ Central atom without lone pair and attached with same side atoms Example: XeO_4, $BeCl_2$, BF_3	❑ CA with LP or attached with different side atoms Example: $H_2\ddot{O}$, $\ddot{N}H_3$, CH_2F_2

Bond Dissociation Energy

Minimum energy required to break a bond is known as bond dissociation energy.

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(1) **BDE ∝ Bond Order**

BDE : $C \equiv C > C = C > C - C$ ← Easy to break : BDE ↓

(2) **BDE ↓ : Bond Length ↑** (Size : C > N)

- Bond length : $C \equiv C > C \equiv N > N \equiv N$

BDE : $C \equiv C < C \equiv N < N \equiv N$

- $-\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}- > -\overset{|}{\underset{|}{Si}}-\overset{|}{\underset{|}{Si}}-$

Order of bond length: $C-C < Si-Si$

Bond dissociation energy: $C-C > Si-Si$

(3) **BDE ↓ : LP-LP Repulsion ↑**

$-\ddot{N}-\ddot{N}- < -\ddot{P}-\ddot{P}-$

$-\ddot{O}-\ddot{O}- < -\ddot{S}-\ddot{S}-$

$:\ddot{F}-\ddot{F}: < :\ddot{Cl}-\ddot{Cl}:$

- In these 3 cases, bonded atoms has/have lone pair(s) so, we are going to consider lone pair-lone pair repulsion as a major factor in order to compare the bond dissociation energy.
- Lone pair-lone pair repulsion is more when bonded atoms are small in size, so it will be easy to break the bond.

Note

BDE : $Cl-Cl > Br-Br > F-F > I-I$

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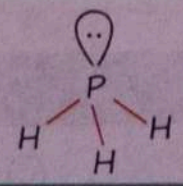
PH_3 HAS NO HYBRIDISATION BUT PF_3 & PH_4^+ HAS ? WHY

Complete or partial +ve charge on central atom supports ORBITAL CONTRACTION, which hybridisation of atomic orbitals.

PH_3

P : | ⇅ | 1 | 1 | 1 |
3s 3p

⇒ P in PH_3 has no hybridisation



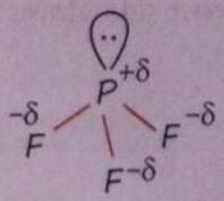
PF_3

$+\delta$ 3s 3p

P : | ⇅ | 1 | 1 | 1 |

| ⇅ | 1 | 1 | 1 |

4 sp^3 hybrid orbitals



⇒ P in PF_3 is sp^3 hybridised

PH_4^+

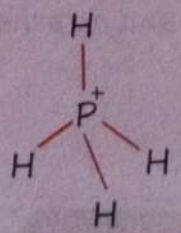
3s 3p

P^+ : | ⇅ | 1 | 1 |

P^+_{es} : | 1 | 1 | 1 | 1 |

| 1 | 1 | 1 | 1 |

4 sp^3 hybrid orbitals



⇒ P in PH_4^+ is sp^3 hybridised

Bond Angle: $PH_3 < PF_3 < PH_4^+$

$[\theta \cong 90°]$ $[\theta > 90°]$ $[\theta = 109°28']$

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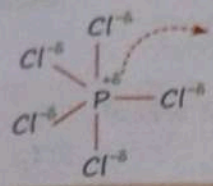
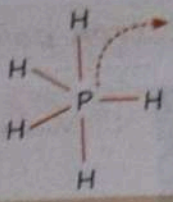
25. PCl_5 exists but not PH_5 ?

Sol. d-orbital contraction is possible in PCl_5 but not in PH_5



Partial (+)ve charge on P helps in d orbital contraction



No charge on P.

26. SF_6 exists but not SH_6 ?

Sol. d-orbital contraction is possible in SF_6 but not in SH_6.

EXISTENCE & NON-EXISTENCE

- 2^{nd} Period elements → B/C/N/O/F → Can have maximum $8e^-$ in valence shell
  MANOJ Yogi
  → max. covalency = 4
  → d orbitals are not available
  → Possible hybridisation are - sp, sp^2, sp^3
- $3^{rd}/4^{th}/5^{th}/6^{th}/7^{th}$ Period elements → Al/Si/P/S etc.
  → d orbitals are available of bonding
  → Possible hybridisation are sp, sp^2, sp^3, sp^3d, sp^3d^2, sp^3d^3
  → These elements can form more than 4 bonds.

NCl_5, BCl_6^{3-} → N and B can not form more than 4 bonds

PCl_5, $AlCl_6^{3-}$ → P and Al can have covalency more than 4.

Note

Molecules That Do Not Exist

(1) SF_4, SF_6 & PF_5, exist while OF_4, OF_6, NF_5 do not exist

(2) (a) PI_5(vap) & SCl_6 do not exist

(b) SCl_6 does not exist while $TeCl_6$ exist due to steric effect.

(c) PI_5 (solid) exist as PI_3 and I_2.

$$PI_{5\ (Solid)} \rightarrow PI_3 + I_2$$

(3) SF_6, PF_5, XeF_6, XeF_4 & XeF_2 exist while SH_6, PH_5, XeH_6, XeH_4, XeH_2 do not exist.

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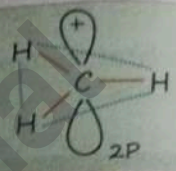
LEWIS BASE AND LEWIS ACID

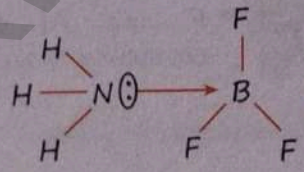
Lewis Acid → Electron pair acceptor

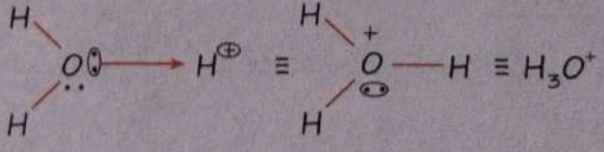
CA having vacant orbital:	(i) Vacant p orbital: BF_3, $B(OH)_3$, $AlCl_3$	(ii) Vacant d orbital: SiF_4, PCl_5
CA with +ve charge having vacant orbital:	(i) Vacant s orbital: H^+ (ii) Vacant p orbital: CH_3^+ (sp^2)	

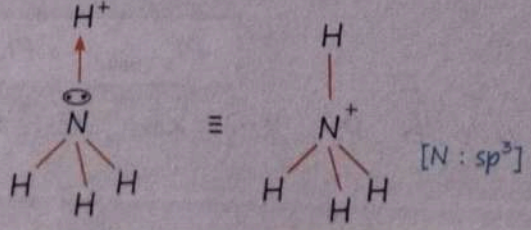
- NH_4^+ ⟶ not a Lewis Acid because there is no vacant orbital in N.

Lewis Base → Electron pair donor

CA with lone pair :	$\ddot{N}H_3$, $\ddot{P}H_3$, $H_2\ddot{O}$
CA with -ve charge :	F^-, Cl^-, OH^-

- $F^- + BF_3 \longrightarrow BF_4^-$; [B : sp^2] ; [B : sp^3]
- $Cl^- + AlCl_3 \longrightarrow AlCl_4^-$; [Al : sp^2] ; [Al : sp^3]
- $\ddot{N}H_3 + BF_3 \longrightarrow$  Adduct ; [N : sp^3] [B : sp^2] ; [N : sp^3] ; [B : sp^3]

- $H_2\ddot{O} + H^+ \longrightarrow$  ; [O : sp^3] ; [O : sp^3]

- $\ddot{N}H_3 + H^+ \longrightarrow$  ; [N : sp^3] ; [N : sp^3]

- $OH^- + B(OH)_3 \longrightarrow$ B(OH ; [B : sp^2] ; [B : sp
- $Cl^- + PCl_5 \longrightarrow PCl_6^-$; [P : sp^3d] ; [P : sp^3d^2]
- $2F^- + SiF_4 \longrightarrow SiF_6^{2-}$; [Si : sp^3] ; [Si → sp^3d^2]

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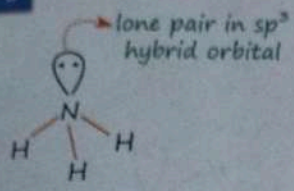
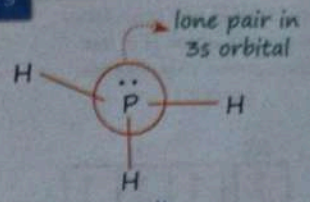
Lewis Basicity NH_3 vs PH_3

NH_3



❑ $H_3N: \xrightarrow{H^+} \underset{K_{b1}}{\rightleftharpoons} NH_4^+$

PH_3



❑ $H_3P: \xrightarrow{H^+} \underset{K_{b2}}{\rightleftharpoons} PH_4$

- $K_{b1} >> K_{b2}$
- Lewis basicity : $NH_3 > PH_3$

Order of Lewis basicity is due to fact that lone pair on N is in sp^3 hybrid orbital which is more diffused and directional. But lone pair on P is present in a 's' orbital which is contracted and non-directional.

(IIT JEE 2002)

27. Specify the coordination geometry around and hybridising N and B atoms in a 1 : 1 complex of BF_3 and NH_3.

(a) N: tetrahedral, sp^3; B: tetrahedral, sp^3 (b) N: pyramidal, sp^3; B: pyramidal, sp^3

(c) N: pyramidal, sp^3; B: planar, sp^2 (d) N: pyramidal, sp^2; B: tetrahedral, sp^2

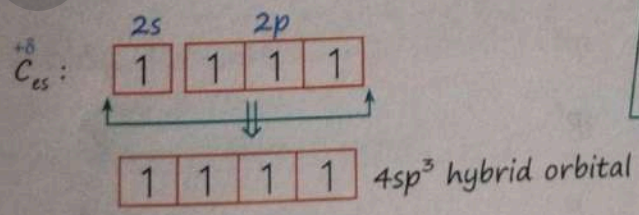
Sol. (a)

HYBRIDISATION IN ODD ELECTRON SPECIES

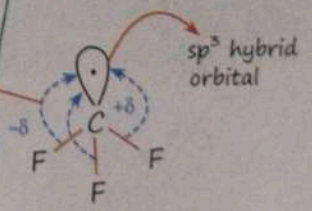
❑ When side atoms are more electronegative than central atom then odd electrons are considered in steric number to find hybridisation.

$\dot{C}F_3$ $SN = 1 + 3 = 4 \Rightarrow sp^3$ hybridisation

(1 ↓ odd electron; 3 → side atoms)



Flow of %s character from C—F bonds to the orbital having an odd electron.



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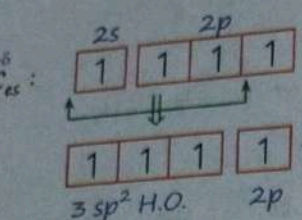
$\dot{C}H_3$

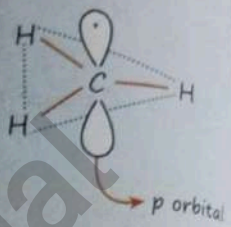
SN = 0 + 3 = 3 ⇒ sp^2 hybridisation

3 → side atoms

[odd electron is not considered because side atom H is less electronegative than central atom C]

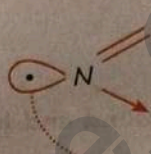


Odd e^- is present in 2p orbital of C



$\dot{N}O_2$

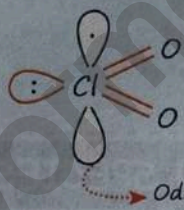
- SN = 1 + 2 = 3
- sp^2 hybridisation



Odd e^- is present in sp^2 hybrid orbital

ClO_2

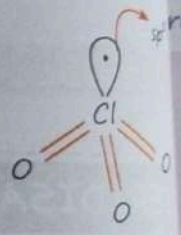
- SN = 1 + 2 = 3
- sp^2 hybridisation



Odd electron in p orbital

$\dot{C}lO_3$

- SN = 1 + 3 = 4 = sp^3



Odd e^- molecule	Orbital having odd electron	hyb^n of central atom
$\dot{C}H_3$	p	sp^2
$\dot{C}F_3$	sp^3	sp^3
$\dot{N}O_2$	sp^2	sp^2
$\dot{C}lO_3$	sp^3	sp^3
$\dot{C}lO_2$	p	sp^2

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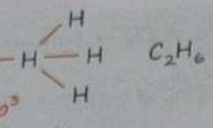
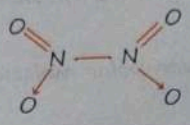
Dimer Formation- Hybridisation Change & Magnetic Behaviour

2 unit

- $\dot{C}H_3 + \dot{C}H_3 \longrightarrow C_2H_6$
  sp^2 (Paramagnetic) → sp^3 (Diamagnetic)



- $NO_2 + NO_2 \longrightarrow N_2O_4$
  sp^2 (paramagnetic) → sp^2 (Diamagnetic)



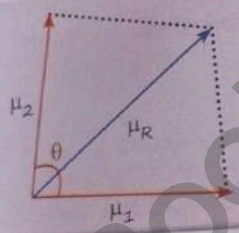
DIPOLE MOMENT

Product of the magnitude of the charge and the distance between the centers of the positive and negative charges.

Dipole moment = charge . distance of separation

Dipole moment $\mu = q \cdot l$

Dipole moment is a vector quantity when 2 vectors (μ_1 & μ_2) are separated by θ then we can calculate resultant μ_R.



- $\mu_R^2 = \mu_1^2 + \mu_2^2 + 2\mu_1 \cdot \mu_2 \cos\theta$
- $\cos\theta \uparrow : \mu_R \uparrow$
- $\theta \uparrow : \cos\theta \downarrow : \mu_R \downarrow$

Unit ≡ C . m
≡ esu . cm
≡ Deby [D]
C → Coulomb
m → metre
esu → Electrostatic unit of charge

1 Debye = 10^{-18} esu cm
Charge on e^- → $q_e = 1.6 \times 10^{-19}$ C
$= 4.8 \times 10^{-10}$ esu

Polar Molecule ($\mu_R \neq 0$)	Non-polar Molecule ($\mu_R = 0$)
Hetero-nuclear diatomic	Homonuclear diatomic
Molecule having 2 different atoms	Molecule having 2 same atoms
HCl $H^{+\delta}-Cl^{-\delta}$ (Polar bond)	H_2, F_2, O_2, N_2 $F-F$ (Non polar bond)

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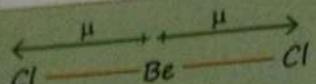
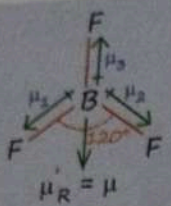
Symmetry and Dipole Moment

A polyatomic molecule having polar covalent bonds and symmetrical structure indicates the zero moment of the molecule.



- $BeCl_2$ → Symmetrical molecule
- $\mu_{BeCl_2} = 0$



- BF_3 → Symmetrical molecule
- $\mu_{BF_3} = 0$
- BF_3 → Non polar molecule

$\mu_R'^2 = \mu_1^2 + \mu_2^2 + 2\mu_1.\mu_2 \cos 120°$

$\mu_R' = \mu \quad [\mu_1 = \mu_2 = \mu]$

→ μ_R' is cancelled by $\mu_3 (= \mu)$.

Symmetrical Molecule

- Symmetrical molecules are non-polar molecules.
- $(\mu_R)_{molecule} = 0$

1. Molecule having no lone pair on central atom and having same side atoms.

CO_2, CS_2, $BeCl_2(g)$

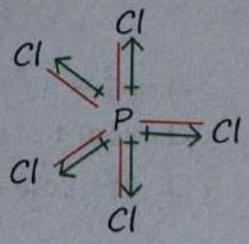
O = C = O

S = C = S

Cl — Be — Cl

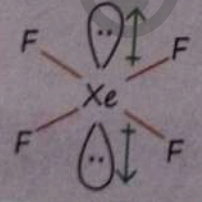
→ Linear $(\mu_R)_{mol} = 0$

BF_3, CH_4, PCl_5, SF_6, IF_7 → $\mu_R = 0$ → Non-Polar mol



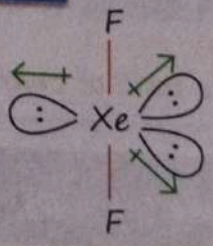
2. Molecule having lone pairs on central atom but in such a way that they can cancel each others moment.

XeF_4



- Non polar molecule
- $\mu_R = 0$

XeF_2



Unsymmetrical Molecule

Molecule having lone pairs on central atom but in such a way that they can not cancel each others moment.

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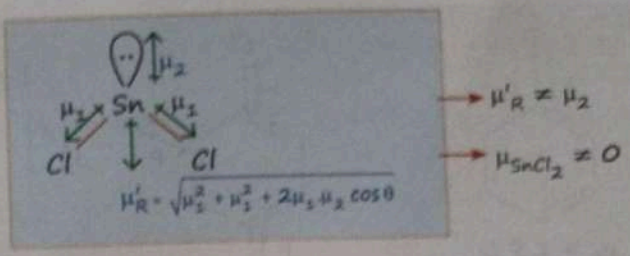
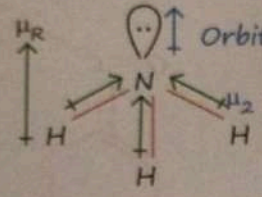
$\ddot{S}nCl_2$, $\ddot{P}bCl_2$, $\ddot{S}O_2$: Angular mol. : Polar

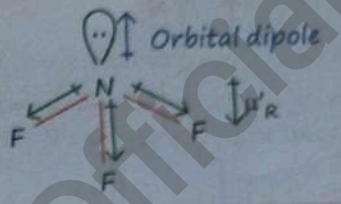
$\ddot{N}H_3$, $H_2\ddot{O}$, $\ddot{N}F_3$, $\ddot{S}F_4$, $H_2\ddot{S}$: : Polar molecule



NH_3 > NF_3 : Dipole moment



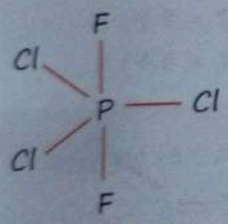
In case of NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N – H bonds.



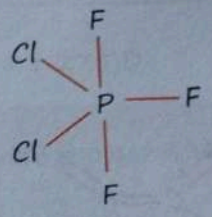
In NF_3, the orbital dipole is in the direction opposite to the resultant dipole moment of the three N–F bonds.

The orbital dipole because of lone pair decreases the effect of the resultant N – F bond moments, which results in the low dipole moment of NF_3.

- PCl_3F_2 < PCl_2F_3

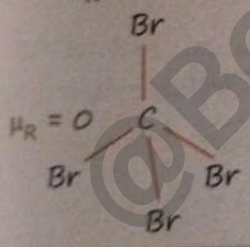


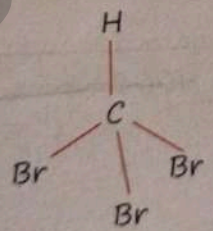
$\mu_R = 0$ (Non Polar)

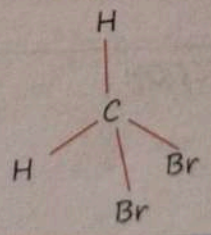


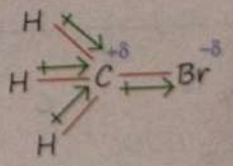
$\mu_R \neq 0$ (Polar molecule)

- μ_R → CBr_4 < $CHBr_3$ < CH_2Br_2 < CH_3Br

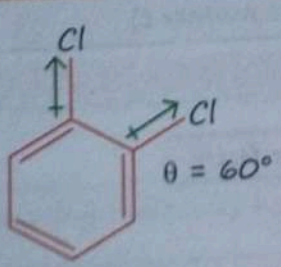


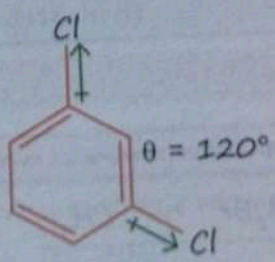


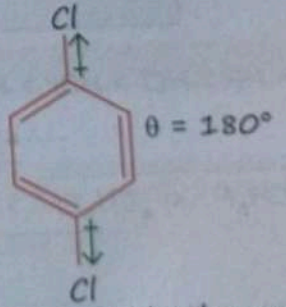




- μ_R ↑ : θ ↓ : cosθ ↑







μ_R : ortho – dichlorobenzene > meta-dichlorobenzene > para-dichlorobenzene

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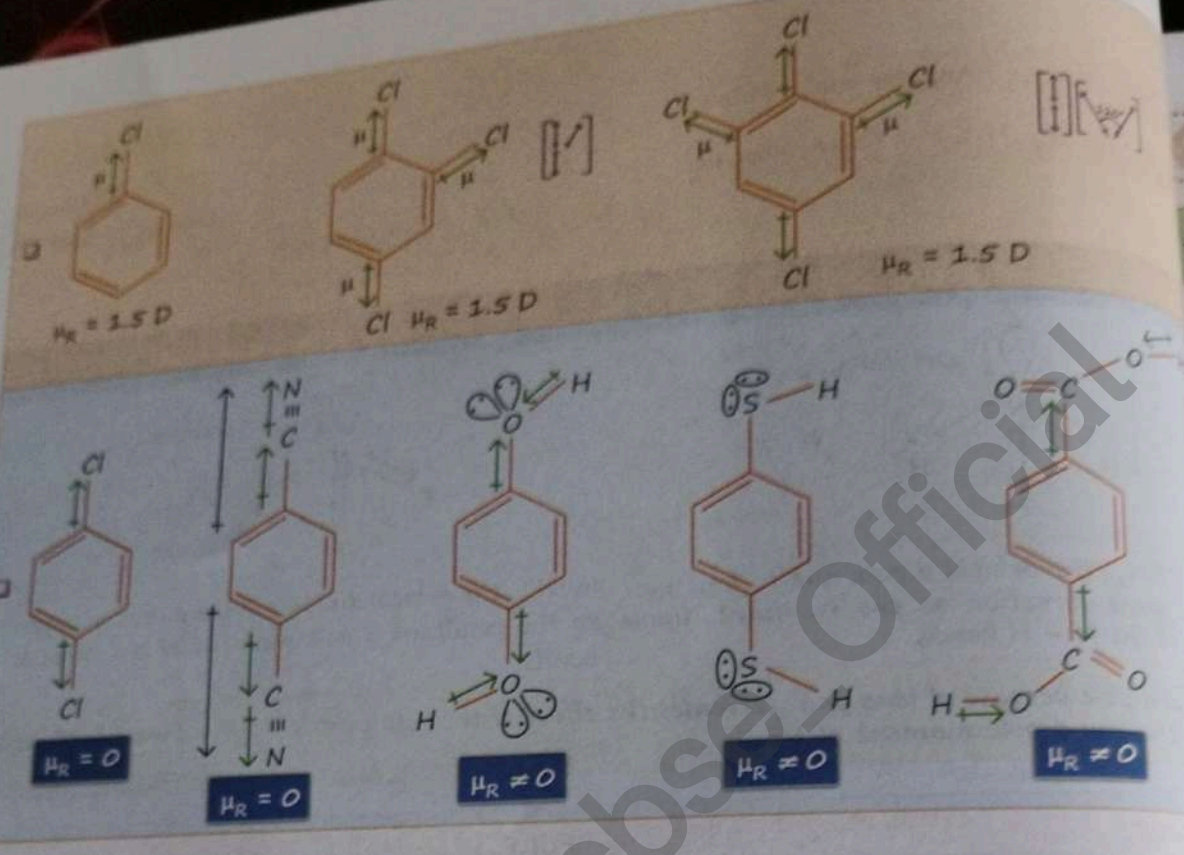
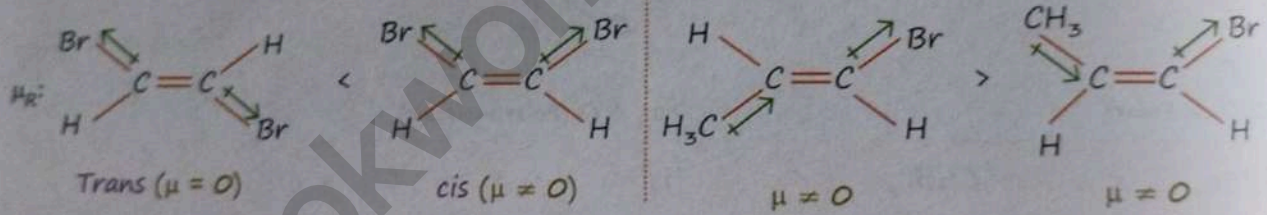
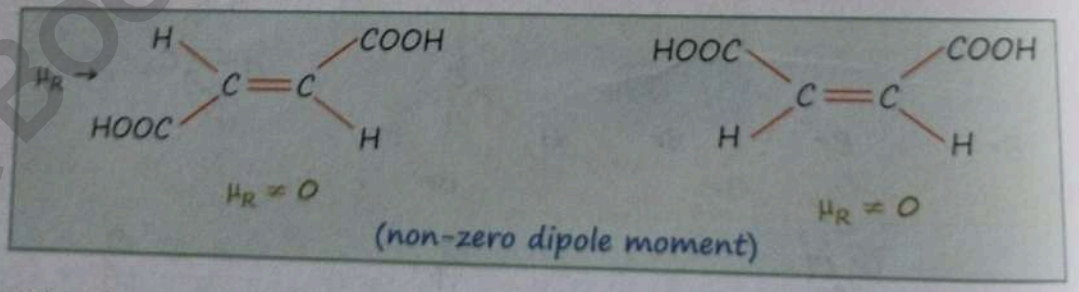


Cis & Trans





$\mu_R \longrightarrow HF > HCl > HBr > HI$

$\mu_R \longrightarrow CH_3F < CH_3Cl > CH_3Br > CH_3I$

length factor dominates over charge factor.

charge factor dominates over length factor.

$CH_3 - Cl > CH_3F > CH_3Br > CH_3I$

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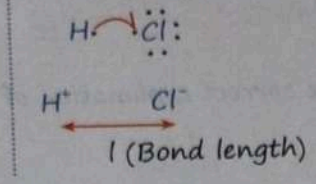
% Ionic Character in Covalent Bond

$$\% \text{ ionic character} = \frac{\mu_{(exp/obs)}}{\mu_{(cal/theo)}} \times 100$$

$\mu_{experiment}$ = given ← $\overset{+\delta}{H} - \overset{-\delta}{Cl}$ → $\mu_{calculated} = q.l = (4.8 \times 10^{-10}$ esu). Bond length



❑ For 100% ionic character–
It is assumed that 1 electron is completely transferred from one atom to another.

28. For HCl gas molecule μ = 1.03 D and bond distance = 1.275 Å. Calculate the % ionic character.

Sol. $\mu_{exp} = 1.03$ D

$l = 1.275 \times 10^{-8}$ cm

$\mu_{cal} = ?$

❑ $\mu_{cal} = q_e \cdot l$

$= (4.8 \times 10^{-10}$ esu$) \cdot (1.275 \times 10^{-8}$ cm$)$

$= (4.8 \times 1.275) \cdot (10^{-18}$ esu $-$ cm$) = (4.8 \times 1.275)$D

$$\% \text{ Ionic character} = \frac{1.03}{(4.8 \times 1.275)} \times 100\%$$

$$= 0.1683 \times 100 = 16.83\%$$

29. Among the following, the molecule with the highest dipole moment is (IIT JEE 2003)

(a) CH_3Cl (b) CH_2Cl_2 (c) $CHCl_3$ (d) CCl_4

Sol. (a)

30. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. (JEE Adv. 2019)

(a) SO_2, C_6H_5Cl, H_2Se, BrF_5

(b) $BeCl_2$, CO_2, BCl_3, $CHCl_3$

(c) NO_2, NH_3, $POCl_3$, CH_3Cl

(d) BF_3, O_3, SF_6, XeF_6

Sol. (a, c)

31. Which one of the following molecules has maximum dipole moment? [04 April, 2024 (Shift-I)]

(a) NF_3 (b) CH_4 (c) NH_3 (d) PF_5

Sol. (c)

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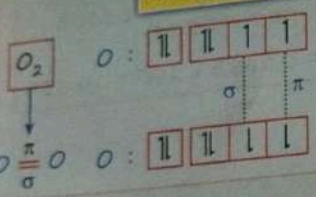
32. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3.
Reason (R): In NH_3, the orbital dipole moment of NH_3 is greater than that of NF_3. The dipole moment of the N–H bonds. F is the most electronegative element.
In the light of the above statements, choose the correct answer from the options given be [05 April, 2024 (Shift

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) (A) is false but (R) is true
(c) (A) is true but (R) is false
(d) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

Sol. (a)

MOLECULAR ORBITAL THEORY

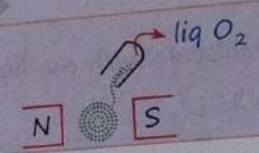
According to VBT



- All electrons are paired.
- O_2 is a diamagnetic molecule

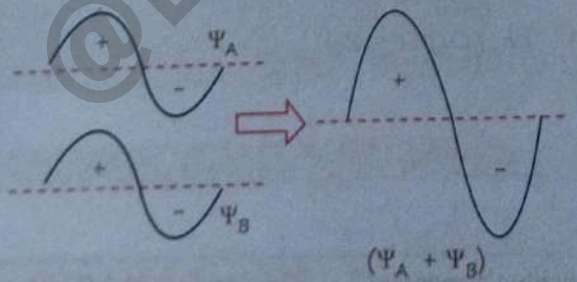
Need of new theory

- Experimental fact → (i) O_2 is a paramagnetic molecule
  (ii) O_2 has 2 unpaired electrons

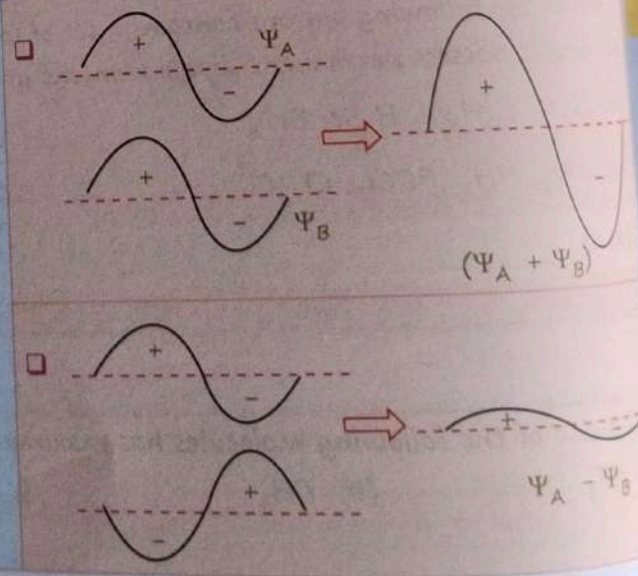


- Interaction between liq. O_2 and magnetic field.

Valence Bond Theory



Molecular Orbital Theory



- VBT & MOT both are quantum mechanical theories of chemical bonding.

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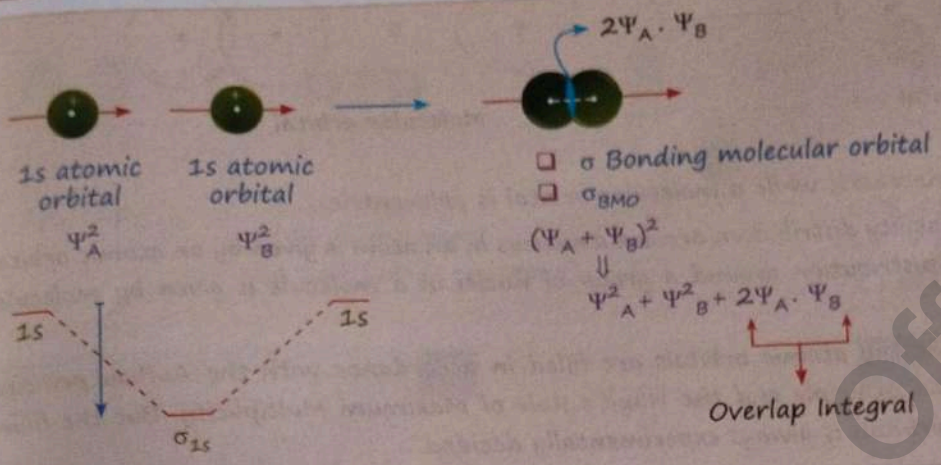
VBT

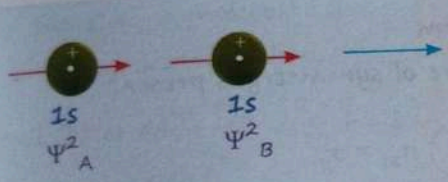


- σ Bonding molecular orbital
- σ_{BMO}

$$(\Psi_A + \Psi_B)^2$$
$$\Downarrow$$
$$\Psi^2_A + \Psi^2_B + 2\Psi_A \cdot \Psi_B$$

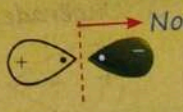
Overlap Integral

MOT



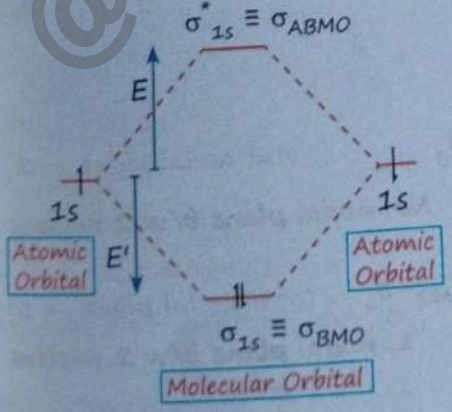
$$(\Psi_A + \Psi_B)^2 = \Psi^2_A + \Psi^2_B + 2\Psi_A \cdot \Psi_B$$

♦ Energy ↓ : Because electron density ↑ between the nucleus, so force of attraction ↑.



$$(\Psi_A - \Psi_B)^2 = \Psi^2_A + \Psi^2_B - 2\Psi_A \cdot \Psi_B$$

♦ Energy ↑ : Because of no electron density between the nucleus, so force of repulsion ↑ between 2 nucleus



$E > E'$

Energy Order: $\sigma_{1s} > \sigma^*_{1s}$

Electronic configuration: $\sigma^2_{1s}\sigma^{*0}_{1s}$

Bond Order = $\frac{N_b - N_{ab}}{2}$

[N_b = no. of electrons in bonding molecular orbital
N_{ab} = no. of electrons in anti bonding molecular orbital]

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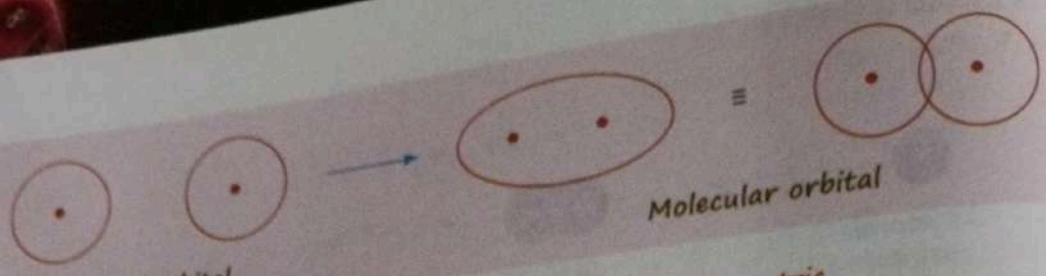
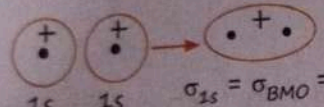


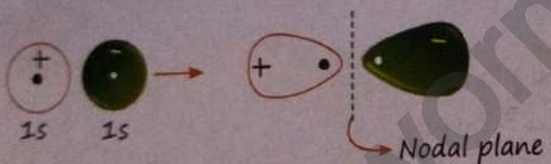
- An atomic orbital is **monocentric** while a molecular orbital is **polycentric**.
- Just as the electron probability distribution around a nucleus in an atom is given by an atomic orb... the electron probability distribution around a group of nuclei in a molecule is given by mole... orbital.
- The molecular orbitals like the atomic orbitals are filled in accordance with the **Aufbau prin...** obeying the **Pauli Exclusion principle** and the **Hund's Rule** of Maximum Multiplicity. But the ... order of these molecular orbitals is always experimentally decided.
- To construct such molecular orbitals we need to combine the atomic orbitals of the atoms that ... up the molecule. This approach is known as the Linear Combination of Atomic Orbitals [LCAO].

➢ **s-s overlapping**



[No, nodal plahe]

$\sigma_{1s} = \sigma_{BMO} = \sigma_g$

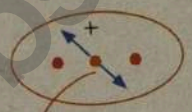


$\sigma^*_{1s} = \sigma_{ABMO} = \sigma_u$

[1 nodal plane]

➢ **Symmetry Term**

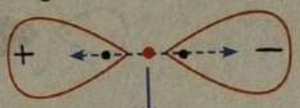
- Gerade : Centre of symmetry is present



$\sigma_{1s} = \sigma_g$

COS is present

- Ungerade: COS is not present

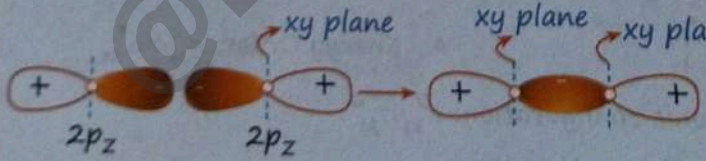


$\sigma^*_{1s} = \sigma_u$

COS is not present

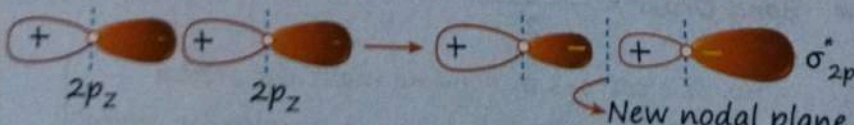
COS : A molecular orbi... has a centre of symme... when, for any molecula... orbital in the molecule ... identical lobes (phase ... + or -) exists diametri... opposite to this centr... an equal distance.

➢ **p-p overlapping**



$\sigma_{2p_z} = \sigma_{BMO} = \sigma_g$ [Total nodal plane →

No, nodal plane b/w 2 nucle...



$\sigma^*_{2p_z} \equiv \sigma_{ABMO} = \sigma_u$ [Total nodal plane →

1, nodal plane b/w 2 nucle...

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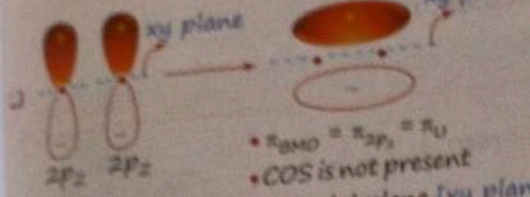
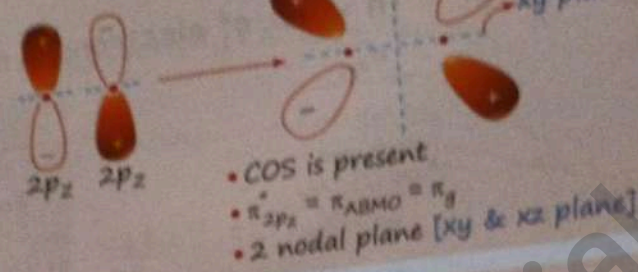
- p-p overlapping

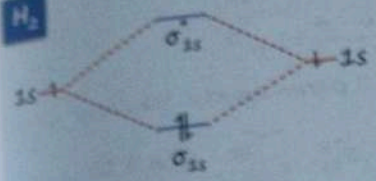


- $\pi_{BMO} = \pi_{2p_z} = \pi_u$
- COS is not present
- 1 nodal plane [xy plane]



- COS is present
- $\pi^*_{2p_z} = \pi_{ABMO} = \pi_g$
- 2 nodal plane [xy & xz plane]

H_2

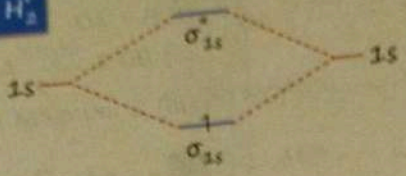


No. of e^- in molecule : 2

Electronic Configuration: σ_{1s}^2

Bond Order $= \frac{N_b - N_{ab}}{2} = \frac{2-0}{2} = 1$

H_2^+

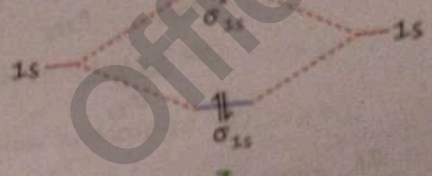


1

σ_{1s}^1

$BO = \frac{1-0}{2} = 0.5$

H_2^-



3

$\sigma_{1s}^2\ \sigma_{1s}^{*1}$

$BO = \frac{2-1}{2} = 0.5$

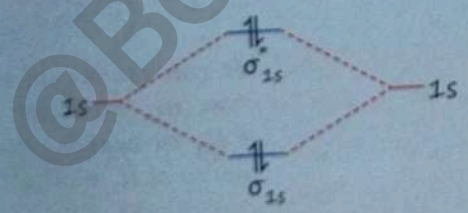
Bond Length: $H_2 < H_2^+ < H_2^-$

Magnetic Behaviour: Diamagnetic Paramagnetic Paramagnetic

Note

- If bond order is same → $N_{ab}\uparrow$: BL↑

He_2



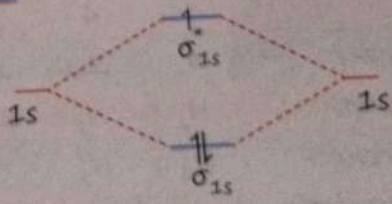
No. of e^- in a molecule → 4

Electronic configuration → $\sigma_{1s}^2 \sigma_{1s}^{*2}$

Bond Order → $\frac{N_b - N_{ab}}{2} = \frac{2-2}{2} = 0$

Existence → He_2 molecule does not exist because He atoms are not bonded to each other.

He_2^+



3

$\sigma_{1s}^2\ \sigma_{1s}^{*1}$

$BO = \frac{2-1}{2} = 0.5$

He_2^+ exists

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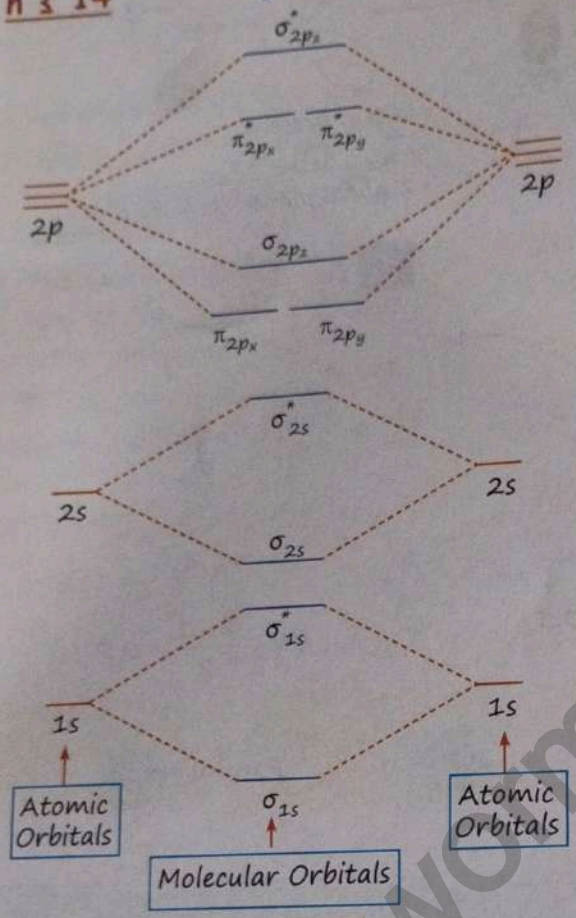
Electronic Configuration

n ≤ 14 [n → no. of electrons in molecules, ions]



Energy Increases

- Energy Order: $\sigma_{1s}\ \sigma^*_{1s}\ \sigma_{2s}\ \sigma^*_{2s}\ \begin{bmatrix}\pi_{2p_x}\\ \pi_{2p_y}\end{bmatrix}\ \sigma_{2p_z}\ \begin{bmatrix}\pi^*_{2p_x}\\ \pi^*_{2p_y}\end{bmatrix}\ \sigma^*_{2p_z}$
- Due to s-p mixing energy order: $\pi_{2p} < \sigma_{2p}$

B_2 → $\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \begin{bmatrix}\pi^1_{2p}\\ \pi^1_{2p}\end{bmatrix}$

(i) n ≡ 10

(ii) BO ≡ $\frac{N_b - N_a}{2} = \frac{6-4}{2} = 1$

(iii) 2 unpaired e^- ≡ paramagnetic behaviour

C_2 → $\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \begin{bmatrix}\pi^2_{2p}\\ \pi^2_{2p}\end{bmatrix}$

(i) n ≡ 12

(ii) BO ≡ 2

(iii) Diamagnet

❑ 4 electrons are present in π_{BMO} which indicates the presence of two π bonds between carbon atoms with no σ bond.

N_2 → $\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \begin{bmatrix}\pi^2_{2p}\\ \pi^2_{2p}\end{bmatrix}\ \sigma^2_{2p}$

(i) n ≡ 14

(ii) BO ≡ 3

(iii) Diamagnetic

(iv) HOMO → Highest occupied MO → σ_{2p}

(v) LUMO → Lowest unoccupied MO → π^*_{2p}

	N_2	N_2^+	N_2^-
N_b	10	9	10
N_{ab}	4	4	5
Bond Order	3	2.5	2.5
Bond Length	N_2 <	N_2^+	< N_2^-
Bond Strength	N_2 >	N_2^+	> N_2^-
Bond Dissociation Energy	N_2 >	N_2^+	> N_2^-
Magnetic Behaviour	Dia	Para	Para
Unpaired e^- is present in	–	σ_{2p}	π^*_{2p}

N_2 —Remove 1 e^- from σ_{2p}→ **N_2^+**

Energy Order → $\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \begin{bmatrix}\pi^2_{2p}\\ \pi^2_{2p}\end{bmatrix}\ \sigma^*_{2p}$

❑ We can remove an electr from σ_{2p} molecular orbi of N_2 by applying ionisat energy of N_2 molecule.

N_2 —Add 1 e^- in π^*_{2p}→ **N_2^-**

Energy Order: $\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \begin{bmatrix}\pi^2_{2p}\\ \pi^2_{2p}\end{bmatrix}\ \sigma^2_{2p}\ \begin{bmatrix}\pi^{*1}_{2p}\\ \pi^{*0}_{2p}\end{bmatrix}$

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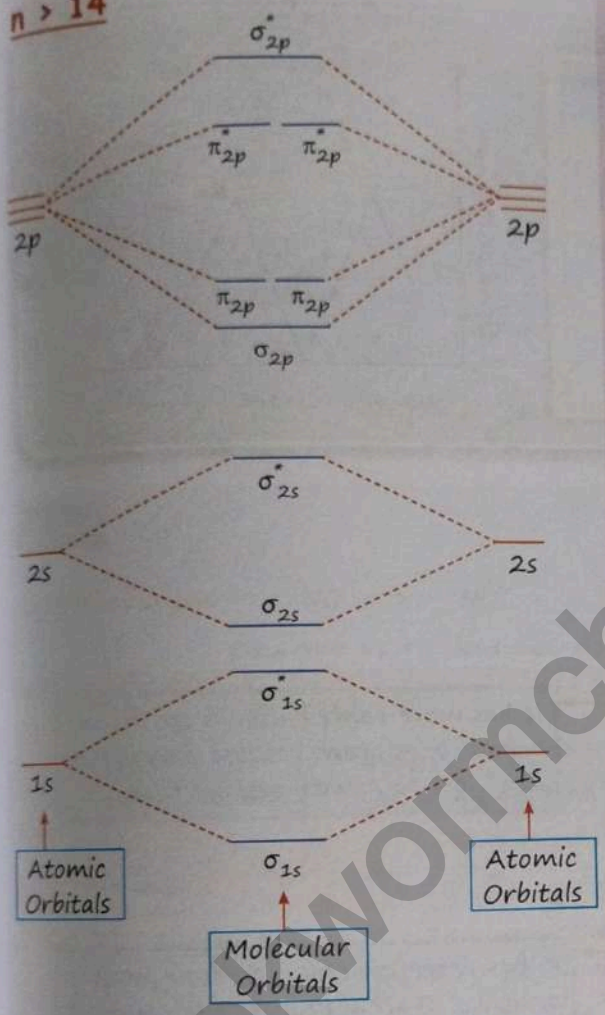
n > 14



Energy increases →

- Energy Order: $\sigma_{1s}\ \sigma^*_{1s}\ \sigma_{2s}\ \sigma^*_{2s}\ \sigma_{2p_z}\ \begin{bmatrix}\pi_{2p_x}\\ \pi_{2p_y}\end{bmatrix}\ \begin{bmatrix}\pi^*_{2p_x}\\ \pi^*_{2p_y}\end{bmatrix}\ \sigma^*_{2p_z}$
- No, s-p mixing ⟶ Energy: $\sigma_{2p} < \pi_{2p}$

$F_2 \longrightarrow \sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \sigma_{2p}^2\ \begin{bmatrix}\pi_{2p}^2\\ \pi_{2p}^2\end{bmatrix}\ \begin{bmatrix}\pi_{2p}^{*2}\\ \pi_{2p}^{*2}\end{bmatrix}$

(i) n ≡ 18

(ii) $BO \equiv \frac{10-8}{2} = 1$

(iii) HOMO ⟶ Highest occupied MO ⟶ π^*_{2p}

(iv) LUMO ⟶ Lowest unoccupied MO ⟶ σ^*_{2p}

(v) Colour ⟶ Pale yellow [HOMO to LUMO electronic transition is the reason behind colour]

(vi) Magnetic Behaviour : Diamagnetic

- F_2, Cl_2, Br_2 and I_2, all 4 are diamagnetic in nature and are colored (due to transition of electron from HOMO to LUMO ≡ π* to σ*).

	O_2^{2+}	O_2^+	O_2	O_2^-	O_2^{2-}
N_b	10	10	10	10	10
N_{ab}	4	5	6	7	8
Bond order	3	2.5	2	1.5	1
Bond length	$O_2^{2+} <$	$O_2^+ <$	$O_2 <$	$O_2^- <$	O_2^{2-}
Bond strength	$O_2^{2+} >$	$O_2^+ >$	$O_2 >$	$O_2^- >$	O_2^{2-}
Bond energy	$O_2^{2+} >$	$O_2^+ >$	$O_2 >$	$O_2^- >$	O_2^{2-}
Magnetic Behaviour	Dia	Para	Para	Para	Dia
No. of Unpaired e^-	0	1	2	1	0

$O_2^{2+} \xleftarrow{-e^-} O_2^+ \xleftarrow{-e^-} O_2 \xrightarrow{+e^-} O_2^- \xrightarrow{+e^-} O_2^{2-}$

Electronic Configuration of O_2

$\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \sigma_{2p}^2\ \begin{bmatrix}\pi_{2p}^2\\ \pi_{2p}^2\end{bmatrix}\ \begin{bmatrix}\pi^{*1}_{2p}\\ \pi^{*1}_{2p}\end{bmatrix}$

(i) n = 16

(ii) Bond order = 2

(iii) 2 unpaired electrons are present in π^*_{2p} ⟶ O_2 is paramagnetic in nature

- We can remove an electron from π^*_{2p} molecular orbital of O_2 by applying 1st ionisation energy of O_2 molecule.

$$O_2 \xrightarrow{IE_1} O_2^+$$

- S_2 molecule is also paramagnetic with 2 unpaired electrons as O_2 molecule.

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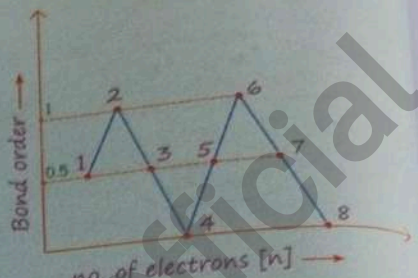
Note

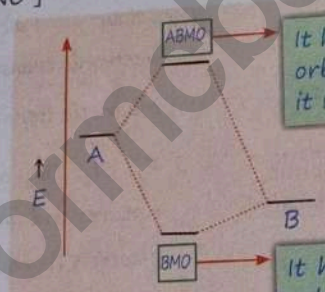
n	Bond Order	n
14	3	15
13	2.5	16
12	2	17
11	1.5	18
10	1	19
9	0.5	20
8	0	



MOT for Heteronuclear Diatomic

AB molecule Type [CO, NO, CN^-, NO^+]

- Electronegativity: A < B
- Energy of same orbital: A > B



It has more contribution from atomic orbital of A atom because energywise it is close to A with respect to B.

It has more contribution from atomic orbital of B atom because energywise it is close to B with respect to A.

Isoelectronic Species

N_2 ($7+7=14e^-$) | CO ($6+8=14e^-$) | NO^+ ($7+8-1=14e^-$) | CN^- ($6+7+1=14e^-$) → Bond order = 3

N_2^- ($14+1=15e^-$) | NO ($7+8=15e^-$) → Bond order = 2.5

Box-1a

	BaO_2	O_2	NaO_2
	$Ba^{2+}\ O_2^{2-}$ ($18e^-$)	($16e^-$)	$Na^+\ O_2^-$ ($17e^-$)
Bond Order →	1	2	1.5

Bond Length → $O_2^{2-} > O_2^- > O_2$ [O – O]

Box-1b O_2 ($16e^-$) | O_3 | $O_2[PtF_6] \rightarrow O_2^+[PtF$

O_2^+ ($15e^-$)

2 $BO = \frac{\sigma+\pi}{\sigma} = \frac{2+1}{2} = 1.5$ 2.5 ← Bond Order

$O_3 > O_2 > O_2^+$: Bond Len

Box-2

No. of e^- in oxygen c

Bond Order → 1.5

Bond Length → O [O – O]

Box-3

Bond Order →

C–O Bond Leng

Box-4

Bond Length

33. Assuming

(a) Be_2

Sol. (c) Assu

(a) Be_2

(b) B_2

(c) C_2

(d) N_2

34. The co

(a)

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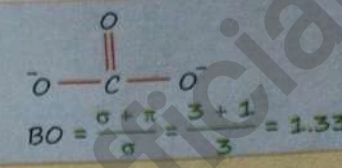
Box-2

KO_2 $[K^+O_2^-]$	O_2	$O_2[BF_4] \rightarrow O_2^+[BF_4]^-$
No. of e^- in oxygen containing species → O_2^- (16+1)	16	O_2^+ (16-1)
Bond Order → 1.5	2	2.5
Bond Length → $O_2^+ < O_2 < O_2^-$ [O – O]		

Box-3

CO (6+8 = 14)	CO_2	CO_3^{2-}
Bond Order → 3	2	1.33



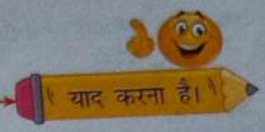
$BO = \frac{\sigma + \pi}{\sigma} = \frac{3+1}{3} = 1.33$

C–O Bond Length → $CO < CO_2 < CO_3^{2-}$

Box-4

- Bond Length: NO (Bond order = 2.5) > NO^+ (Bond order = 3)

$CO > CO^+$: Bond length



33. Assuming 2s-2p mixing is not operative, the paramagnetic species among the following is (JEE Adv. 2014)

(a) Be_2 (b) B_2 (c) C_2 (d) N_2

Sol. (c) Assuming that no 2s – 2p mixing takes place the MO electronic configuration can be written as

(a) $Be_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2$ (diamagnetic)

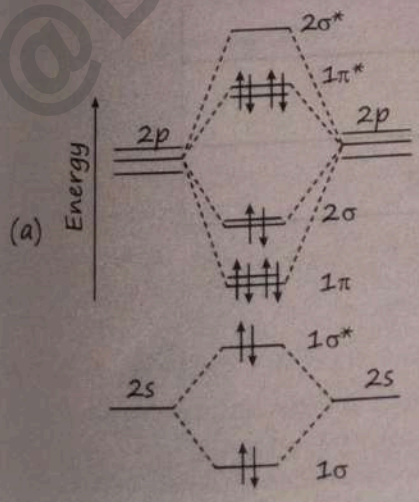
(b) $B_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2$, (diamagnetic)

(c) $C_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^1, \pi 2p_y^1$ (paramagnetic)

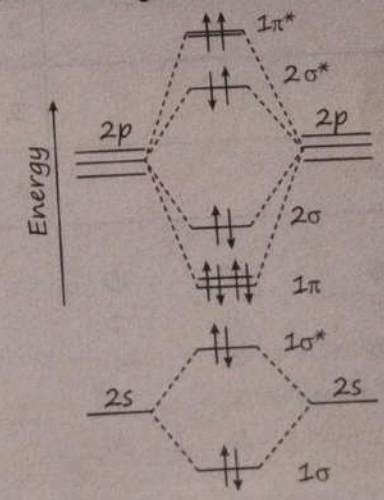
(d) $N_2 \rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2$ (diamagnetic)

34. The correct molecular orbital diagram for F_2 molecule in the ground state is (JEE Adv. 2023)

(a)



(b)



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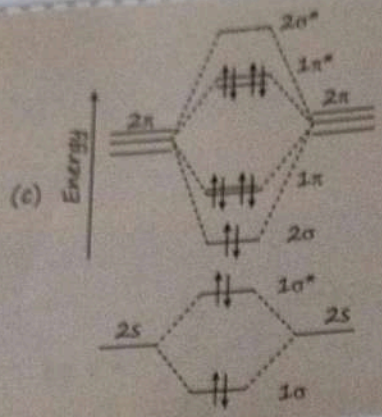
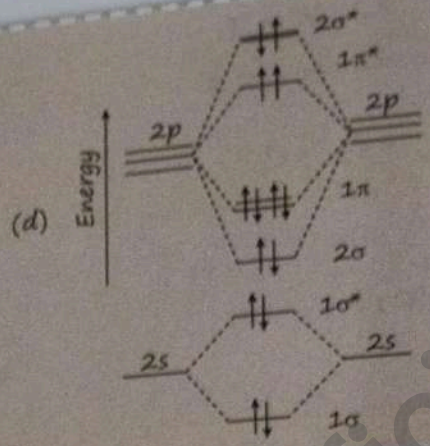




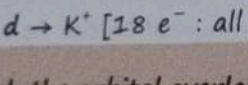
Sol. (c)

35. The species having bond order different from that in CO is- (IIT JEE 200
(a) NO^- (b) NO^+ (c) CN^- (d) N_2

Sol. (a)

36. Among the following, the paramagnetic compound is (IIT JEE 200
(a) Na_2O_2 (b) O_3 (c) N_2O (d) KO_2

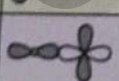
Sol. (d) a → $2Na^+$ [10 e^- : all paired], O_2^{2-} [18 e^- : all paired],

b → O_3  All electrons are paired

c → N_2O [N≡N⟶O] All electrons are paired

d → K^+ [18 e^- : all paired], O_2^- [17 e^- : 1 unpaired]

37. Match the orbital overlap figures shown in Column-I with the description given in Column-II a select the correct answer using the codes given below the Columns. (JEE Adv. 20

Column-I		Column-II	
A.		1.	p–d π antibonding
B.		2.	d–d σ bonding
C.		3.	p–d π bonding
D.		4.	d–d σ antibonding

Codes:

	A	B	C	D
(a)	4	3	2	1
(b)	1	2	3	4
(c)	2	3	1	4
(d)	4	1	2	3

Sol. (c)

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38. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is(are) (JEE Adv 2022)

(a) σ orbital has a total of two nodal planes.

(b) σ* orbital has one node in the xz-plane containing the molecular axis.

(c) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.

(d) π* orbital has one node in the xy-plane containing the molecular axis.

Sol. (a, d)

39. According to molecular orbital theory, which of the following statements is(are) correct? (JEE Adv. 2016)

(a) C_2^{2-} is expected to be diamagnetic

(b) O_2^{2+} is expected to have a longer bond length than O_2

(c) N_2^+ and N_2^- have the same bond order

(d) He_2^+ has the same energy as two isolated He atoms

Sol. (a, c)

40. Among H_2, He_2^+, Li_2, Be_2, B_2, C_2, N_2, O_2^- and F_2, the number of diamagnetic species is (Atomic numbers : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9) (JEE Adv. 2017)

Sol. [6] (H_2, Li_2, Be_2, C_2, N_2, F_2) are diamagnetic.

41. During the change of O_2 to O_2^-, the incoming electron goes to the orbital: [10 April, 2019 (Shift-I)]

(a) σ^*2p_z (b) $\pi 2p_y$ (c) π^*2p_x (d) $\pi 2p_x$

Sol. (c)

42. In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic? [9 Jan, 2019 (Shift-II)]

(a) $NO \rightarrow NO^+$ (b) $N_2 \rightarrow N_2^+$ (c) $O_2 \rightarrow O_2^+$ (d) $O_2 \rightarrow O_2^{2-}$

Sol. (a)

$NO \longrightarrow NO^+$	
B.O-2.5	3
Para	Dia

$N_2 \longrightarrow N_2^+$	
B.O-3	2.5
Dia	Para

$O_2 \longrightarrow O_2^+$	
B.O-2	2.5
Para	Para

$O_2 \longrightarrow O_2^{2-}$	
B.O-2	1
Para	Dia

43. AX is a covalent diatomic molecule where A and X are second row elements of periodic table. Based on molecular orbital theory, the bond order of AX is 2.5. The total number of electrons in AX is _____. (Round off to the nearest integer). [18 March, 2021 (Shift-I)]

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Sol. [15] Given: AX ⇒ Diatomic molecule
Bond order is 2.5
According to the given data, the compound is NO.
Total number of electrons in it = 15
(∵ Number of electrons in N = 7, Number of electron in O = 8)

44. Amongst the following, the number of oxide(s) which are paramagnetic in nature is
Na_2O, KO_2, NO_2, N_2O, ClO_2, NO, SO_2, Cl_2O (Nearest Integer) [27 July, 2022 (Shift-

Sol. [4] Paramagnetic species = KO_2, NO_2, ClO_2, NO [27 July, 2022 (Shift-

45. Match List-I with List-II.

List-I		List-II	
A.	$\psi_{MO} = \psi_A - \psi_B$	I.	Dipole moment
B.	$\mu = Q \times r$	II.	Bonding molecular orbital
C.	$\frac{N_b - N_a}{2}$	III.	Anti-bonding molecular orbital
D.	$\psi_{MO} = \psi_A + \psi_B$	IV.	Bond order

Choose the correct answer from the options given below
(a) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(b) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(c) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(d) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Sol. (c)

46. The linear combination of atomic orbitals to form molecular orbitals takes place only when t combining atomic orbitals

A. have the same energy
B. have the minimum overlap
C. have same symmetry about the molecular axis
D. have different symmetry about the molecular axis

Choose the most appropriate from the options given below: [31 Jan, 2024 (Shift-
(a) A, B, C only (b) A and C only (c) B, C, D only (d) B and D only

Sol. (b)

47. Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of M Theory, by removal of an electron? [25 June, 2022 (Shift-

(A) NO (B) N_2 (C) O_2 (D) C_2 (E) B_2

Choose the most appropriate answer from the options given below:
(a) (A), (B), (C) only
(b) (B), (C), (E) only
(c) (A), (C) only
(d) (D) only

Sol. (c)

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48. Given below are two statements:

Statement-I: A π bonding MO has lower electron density above and below the inter-nuclear axis.

Statement-II: The π* antibonding MO has a node between the nuclei.

In the light of the above statements, choose the most appropriate answer from the options given below:

[01 Feb, 2024 (Shift-II)]

(a) Both Statement-I and Statement-II are false

(b) Both Statement-I and Statement-II are true

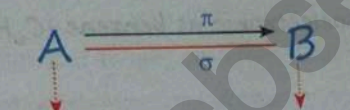
(c) Statement-I is false but Statement-II is true

(d) Statement-I is true but Statement-II is false

Sol. (c) A pi bonding molecular orbital exhibits higher electron density both above and below the internuclear axis.

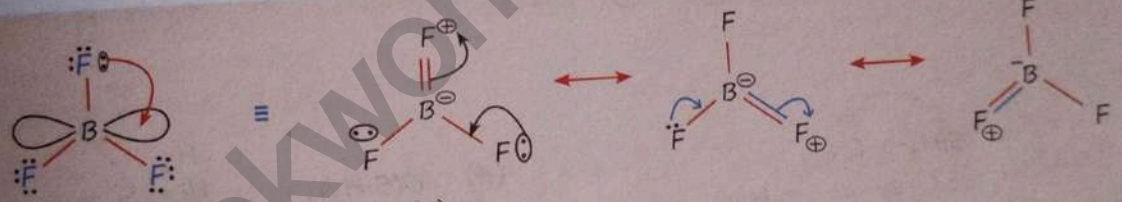
BACK BONDING

- Coordinate pi bond



non-bonding e^- pair donor for co-ordinate pi bond

e^- pair acceptor having vacant orbital

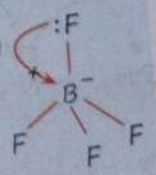


Back bond → π bond (Pπ – Pπ)

- Back bonding increases the bond strength and decreases the bond length.

B—F Bond length: $BF_3 < BF_4^-$

No back bonding



Condition for Excellent Back Bonding

1. One atom from 2nd period
2. Second atom from 2nd/3rd period

π bond strength [back bond strength] : 2p – 2p > 2p – 3d > 2p – 3p > 3p – 3p

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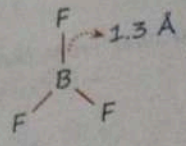
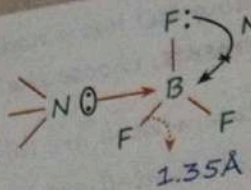
Structure

49. The B – F bond length in $Me_3N.BF_3$ is 1.35 Å, much longer than 1.30 Å in BF_3. Explain?

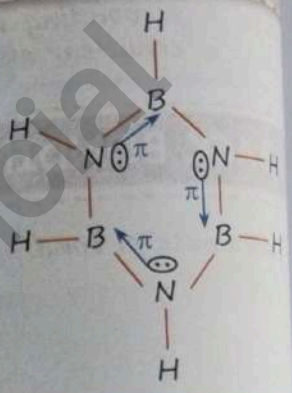
Sol.



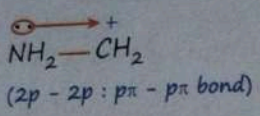


- back bonding from F to B → small B – F bond
- No back bonding → large B – F bond

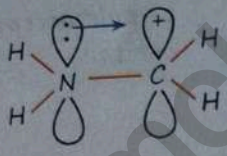
Inorganic Benzene ($B_3N_3H_6$)

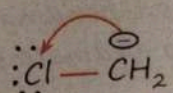


- Back bonding from N to B.
- 3π bonds [6π electrons]
- All nitrogen and boron are sp^2 hybridised.
- Molecule is planar.
- It is an aromatic compound and isoelectronic as benzene [C_6H_6].

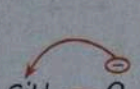
(1) 

(2p – 2p : pπ – pπ bond)

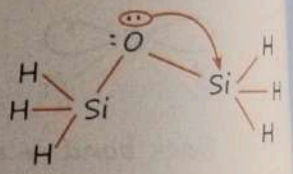


(2) 

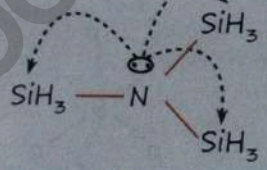
(3d – 2p : dπ – pπ bond)

(3) 

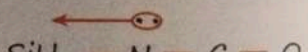
(3d – 2p : dπ – pπ bond)

(4) $O(SiH_3)_2$ 

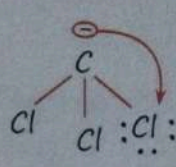
(dπ – pπ back bonding)

(5) 

(pπ – dπ back bonding)

(6) 

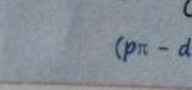
(dπ – pπ back bonding)

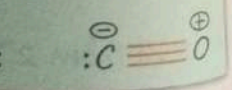
(7) 

(pπ – dπ back bonding)

(8)

CO Molecule





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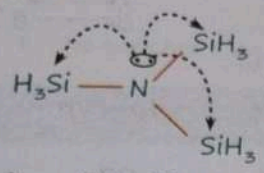
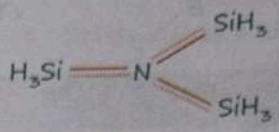
(JEE Adv. 2005)

50. Trisilylamine is a planar molecule and does not act as a Lewis base while trimethylamine is pyramidal and act as Lewis base !

Sol. $N(SiH_3)$ → Trisilylamine

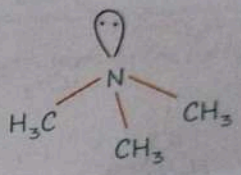




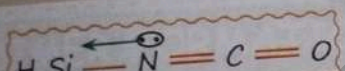
- Back bonding from N to Si
- Lone pair on nitrogen is delocalised as it is participating in back bonding with Si, so does not act as lewis base.
- Steric no = LP + SA = 0 + 3 = 3
- sp^2 hybridisation (planar molecule)

$N(CH_3)_3$ → Trimethylamine

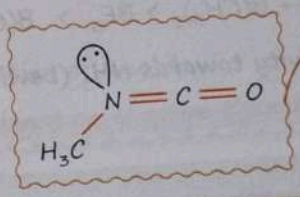
- No back bonding
- Lone pair is localised on nitrogen, so act as lewis base
- Steric No = LP + SA = 1 + 3 = 4
- sp^3 hybridisation (pyramidal shape)



51. Silyl isocyanate (SiH_3NCO) is linear but methyl isocyanate (CH_3NCO) is bent explain!

Sol. 

- Steric number of N : LP + SA = $\underset{\text{LP}}{0} + \underset{\text{SA}}{2} = 2$
- sp hyb^n : Linear
- Back bonding from N to Si



- No back bonding
- SN ≡ 1 + 2 ≡ 3
- sp^2 hyb^n
- Bent shape

Lewis Basicity

$N(SiH_3)_3$ <	$P(SiH_3)_3$ <	$N(CH_3)_3$: Lewis Basicity
❑ Lone pair on nitrogen is not available for donation due to back bonding, so it is less basic.	❑ Lone pair on P is present in a pure s orbital (near to the nucleus), so donation of e^- pair is tough.	❑ Lone pair on N is present in sp^3 hybrid orbital (which is directional in nature). sp^3 hybrid orbital has only 33.3 %s character. ❑ Donation of e^- pair is easy from sp^3 hybrid orbital w.r.t. pure s orbital.

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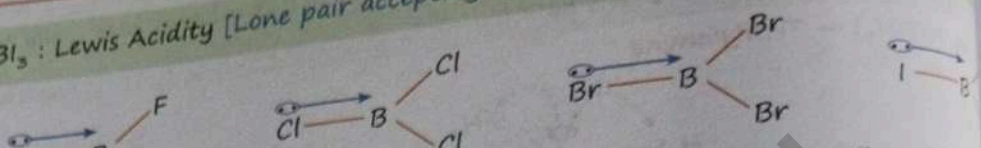
Lewis Acidic Character

$BF_3 < BCl_3 < BBr_3 < BI_3$: Lewis Acidity [Lone pair accepting tendency]



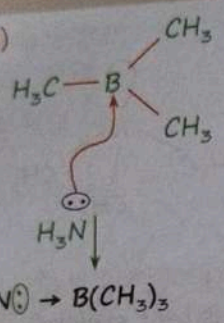
Extent of overlapping: 2p - 2p > 3p - 2p > 4p - 2p > 5p - 2p
(Back Bonding)

Lewis acidity: BF_3 < BCl_3 < BBr_3 < BI_3

52. Find the order of reactivity towards NH_3 in following molecules?

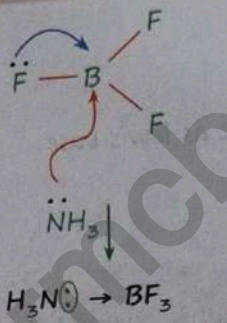
(a) $B(CH_3)_3$ (b) BF_3 (c) $B(OH)_3$ (d) $B(NH_2)_3$

Sol. (a)



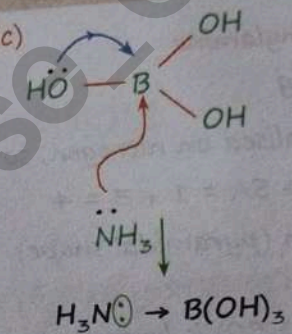
$H_3N: \rightarrow B(CH_3)_3$

(b)



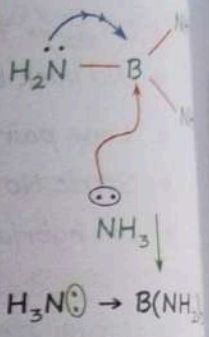
$H_3N: \rightarrow BF_3$

(c)



$H_3N: \rightarrow B(OH)_3$

(d)

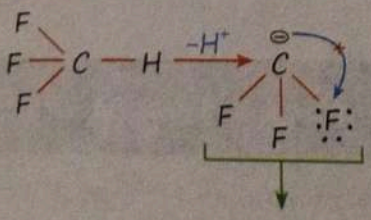


$H_3N: \rightarrow B(NH_2)_3$

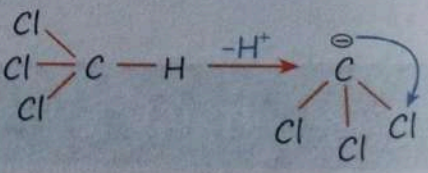
- Extent of back bonding → $B(CH_3)_3 < BF_3 < B(OH)_3 < B(NH_2)_3$
- Lewis Acidity → $B(CH_3)_3 > BF_3 > B(OH)_3 > B(NH_2)_3$
- Order of reactivity towards NH_3 (Lewis base): a > b > c > d (Rate of reaction)

Acidic Nature

Fluoroform CHF_3 < Chloroform $CHCl_3$



- No back bonding
- Carbanion is stabilised by -I effect of F.



- Conjugate base is stable due to back bonding

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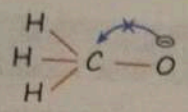
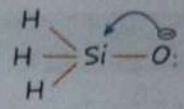
Acidic Nature

Methanol CH_3OH < Silanol SiH_3-OH



♦ No back bonding

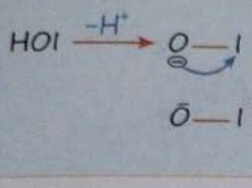
<



Stability of conjugate base

♦ Stable due to back bonding

HOI < HOBr < HOCl



Stability of conjugate base

(IIT JEE. 1996)

53. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order.

ClOH (I), BrOH (II), IOH (III)

(a) I > II > III (b) II > I > III (c) III > II > I (d) I > III > II

Sol. (a)

DIMER FORMATION

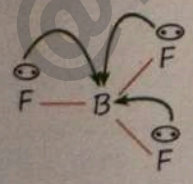
mer ≡ Unit,

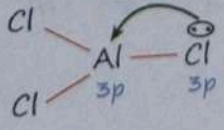
mono-mer ≡ Single Unit

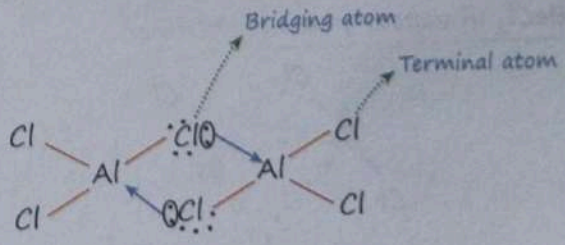
Dimer ≡ Combination of 2 unit

Poly-mer ≡ Combination of more than 2 unit

BF_3 vs $AlCl_3$







- monomer is stable due to excellent back bonding [2p - 2p]
- Monomer is not stable because back bonding from Cl to Al [3p - 3p] is not so good
- Dimer formation takes place through bridge bonding to attain stability
- $[AlCl_3]_2 \equiv Al_2Cl_6$

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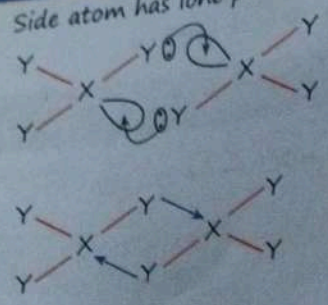
BRIDGE BONDING

Type-1

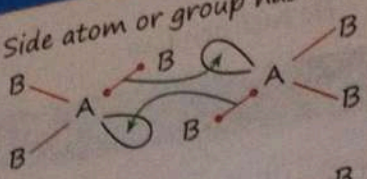
- Side atom has lone pair (Br, Cl)

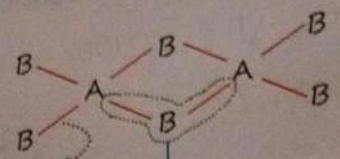


- All bonds have $2e^-$ in it
  [2 centre - $2e^-$ bond]

Type-2

- Side atom or group has no lone pair (-H, $-CH_3$, $-C_6H_5$)

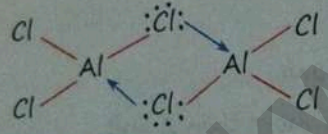
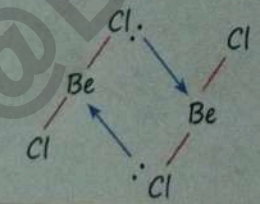
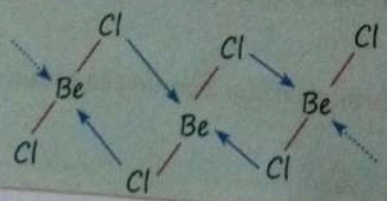




$2C - 2e^-$ bond

- Banana bond
- $3C - 2e^-$ bond
- Electron deficient bond

Type-1

Type-1	Hybridisation	Max. No. of atoms in same plane
Al_2Cl_6 	Al – Cl → Al $sp^3 - sp^3 - sp^3$	6 [2Al , 4Cl] Terminal
$BeCl_2$ at high temperature → Monomer Cl – Be – Cl	Be → sp (Linear)	3
$BeCl_2$ in vapour phase → Dimer 	❑ Be → sp^2 (planar) ❑ Be – Cl → Be $sp^2 - sp^3 - sp^2$	6
$BeCl_2$ in solid phase → Polymer 	❑ Be → sp^3 (Tetrahedral) ❑ Be – Cl → Be $sp^3 - sp^3 - sp^3$	–

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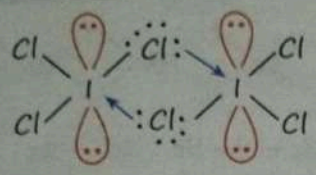
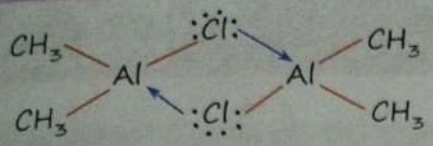
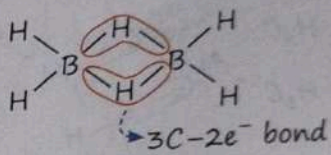
I_2Cl_6 in solid phase → Planar molecule 	❑ I − Cl − I $sp^3d^2 - sp^3 - sp^3d^2$	8 [2I and 6Cl]
$Al_2Cl_2(CH_3)_4$ 	Al − Cl → Al $sp^3 - sp^3 - sp^3$	—

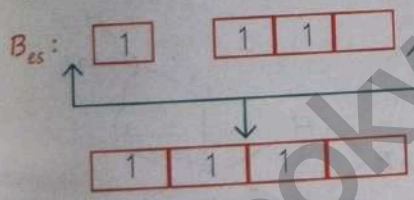
Type-2

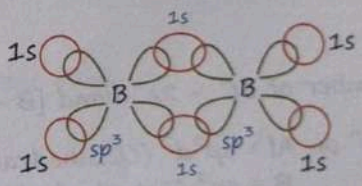
Diborane: $[BH_3]_2 = B_2H_6$

❑ B—H—B bond ≡ banana bond ≡ $3C - 2e^-$ bond
≡ Electron deficient bond

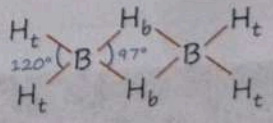


❑ Hybridisation of B → sp^3





❑ B—H—B ⇒ $sp^3 - 1s - sp^3$



$< H_tBH_t = 120°$ → θ↑ : %s↑ : % p↓: Bond length↓

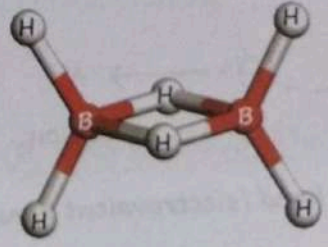
$< H_bBH_b = 97°$ → θ↓ : %s↓ : % p↑: Bond length↑

❑ Terminal B-H bonds have less p-character when compared to bridging bonds.

❑ Bond Length: $B—H_t < B - H_b$

❑ Bond Angle: $H_tBH_t > H_bBH_b$

❑ Max. No. of atoms in same plane: 6[2B, $4H_t$]



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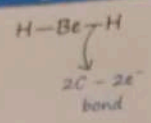
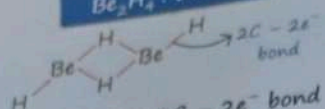
BeH_2 : Monomer

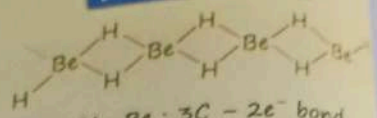


Be_2H_4 : Dimer



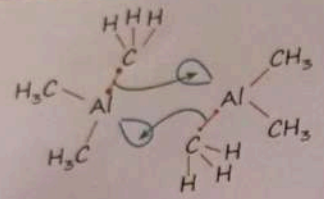
- Be—H—Be : 3C – 2e⁻ bond

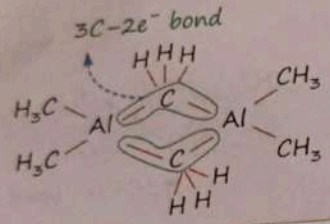
$[BeH_2]_n$: Polymer



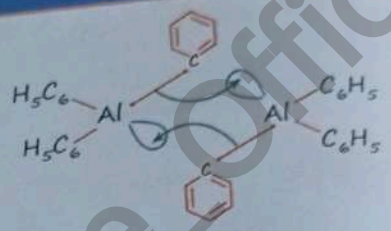
- Be—H—Be : 3C – 2e⁻ bond

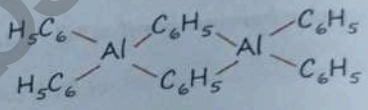
$Al_2(CH_3)_6$ Al—C—Al ≡ $sp^3-sp^3-sp^3$





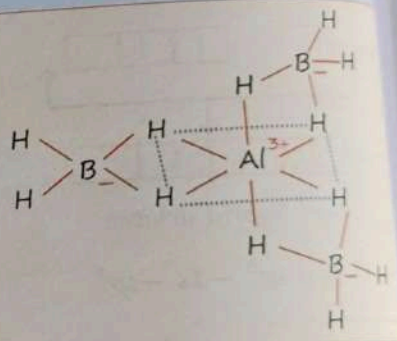
$Al_2(C_6H_5)_6$ Al—C—Al ≡ $sp^3-sp^3-sp^3$ ≡ 3C–2e⁻ bond





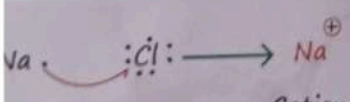
$Al(BH_4)_3$

- Number of 3C – 2e⁻ bond [B—H—Al] in $Al(BH_4)_3$ ≡ 6
- Hyb^n on Al ≡ sp^3d^2 (Octahedral)
  B ≡ sp^3 (Tetrahedral)



IONIC BONDING

The chemical bond formed between two or more atoms as a result of transfer of one or more electrons between them.



Electrostatic force of attraction between cation and anion ≡ Ionic Bond
≡ Electrovalent bond

Ionic bond (electrovalent bond) is non-directional in nature but covalent bond is directional in nature.

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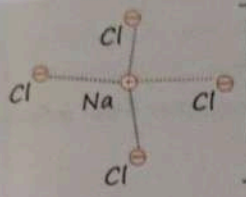
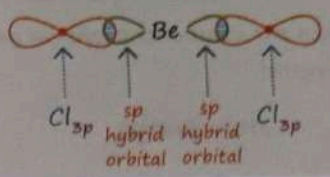


→ Here force of attraction does not depend on direction of interaction between anion and cation

$BeCl_2$ → Covalent molecule



→ Direction of overlapping plays an important role to form covalent bond

Ionic Compounds

A^+ Cation	B^- Anion
(a) Metal (Na, Mg)	(a) Non-Metal (B, C, N, S, O, F)
(b) Group of atoms having (+)ve charge $NH_4^{\oplus}$	(b) Group of atoms having (−)ve charge SO_4^{2-}, PO_4^{3-}

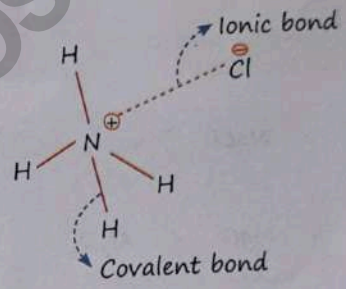
$NaF \longrightarrow Na^+ \quad F^-$

$KCl \longrightarrow K^+ \quad Cl^-$

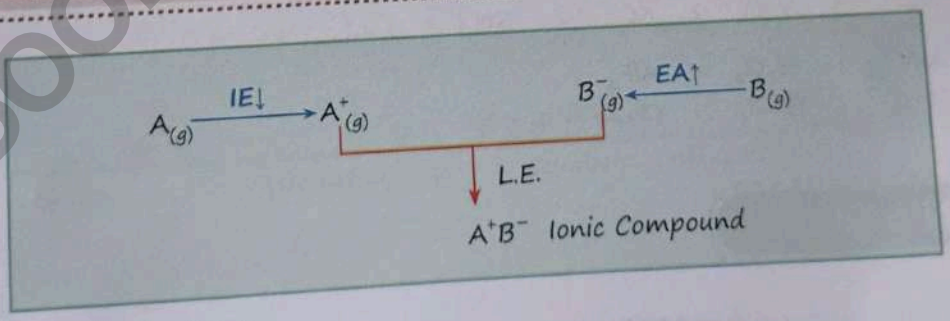
$CaO \longrightarrow Ca^{2+} \quad O^{2-}$

$NH_4Cl \longrightarrow NH_4^+ \quad Cl^-$

$Na_2SO_4 \longrightarrow 2Na^+ \quad SO_4^{2-}$



Favourable Condition for Ionic Bond



(i) Less Ionisation Energy for A

(ii) More Electron Affinity for B

(iii) Energy released because of the combination of cation (A^+) and anion (B^-) should be high. This e is also defined in terms of lattice enthalpy.

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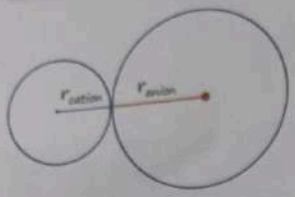
LATTICE ENERGY

- Energy which is released when the component ions (in gas phase) were brought together from infi... to make the lattice of the crystal.

$$Na^+_{(g)} + Cl^-_{(g)} \longrightarrow NaCl_{(s)} + \text{Lattice Energy}$$



$r = r_{cation} + r_{anion} \equiv$ Internuclear Distance

$$\text{Lattice Energy} \propto \frac{q_1 \cdot q_2}{r}$$

Factors Affecting LE

(i) Charge ↑ : LE ↑

(ii) Size ↑ : LE ↓

1. $LE \propto q_1 \cdot q_2$
2. $LE \propto \frac{1}{r_{cation} + r_{anion}}$

पहले charge फिर size

LE → NaF > NaCl [size: $F^- < Cl^-$]
(1×1) (1×1) → $q_1 \cdot q_2$

LE → Na_2O < MgO < Al_2O_3
(1×2) (2×2) (3×2) → $q_1 \cdot q_2$

LE → NaCl < $MgCl_2$ < $AlCl_3$
(1×1) (2×1) (3×1) → $q_1 \cdot q_2$

LE → LiCl > NaCl > KCl > RbCl > CsCl (Size: $Li^+ < Na^+ < K^+ < Rb^+$)

LE → AlN > Al_2O_3 > AlF_3
(3×3) (3×2) (3×1) → $q_1 \cdot q_2$

FAJAN'S RULE

- There is no compound which is 100% ionic.
- Covalent character in ionic compound can be explained with the help of Fajan's rule.
- The negative charge of an isolated anion is evenly distributed, but in the presence of a cation anion's negative charge density is distorted. This distortion of e^- cloud is known as polarisation

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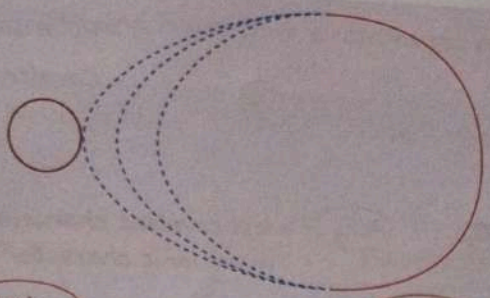
- **Polarising power** → a cation's ability or power to distort the electron density of an anion towards itself, or to pull the electron density of an anion towards itself.
- **Polarisability** → tendency for an anion to become polarised.



Cation

Polarising Power ↑

(i) Small size
(ii) More charge
(iii) Pseudo inert gas configuration of cation

Anion

Polarisability ↑

(i) Large size
(ii) More charge

Polarising power of cation ↑
Polarisability of anion ↑
→ Polarisation ↑ → Covalent character ↑

Polarising power of cation → $Li^+ > Na^+ > K^+$
- $Be^{2+} > Mg^{2+} > Ca^{2+}$
- $Na^+ < Mg^{2+}$

Polarisability of anion → $F^- < Cl^- < Br^- < I^-$
- $N^{3-} > O^{2-} > F^-$

Covalent character → $LiCl > NaCl > KCl > RbCl > CsCl$

Size of cation ↑ : Polarisation ↓ : Covalent character ↓

- $BeCl_2$ $MgCl_2$ $CaCl_2$ $SrCl_2$ $BaCl_2$ →

Size of cation ↑ : Polarisation ↓ : Covalent character ↓

- LiF LiCl LiBr LiI →
  - Size of anion ↑
  - Polarisation ↑
  - Covalent character ↑

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- $NaCl < MgCl_2 < AlCl_3$ — Covalent character
  Na^+ Mg^{+2} Al^{+3}
  → – Charge of cation ↑
  – Polarisation ↑
  – Covalent character ↑

- $AlF_3 < Al_2O_3 < AlN$ — Covalent character
  F^- O^{-2} N^{-3}
  → – Charge on anion ↑
  – Polarisation ↑
  – Covalent Character ↑

Ionic Character

- $AlN < Al_2O_3 < AlF_3$ → less covalent character, more ionic character
- AlF_3 (Ionic compound $(Al^{3+}\ 3F^-)$) > $AlCl_3$ (Covalent compound)
- $BeCl_2$ (Covalent molecule) < $MgCl_2$ (Ionic compound $(Mg^{2+}\ 2Cl^-)$)
- BeF_2 $(F-Be-F)$ > $BeCl_2$ $(Cl-Be-Cl)$

[BeF_2 has more ionic character than $BeCl_2$ but both are covalent molecules]

Note

- Cation having pseudo-inert gas configuration, has more polarising power than other cations

$CuCl$ > $NaCl$: Covalent character

[d block metal] > [s block metal]

- Z_{eff} : Cu^+ $[Ar]3d^{10}$ (Pseudo inert gas configuration) > Na^+ $[Ne]$ (Inert gas configuration)
- Polarising power : $Cu^+ > Na^+$

Melting point ↑ : Ionic character ↑

- Melting point of covalent compound < melting point of ionic compound.

r. ↑ : Ionic character ↑	$BeCl_2 < MgCl_2 < CaCl_2$	: M.P.
r. ↑ : Ionic character ↓	$CaF_2 > CaCl_2 > CaBr_2 > CaI_2$	: M.P.

Colour

- The colour of
  anion.
- The bigger an
  visible light.

1. AgCl is
2. $SnCl_2$
3. $PbCl_2$

54. The numbe
(A) Polari
(C) Polari
Sol. [3] Acc. t
having
of the

55. Which of
(a) $BaCl_2$
Sol. (b) AgCl
Reason: A

56. Arrange
ClF_3.
(a) ClF_3
(c) N_2 <
Sol. (c) Incre
N_2 < SO
The deg
determ

57. Order o
(A) KF
(C) Sn
(E) KF
(a) C,
Sol. (c)

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Colour

- The colour of some compounds can be explained on the basis of polarisation of their bigger negative anion.
- The bigger anions are more polarised and hence their electrons get excited by partial absorption of visible light.

1. AgCl is white but AgBr, AgI is yellow.	⇒	Polarisation : AgCl < AgBr < AgI (more polarisation, more intensity of colour)
2. $SnCl_2$ is white but SnI_2 is red.	⇒	Polarisation : $SnCl_2 < SnI_2$
3. $PbCl_2$ is white but PbI_2 is yellow.	⇒	Polarisation : $PbCl_2 < PbI_2$

54. The number of following factors which affect the percent covalent character of the ionic bond is___.

[8 April, 2023 (Shift-I)]

(A) Polarising power of cation
(B) Extent of distortion of anion
(C) Polarisability of the anion
(D) Polarising power of anion

Sol. [3] Acc. to Fajan's rule a compound is said to be covalent having smaller cation, larger anion and having high charge density. A, B and C are factors which affect the percent covalent character of the ionic bond according to Fajan's rule

55. Which of the following is least ionic? [31 Jan, 2024 (Shift-II)]

(a) $BaCl_2$ (b) AgCl (c) KCl (d) $CaCl_2$

Sol. (b) $AgCl < CaCl_2 < BaCl_2 < KCl$ (ionic character)

Reason: Ag^+ has pseudo inert gas configuration.

56. Arrange the bonds in order of increasing ionic character in the molecules. LiF, K_2O, N_2, SO_2 and ClF_3. [01 Feb, 2024 (Shift-I)]

(a) $ClF_3 < N_2 < SO_2 < K_2O < LiF$
(b) $LiF < K_2O < ClF_3 < SO_2 < N_2$
(c) $N_2 < SO_2 < ClF_3 < K_2O < LiF$
(d) $N_2 < ClF_3 < SO_2 < K_2O < LiF$

Sol. (c) Increasing order of ionic character

$N_2 < SO_2 < ClF_3 < K_2O < LiF$

The degree of ionic character in a bond depends on the difference in electronegativity, which determines the bond polarity.

57. Order of Covalent bond; [24 Jan, 2023 (Shift-I)]

(A) KF > KI; LiF > KF
(B) KF < KI; LiF > KF
(C) $SnCl_4 > SnCl_2$; CuCl > NaCl
(D) LiF > KF; CuCl < NaCl
(E) KF < KI; CuCl > NaCl

(a) C, E only (b) B, C only (c) B, C, E only (d) A, B only

Sol. (c)

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THERMAL STABILITY

- $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2 \uparrow$
- $Na_2CO_3 \xrightarrow{\Delta}$ No Reaction

Li_2CO_3 is thermally unstable
Na_2CO_3 is thermally stable

- Thermally stable → Strong lattice → (i) Lattice Energy ↑ (ii) Ionic Character ↑

Anions

Monoatomic anions

F^- Cl^- Br^- I^- O^{2-} H^-

Apply Lattice Energy concept to get order for thermal stability

LiCl > NaCl > KCl > RbCl > CsCl

size of cation ↑
Lattice Energy ↓
Thermal stability ↓

LiH > NaH > KH > RbH > CsH

- BeO > MgO > CaO > SrO > BaO

Be^{2+} O^{2-} — Monoatomic anion

Lattice Energy ↓
Thermal stability ↑

Polyatomic anions

SO_4^{2-}, NO_2^-, CO_3^{2-}, OH^-, NO_3^-

Apply ionic character concept to get order for thermal stability

Li_2SO_4 < Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4

size of cation ↑
Ionic character ↑
Thermal stability ↑

Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3

- $BeCO_3$ < $MgCO_3$ < $CaCO_3$ < $SrCO_3$ < $BaCO_3$

Be^{2+} CO_3^{2-} — Polyatomic anion

Ionic character ↑
Thermal stability ↑

LiOH < NaOH < KOH < RbOH < CsOH

Li^+ $\bar{O}H$ — Polyatomic anion

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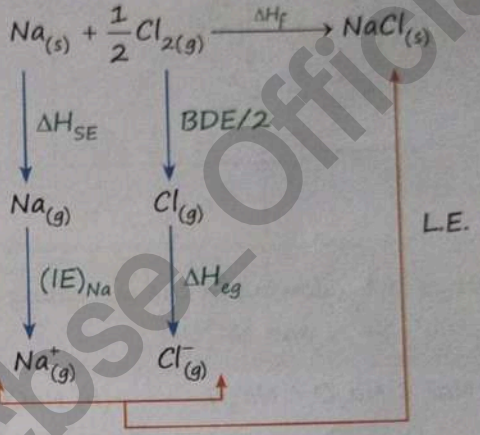
HEAT OF FORMATION

- Amount of heat released when 1 mole of substance is formed by its elements in pure and stable state.

$$Na_{(s)} + \frac{1}{2}Cl_{2(g)} \longrightarrow NaCl_{(s)} + \Delta H_f$$

BORN-HABER CYCLE

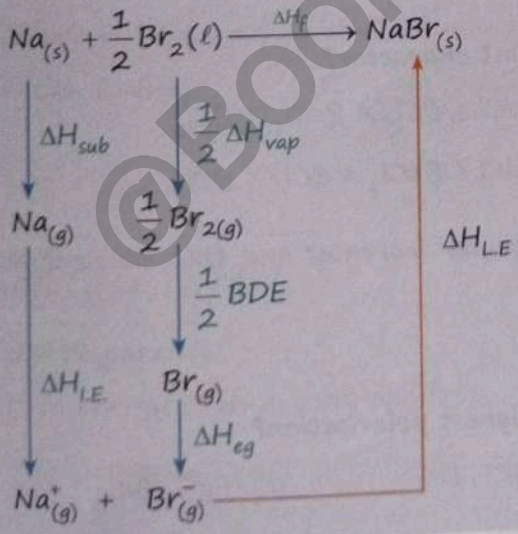
- ΔH_f → Heat of formation of NaCl
- $[\Delta H_{SE}]_{Na}$ → Sublimation energy of Na
- BDE → Bond dissociation energy of 1 mole Cl_2 molecule
- ΔH_{eg} → Electron gain enthalpy of $Cl_{(g)}$
- ΔH_{IE} → Ionisation Energy
- ΔH_{LE} → Lattice Energy



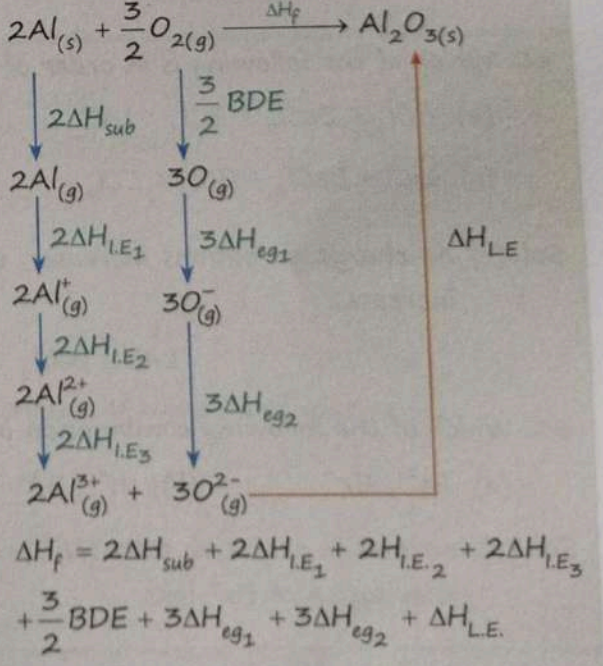
$$\Delta H_f \equiv (\Delta H_{SE})_{Na} + (IE)_{Na} + \left(\frac{BDE}{2}\right)_{Cl_2} + (\Delta H_{eg})_{Cl} + (L.E.)_{NaCl}$$

(Exo) → ΔH → ⊖ve; (Endo) → ⊕ve; (Endo) → ⊕ve; (Endo) → ⊕ve; (Exo) → ⊖ve; (Exo) → ⊖ve

Born Haber Cycle for NaBr



Born Haber Cycle for $Al_2O_{3(s)}$



$$\Delta H_f = 2\Delta H_{sub} + 2\Delta H_{I.E_1} + 2H_{I.E._2} + 2\Delta H_{I.E_3} + \frac{3}{2}BDE + 3\Delta H_{eg_1} + 3\Delta H_{eg_2} + \Delta H_{L.E.}$$

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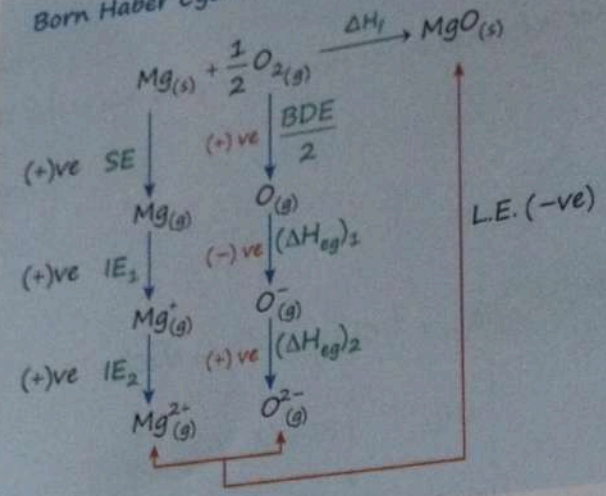
Born Haber Cycle for MgO



58. Arrange the following in the increasing order of their covalent character.

(a) NaF, Na_3N and Na_2O (b) NaCl, $MgCl_2$, $AlCl_3$, $SiCl_4$ and PCl_5

Sol. (a) $NaF < Na_2O < Na_3N$ (b) $NaCl < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$: Covalent character

[$Na^+ < Mg^{2+} < Al^{3+} < Si^{4+} < P^{5+}$: Charge]

59. $SnCl_4$ has melting point −15°C whereas $SnCl_2$ has melting point 535°C. Why?

Sol. Covalent character : $SnCl_4 > SnCl_2$ [Charge : $Sn^{+4} > Sn^{+2}$]

According to Fajan's rule, as charge on cation increases, its polarising power increases resulting in the greater polarisation of anion. Thus, covalent character increases and melting point decrease

60. Which of the following is in order of increasing covalent character?

(a) $CCl_4 < BeCl_2 < BCl_3 < LiCl$ (b) $LiCl < CCl_4 < BeCl_2 < BCl_3$

(c) $LiCl < BeCl_2 < BCl_3 < CCl_4$ (d) $LiCl < BeCl_2 < CCl_4 < BCl_3$

Sol. (c) As charge on cations increases, their polarising power increases and thus covalent character increases.

$$\overset{+1}{LiCl} < \overset{+2}{BeCl_2} < \overset{+3}{BCl_3} < \overset{+4}{CCl_4}$$

61. Which of the following combination of ion will have highest polarisation?

(a) Fe^{2+}, Br^- (b) Ni^{4+}, Br^- (c) Ni^{2+}, Br^- (d) Fe^{3+}, Br^-

Sol. (b) Increase in oxidation state (Ni^{4+}) increases the polarising power of cation and thus increases polarisation of Br^- ions.

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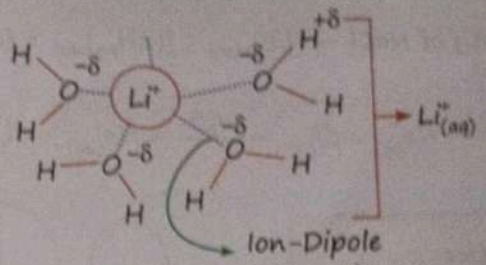
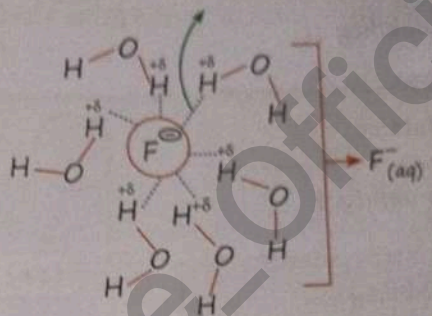
HYDRATION ENERGY

Hydration energy is the amount of energy released when one mole of ions undergoes hydration.

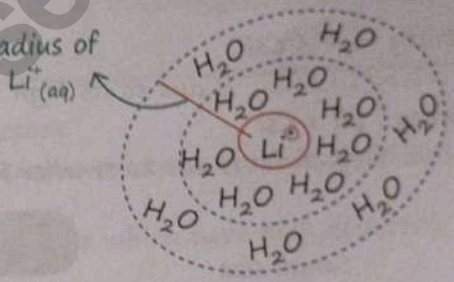
- $Li^+_{(g)} \xrightarrow{H_2O} Li^+_{(aq)} + (\Delta H_{hy})_{Li^+}$



- $F^-_{(g)} \xrightarrow{H_2O} F^-_{(aq)} + (\Delta H_{hy})_{F^-}$



$$\text{Hydration Energy} \propto \frac{\text{charge } (q)}{\text{size } (r)}$$



$Radius_{(g)}$: $Li^+ < Na^+$

$Radius_{(aq)}$: $Li^+_{(aq)} > Na^+_{(aq)}$

Mobility : $Li^+_{(aq)} < Na^+_{(aq)}$

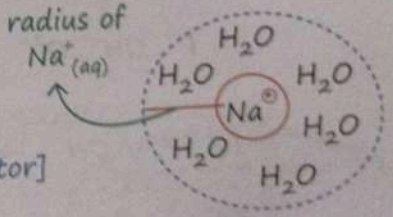
Ion Dipole Interaction : $Li^+ > Na^+$

Hydration Energy : $Li^+ > Na^+$

Hydration Energy : $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ [Size factor]

: $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$ [Size factor]

: $Na^+ < Mg^{2+}$ [Charge factor]



Solubility in Water

2 factors are responsible:

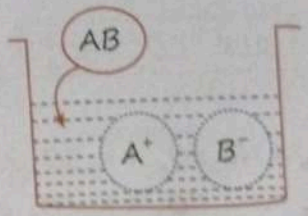
(i) Lattice energy of ionic compounds (ii) Hydration energy of component ions

$$AB_{(s)} \xrightarrow{L.E.} A^+_{(g)} + B^-_{(g)}$$

Ionic Comp.

$$A^+_{(g)} \xrightarrow{H_2O} A^+_{(aq)} + (\Delta H_{hy})_{A^+}$$

$$B^-_{(g)} \xrightarrow{H_2O} B^-_{(aq)} + (\Delta H_{hy})_{B^-}$$



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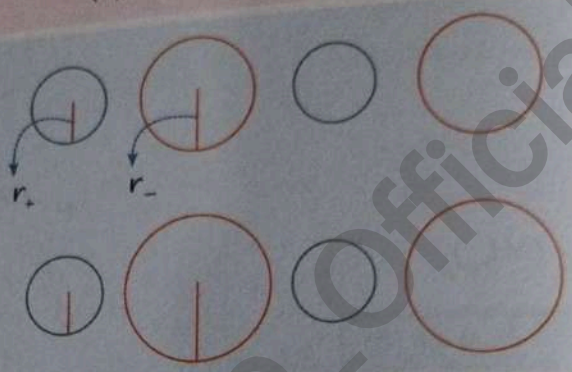
- Ionic Compound is soluble: $LE < [(\Delta H_{hy})_{A^+} + (\Delta H_{hy})_{B^-}]$
- Ionic Compound is insoluble: $LE > [(\Delta H_{hy})_{A^+} + (\Delta H_{hy})_{B^-}]$
- High solubility of NaCl → $(LE)_{NaCl} < [(\Delta H_{hy})_{Na^+} + (\Delta H_{hy})_{Cl^-}]$

$$NaCl_{(s)} \xrightarrow{L.E.} Na^+_{(g)} + Cl^-_{(g)} \xrightarrow{H_2O} Na^+_{(aq)} + Cl^-_{(aq)}$$

- Perfect lattice: $r_+ = r_-$
  → Solubility ↓↓
- Imperfect lattice: $r_+ < r_-$
  → Solubility ↑



- Extent of solubility $\propto |r_- - r_+|$
- Solubility : LiF < CsF
  Low solubility of LiF is due to its high lattice enthalpy
- Solubility : LiI > CsI
  Low solubility of CsI is due to smaller hydration enthalpy of its two ions (Cs^+, I^-)

Anion

Small size → F^-, OH^-, O^{2-}

Large size → I^-, Br^-, SO_4^{2-}, NO_3^-, NO_2^-, CO_3^{2-}

- Li^+ F^- / OH^- → $r_+ \approx r_-$ → less soluble
- Cs^+ I^-, Br^-, NO_3^-, NO_2^- → $r_+ \approx r_-$ → less soluble

- $LiOH < NaOH < KOH < RbOH < CsOH$
- $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2 <$ $Be(OH)_2$ → Be compounds are generally mor than others due to high hydr Be^{2+} ion.

- $2Li^+$ CO_3^{2-}/SO_4^{2-} → $2r_+ \approx r_-$ → Solubility ↓
- $2Cs^+$ CO_3^{2-}/SO_4^{2-} → $2r_+ > r_-$ → Solubility ↑

- $Li_2SO_4 < Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4$
- $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

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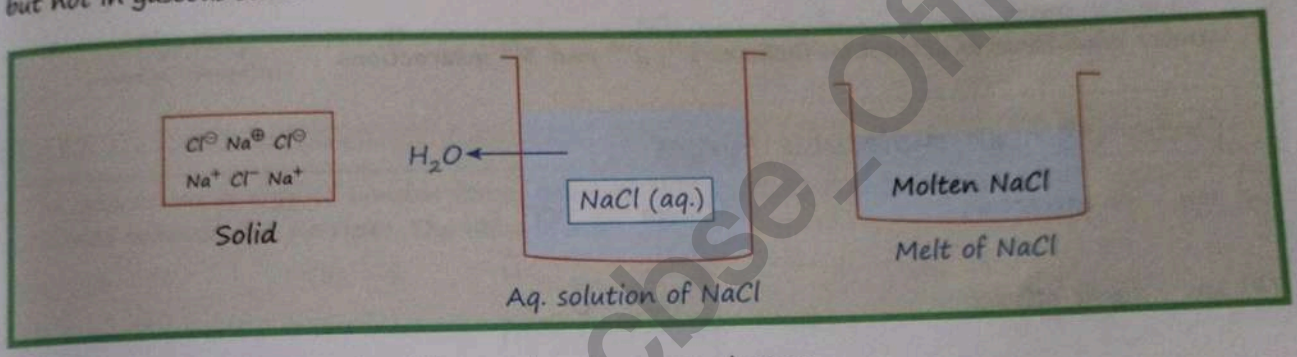
- $LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$ | Same order for : $CO_3^{2-} = HCO_3^-$

Be^{2+} CO_3^{2-}/SO_4^{2-} $\longrightarrow r_+ < r_- \rightarrow$ Solubility ↑ | Ba^{2+} CO_3^{2-}/SO_4^{2-} $\longrightarrow r_+ \approx r_- \rightarrow$ Solubility ↓

- $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
- $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

Properties of Ionic Compound

- **Physical state :** At room temperature, ionic compounds exist either in solid state or in solution phase but not in gaseous state.



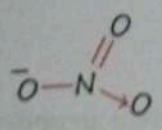
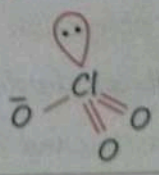
- **Electrical conductivity :** Ionic solids are almost non-conductors.
  All ionic solids are good conductors in molten state & in aqueous solutions because their ions are free to move.
- **Solubility of ionic compounds :** Soluble in polar solvents like water which have high dielectric constant (like H_2O, C_2H_5OH)

ISOMORPHISM

- Crystals of different ionic compounds having similar crystal structures are known to be isomorphs to each other and the phenomenon is known as isomorphism.
  Example: $FeSO_4 \cdot 7H_2O$ & $MgSO_4 \cdot 7H_2O$ are isomorphs.

Crystals having same shape of anion are isomorphs

- $BaSO_4$ & $KMnO_4$ → isomorphs
  → Both anions SO_4^{2-} and MnO_4^- are tetrahedral in shape.
- $NaNO_3$ & $NaClO_3$ → not isomorphs
  → Because anions have different shape.

- $Na^+ NO_3^-$  (Planar)
- $Na^+ ClO_3^-$  (Pyramidal)

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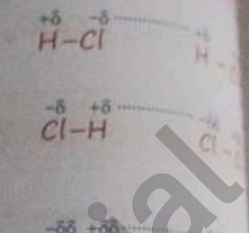
INTER MOLECULAR FORCE OF ATTRACTION

Molecules → Polar (H_2O, HCl)

Molecules → Non-polar (Cl_2, O_2, CO_2)

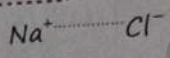


1. Interaction between Polar molecules
   - Dipole - dipole attraction
2. Interaction between Polar & non polar molecules
   - Dipole - induced dipole attraction
3. Interaction between non-polar molecules
   - Instantaneous induced dipole - induced dipole attraction [IID-ID]
   - Dispersion force
   - London force

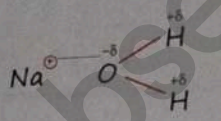
❑ Vander Waal force of attraction includes 1st, 2nd and 3rd interactions.

Order of Inter-molecular Forces

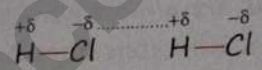
(a) Ion - ion attraction



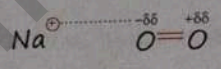
(b) Ion - dipole attraction



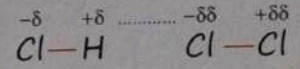
(c) Dipole - dipole attraction



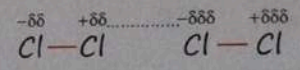
(d) Ion-induced dipole attraction



(e) Dipole-induced dipole attraction



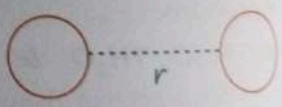
(f) Ins. Induced dipole - induced dipole



Energy & Distance Relationship between 2 Interacting Particles

1. Ion-ion attraction $E \propto \frac{1}{r}$
2. Ion-dipole attraction $E \propto \frac{1}{r^2}$
3. Dipole-dipole attraction $E \propto \frac{1}{r^3}$
4. Ion-induced dipole attraction $E \propto \frac{1}{r^4}$
5. Dipole - induced dipole attraction
6. Ins. Induced dipole - induced dipole

→ $E \propto \frac{1}{r^6}$



These forces are always attractive and interaction energy is inversely proportional to the sixth of the distance between two interacting particles ($1/r^6$ where r is the distance between two par

62. Identify type of
(1) Xe and H_2O
(6) K^+ and Cl^-

Sol. 1. Xe , H_2O
[non polar-pol

2. Na^+, Br_2
[ion-nonpolar

3. Br_2, E
[nonpolar -

4. CO_2,
[nonpolar -

5. CCl_4,
[nonpolar -

63. The interacti
distance betw
(a) −3

Sol. (b) $E \propto r^{-6}$
Thus, x = −

Vander W

Molecular m

Size of the

Lowest B.P
(4.2 K) of
known su

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62. Identify type of interaction present between following species

(1) Xe and H_2O (2) Na^+ and Br_2 (3) Br_2 and Br_2 (4) CO_2 and CO_2 (5) CCl_4 and CCl_4
(6) K^+ and Cl^- (7) Li^+ and H_2O (8) H_2S and H_2S (9) Ne and Ne (10) I^- and I_2

Sol. 1. Xe, H_2O → Dipole-induced dipole
[non polar-polar]

2. Na^+, Br_2 → Ion-induced dipole
[ion-nonpolar]

3. Br_2, Br_2 → London force/IID-ID
[nonpolar - nonpolar]

4. CO_2, CO_2 → IID-ID
[nonpolar - nonpolar]

5. CCl_4, CCl_4 → IID-ID
[nonpolar - nonpolar]

6. $K^{\oplus}$, $Cl^{\ominus}$ → Ion-ion attraction
[ion - ion]

7. Li^+, H_2O → Ion-dipole attraction
[ion - polar molecule]

8. H_2S, H_2S → Dipole-dipole attraction
[polar - polar molecule]

9. Ne, Ne → London force
[nonpolar - nonpolar]

10. I^-, I_2 → Ion-induced dipole attraction
[ion - nonpolar molecule]
$I^{\ominus} \cdots I^{+\delta} - I^{-\delta} \Rightarrow I_3^-$ ion

63. The interaction energy of London forces between two particles is proportional to r^x, where r is the distance between the particles. The value of x is: [26 Aug, 2021 (Shift-II)]

(a) −3 (b) −6 (c) 6 (d) 3

Sol. (b) $E \propto r^{-6}$ where E is the interaction energy of London forces.
Thus, $x = -6$.

Vander Waal Force of Attraction

Molecular mass ↑ : Vander Waal force ↑
: Boiling Point ↑

Order of Boiling Point		
PH_3	H_2S	HCl
^	^	^
AsH_3	H_2Se	HBr
^	^	^
SbH_3	H_2Te	HI

Size of the atom ↑ : Vander Waal force ↑ : BP ↑
He < Ne < Ar < Kr < Xe : BP/MP
Lowest B.P. (4.2 K) of any known substance

	Boiling Point (K)	Melting Point (K)
He	4.216	0.95
Ne	27.1	24.7
Ar	87.29	83.6
Kr	120.85	115.8
Xe	166.1	161.7
Rn	211.5	202.2

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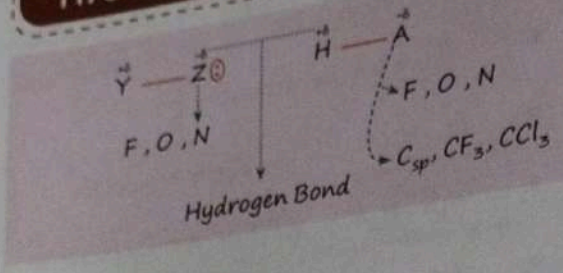
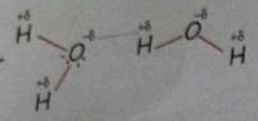
HYDROGEN BONDING

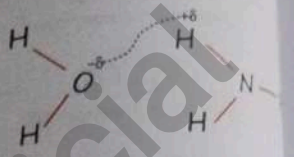


Force of attraction : H– bonding
↓
Van der Waals force of attraction

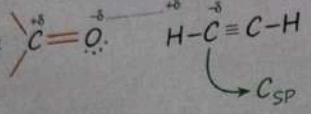
Water



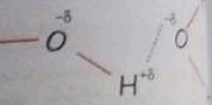
Water & Amine



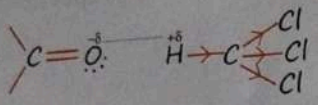
Acetone and Acetylene

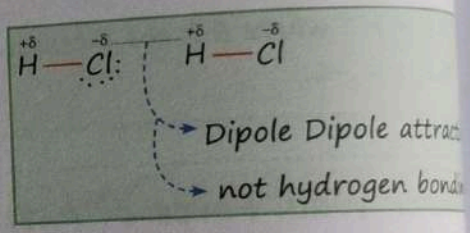


Water & Alcohol

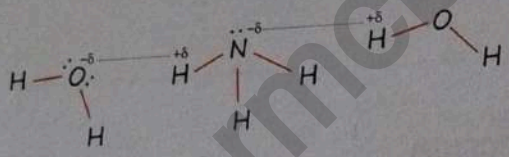


Acetone & Chloroform





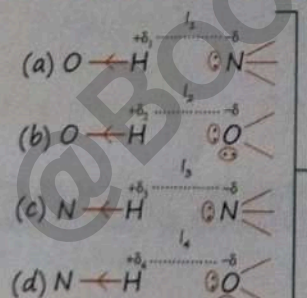
Water & Ammonia



Extent of H-bonding

Extent of H-bonding ↑ : (i) Partial (+)ve charge on H ↑

(ii) Electron density on Z ↑



Extent of H-bonding : a > b > c > d

Bond length : $l_1 < l_2 < l_3 < l_4$

Order for partial charge : $(\delta_1 = \delta_2) > (\delta_3 = \delta_4)$

- $K[HF_2]$

$K^+[HF_2]^- \equiv K^+F^- \cdots H\text{—}F$ [Bond length : x = y]

(y, x)

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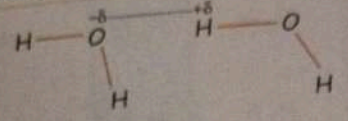
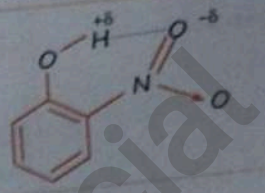
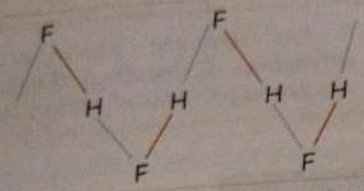
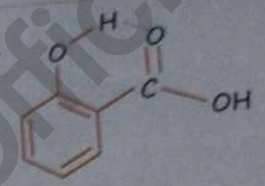
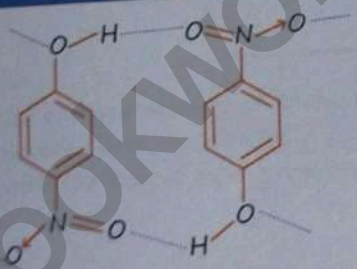
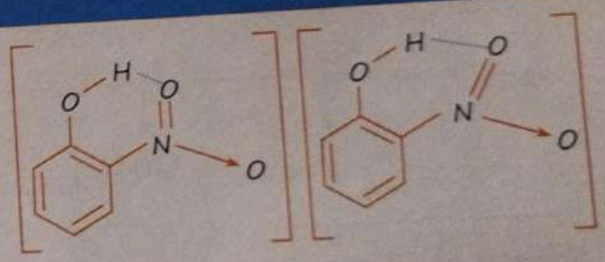
Types of Hydrogen Bonding

Inter-molecular H-bonding		Intra-molecular H-bonding	
H-bonding between molecules		H-bonding within a molecule	
H_2O		o-Nitrophenol	
$(HF)_n$ Solid or liquid		Salicylic Acid	

- The hydrogen bonds in HF link the F atom of one molecule with the H-atom of another molecule, thus forming a zig-zag chain $(HF)_n$ in both the solid and also in the liquid.

Boiling Point

p-Nitrophenol	>	o-Nitrophenol
		
Association of molecules due to inter-molecular H-bonding ⇒ BP ↑		It exists as discrete molecules which are held together by weak van der waals forces of attraction. A small amount of energy is required to break these forces of attraction. So, less boiling point.
The intermolecular hydrogen bonding in p-nitrophenol requires more energy to break the intermolecular forces, resulting in a higher boiling point.		

Order of Boiling Point

NH_3, H_2O and HF → Inter-molecular hydrogen bonding ⇒ BP ↑

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Top → Bottom

: Molecular Mass ↑
: Vander Waal Force of attraction ↑
: BP ↑

CH_4 ^ SiH_4 ^ GeH_4 ^ SnH_4

NH_3 | PH_3 | AsH_3 | SbH_3

PH_3 ^ AsH_3 ^ NH_3 ^ SbH_3

H_2O | H_2S | H_2Se | H_2Te

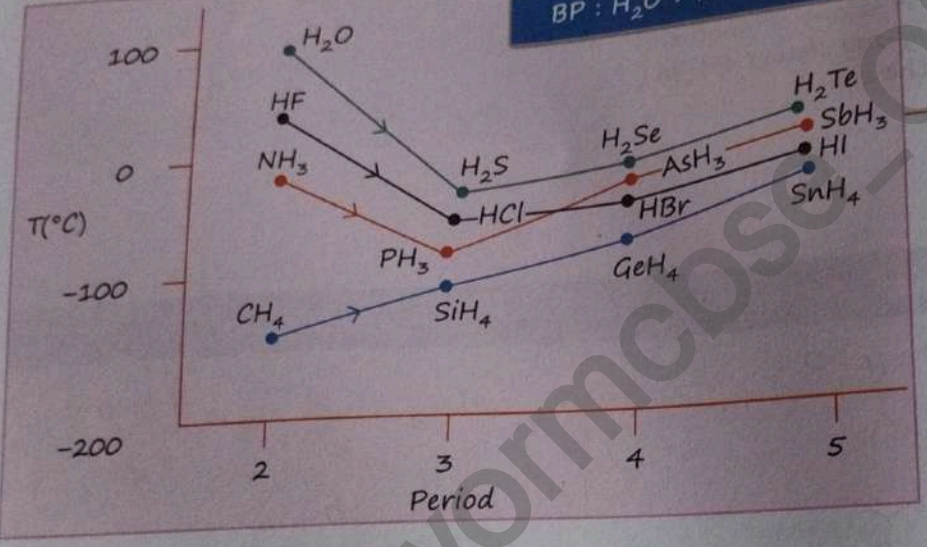
H_2S ^ H_2Se ^ H_2Te ^ H_2O

HF | HCl | HBr | HI

HCl ^ HBr ^ HI ^ HF

It has stronger intermolecular force of attraction (H-bonding) than others (hav... vander Waal force of attraction).

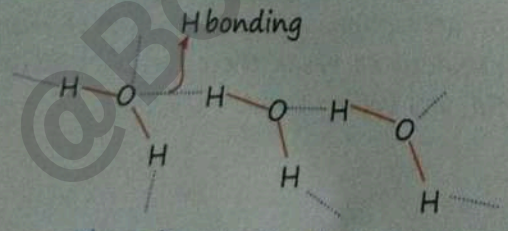
BP : $H_2O > HF > NH_3$



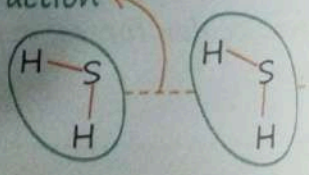
- Sudden increase... boiling point of ... H_2O and HF is du... hydrogen bondi... $H_2O > HF > NH_3$

Physical State

(i) H_2O is liquid but H_2S is gas at room temperature.



Vander Waal force of attraction



There is extensive hydrogen bonding in H_2O, which is absent in H_2S. Molecules of H_2S are held together only by weak van der Waal's forces of attraction.

Hence, H_2O exists as a liquid while H_2S as a gas.

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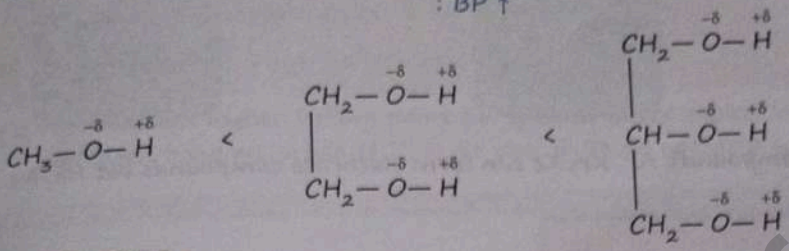
(ii) HF is liquid but HCl is gas. Think why?

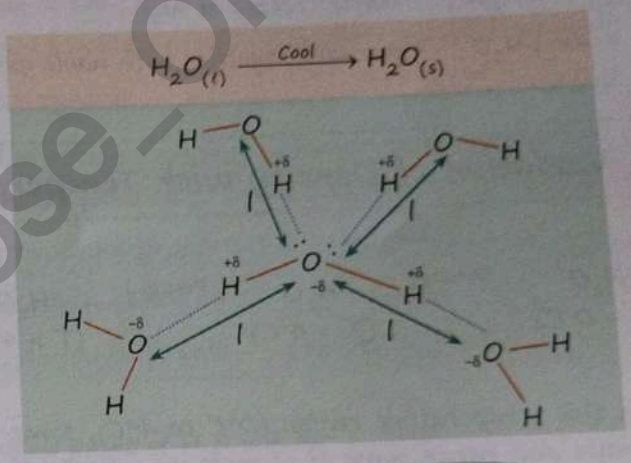
B.P. & Viscosity

→ No. of Hydrogen bonds per molecule ↑ : viscosity ↑ : BP ↑



WATER

- No. of water molecules attached to 1 H_2O molecule = 4

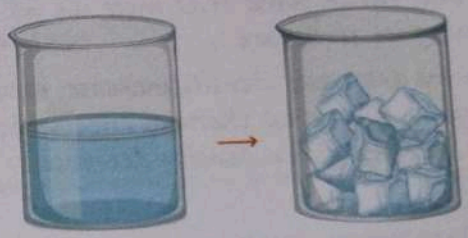


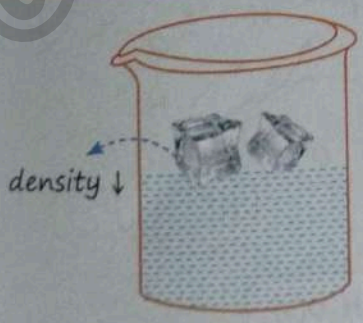
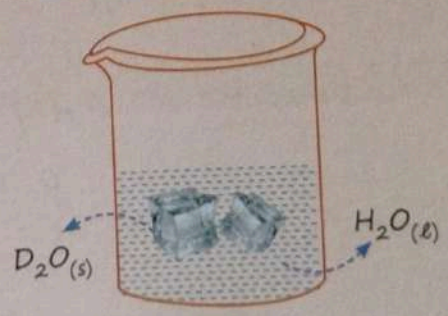
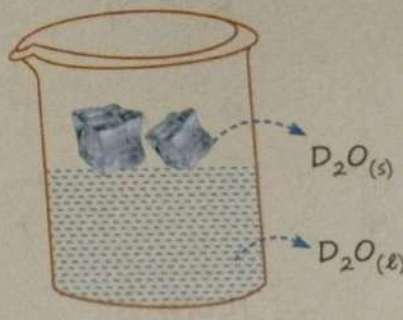
- Tetrahedrally bonded to a water molecule in solid state.

  Density : $H_2O_{(s)} < H_2O_{(\ell)}$

  Volume : $H_2O_{(s)} > H_2O_{(\ell)}$

- Volume of ice is more because of open cage like crystal structure, form by association of water molecules with the help of hydrogen bond.



$H_2O_{(s)}$ in $H_2O_{(\ell)}$	$D_2O_{(s)}$ in $H_2O_{(\ell)}$	$D_2O_{(s)}$ in $D_2O_{(\ell)}$
		
Ice [H_2O (s)] floats on water	D_2O (s) sinks into water	D_2O (s) floats on heavy wate

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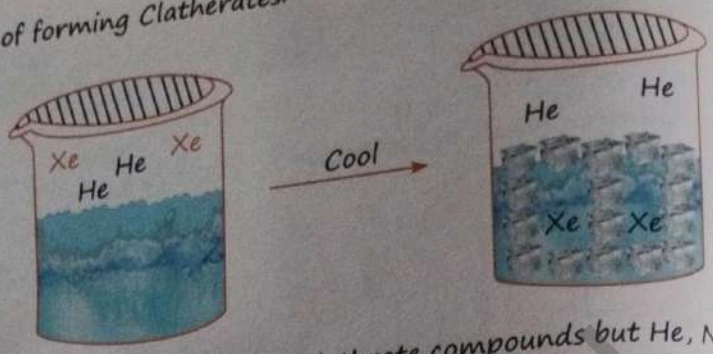
- Due to open structure, ice is capable of forming Clatherates.

Clatherates compounds
↓
Cage like compounds

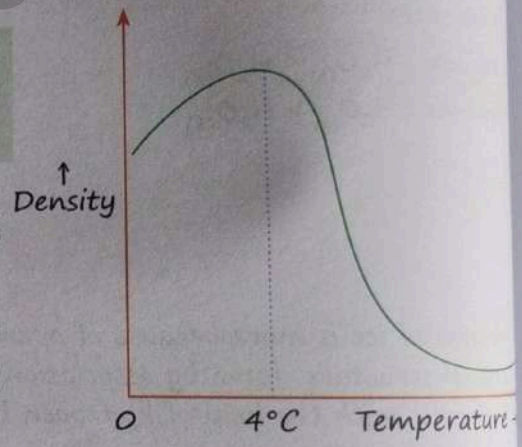


- **Noble gas hydrate (clathrate compound):** Ar, Kr, Xe can form clathrate compounds but He, Ne cannot due to their smaller size.

$Xe \cdot 6H_2O$ $Ar \cdot 6H_2O$ $Kr \cdot 6H_2O$	formed only where water freezes at high pressure together with noble gas

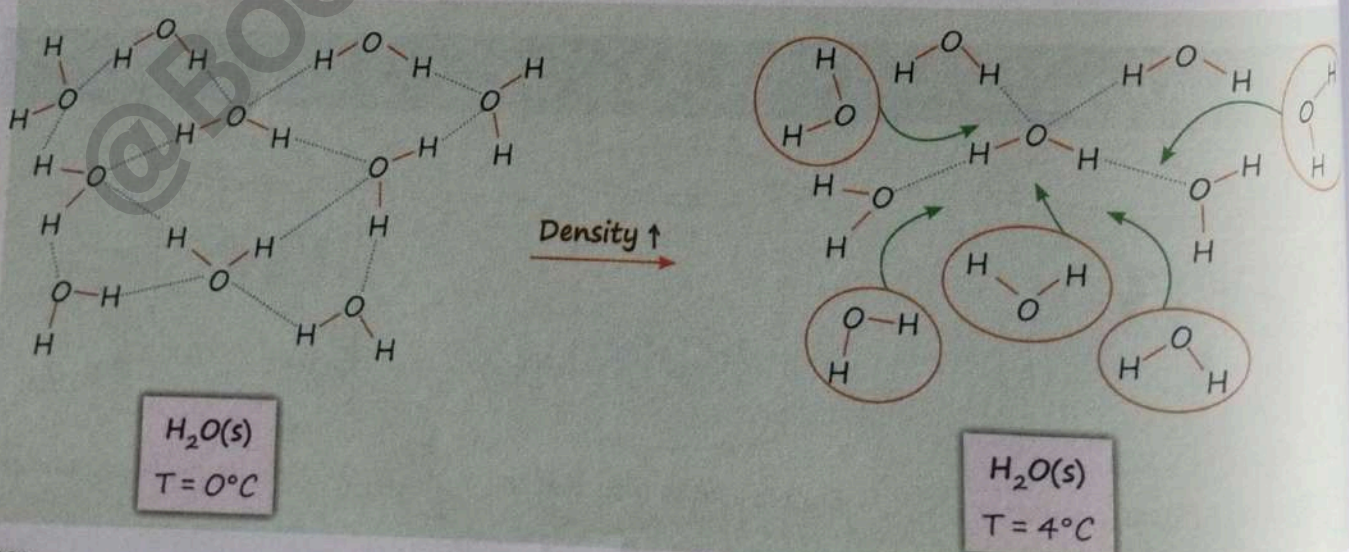
Variation of Density with Temperature

H_2O ($T = 0°C$) $\xrightarrow{\text{Density}\uparrow}$ H_2O ($T = 4°C$) $\xrightarrow{\text{Density}\downarrow}$ H_2O ($T > 4°C$)



- In this temperature region (0°C to 4°C), some of the ice melts and hence some H_2O molecule go into the cages of remaining ice structure.

  So, volume decreases, density increases becoming max at 4°C but beyond this temp thermal effects become dominating volume increases then density decreases.



$H_2O(s)$
$T = 0°C$

$H_2O(s)$
$T = 4°C$

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(IIT JEE 2000)

64. Amongst H_2O, H_2S, H_2Se and H_2Te, the one with the highest boiling point is
(a) H_2O because of hydrogen bonding
(b) H_2Te because of higher molecular weight
(c) H_2S because of hydrogen bonding
(d) H_2Te because of lower molecular weight

Sol. (a) H_2O displays higher boiling point (in spite of lower molecular weight) on account of strong hydrogen bonding, while H_2S, H_2Se and H_2Te do not display H-bonding.

65. The correct statement/s about Hydrogen bonding is/are : [04 April, 2024 (Shift-II)]
A. Hydrogen bonding exists when H is covalently bonded to the highly electronegative atom.
B. Intermolecular H bonding is present in o-nitrophenol
C. Intramolecular Hs bonding is present in HF.
D. The magnitude of H bonding depends on the physical state of the compound.
E. H-bonding has powerful effect on the structure and properties of compounds.
Choose the correct answer from the options given below:
(a) A only (b) A, D, E only (c) A, B, D only (d) A, B, C only

Sol. (b)

66. If the boiling point of ssssH_2O is 373 K, the boiling point of H_2S will be: [3 Sept, 2020 (Shift-I)]
(a) more than 373 K (b) less than 300 K
(c) greater than 300 K but less than 373 K (d) equal to 373 K

Sol. (b) Boiling point of H_2S is 213K.

"Chalo, NEET ke PYQs practice karte hain, QR code scan karo aur apni preparation ko boost karo!"

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3 Coordination Compounds

Syllabus (JEE Advanced)

Werner's theory; Nomenclature, cis-trans and ionization isomerism, hybridization and geom (linear, tetrahedral, square planar and octahedral) of mononuclear coordination compounds; Bo [VBT and CFT (octahedral and tetrahedral fields)]; Magnetic properties (spin-only) and co 3d-series coordination compounds; Ligands and spectrochemical series; Stability; Importan applications; Metal carbonyls.

ADDITION COMPOUNDS

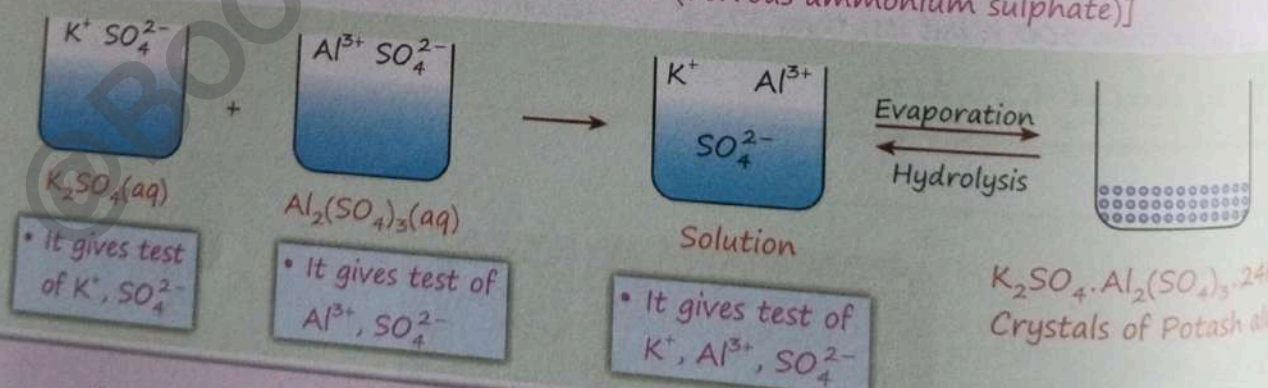
- Combination of two or more stable compounds
- There are 2 types of addition compounds : (i) Double salts and (ii) Coordination Compounds

1. Double Salts: Ions in addition compound, don't lose their identity in aq. solution.

(a) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ [Potash Alum]

(b) $KCl \cdot MgCl_2 \cdot 6H_2O$ [Carna

(c) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ [Mohr's salt/FAS (Ferrous ammonium sulphate)]



$KCl_{(aq.)} + MgCl_{2(aq.)} \xrightarrow{Cool} KCl \cdot MgCl_2 \cdot 6H_2O$ (Carnallite (solid)) $\xrightarrow[Solution]{Aq.}$ Test of K^+, Mg^{2+}, Cl^- ions

$FeSO_{4\,(aq)} + (NH_4)_2SO_{4\,(aq)} \xrightarrow{Cool} FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ (Mohr's Salt (solid)) $\xrightarrow[Solution]{Aq.}$ Test of Fe^{2+}, SO_4^{2-}, NH_4

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8 Hydrogen

(JEE Advanced)

Syllabus Position of hydrogen in periodic table, occurrence, isotopes, preparation, properties and uses of hydrogen; hydrides – ionic, covalent and interstitial; physical and chemical properties of water, heavy water; hydrogen peroxide preparation, reactions, use and structure; hydrogen as a fuel.

Hydrogen, the most abundant element in the universe and the third most abundant on the surface of the globe, is being visualised as the major future source of energy.

PROPERTIES OF HYDROGEN

- Hydrogen resembles both alkali metals and halogens

Alkali Metal	Halogens
❑ Electronic Configuration : ns^1 $H \rightarrow 1s^1$ $Li \rightarrow 2s^1$ ❑ Reducing Agent: Reduction: Addition of Hydrogen $Na \rightarrow Na^+ + e^-$ (Reducing agent) $H \rightarrow H^+ + e^-$	❑ Requires only one electron to have the configuration of the nearest noble gas. $H \xrightarrow{e^-} H^- [1s^2 \rightarrow He]$ $F \xrightarrow{e^-} F^- [1s^2\ 2s^2\ 2p^6 \rightarrow Ne]$ ❑ Diatomic Molecule – H_2 | F_2

ISOTOPES OF HYDROGEN

$^{A}_{Z}X \longrightarrow$ no. of neutrons = $A - Z$

	Protium ^1_1H	Deuterium ^2_1H, D	Tritium ^3_1H, T
Representation	Ordinary hydrogen	Heavy hydrogen	Radioactive
Neutrons	$1 - 1 = 0$	$2 - 1 = 1$	$3 - 1 = 2$
Occurrence	99.98%	0.016%	10^{-15}%

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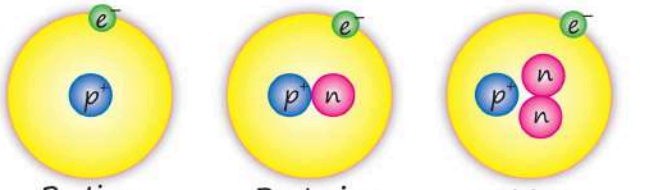
Note

- Tritium is radioactive and emits low energy β – particles.
- Bond Strength : H–H < D–D < T – T
- Bond Energy : $H_2 < D_2 < T_2$
- Reactivity : $H_2 > D_2 > T_2$



Protium Mass number : 1 | Deuterium Mass number : 2 | Tritium Mass number : 3

Isotopes of Hydrogen

1. The total number of isotopes of hydrogen and number of radioactive isotopes among them, respectively, are:
   (a) 3 and 1 (b) 3 and 2 (c) 2 and 1 (d) 2 and 0

Sol. (a) Hydrogen has three main isotopes: Protium, Deuterium, Tritium. Out of these three isotopes of hydrogen, only tritium is radioactive in nature which emits low-energy beta particles.

2. Hydrogen has three isotopes (A), (B) and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x), (y) and (z) is:
   (a) 3 (b) 2 (c) 4 (d) 1

Sol. (a) H D T
0 + 1 + 2 = 3

PREPARATION OF H_2

1 By Active metals (Na, K)

$$Na + H_2O \longrightarrow \frac{1}{2} H_2\uparrow + NaOH$$

- $Na \rightarrow Na^+ + e^-$
- $H_2O \rightarrow H^+ + OH^-$

2 By less active metals (Ca, Zn, Mg, Al)

$$Al + 3H_2O \xrightarrow{\Delta} Al(OH)_3 + \frac{3}{2}H_2\uparrow$$

3 By Reaction of NaOH

$$[Sn, Al, Pb, Zn, B, Si, Be] \xrightarrow{NaOH} H_2\uparrow$$

$$Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2\uparrow$$

4 Gassing Reaction

$$Fe + H_2O \longrightarrow Fe_3O_4 + H_2\uparrow$$

$$[Fe_3O_4 \equiv \overset{+2}{Fe}O \cdot \overset{+3}{Fe_2}O_3]$$

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Steam is passed over hot iron and Iron decomposes steam with the formation of magnetic oxide (Fe_3O_4) and hydrogen.

5 Bosch Process

Industrial Preparation

$$\underset{\text{[Hot Coal]}}{C_{(s)}} + H_2O_{(l)} \xrightarrow{1000°C} \boxed{CO_{(g)} + H_{2(g)}} \begin{matrix}\rightarrow \text{Water gas} \\ \rightarrow \text{Syn gas}\end{matrix}$$

Syn gas [Synthesis gas] can synthesize methanol (CH_3OH) in presence of catalyst ($ZnO \mid Cr_2O_3$) and from methanol we can synthesize many other organic compounds.

$$CO + 2H_2 \xrightarrow[ZnO/Cr_2O_3]{\text{Catalyst}} CH_3 - OH$$

- The process of producing 'syngas' from coal is called 'coal gasification'.

$$CO + H_2O \xrightarrow[Fe_2O_3 \cdot Cr_2O_3]{500°C} CO_2 \uparrow + H_2 \uparrow \xrightarrow[\text{absorbs } CO_2 \text{ gas}]{NaAsO_2 \text{ solution}} H_2\uparrow$$

Above reaction is known as water gas shift reaction.

CO_2 gas is removed by sodium arsenite solution, and this process is called as scrubbing.

6 Reaction of Steam on Hydrocarbon

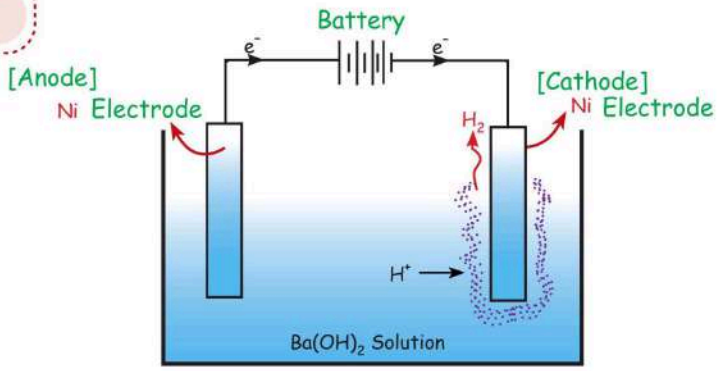
$$\underset{\text{methane}}{CH_4} + H_2O \xrightarrow[Ni]{1000°C} CO + 3H_2$$

What is coal gas?

- Coal Gas - $[CO + H_2 + CH_4]$

Preparation of Pure Hydrogen

1. The electrolysis of a solution of $Ba(OH)_2$ using Ni electrodes gives extra pure H_2.
2. $NaH + H_2O \longrightarrow H_2\uparrow + NaOH\ (aq)$.



Note

Hydrogen Economy: In space programmes, it is used as a rocket fuel. In fact, it has promising potential for use as a non-polluting fuel of the near future.

- The basic principle of hydrogen economy is the transportation and storage of energy in the form of liquid or gaseous dihydrogen. Advantage of hydrogen economy is that energy is transmitted in the form of dihydrogen and not as electric power.

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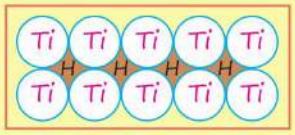
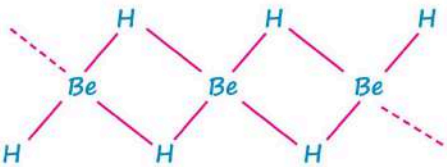
- Dihydrogen gas is converted into liquid state by cooling to 20 K. This would require expensive insulated tanks.
- Tanks of metal alloy like $NaNi_5$, $Ti\text{-}TiH_2$, $Mg\text{-}MgH_2$ etc. are in use for storage of dihydrogen in small quantities.

Energy released on combustion in kJ state	Dihydrogen (in gaseous state)	Dihydrogen (in liquid)	LPG	CH_4 gas	Octane (in liquid state)
per mole	286	285	2220	880	5511
per gram	143	142	50	53	47
per litre	12	9968	25590	35	34005

TYPE OF HYDRIDES

Saline / Salt Hydride [Ionic Hydride]	Covalent / Molecular Hydrides	Interstitial Hydrides [Metallic Hydrides]
❑ s block Metal + Hydrogen ❑ NaH, KH, CaH_2 ❑ Crystalline solids ❑ High M.P.& B.P. due to strong ionic bond ❑ Non-volatile and Non-conducting in solid state ❑ Good conductor of Electricity in molten & liquid state.	❑ p block (Non-metal) + Hydrogen ❑ CH_4, NH_3, PH_3, HF, H_2S ❑ Less M.P. & B.P. due to weak intermolecular force of attraction. ❑ Poor conductor of electricity	❑ d & f block Metal + Hydrogen ❑ Fe_3H, $VH_{0.56}$ and $TiH_{1.7}$  $TiH_{1.7}$ (Interstitial compound) ❑ High M.P. & B.P. ❑ Non-stoichiometric compound

Ionic Hydride



- Significant covalent character is found in the lighter metal hydrides (due to small size of ion) such as LiH, BeH_2 and MgH_2
- In fact, BeH_2 and MgH_2 are Polymeric in structure.
- On electrolysis liberate dihydrogen gas at anode, which confirms the existence of H^- ion.

$$2H^-(melt) \xrightarrow{Anode} H_2(g) + 2e^-$$

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- Saline hydrides react violently with water producing dihydrogen gas.

$$NaH\,(s) + H_2O\,(aq) \longrightarrow NaOH\,(aq) + H_2\uparrow$$

$$CaH_2(s) + 2H_2O(aq) \longrightarrow Ca(OH)_2(aq) + H_2\uparrow$$

Important

CaH_2 (Hydrolith): Used as a reducing agent and is also a source for Hydrogen. It is called Hydrolith because it is used in industries to prepare Dihydrogen.

Covalent / Molecular Hydrides

Based on relative numbers of electrons and bonds in their Lewis structure into:

(i) Electron - deficient hydrides → BH_3/B_2H_6 [Group 13 Hydrides]
These hydrides have less than 8 electrons on central atom and therefore they are electron deficient.

(ii) Electron - precise hydrides → CH_4 [Group 14 Hydrides]
These hydrides have 8 electrons on its central atom with tetrahedral geomety.

(iii) Electron - rich hydrides → $\ddot{N}H_3$ / $H_2\ddot{O}$ / $H-\ddot{F}:$ [Group 15/16/17 Hydrides]
These hydrides have excess of electrons which are present as lone pairs.

Interstitial Hydrides

- Metals of group 7,8 and 9 do not form hydride. Even from group 6, only chromium forms CrH.

6^{th}	7^{th}	8^{th}	9^{th}
Cr	Mn	Fe	Co
CrH	X	X	X

- Strong Media For Hydrogen: Some of the metals (Pd, Pt) can accommodate a very large volume of hydrogen.

H_2O VS D_2O (HEAVY WATER)

Property	H_2O		D_2O
Molecular mass (g mol^{-1})	18.0151	<	20.0276
Melting point/K	273.0	<	276.8
Boiling point/K	373.0	<	374.4
Temp of max. density/K	276.98	<	284.2
Density (298K)/g cm^{-3}	1.0000	<	1.1059

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Viscosity / centipoise	0.8903	<	1.107
Dielectric constant/ C^2 / $N.m^2$	(Better solvent) 78.39	>	78.06

- D_2O is extensively used as a moderator in nuclear reactors and in exchange reaction for the study of reaction mechanisms.

Reaction with halogens: While the reaction with fluorine occurs even in the dark, with iodine it requires a catalyst.

- $H_2 + F_2 \longrightarrow 2\,HF$
- $H_2 + I_2 \xrightarrow[\text{Catalyst}]{} 2\,HI$

Reactivity : $F_2 > Cl_2 > Br_2 > I_2$

DIFFERENT FORM OF HYDROGEN

(i) **Simple atomic hydrogen:** It is formed by simple dissociation of hydrogen.

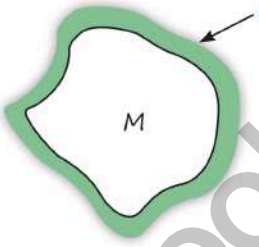
$H_2 \overset{3000\text{-}5000}{\rightleftharpoons} 2H$

- Favourable condition are high temp & low pressure.

(ii) **Nascent hydrogen:** Hydrogen at the moment of its birth it called nascent hydrogen. It is formed only by some specific chemical reaction.

(a) Acid + Metals	:	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + 2[H]$
(b) Base + Amphoteric Metal (Be, Zn, Al, Ga, Pb, Sn)	:	$2NaOH + Be \longrightarrow Na_2BeO_2 + 2[H]$
(c) C_2H_5OH + Alkali metal	:	$C_2H_5OH + Na \longrightarrow C_2H_5ONa + [H]$

(iii) **Adsorbed/Occluded hydrogens**



Occlusion: The property of metal to adsorb any gas is called occlusion.

Adsorbed H is hydrogen present at the outer surface of metal.

DIHYDROGEN (H_2)

Dihydrogen (H_2) is the most abundant element in the universe.

Types of Nuclear Spin H_2

- 2 Types of Nuclear Spin in H_2: ortho and para both are nuclear spin isomer.

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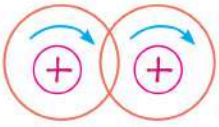
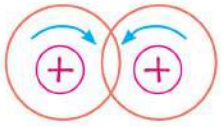


Ortho – H_2 molecule



Para – H_2 molecule

- **Ortho H_2 molecule:** spins of nuclei in the same direction
- **Para H_2 molecule:** spins of nuclei in the opposite direction
- **Room Temperature:** Ortho (75 %) + Para (25%)
- **0°K Temperature:** Para (mainly)
- Ortho and para both H_2 molecule have same chemical properties but different physical properties (thermal conductivity, boiling point).

We can obtain nearly 100% pure para-H_2 at low temperatures, as it is the more stable form (the nuclear spins are aligned oppositely). But pure ortho-H_2 is unattainable because Ortho-H_2 is dominant at higher temperatures, with para H_2.

HARD WATER & SOFT WATER

Hardness is due to presence of the bicarbonates [HCO_3^-], sulphates [SO_4^{2-}] and chlorides [Cl^-] of Ca and Mg.

$$\text{Hardness} \longrightarrow [HCO_3^-, SO_4^{2-}, Cl^-] \longrightarrow Ca^{+2} / Mg^{2+}$$

Soft water : It produces sufficient froth (lather) with the soap

Hard water : It forms an insoluble precipitate (scum) before it form a lather with soap.

$$\underset{\text{[Soap] Sodium stearate}}{2C_{17}H_{35}COO^-Na^+} + \underset{\text{Hard Water}}{CaCl_{2(aq.)}} \longrightarrow Ca(C_{17}H_{35}COO)_2\downarrow + 2NaCl$$

$$2C_{17}H_{35}COO^-Na^+ + MgSO_{4(aq.)} \longrightarrow Mg(C_{17}H_{35}COO)_2\downarrow + Na_2SO_4$$

Hard water is unsuitable for laundry & this reduces the efficiency of the boiler because of deposition of salts.

Temporary Hardness

- This hardness is due to bicarbonates of Ca and Mg [$Ca(HCO_3)_2$ or $Mg(HCO_3)_2$].

(1) **By Boiling** Bicarbonates decompose in insoluble carbonates (ppt).

$$2HCO_3^- \longrightarrow H_2O + CO_2\uparrow + CO_3^{2-}\downarrow$$

$$Ca(HCO_3)_{2\,(aq.)} \xrightarrow{\text{Boil}} CaCO_3\downarrow + H_2O + CO_2\uparrow$$

$$Mg(HCO_3)_{2\,(aq.)} \xrightarrow{\text{Boil}} Mg(OH)_2\downarrow + CO_2\uparrow$$

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(2) **Clark's Method** It can be removed by addition of slaked lime $[Ca(OH)_2]$.

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow H_2O + 2CaCO_3\downarrow$$

$$Mg(HCO_3)_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaCO_3\downarrow + H_2O$$

$$HCO_3^- \xrightarrow[-H_2O]{OH^-} CO_3^{2-}$$

Permanent Hardness

- This hardness is due to Sulphates, Chlorides of Ca, Mg [SO_4^{2-} and $Cl^- \rightarrow Ca^{+2}$ / Mg^{2+}]

This type of hardness cannot be removed by boiling or by the addition of slaked lime.

Water Softeners

(1) Washing Soda → $Na_2CO_3 \cdot 10\,H_2O$

(2) Calgon → $[NaPO_3]_6 \equiv Na_6P_6O_{18}$ ≡ Sodium hexa meta phosphate

(3) Permutit/Hydrated Zeolite

(4) Synthetic Resin Method

- H_3PO_4
  Ortho – Phosphoric acid
- $H_3PO_4 - H_2O \equiv HPO_3$ ($-H^+$)
  meta – phosphoric acid
  meta phosphate ion PO_3^-

Washing Soda

It removes both the temporary and permanent hardness by converting soluble Ca & Mg compounds into insoluble carbonates.

$$CaCl_2 + Na_2CO_3 \longrightarrow 2NaCl + CaCO_3\downarrow$$

$$CaSO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + CaCO_3\downarrow$$

$$Ca(HCO_3)_2 + Na_2CO_3 \longrightarrow 2NaHCO_3\downarrow + CaCO_3\downarrow$$

Calgon

Sodium Hexametaphosphate = $Na_4[Na_2(PO_3)_6] \equiv Na_6P_6O_{18} \equiv [NaPO_3]_6$

$$CaSO_4 + Na_4[Na_2(PO_3)_6] \longrightarrow Na_4[Ca(PO_3)_6] + Na_2SO_4$$

$$MgSO_4 + Na_4[Na_2(PO_3)_6] \longrightarrow Na_4[Mg(PO_3)_6] + Na_2SO_4$$

The complex anion keeps the Mg^{2+} and Ca^{2+} ions in solution.

- Polyphosphates act as water softening agents by forming soluble complexes with cations like Ca^{2+} and Mg^{2+}, which are responsible for water hardness.

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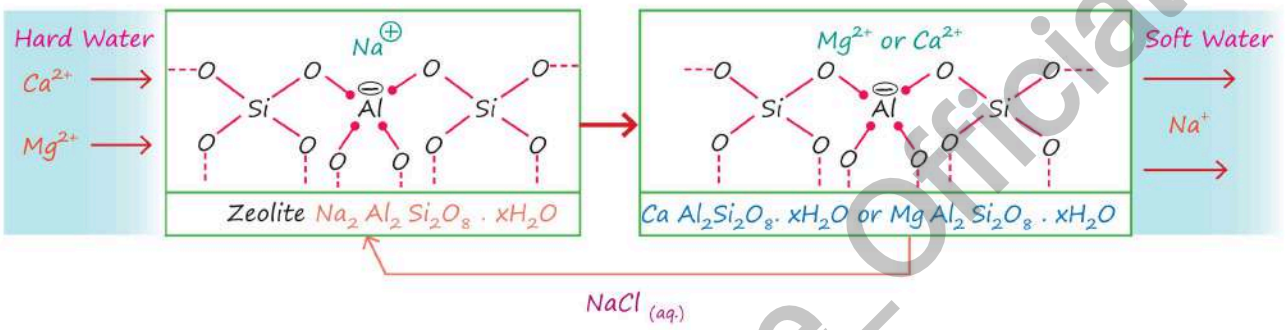
Permutit/Hydrated Zeolite (Ion-exchange method)

$Na_2O \cdot Al_2O_3 \cdot 2SiO_2 \cdot xH_2O$ [Alumino–silicate]

- $\rightarrow Na_2Al_2Si_2O_8$
- $\rightarrow 2\ NaAlSiO_4$
- $\rightarrow 2\ NaZ$ $\quad [Z \equiv AlSiO_4]$



- Permutit/zeolite is said to be exhausted when all the sodium in it is used up. It is regenerated for further use by treating with an aqueous sodium chloride solution.
- This method is useful for the removal of both temporary and permanent hardness of water.

3. The one that is NOT suitable for the removal of permanent hardness of water is:

(a) Clark's method (b) Ion-exchange method

(c) Calgon's method (d) Treatment with sodium carbonate

Sol. (a) Clark's method uses calcium hydroxide to remove temporary hardness, which is caused by calcium and magnesium bicarbonates.

Therefore, the method that is NOT suitable for the removal of permanent hardness of water is Clark's method.

4. The temporary hardness of water is due to:

(a) Na_2SO_4 (b) $NaCl$ (c) $Ca(HCO_3)_2$ (d) $CaCl_2$

Sol. (c) The temporary hardness of water is primarily caused by the presence of dissolved calcium and magnesium bicarbonates.

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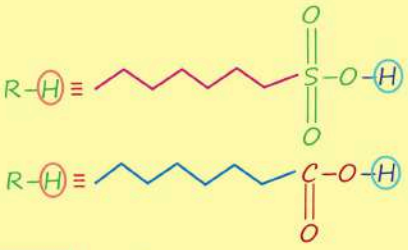
Synthetic Resins Method

Cation Exchange Resin [R–H]	Anion Exchange Resin [$R'-\overset{+}{N}H_3\bar{O}H$]
❑ Insoluble organic acid resin having giant molecules with $-SO_3H$ or $-COOH$ groups	❑ Giant organic molecules with basic group derived from amines

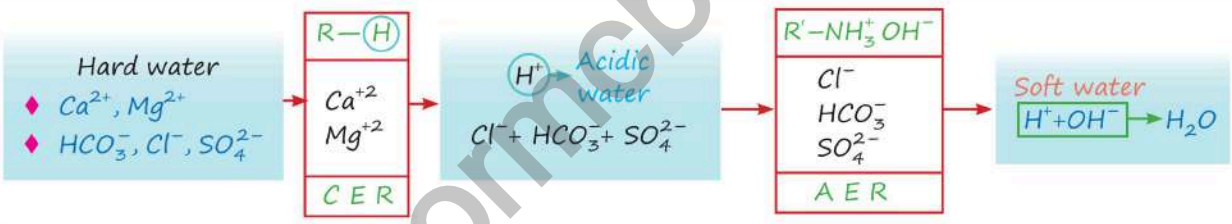


$$2R-H + Ca^{2+} \longrightarrow R_2Ca + 2H^+$$

$$2R^- + 2H^+$$

❑ $R'-\overset{+}{N}H_3\bar{O}H + Cl^- \longrightarrow R-\overset{+}{N}H_3Cl^- + OH^-$

❑ Pure de- mineralised (de- ionized) water free from all soluble mineral salts is obtained by passing water successively through a cation exchange (in the H^+ form) and an anion- exchange (in the OH^- form) resins–



❑ The hard water is first passed through a bed of CER [Cation Exchange Resin] which removes the cations like Na^+, Mg^{2+}, Ca^{2+} by exchanging with H^+ ions.

❑ The water coming from cation exchanger is acidic on account of free H^+ ions.

❑ This water is then passed through another bed containing anion exchanger.

❑ AER [Anion Exchange Resin] removes anions like Cl^-, SO_4^{2-}, NO_3^- by exchanging with OH^- ions.

❑ The OH^- ions neutralize the H^+ ions and form H_2O.

Note

In the zeolite process, it utilizes the mineral zeolites as the exchanging resins for only cations in hard water.

❑ In synthetic resins method, it utilizes many different resins for ion exchange for both cations and anions.

So, the synthetic resins method is more efficient compared to the zeolite process.

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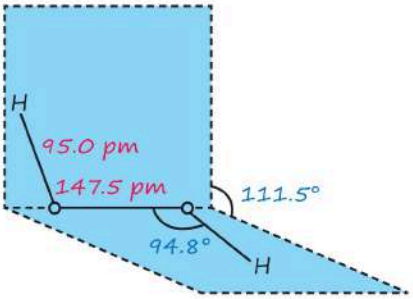
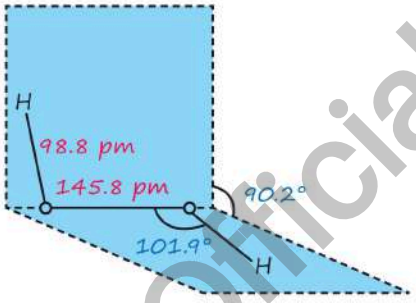
HYDROGEN PEROXIDE (H_2O_2)

Structure of H_2O_2



(a) Gas phase

H_2O_2 structure in gas phase, dihedral angle is 111.5°.



(b) Solid phase

H_2O_2 structure in solid phase at 110K, dihedral angle is 90.2°.

Preparation of H_2O_2

Laboratory Method

1. $\overset{Ba^{2+}O_2^{2-}}{BaO_2} \cdot 8H_2O + \overset{2H^+SO_4^{2-}}{H_2SO_4} \longrightarrow BaSO_4 + 8H_2O + H_2O_2$

2. Peroxodisulphate, obtained by electrolytic oxidation of acidified sulphate solutions at high current density, on hydrolysis yields hydrogen peroxide.

- $SO_4^{2-} \xrightarrow{H^+} HSO_4^-$ [acidified sulphate solution]
- $H-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\ddot{O}: \xrightarrow[\text{on anode}]{\text{Oxidation}} H-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\dot{O}:$
- $H-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\ddot{O}: + :\ddot{O}-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OH \longrightarrow HO-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-O-O-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OH \xrightarrow{\text{Hydrolysis}} 2HO-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OH + H_2O_2$

This method is now used for the laboratory preparation of D_2O_2.

- $K_2S_2O_{8\,(s)} + 2D_2O_{(l)} \longrightarrow D_2O_2$

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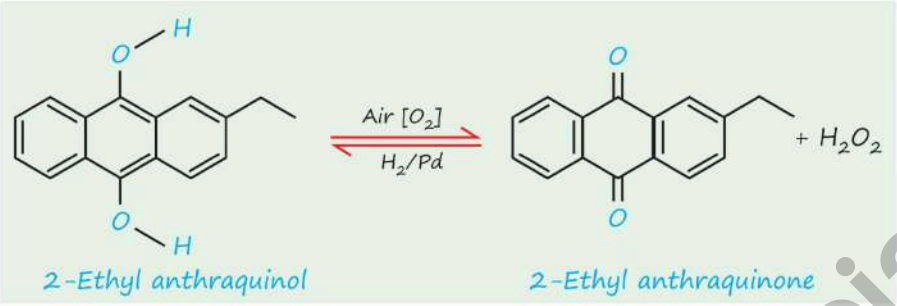
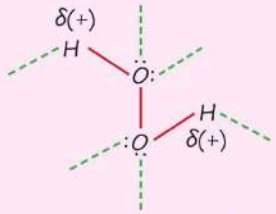
Industrial Method → Auto oxidation of 2-Ethyl anthraquinol



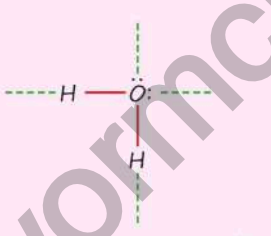
In this case 1% H_2O_2 is formed. It is extracted with water and concentrated to 30% (by mass) by distillation under reduced pressure. It can be further concentrated to 85% by careful distillation under low pressure. The remaining water can be frozen out to obtain pure H_2O_2.

Physical Properties of H_2O_2

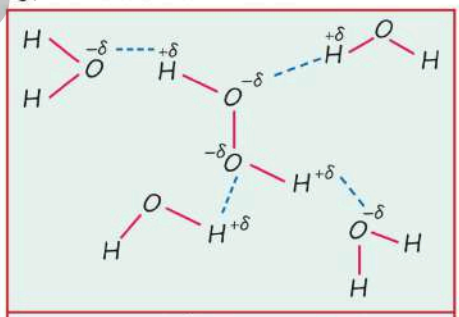
- Colourless viscous liquid which appears blue in the larger quantity and is soluble in water.
- The extent of hydrogen bonding in H_2O_2 is more than in H_2O due to availability of greater number of sites per molecule.



Six sites are available per molecule



Four sites are available per molecule



H_2O_2 is soluble in water due to intermolecular hydrogen bonding between H_2O and H_2O_2.

- Density : $H_2O < H_2O_2$
- Viscosity : $H_2O < H_2O_2$
- Boiling point : $H_2O < H_2O_2$
- Dielectric constant : $H_2O >> H_2O_2$
- H_2O_2 cannot be used as solvent as it dissociates into H_2O.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

- In the presence of metal surfaces or traces of alkali (present in glass containers Na_2SiO_3), the above reaction is catalysed. It is, therefore, stored in wax-lined glass or plastic vessels in dark.

Note

Hydrazine (N_2H_4) and hydrogen peroxide (H_2O_2) can be used together as a rocket propellant-

$$N_2H_{4(\ell)} + 2H_2O_{2(\ell)} \longrightarrow N_{2(g)} + 4H_2O_{(g)}$$

- Highly exothermic reaction with large increase in volume.

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Uses of Hydrogen Peroxide

1. It is commonly used as a hair bleach and a mild disinfectant. As an antiseptic, it is available in the market as perhydrol.
2. It is used to produce chemicals like sodium perborate and sodium percarbonate, which are key ingredients in high-quality detergents.
3. It plays a role in the preparation of hydroquinone, tartaric acid, and several food products and medicines, including some antibiotics like cephalosporins.
4. In industries, it serves as a bleaching agent for textiles, paper, leather, oils, and fats.
5. It is also used in environmental (green) chemistry, such as in:
   - Treating domestic and industrial waste,
   - Oxidizing harmful cyanides, and
   - Restoring oxygen levels in sewage treatment.

ALUM

Alums $M_2SO_4.M'_2(SO_4)_3.24H_2O$

M Univalent positive cation → Na^+ or K^+ or NH_4^+

M' Trivalent positive cation → Fe^{3+} or Cr^{+3} or Al^{+3}

Examples

$K_2SO_4.Al_2(SO_4)_3.24H_2O$ (Potash alum)

$(NH_4)_2SO_4.Al_2(SO_4)_3.24H_2O$ (Ammonium alum)

PORTLAND CEMENT

- Raw material: limestone and clay
- clay + lime ⟶ cement clinker
- cement clinker + 2-3% of gypsum = cement
- Setting of Cement: The purpose of adding gypsum is only to slow down the process of setting of the cement so that it gets sufficiently hardened.
- **Composition**

CaO : 50 - 60%	SiO_2 : 20 - 25%	Al_2O_3 : 5 - 10%
MgO : 2 - 3%	Fe_2O_3 : 1 - 2%	SO_3 : 1 - 2%

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❑ Important Ingredients

Dicalcium silicate ($2CaO \cdot SiO_2$) – 26% Tricalcium silicate ($3CaO \cdot SiO_2$) – 51%

Tricalcium aluminate ($3CaO \cdot Al_2O_3$) – 11%

❑ For a good quality cement–

$SiO_2 / Al_2O_3 = 2.5 - 4$ $CaO / (Fe_2O_3 + Al_2O_3 + SiO_2) = 2$

5. Polyphosphates are used as water softening agent becuase they :- (IIT JEE 2002)

(a) Form soluble complexes with anionic species
(b) Precipitate anionic species
(c) Form soluble complexes with cationic species
(d) Precipitate cationic species.

Sol. (c)

6. Hydrogen peroxide in its reaction with KlO_4 and NH_2OH respectively, is acting as a (JEE Adv. 2014)

(a) reducing agent, oxidising agent
(b) reducing agent, reducing agent
(c) oxidising agent, oxidising agent
(d) oxidising agent, reducing agent

Sol. (a) $H_2O_2 + KIO_4 \rightarrow KIO_3 + H_2O + O_2$
Here, H_2O_2 acts as a reducing agent.
$H_2O_2 + NH_2OH \rightarrow HNO_3 + H_2O$
Here, H_2O_2 acts as an oxidizing agent.

7. Which of the following combination will produce H_2 gas? (JEE Adv. 2017)

(a) Zn metal and NaOH (aq.)
(b) Au metal and NaCN (aq.) in the presence of air
(c) Cu metal and conc. HNO_3
(d) Fe metal and conc. HNO_3

Sol. (a)

8. In comparison to the zeolite process for the removal of permanent hardness, the synthetic resins method is: [JEE (Main) 2020]

(a) less efficient as it exchange only anions
(b) more efficient as it can exchange only cations
(c) less efficient as the resins cannot be regenerated
(d) more efficient as it can exchange both cations as well as anions

Sol. (d)

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9. **Among the statements (1) – (4), the correct ones are–** [JEE (Main) 2020]

(1) Decomposition of hydrogen peroxide gives dioxygen

(2) Like hydrogen peroxide, compounds, such as $KClO_3$, $Pb(NO_3)_2$ and $NaNO_3$ when heated liberated dioxygen

(3) 2-Ethylanthraquinone is useful for the industrial preparation of hydrogen peroxide.

(4) Hydrogen peroxide is used for the manufacture of sodium perborate

(a) (1), (2) and (3) only (b) (1) and (3) only

(c) (1), (2), (3) and (4) (d) (1), (3) and (4) only

Sol. (c)

10. **The correct statements among (1) to (4) regarding H_2 as a fuel are:** [JEE (Main) 2019]

(1) It produce less pollutant than petrol

(2) A cylinder of compressed dihydrogen weighs ~30 times more than a petrol tank producing the same amount of energy.

(3) Dihydrogen is stored in tanks of metal alloys like $NaNi_5$

(4) On combustion, values of energy released per gram of liquid dihydrogen and LPG are 50 and 142 kJ, respectively.

(a) 2 and 4 only (b) 1, 2 and 3 only

(c) 2, 3 and 4 only (d) 1 and 3 only

Sol. (b)

11. **The correct statements among (1) to (4) are:** [JEE (Main) 2019]

(1) saline hydrides produce H_2 gas when reacted with H_2O.

(2) reaction of $LiAlH_4$ with BF_3 leads to B_2H_6.

(3) PH_3 and CH_4 are electron – rich and electron-precise hydrides, respectively.

(4) HF and CH_4 are called as molecular hydrides.

(a) (3) and (4) only (b) (1), (2) and (3) only

(c) (1), (2), (3) and (4) (d) (1), (3) and (4) only

Sol. (c)

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9 The s-Block Elements

(JEE Advanced)

Syllabus

Alkali and alkaline earth metals-reactivity towards air, water, dihydrogen, halogens, acids; their reducing nature including solutions in liquid ammonia; uses of these elements; general characteristics of their oxides, hydroxides, halides, salts of oxoacids; anomalous behaviour of lithium and beryllium; preparation, properties, and uses of compounds of sodium (sodium carbonate, sodium chloride, sodium hydroxide, sodium hydrogen carbonate) and calcium (calcium oxide, calcium hydroxide, calcium carbonate, calcium sulphate).

s-BLOCK ELEMENTS

s Block

Group – 1 Alkali Metals

Lithium	Li : [He] $2s^1$
Sodium	Na : [Ne] $3s^1$
Potassium	K : [Ar] $4s^1$
Rubidium	Rb : [Kr] $5s^1$
Caesium	Cs : [Xe] $6s^1$
Francium	Fr : [Rn] $7s^1$

→ MOH is alkaline in nature.

- Form hydroxides on reaction with water which are strongly alkaline in nature

$$Na \xrightarrow{H_2O} NaOH + \frac{1}{2}H_2\uparrow$$

- Fr is highly radioactive → its longest-lived isotope ^{223}Fr has a half-life of only 21 minutes.

Group – 2 Alkaline Earth Metals

Beryllium	Be : [He] $2s^2$
Magnesium	Mg : [Ne] $3s^2$
Calcium	Ca : [Ar] $4s^2$
Strontium	Sr : [Kr] $5s^2$
Barium	Ba : [Xe] $6s^2$
Radium	Ra : [Rn] $7s^2$

→ BeO & $Be(OH)_2$: Amphoteric

→ MO & $M(OH)_2$: Basic in nature

- Oxides and hydroxides of alkaline earth metals are alkaline (basic) in nature and these metal oxides are found in the earth's crust*.
- Be is not a alkaline earth metal because their hydroxides are amphoteric, not alkaline.
- Ra is radioactive in nature.
- Beryllium is rare and radium is the rarest of all comprising only 10^{-10} percent of igneous rocks#.

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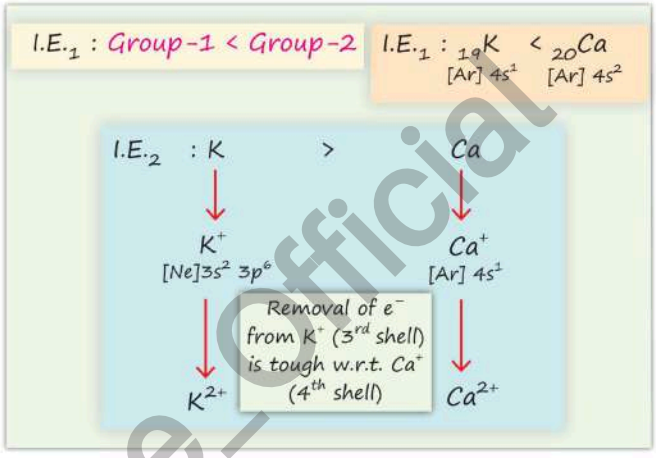
Note

- *The thin, rocky outer layer of the Earth is crust.
- #A type of rock formed from magma (molten rock) that has cooled and hardened.

Oxidation State

- Oxidation state – Group – 1 : + 1
  Group – 2 : + 2
- As we move from top to bottom in group – 1 and group – 2, atomic size ↑ and ionisation energy ↓.
- Metallic character ↑ because IE ↓.
  [$M \longrightarrow M^+ + e^-$]
  Metallic character : Li < Na < K < Rb < Cs < Fr
- Photoelectric Effect:
  - Cs & K useful as electrodes in photoelectric cells
  - Cs is used in devising photoelectric cells.



1. The metal mainly used in devising photoelectric cells is: [02 Sept, 2020 (Shift-I)]

(a) Rb (b) Cs (c) Li (d) Na

Sol. (b) The metal mainly used in devising photoelectric cells is Cs (Cesium).

Reducing Nature of Alkali Metals

- E°_{RP} = high (+) value means M^+ has greater tendency to get reduced in Metal.
- E°_{RP} = high (–) value means M^+ has low tendency to get reduced in metal.
- Reduction Potential depend on 3 following factors.

(a) $M_{(s)} \longrightarrow M_{(g)}$		Sublimation energy
(b) $M_{(g)} \longrightarrow M^+_{(g)} + e^-$		Ionisation enthalpy [IE_1]
(c) $M^+_{(g)} + H_2O \longrightarrow M^+_{(aq)}$		Hydration enthalpy
$M_{(s)} \longrightarrow M^+_{(aq)}$		

Elements	Standard potentials $E^\circ_{M^+/M}$
Li	– 3.04
Na	–2.714
K	–2.925
Rb	–2.93
Cs	–2.927

- $M_{(s)} \xrightarrow{\text{Oxidation}} M^+_{(aq.)}$ becomes very spontaneous [high (–)ve E° value] when –
  1. Sublimation energy of metal $M_{(s)}$ is low.
  2. 1st ionisation enthalpy of $M_{(g)}$ is low.
  3. Hydration enthalpy of $M^+_{(g)}$ is high.

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- With the small size of its ion, Li^+ has the highest hydration enthalpy which accounts for its high negative E^θ value and its high reducing power.

Reducing Nature of Alkaline Earth Metals –

- Reduction Potential [$E^\circ_{M^{2+}/M}$] depend on 3 following factors.

(a) $M_{(s)} \longrightarrow M_{(g)}$ Atomisation enthalpy

(b) $M_{(g)} \longrightarrow M^{2+}_{(g)} + 2e^-$ Ionisation enthalpy $[IE_1+IE_2]$

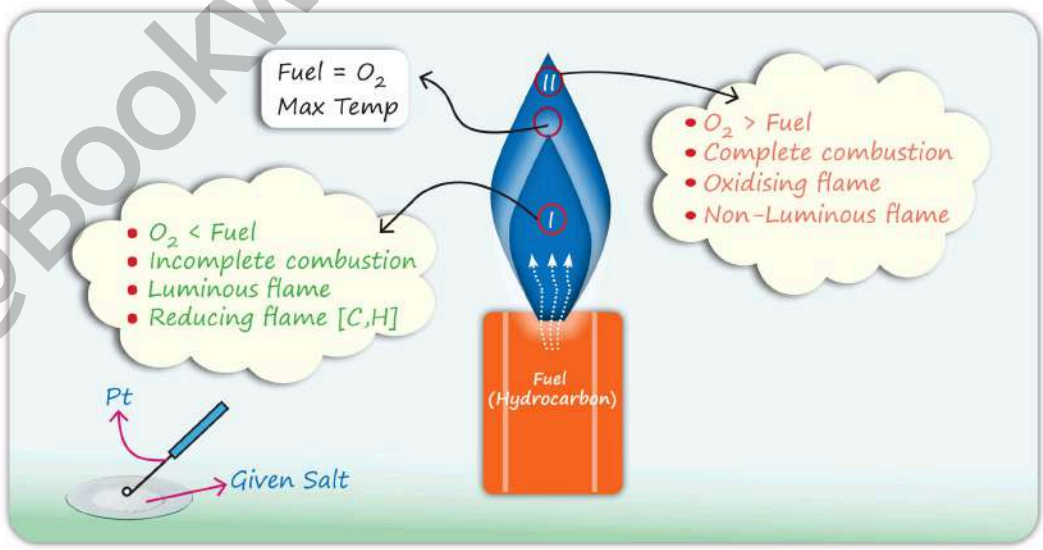
(c) $M^{2+}_{(g)} + H_2O \longrightarrow M^{2+}_{(aq)}$ Hydration enthalpy

$M_{(s)} \longrightarrow M^{2+}_{(aq)}$

Elements	Standard potentials $E^\circ_{M^{+2}/M}$
Be	–1.97
Mg	–2.36
Ca	–2.84
Sr	–2.89
Ba	–2.92
Ra	–2.92

- $M_{(s)} \xrightarrow{\text{Oxidation}} M^{2+}_{(aq.)}$ becomes very spontaneous [high (–)ve E° value] when –
  1. Sublimation energy/Atomisation energy of metal $M_{(s)}$ is low.
  2. Sum of 1st and 2nd ionisation enthalpy of $M_{(g)}$ is low.
  3. Hydration enthalpy of $M^{2+}_{(g)}$ is high.
- Be has less negative value compared to other alkaline earth metals. However, its reducing nature is due to large hydration energy associated with the small size of Be^{2+} ion and relatively large value of the atomization enthalpy of the metal.

FLAME TEST



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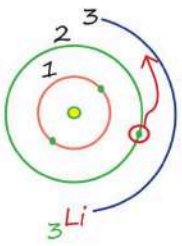

Li	Crimson Red	Be	No, Colour
Na	Yellow	Mg	
K	Violet	Ca	Brick Red
Rb	Red Violet	Sr	Crimson Red
Cs	Blue	Ba	Apple Green

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- Sr Li - Crimson Red
- K.V. - Yello Na
- Rb - Red violet
- Cs is blue
- Baag is apple green

The electrons in Be and Mg are too strongly bound to get excited by flame. Hence, these elements do not impart any colour to the flame.

2. Match List-I with List-II.

	List-I (Elements)		List-II (Colour imparted to flame)
(A)	K	(I)	Brick Red
(B)	Ca	(II)	Violet
(C)	Sr	(III)	Apple Green
(D)	Ba	(IV)	Crimson Red

[25Jan, 2023 (Shift-I)]

(a) (A)-(II), (B)-(I), (C)-(III), (D)-(IV) (b) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(c) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) (d) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Sol. (c)

3. Match List-I with List-II.

	List-I (Metal)		List-II (Emitted light wavelength (nm))
(A)	Li	(I)	670.8
(B)	Na	(II)	589.2
(C)	Rb	(III)	780.0
(D)	Cs	(IV)	455.5

Choose the most appropriate answer from the options given below : [29 June,2022 (Shift-I)]

(a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV) (b) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(c) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) (d) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Sol. (a)

Metal	Li	Na	K	Rb	Cs
Colour	Crimson red	Yellow	Violet	Red violet	Blue
λ/nm	670.8	589.2	766.5	780.0	455.5

Metallic Bonding ⟶ Electron Sea Model

- Metal Cations (Kernel) are fixed within a sea of electrons.

$M \longrightarrow M^+ + e^-$ [No of e^- in Valence shell ↑ : Metallic bonding ↑]

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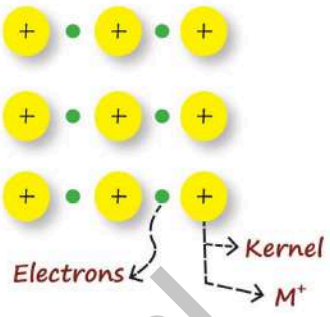
No. of valence shell electrons	:	Group I < Group II
Metallic Bonding	:	Group I < Group II
Melting point	:	Group I < Group II
Boiling point	:	Group I < Group II
Soft	:	Na [Knife Cutting] > Ca



PROPERTIES

- Metals have high Electrical and Thermal conductivity.
- Metals are Silvery White, Soft & Light, but Be / Mg → Greyish in colour.
- Hydration Enthalpy $\alpha \frac{1}{r} \Rightarrow$ $Li^+ > Na^+ > K^+ > Rb^+$: HE
- M.P / B.P. : Li > Na > K > Rb > Cs
- Density : Li < K > Na < Rb < Cs

Property	Lithium Li	Sodium Na	Potassium K	Rubidium Rb	Caesium Cs
Ionization enthalpy / kJ mol^{-1}	520	496	419	403	376
Hydration enthalpy / kJ mol^{-1}	-506	-406	-330	-310	-276
Metallic radius / pm	152	186	227	248	265
Ionic radius M^+ / pm	76	102	138	152	167
m.p. / K	454	371	336	312	302
b.p / K	1615	1156	1032	961	944
Density / g cm^{-3}	0.53	0.97	0.86	1.53	1.90

Property	Beryllium Be	Magnesium Mg	Calcium Ca	Strontium Sr	Barium Ba
Ionization enthalpy (I) / kJ mol^{-1}	899	737	590	549	503
Ionization (II) / kJ mol^{-1}	1757	1450	1145	1064	965
Hydration enthalpy (kJ / mol)	– 2494	– 1921	– 1577	– 1443	– 1305
Metallic radius / pm	111	160	197	215	222
Ionic radius M^{2+} / pm	31	72	100	118	135
m.p. / K	1560	924	1124	1062	1002
b.p / K	2745	1363	1767	1655	2078
Density / g cm^{-3}	1.84	1.74	1.55	2.63	3.59

Density : Be > Mg > Ca (lowest) < Sr < Ba : Ba > Sr > Be > Mg > Ca

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4. The density of alkali metals is in the order [12 April, 2023 (Shift-I)]

(a) Na < K < Cs < Rb (b) K < Na < Rb < Cs

(c) K < Cs < Na < Rb (d) Na < Rb < K < Cs

Sol. (b) Density : Li < K < Na < Rb < Cs

5. The correct order of density is [25 July, 2022 (Shift-II)]

(a) Be > Mg > Ca > Sr (b) Sr > Ca > Mg > Be

(c) Sr > Be > Mg > Ca (d) Be > Sr > Mg > Ca

Sol. (c) Density: Ba > Sr > Be > Mg > Ca

6. Alkali metal from the following with least melting point is : [20 July, 2021 (Shift-I)]

(a) Rb (b) K (c) Na (d) Cs

Sol. (d) The alkali metal with the least melting point is Caesium (Cs).

7. The number of alkali metal(s), from Li, K, Cs, Rb having ionization enthalpy greater than 400 kJ mol^{-1} is _____. [31 Jan, 2023 (Shift-II)]

Sol. [3] Li, K, and Rb have ionisation enthalpy greater than 400 kJ. but Cs has ionisation enthalpy less than 400 kJ.

Reaction with Water

Li, Na, K, Rb, Cs $\xrightarrow{H_2O}$ $MOH + \frac{1}{2}H_2$ + Energy

Be, Mg $\xrightarrow[\text{water}]{\text{Hot}}$ $M(OH)_2 + H_2$

Ca, Sr, Ba $\xrightarrow{\text{water}}$ $M(OH)_2 + H_2$

Note

- Na to Cs : All metals are kept in Kerosene.
- Li metal because of low density is a light metal and therefore floats on the surface of the kerosene oil. To prevent its contact with air, it is kept wrapped in paraffin wax.

Reaction with O_2

Li $\xrightarrow{O_2}$ Oxide

Na $\xrightarrow{O_2}$ peroxide

K, Rb, Cs $\xrightarrow{O_2}$ Superoxide

$Li + O_2 \longrightarrow \underset{[major]}{Li_2O} + \underset{[minor]}{Li_2O_2}$

$Na + O_2 \longrightarrow \underset{[minor]}{NaO_2} + \underset{[major]}{Na_2O_2}$

$K / Rb / Cs \xrightarrow{O_2} KO_2 / RbO_2 / CsO_2$

The oxides and the peroxides are colourless when pure, but the superoxides are yellow or orange in colour. The increasing stability of the peroxide or superoxide, as the size of the metal ion increases, is due to the stabilisation of large anions by larger cations through lattice energy effects.

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Oxides

$Na_2O \xrightarrow[H^+\bar{O}H]{H_2O} 2NaOH$

$2Na^+ O^{2-}$

$[O^{2-} + H^+ \longrightarrow O^- - H]$

oxide — hydroxide

Peroxides

$[O_2^{2-}: O^- - O^-]$

$Na_2O_2 \xrightarrow[2H^+ 2OH^-]{2H_2O} H_2O_2 + 2NaOH$

$2Na^+ O_2^{2-}$

Hydrogen peroxide

Superoxides

$2KO_2 \xrightarrow[H^+ /\bar{O}H]{H_2O} O_2\uparrow + H_2O_2 + 2KOH$

$2O_2^-$ (Super–oxide)

Oxidation → $O_2\uparrow$ oxygen gas

Reduction → O_2^{2-} peroxide

8. A s-block element (M) reacts with oxygen to form an oxide of the formula MO_2. The oxide is pale yellow in colour and paramagnetic. The element (M) is: [20 July, 2021 (Shift–I)]

(a) Na (b) K (c) Mg (d) Ca

Sol. (b) The element (M) is K and oxide is KO_2.

- Oxide (O^{2-}) & peroxide (O_2^{2-}) ⟶ Colourless
  (O_2^-) (Superoxide) ⟶ Yellow
- O_2^{2-} ($18e^-$) ⟶ MOT (0 unpaired electron) ⟶ Diamagnetic
  O_2^- ($17e^-$) ⟶ paramagnetic

9. On combustion of Li, Na and K in excess of air, the major oxides formed, respectively, are: [04 Sept, 2020 (Shift–I)]

(a) Li_2O, Na_2O_2 and K_2O (b) Li_2O, Na_2O_2 and KO_2

(c) Li_2O, Na_2O and K_2O_2 (d) Li_2O, Na_2O_2 and K_2O_2

Sol. (b)

Reaction with N_2

- Li $\xrightarrow{N_2}$ Li_3N
- Mg $\xrightarrow{N_2}$ Mg_3N_2
- Na $\xrightarrow{N_2}$ No reaction

Li — Diagonal Relationship — Mg

Na, K, Rb, Cs: $\xrightarrow{N_2}$ No Reaction

Be, Mg, Ca, Sr, Ba: $\xrightarrow{N_2}$ Metal nitride

Reaction with Air

❒ Air (78% N_2 + 21% O_2)

Li $\xrightarrow{\text{Air}}$ $\xrightarrow{O_2}$ Li_2O; $\xrightarrow{N_2}$ Li_3N

Mg $\xrightarrow{\text{Air}}$ $\xrightarrow{O_2}$ MgO; $\xrightarrow{N_2}$ Mg_3N_2

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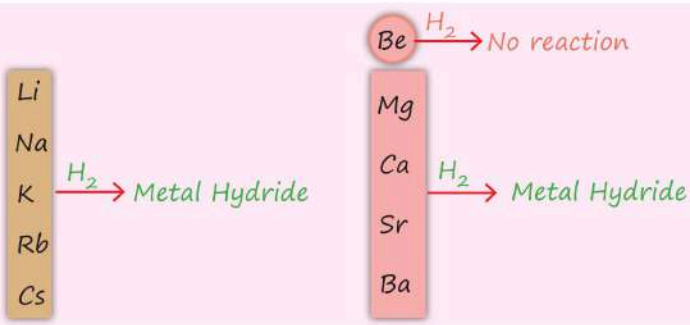
- Be and Mg are kinetically inert to oxygen and water because of the formation of an oxide film on their surface.
- However, powdered Be burns brilliantly on ignition in air to give BeO and Be_3N_2.
- Mg is more electropositive and burns with dazzling brilliance in air to give MgO and Mg_3N_2.
- Ca, Sr and Ba are readily attacked by air to form the oxide and nitride.

Reaction with H_2



- $2\ Na + H_2 \longrightarrow 2\ Na^+H^-$
- $Mg + H_2 \longrightarrow Mg^{2+}.2H^- \equiv MgH_2$

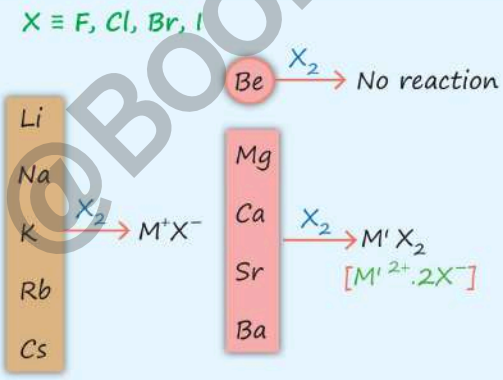
BeH_2

BeH_2 can be prepared by the reaction of $BeCl_2$ with $LiAlH_4$.

$$Cl - Be - Cl \xrightarrow{LiAlH_4} BeH_2 + LiCl + AlCl_3$$

($LiAlH_4$ → $LiH + AlH_3$; H^- Donors)

Reaction with X_2



- $Na_2O \xrightarrow{2HCl} 2NaCl + H_2O$; $[2Na^+O^{2-}]$; $O^{2-} + 2H^+ \longrightarrow H_2O$
- $NaOH \xrightarrow{HCl} NaCl + H_2O$
- $Na_2CO_3 \xrightarrow{2HCl} 2NaCl + H_2CO_3$; $[2Na^+ CO_3^{2-}]$; $H_2CO_3 \longrightarrow H_2O + CO_2\uparrow$

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Note

$BeF_2 + 2F^- \longrightarrow [BeF_4]^{2-}$ tetrafluoroberylate

❒ **Formation of BeF_2 :**

$(NH_4)_2\,[BeF_4]$ Ammonium-tetrafluoroberylate $\xrightarrow{\Delta}$ $BeF_2 + 2NH_3 \uparrow + 2HF$

$2NH_4^+ \cdot [BeF_4]^{2-}$

$2H^+ + 2\ddot{N}H_3$ ⟵ ⟶ $BeF_2 + 2F^-$

❒ **Formation of $BeCl_2$:** $2Be \xrightarrow{O_2} BeO$

$BeO + C + Cl_2 \underset{}{\overset{\Delta}{\rightleftharpoons}} CO \uparrow + BeCl_2$

ENTHALPY OF FORMATION

$Na_{(s)} + \frac{1}{2} Cl_{2\,(g)} \longrightarrow NaCl_{(s)} + \Delta H_f$

ΔH_f → Exothermic = ⊖ ve

❒ For a given metal ΔH_f° always becomes less negative from fluoride to iodide –

❒ $Na + \frac{1}{2} F_2 \longrightarrow NaF$; ΔH_f = most ⊖ ve

$\frac{1}{2} I_2$ ↓

NaI ; ΔH_f = less ⊖ ve

❒ ΔH_f° : NaF > NaCl > NaBr > NaI ⟶ less -ve ΔH_f

❒ M.P. / B.P. : NaF > NaCl > NaBr > NaI

: Fluoride > chloride > bromide > iodide

❒ ΔH_f° values for fluorides become less negative as we go down the group.

ΔH_f°: LiF > NaF > KF (less ⊖ ve)

ΔH_f°: LiCl < NaCl < KCl (most ⊖ ve)

LiBr < NaBr < KBr

LiI < NaI < KI

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HEATING EFFECT

Metal Carbonate

$$\text{Metal Carbonate} \xrightarrow{\Delta} \text{Metal Oxide} + CO_2\uparrow$$

M = Li & Be to Ba
M ≠ Na / K / Rb / Cs

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2\uparrow$$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$$

$$Na_2CO_3 \xrightarrow{\Delta} \text{No reaction}$$

Metal carbonates of Na, K, Rb and Cs are thermally stable

Metal Sulphate

$$\text{Metal Sulphate} \xrightarrow{\Delta} \text{Metal Oxide} + SO_3\uparrow$$

M = Li & Be to Ba
M ≠ Na / K / Rb / Cs

$$CaSO_4 \xrightarrow{\Delta} CaO + SO_3\uparrow$$

$$Li_2SO_4 \xrightarrow{\Delta} Li_2O + SO_3\uparrow$$

$$Na_2SO_4 \xrightarrow{\Delta} \text{No reaction}$$

Metal Bicarbonate

$$\text{Metal bicarbonate} \xrightarrow{\Delta} \text{Metal Carbonate} + CO_2\uparrow + H_2O$$

$$2HCO_3^- \xrightarrow{\Delta} CO_3^{2-} + CO_2\uparrow + H_2O$$

$$MHCO_{3(s)} \xrightarrow{\Delta} M_2CO_3 + CO_2\uparrow + H_2O$$

(Na^+, K^+, Rb^+, Cs^+)

$$LiHCO_{3(aq)} \longrightarrow Li_2CO_3 + CO_2\uparrow + H_2O$$

$$NaHCO_{3(s)} \xrightarrow{\Delta} Na_2CO_3 + CO_2\uparrow + H_2O$$

Bicarbonates exist in aqueous medium

M = Li & Be to Ba

M = Na / K / Rb / Cs

Bicarbonates of these 4 metals exist in soild state.

Metal Nitrate

$$\text{Metal Nitrate} \xrightarrow{\Delta} \text{Metal Oxide} + NO_2 + O_2$$

M = Li & Be to Ba

$$\text{Metal Nitrate} \xrightarrow{\Delta} \text{Metal Nitrite} + O_2$$

M = Na, K, Rb, Cs

$$2LiNO_3 \xrightarrow{\Delta} Li_2O + 2NO_2\uparrow + \frac{1}{2}O_2\uparrow$$

$$NaNO_3 \xrightarrow{\Delta} NaNO_2 + \frac{1}{2}O_2\uparrow$$

$$Mg(NO_3)_2 \xrightarrow{\Delta} MgO + 2NO_2\uparrow + \frac{1}{2}O_2$$

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10. Which of the following reaction is correct? [30 Jan, 2023 (Shift–II)]

(a) $2LiNO_3 \xrightarrow{\Delta} 2LiNO_2 + O_2$

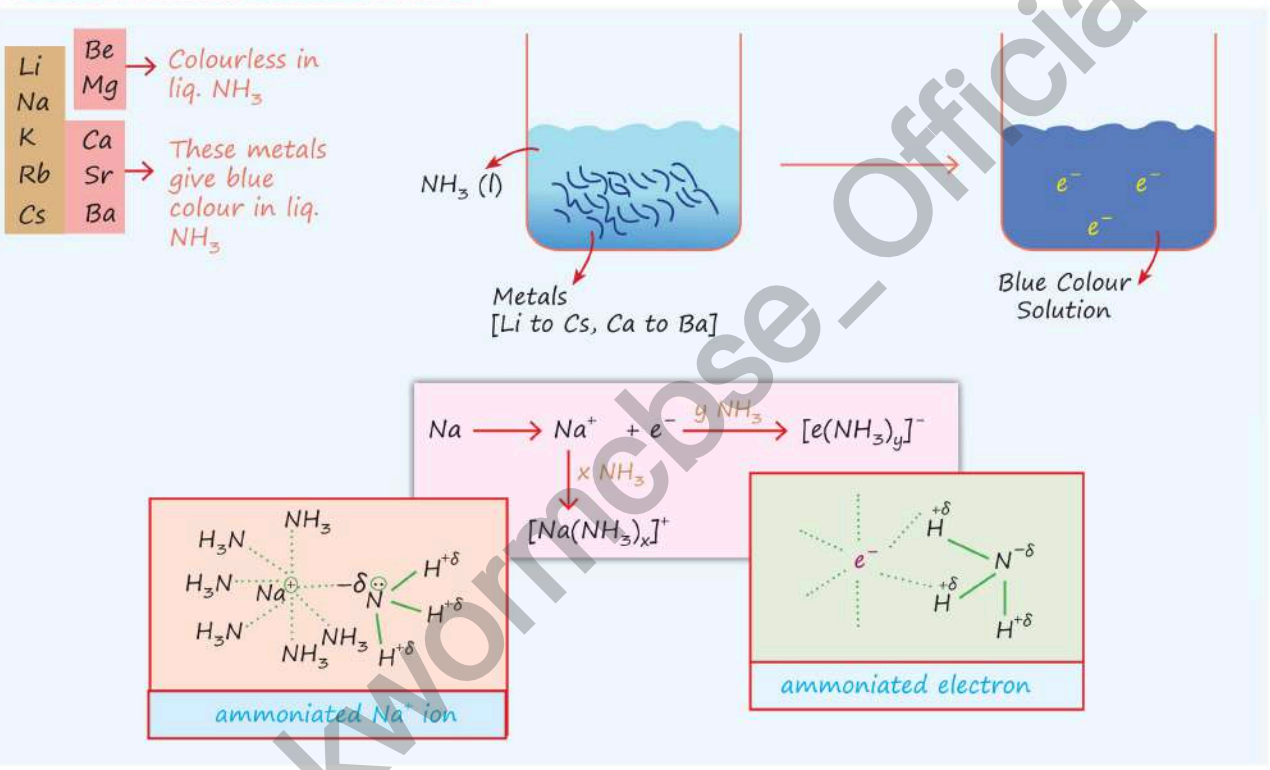
(b) $4LiNO_3 \xrightarrow{\Delta} 2LiNO_2 + 2N_2O_4 + O_2$

(c) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

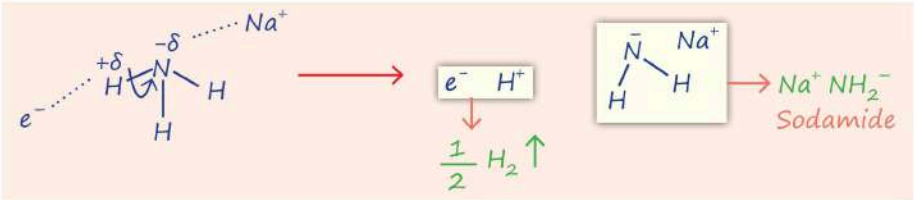
(d) $2LiNO_3 \xrightarrow{\Delta} 2Li + 2NO_2 + O_2$

Sol. (c) $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$

METAL IN NH_3 (ℓ)



- Ammoniated electrons are responsible for (i) reducing nature (ii) electrical conductivity (iii) paramagnetic nature (iv) blue colour of the solution.
- Electrical Conductivity of the solution is due to both –
  (i) ammoniated electrons (ii) ammoniated metal ions
- The blue colour of the solution is due to the ammoniated electron which absorbs energy in the visible region of light and thus imparts blue colour to the solution.
- The solutions are paramagnetic and on standing slowly liberate hydrogen gas resulting in the formation of amide.



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- $M^+_{(am)} + e^- + NH_3\ (l) \rightarrow MNH_{2(am)} + 1/2H_2\ (g)$ (where 'am' denotes solution in ammonia.)
- In concentrated solution, the blue colour changes to bronze colour and becomes diamagnetic.

11. Sodium metal on dissolution in liquid ammonia gives a deep blue solution due to the formation of:
[10 Jan, 2019 (Shift–II)]

(a) Sodium – ammonia complex
(b) Sodamide
(c) Sodium ion-ammonia complex
(d) Ammoniated electrons

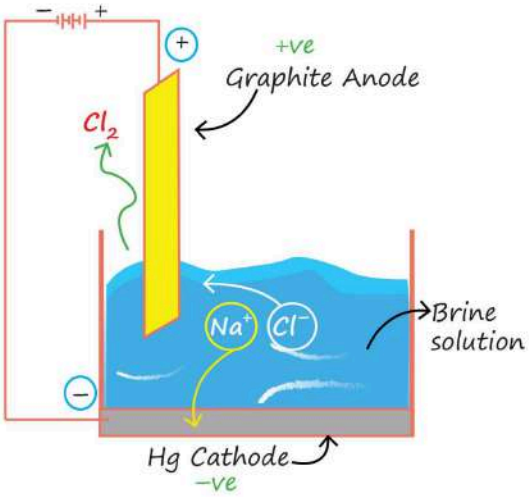
Sol. (d) Sodium metal dissolve in liquid ammonia giving deep blue solutions which are conducting in nature. Blue colour of the solution is due to ammoniated electron.

IMPORTANT COMPOUNDS OF SODIUM

1. **Caustic Soda**
   $NaOH$
   Sodium Hydroxide
2. **Baking Soda**
   $NaHCO_3$
   Sodium bi-carbonate
3. **Washing Soda**
   $Na_2CO_3.10H_2O$
   Sodium carbonate decahydrate

Caustic Soda (NaOH) by Castner Kellner's Cell

- Electrolyte : Saturated Sol^n of NaCl (Brine Sol^n)
- Anode [+ ve] ⟶ Oxidation
- Cathode [– ve] ⟶ Reduction



Cathode : $Na^+ + e^- \xrightarrow{Hg} Na - Hg$ Sodium amalgam (alloy)

Anode : $Cl^- \longrightarrow \frac{1}{2} Cl_2 + e^-$

- $Na\text{-}Hg + H_2O \longrightarrow NaOH + \frac{1}{2} H_2 \uparrow + Hg$
- The amalgam is treated with water to give NaOH & H_2.

Dawn Cell

- Fused NaCl is electrolyzed at a 330°C temperature for extraction of sodium.
- A mixture of KCl and KF is added to sodium chloride to decrease melting point of NaCl.

MP : NaCl > [NaCl + Small amount of KCl / KF]

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Electrolysis of Molten NaCl

Reactions:

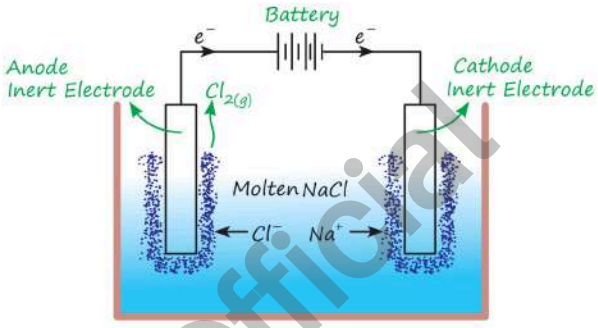
At Cathode : Sodium ion (Na^+) gain electron to form sodium metal.

$$Na^+ + e^- \longrightarrow Na_{(s)}$$

At Anode : Chloride ion (Cl^-) loses electron to form chlorine gas.

$$2Cl^- \longrightarrow Cl_2\,(g) + 2e^-$$

Products : Sodium Metal (Na) at the cathode
Chlorine Gas (Cl_2) at the anode



12. The metal that forms nitride by reacting directly with N_2 of air [09 Jan, 2019 (Shift–II)]

(a) K (b) Li (c) Rb (d) Cs

Sol. (b) Among Alkali metals, Li reacts with air forming nitride.

13. Magnesium powder burns in air to give: [09 April, 2020 (Shift–I)]

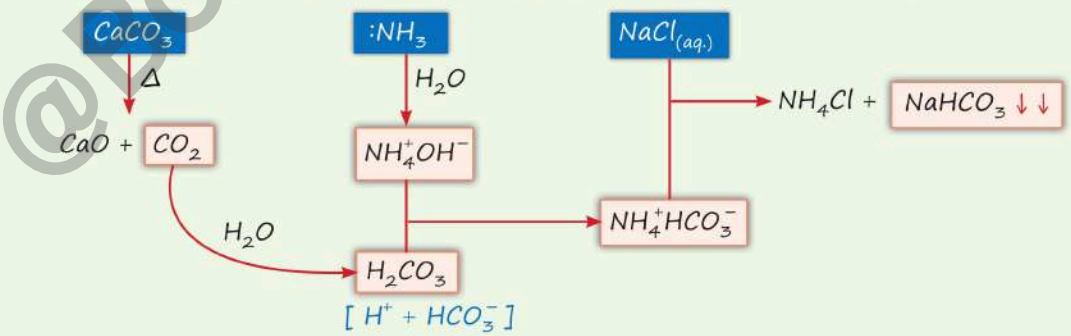
(a) MgO only (b) MgO and $Mg(NO_3)_2$

(c) MgO and Mg_3N_2 (d) $Mg(NO_3)_2$ and Mg_3N_2

Sol. (c) $Mg \xrightarrow[N_2]{O_2} MgO / Mg_3N_2$

Solvay Process—Making of washing soda [$Na_2CO_3.10H_2O$]

❒ Making of Washing Soda [$Na_2CO_3 . 10H_2O$] by 3 raw material – $CaCO_3$, NH_3 and $NaCl_{(aq.)}$



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❒ $2NaHCO_{3\,(s)} \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2\uparrow$

$Na_2CO_3 \xrightarrow{\text{Crystallisation}} Na_2CO_3 . 10H_2O$

❒ **Recovery of NH_3:** $CaO + H_2O \rightarrow \underset{\text{Base}}{Ca(OH)_2} \xrightarrow[[NH_3 + HCl]]{NH_4Cl} CaCl_2 + NH_3\uparrow$

Only by-product in Solvay process

Soda Ash [Na_2CO_3]

$Na_2CO_3 . 10H_2O \xrightarrow[-9H_2O]{100°C} Na_2CO_3 . H_2O \xrightarrow[-H_2O]{\Delta} Na_2CO_3$ (Soda Ash)

14. K_2CO_3 can not be prepared by this method? Why

Sol. K_2CO_3 cannot be prepared by Solvay process because $KHCO_3$ is soluble in water and cannot be separated from NH_4Cl.

Sodium hydrogencarbonate ($NaHCO_3$)– Baking Soda

❒ On heating, it decomposes to release carbon dioxide (CO_2) gas.

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2\uparrow + H_2O$$

- This gas forms bubbles, creating holes in cakes and pastries, making them light and fluffy.

❒ It is prepared by saturating a solution of sodium carbonate (Na_2CO_3) with carbon dioxide (CO_2).

- The less soluble white crystalline powder of sodium hydrogencarbonate separates out.

$$Na_2CO_3 + H_2O + CO_2 \rightarrow 2\,NaHCO_3$$

- It acts as a mild antiseptic for skin infections.
- It is also used in fire extinguishers.

→ Uses

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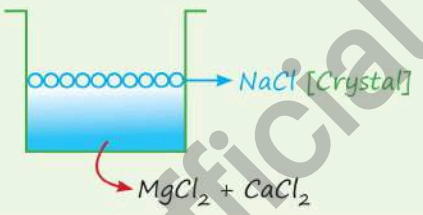
NaCl

- Sea Water $\xrightarrow{\text{Evaporation}}$ Raw Salt ≡ NaCl + $\underbrace{MgCl_2 + CaCl_2}_{\text{Impurities}}$ $\xrightarrow[\text{gas}]{HCl}$ Crystals of $NaCl_{(s)}$ + $MgCl_{2\,(aq.)}$ + $CaCl_{2\,(aq.)}$



- Hydration : $Na^+ < Mg^{2+}$

- To obtain pure NaCl, the crude salt is dissolved in minimum amount of water and filtered to remove insoluble impurities.
- The solution is then saturated with HCl gas. Crystals of pure NaCl separate out.
- $MgCl_2$ & $CaCl_2$ being more soluble than NaCl, remain in solution.

15. HCl was passed through a solution of $CaCl_2$, $MgCl_2$ and NaCl. Which of the following compound(s) crystallize (s) ?

(a) Only NaCl (b) Only $MgCl_2$

(c) NaCl, $MgCl_2$ and $CaCl_2$ (d) Both $MgCl_2$ and $CaCl_2$

Sol. (a) NaCl crystallizes when HCl is passed through the solution of $CaCl_2$, $MgCl_2$, and NaCl.

IMPORTANT COMPOUNDS OF CALCIUM

Lime stone $[CaCO_3]$ $\xrightarrow{\Delta}$ Quick lime $[CaO]$ $\xrightarrow[\text{Water}]{\text{limited}}$ Slaked lime $\xrightarrow{H_2O}$ Milk of lime $\xrightarrow{H_2O}$ Lime water

- Slaked lime → White amorphous solid [powder of $Ca(OH)_2$]
- Milk of lime → Suspension of slaked lime in water
- Lime water → [Clear solution] $[Ca(OH)_2]$

- The addition of limited amount of water breaks the lump of lime. This process is called slaking of lime.
- Slaked lime is a white amorphous powder. It is sparingly soluble in water. The aqueous solution is known as lime water and a suspension of slaked lime in water is known as milk of lime.
- Quick lime slaked with soda gives solid soda – lime. $[Na_2CO_3 + CaO]$

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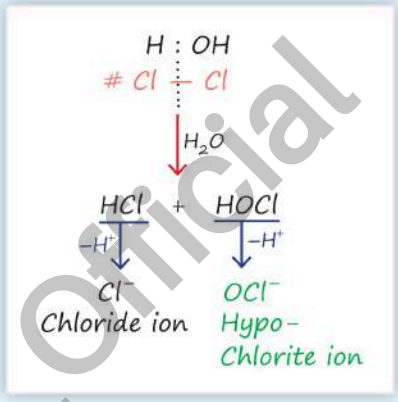
- $Ca(OH)_{2\,(aq)} \xrightarrow{CO_2} CaCO_3 \xrightarrow{CO_2} Ca(HCO_3)_2$
  $CaCO_3$: [milk] → not soluble in water
  $Ca(HCO_3)_2$: Soluble in water [Clear Solution]

- $2\,Ca(OH)_{2\,(aq.)} \xrightarrow{2Cl_2}$ (Bleaching powder) $CaCl_2 + Ca(OCl)_2 \equiv CaCl(OCl) \equiv CaOCl_2$

- $Ca^{2+}CO_3^{2-} \xrightarrow{2HCl} H_2CO_3 \longrightarrow H_2O + CO_2\uparrow + CaCl_2$

- $CaCO_3 \xrightarrow[2H^+ SO_4^{2-}]{H_2SO_4} CaSO_4 + H_2O + CO_2\uparrow$

- $CaCl_2 + Na_2CO_3 \longrightarrow 2NaCl + CaCO_3$



$CaSO_4$

Gypsum $CaSO_4 \cdot 2H_2O$

$CaSO_4 \cdot 2H_2O \xrightarrow{\Delta} CaSO_4 \cdot \frac{1}{2}H_2O \xrightarrow{393\ K} CaSO_4$

Gypsum — [POP] — dead burnt plaster

Plaster of Paris [POP]
Hemi-hydrate of $CaSO_4$
$CaSO_4 \cdot \frac{1}{2}H_2O$

CaO

On exposure to atmosphere, it absorbs moisture and carbon dioxide.

$CaO \xrightarrow{H_2O} Ca(OH)_2 \xrightarrow{CO_2} CaCO_{3(s)}$

$CaO + H_2O \longrightarrow Ca(OH)_2$

$CaO + CO_2 \longrightarrow CaCO_3$ — Calcium Carbonate

$CaO + SiO_2 \longrightarrow CaSiO_3$ — Calcium Silicate

$3CaO + P_2O_5 \longrightarrow Ca_3(PO_4)_2$ — Calcium phosphate

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ANOMALOUS BEHAVIOUR OF Li

Different behaviour of Li w.r.t. other elements of group-1 is due to–

(i) small size of its atom [Li] and ion [Li^+].

(ii) high polarising power [$Li^+ >> Na^+ / K^+$]

- Lithium halides are somewhat covalent, Due to high polarisation capability of Li^+.

  LiF < LiCl < LiBr < LiI : Covalent nature

- Halides of Li are soluble in organic solvent like ethanol [C_2H_5OH], acetone [CH_3COCH_3] and ethylacetate [$CH_3COOC_2H_5$]. LiCl is soluble in pyridine also.
- LiCl is deliquescent.

  Li^+ has maximum degree of hydration, so its salts [$LiCl \cdot 2H_2O$] are mostly hydrated.
- Li unlike other alkali metals forms no ethynide on reaction with ethyne.

$$H-C\equiv C^{-\delta}-H^{+\delta} \xrightarrow{Na} H-C\equiv C^{\ominus}Na^+ + \frac{1}{2}H_2\uparrow$$

$$H-C\equiv C^{-\delta}-H^{+\delta} \xrightarrow{Li} \text{No reaction}$$

$Na \longrightarrow Na^+ + e^-$

$HC\equiv C-H \longrightarrow HC\equiv C^- + H^+$

- **$LiHCO_3$** : not obtained in the solid form but Na, K, Rb, Cs hydrogen carbonates exist in solid form.

ANOMALOUS BEHAVIOUR OF Be

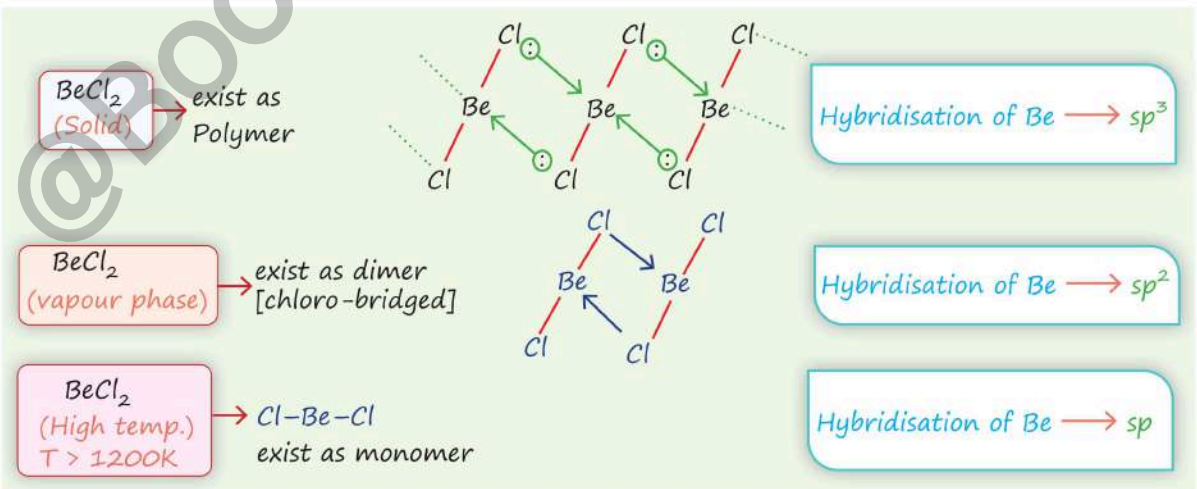
- All other halides of alkaline earth metals are ionic in nature except Be.

  [BeF_2, $BeCl_2$ → Covalent molecule] [$MgCl_2$, MgF_2 → Ionic]

- Beryllium hydroxide, oxide, halides are essentially covalent.

  [BeO, $Be(OH)_2$, $BeCl_2$] → Covalent molecule

- Beryllium halides are soluble in organic solvents.



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$MgCl_2$ and $CaCl_2$ exist as $MgCl_2.8H_2O$ and $CaCl_2.6H_2O$ while NaCl and KCl do not form such hydrates.

$Na^+ << Mg^{+2}$: Hydration

$K^+ << Ca^{2+}$: Hydration

- Tendency to form halide hydrates $[MCl_2 \cdot x\ H_2O]$ gradually decreases as we go down the group.

$MgCl_2 \cdot 6H_2O$, $CaCl_2 \cdot 6H_2O$, $SrCl_2 \cdot 6H_2O \rightarrow BaCl_2 \cdot 2H_2O$

Degree of hydration - $Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

- The dehydration of hydrated chlorides, bromides and iodides of Ca, Sr and Ba can be achieved on heating.

$$CaCl_2 \cdot 6H_2O \xrightarrow{\Delta} CaCl_2 + 6H_2O \uparrow$$

- The corresponding hydrated halides of Be and Mg on heating suffer hydrolysis.

$$BeCl_2 \cdot 4H_2O \equiv [Be(H_2O)_4]Cl_2 \xrightarrow{\Delta} BeO + 3H_2O + 2HCl$$
(white fumes)

$$MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgO + 5H_2O + HCl$$
(White fumes)

DIAGONAL RELATIONSHIP

Lithium	Magnesium
Harder and Lighter than other metals of group-1.	Harder and lighter than other metals of group-2
Oxide : Li_2O [Basic]	Oxide : MgO [Basic]
Hydroxide : LiOH $\xrightarrow{\Delta}$ $Li_2O + H_2O$	Hydroxide : $Mg(OH)_2 \xrightarrow{\Delta} MgO + H_2O$
$Li + Air \rightarrow Li_2O + Li_3N$ Hot Water $\rightarrow LiOH + \frac{1}{2} H_2\uparrow$ Excess $O_2 \rightarrow Li_2O$ (Oxide)	$Mg + Air \rightarrow MgO + Mg_3N_2$ Hot Water $\rightarrow Mg(OH)_2 + H_2\uparrow$ Excess $O_2 \rightarrow MgO$
$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2\uparrow$	$MgCO_3 \xrightarrow{\Delta} MgO + CO_2\uparrow$
$LiHCO_{3(aq)} \xrightarrow{boil} Li_2CO_3 + H_2O + CO_2\uparrow$	$Mg(HCO_3)_{2\,(aq.)} \xrightarrow{boil} Mg(OH)_2\downarrow + H_2O + CO_2\uparrow$
Solid bicarbonates are not formed by Li and Mg.	
LiCl and $MgCl_2$ are deliquescent [Tendency to absorb moisture from the air and dissolve in it] and crystallize from aqueous solution as hydrates. $LiCl \rightarrow LiCl \bullet 2H_2O$ $\quad MgCl_2 \rightarrow MgCl_2 \bullet 8H_2O$	
Both LiCl and $MgCl_2$ are soluble in ethanol.	

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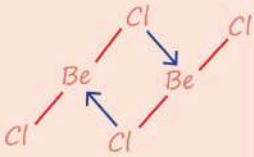
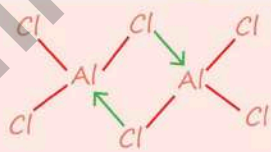
Beryllium	Aluminium
Like aluminimum, beryllium is not readily attacked by acids (for example conc HNO_3) because of the presence of an oxide film on the surface of the metal.	
❒ $Be(OH)_2 \xrightarrow{2Na^+OH^-} Na_2[Be(OH)_4] \equiv Na_2BeO_2.2H_2O$ [Sodium Beryllate]; $\xrightarrow{HCl} BeCl_2$ or $[Be(H_2O)_4]Cl_2$	❒ $Al(OH)_3 \xrightarrow{NaOH} Na^+Al(OH)_4^- \equiv NaAlO_2.2H_2O$ [Sodium aluminate]; $\xrightarrow{HCl} AlCl_3$
❒ $BeO \xrightarrow{NaOH} Na_2[Be(OH)_4]$; $\xrightarrow{HCl} BeCl_2$ BeO, $Be(OH)_2$ → Amphoteric	❒ $Al_2O_3 \xrightarrow{NaOH} NaAlO_2.2H_2O$; $\xrightarrow{HCl} 2\,AlCl_3$ Al_2O_3, $Al(OH)_3$ → Amphoteric
❒ Structure of $BeCl_2$ (Vapour phase) → Exist as dimer 	❒ Structure of $AlCl_3$ 
❒ Both the chlorides are soluble in organic solvents and are strong Lewis acids. [$BeCl_2$ / $AlCl_3$] ❒ Complex formation: $BeF_2 \xrightarrow{2\dot{N}a\bar{F}} Na_2[BeF_4]$ $AlF_3 \xrightarrow{3\,NaF} Na_3[AlF_6]$ Be [2nd Period element] can form maximum 4 bonds but Al [3rd period element] can expand their octet.	

USES

1. **Li** → Electrochemical cell
2. **Liquid sodium metal** → Coolant in fast breeder nuclear reactors.
3. **$Mg(OH)_2$** : Milk of magnesia : Antacid in medicine
4. **Ra** : Radiotherapy (treatment of cancer)
5. **KO_2** : used in oxygen cylinders in space and submarines
6. **Beryllium** is used in the manufacture of alloys. Copper - beryllium alloys are used in the preparation of high strength springs.
7. **Magnesium-aluminium alloys** being light in mass are used in air-craft construction.
8. **Barium-** $BaSO_4$ (White) + ZnS (White) → Lithopone (Used as white pigment)

Biological Importance of Sodium and Potassium

- K^+ ions are the most abundant cations within cell fluids, where they activate many enzymes, participate in the oxidation of glucose to produce ATP and K^+ with Na^+ are responsible for the transmission of nerve signals.
- Sodium ions are found primarily on the outside of cells.

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Sodium-potassium pump

The Na^+-K^+ pump (Na^+/K^+- ATPase enzyme) helps to maintain the right concentrations of ions in cell. The pump uses, three sodium ions are exported and two potassium ions are imported. Failure of the Na^+-K^+ pumps can result in swelling of the cell.

Biological Importance of Magnesium and Calcium

Mg

- All enzymes that utilise ATP in phosphate transfer require magnesium as the cofactor.
- Main pigment for the absorption of light in plants is chlorophyll which contains magnesium.

Ca

- It plays important roles in neuromuscular function, interneuronal transmission, cell membrane integrity and blood coagulation.
- The calcium concentration in plasma is regulated at about 100 mgL^{-1}. It is maintained by two hormones-calcitonin and parathyroid hormone.

Adult body contains

- 25 g of Mg
- 1200 g of Ca
- 5 g of Fe
- 0.06 g of Cu
- 99% of body Ca is present in bones and teeth.

16. Compound A reacts with NH_4Cl and forms a compound B. Compound B reacts with H_2O and excess of CO_2 to form compound C which on passing through or reaction with saturated NaCl solution forms sodium hydrogen carbonate. Compound A, B and C, are respectively.

[25Jan, 2023 (Shift -I)]

(a) $CaCl_2$, NH_3, NH_4HCO_3 (b) $CaCl_2$, NH_4^+, $(NH_4)_2CO_3$

(c) $Ca(OH)_2$, NH_3, NH_4HCO_3 (d) $Ca(OH)_2$, NH_4^+, $(NH_4)_2CO_3$

Sol. (c) $\underset{(A)}{Ca(OH)_2} + 2NH_4Cl \xrightarrow{\Delta} \underset{(B)}{2NH_3} + CaCl_2 + 2H_2O$

$\underset{(B)}{NH_3} + H_2O + \underset{(excess)}{CO_2} \longrightarrow \underset{(C)}{NH_4HCO_3}$

$\underset{(C)}{NH_4HCO_3} + NaCl \longrightarrow NaHCO_3 \downarrow + NH_4Cl$

17. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

(a) Copper (b) Calcium (c) Potassium (d) Iron

Sol. (c) Potassium ions activate many enzymes, participate in the oxidation of glucose to produce ATP and with sodium are responsible for transmission of nerve signals.

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18. The function of "Sodium pump" is a biological process operating in each and every cell of all animals. Which of the following biologically important ions is also a constituent of his pump :

(a) Mg^{2+} (b) K^{+} (c) Fe^{2+} (d) Ca^{2+}

Sol. (b) Potassium ions (K^{+}) are a key constituent and actively transported ion in the sodium-potassium pump.

19. Amongst baking soda, caustic soda and washing soda, carbonate anion is present

[29June, 2022 (Shift -II)]

(a) Washing soda only
(b) Washing soda and caustic soda only
(c) Washing soda and baking soda only
(d) Baking soda, caustic soda and washing soda

Sol. (a)

- Baking soda $\rightarrow NaHCO_3$
- Washing soda $\rightarrow Na_2CO_3 . 10H_2O$
- Caustic soda $\rightarrow NaOH$

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10 General Principles and Processes of Isolation of Elements

(JEE Advanced)

Syllabus Metal ores and their concentration; extraction of crude metal from concentrated ores: thermodynamic (iron, copper, zinc) and electrochemical (aluminium) principles of metallurgy; cyanide process (silver and gold); refining.

MINERAL AND ORE

Mineral: The compound of metal found in nature.
Ore: The minerals from which metal can be economically and conveniently extracted.

Native ores	Oxidised ores	Sulphide ores	Halide ores
Metal in free state	Oxides of metals Oxysalts of metals	Sulphides of metals (Pb/Zn/Hg/Cu/Ag)	Halides of metals (Na/Mg)
Example: Ag/Au/Pt	Fe_2O_3, $FeCO_3$	PbS, ZnS, $CuFeS_2$	NaCl, $MgCl_2$

SOME IMPORTANT ORES OF METALS

Aluminium	Iron	Copper	Zinc
Bauxite $Al_2O_3.xH_2O$	Haematite Fe_2O_3	Copper pyrite (Chalcopyrite) $CuFeS_2$	Zinc blend (Sphalerite) ZnS
Kaolinite (clay) $Al_2(OH)_4Si_2O_5$ → $Al_2O_3.2SiO_2.2H_2O$	Magnetite Fe_3O_4	Copper glance (Chalcocite) Cu_2S	Calamine $ZnCO_3$
	Siderite $FeCO_3$	Cuprite Cu_2O	Zincite ZnO
	Iron pyrite (Fool's gold) FeS_2	Malachite $CuCO_3.Cu(OH)_2$	

Lead	Galena PbS	Mercury	Cinnabar HgS
Silver	Argentite Ag_2S, Horn Silver AgCl	Tin	Tinstone/Casseterite SnO_2

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1. **Calamine and Malachite respectively, are the ores of :**

(a) Nickel and Aluminium (b) Zinc and Copper

(c) Copper and Iron (d) Aluminium and Zinc

Sol. (b)

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1. मैला Cu — काला Zn — सीधा Fe

Malachite — Calamine — Siderite

All 3 are carbonate ores.

2. Hae — Ma — सीधा — Fe — का gold लाया हुँ!

Hae → 23 → Haematite

Ma → 34 → Magnetite

सीधा → Siderite

Fe → Fool's gold

2. **Sulphide ion is soft base and its ores are common for metals.**

(i) Pb (ii) Al (iii) Ag (iv) Mg

(a) (i) and (iii) only (b) (i) and (iv) only

(c) (i) and (ii) only (d) (iii) and (iv) only

Sol. (a) Lead (Pb) and Silver (Ag) are commonly found in the form of sulphide ores, such as PbS (galena) and Ag_2S (argentite). Aluminum (Al) is primarily found in the form of oxide ores, while Magnesium (Mg) is found in the form of halide ores.

3. **Sulphide ores are common for the metals** **(JEE Adv. 2013)**

(a) Ag, Cu and Pb (b) Ag, Cu and Sn (c) Ag, Mg and Pb (d) Al, Cu and Pb

Sol. (a) 1. For Ag, it is Argentite (Ag_2S). 2. For Pb, it is Galena (PbS).
3. For Cu, it is Copper glance (Cu_2S).

4. **Match the anionic species given in Column-I that are present in the ore (s) given in Column-II.** **(JEE Adv. 2015)**

Column-I		Column-II	
A.	Carbonate	p.	Siderite
B.	Sulphide	q.	Malachite
C.	Hydroxide	r.	Bauxite
D.	Oxide	s.	Calamine
		t.	Argentite

Sol. A → p, q, s; B → t; C → q, r; D → r.

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5. Which ore contains both iron and copper? (IIT JEE 2005)

(a) Cuprite (b) Chalcocite (c) Chalcopyrite (d) Malachite

Sol. (c)

METALLURGY

- The entire scientific and technological process used for isolation of the metal from its ore is called metallurgy.

Crushing ⟶ Concentration ⟶ Isolation ⟶ Refining

- Crushing: Break down of ore into smaller pieces [pulverization]
- Concentration: Ore enrichment [dressing] is the process of removing impurities from ore to increase the percentage of the desired metal or mineral
- Isolation: 1. Converting the ore to oxide 2. Then reducing that oxide to obtain the metal
- Refining: Purification of extracted metal

Gangue : Earthy undesired materials which contaminates the ore is called gangue.

Flux, Impurity and Slag : Flux is an additional substance which is added in the ore. It combines with the impurities to form easily fusible product known as slag. Removal of impurities as slag is easy because of low density of slag.

$$\text{Impurities} + \text{Flux} \longrightarrow \text{Slag}$$

Types of Flux

- Acidic flux: SiO_2, P_2O_5
  - Basic Impurity + Acidic Flux ⟶ Slag
  - $FeO + SiO_2 \longrightarrow FeSiO_3$
  - $CaO + SiO_2 \longrightarrow CaSiO_3$
- Basic flux: CaO, MgO
  - Basic Flux + Acidic Impurity ⟶ Slag
  - $CaO + SiO_2 \longrightarrow CaSiO_3$

Slag

$$\text{Ore} + \text{Impurity} + \text{Flux} \longrightarrow \text{Ore} + \text{Slag}$$

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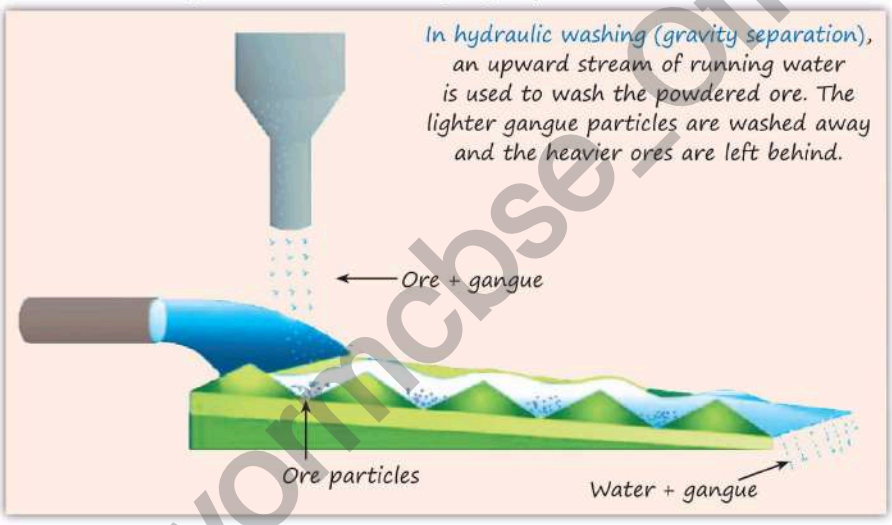
- It is a fusible mass.
- It has low melting point.
- It has less density than molten metal.
- It is immiscible with the molten metal / pure ore.
- Slag floats on the surface of the molten metal, protecting it from oxidation by the atmosphere and keeping it clean.

CONCENTRATION OF ORES

Hydraulic Washing

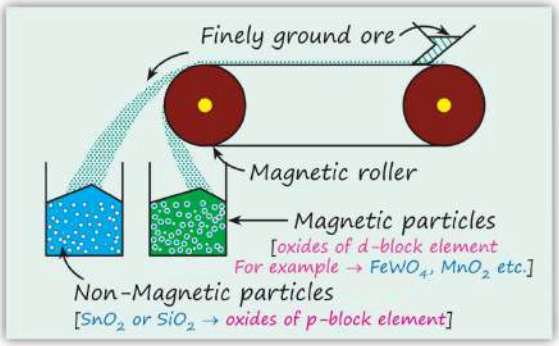
- **Basic Principle:** Difference in gravities of ore and gangue particle.



Magnetic Separation

- **Basic Principle:** Difference in magnetic properties of the ore and gangue.
- A magnetic separator consists of a belt (of leather or brass) moving over two rollers, one of which is magnetic. When the powdered ore is dropped on the belt at the other end, magnetic component of the ore is attracted by the magnetic roller and falls nearer to the roller while the non-magnetic impurities fall away from it.

- **Ore** + Impurity
  Tinstone/Cassiterite (SnO_2) + Wolframite ($FeWO_4 + MnWO_4$)
- **Ore** + Impurity
  1. **Chromite ($FeO.Cr_2O_3$)** + Silica (SiO_2)
  2. **Magnetite (Fe_3O_4)** + Silica (SiO_2)
  3. **Pyrolusite (MnO_2)** + Silica (SiO_2)



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Having a scientific mindset and paying close attention to observations can lead to amazing discoveries. A washerwoman with an innovative mind noticed something interesting while washing a miner's overalls. She saw that sand and other dirt would settle at the bottom of the washtub. However, what stood out was that the copper compounds from the mines would get trapped in the soapsuds and float to the top.

Observation: The washerwoman shared her observation with one of her clients, Mrs. Carrie Everson, who was a chemist.

Idea for Separation: Mrs. Everson realized that this idea could be used to separate copper compounds from rocks and earth on a large scale.

Birth of an Invention: This idea led to the invention of the Froth Flotation Method.

Impact on Copper Mining:

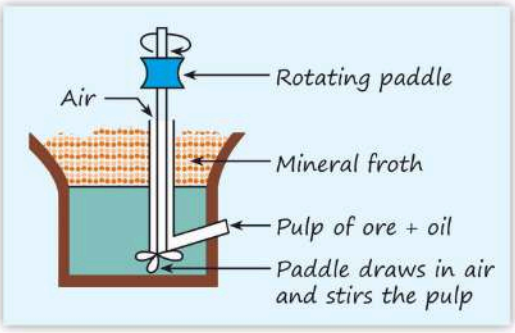
- **Before the Invention:** Only ores with large amounts of copper were used for extraction.
- **After the Invention:** The Froth Flotation Method made it possible to mine copper even from low-grade ores.

  Economic Impact: The world production of copper increased significantly, making the metal cheaper.

Froth Flotation Process

This method has been in use for removing gangue from **sulphide ores**. In this process, a suspension of the powdered ore is made with water. To it, collectors and froth stabilisers are added.

- **Basic Principle:** Difference in Wettability
  - Sulphide ore is wetted by oil and moves upward in the form of froth
  - Gangue particles are wetted by water and settles down at bottom
- **Ore:** Cu_2S, ZnS, FeS_2, PbS
- **Frothers:** Pine oil
  - Pine oil or other similar frothing agents are used to create a stable froth that helps in the separation of hydrophobic (water-repelling) mineral particles.



- **Collectors:** Potassium ethyl xanthate collectors are used to enhance non-wettability of ore in water.
- **Froth stabilizers:** Aniline, Cresol

Selective Froth Flotation Process

- When a mixture of sulphide ores (PbS, ZnS) is present then we use **depressant** ($NaCN$, Na_2CO_3) to depress the floatation property of a sulphide ore.

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- $PbS + NaCN \longrightarrow$ No Reaction
- $ZnS + NaCN \longrightarrow \underset{\text{Complex}}{Na_2[Zn(CN)_4]}$

To increase the adsorption of complex on the surface of ZnS, we can use activator [$CuSO_4$]

- This **complex is wetted with water** and prevented coming into the froth.
- While **PbS comes into froth** because it cannot react with NaCN.

Note

Mechanism behind working of collectors, frothers and activator is beyond our syllabus.

Leaching (Al / Ag / Au)

- **Basic Principle:** Difference in solubility of metal and impurities in a suitable solvent.

Leaching Solvent : Aq. solution of NaCN / KCN

- Generally metal is more soluble than impurities present in its ore.

- **MacArthur-Forrest [MAF] Cyanide Process**

$$\underset{\text{(impure)}}{2\,Ag_{(s)}} \xrightarrow[\text{in presence of } O_2]{CN^-_{(aq)}} \underset{\text{(Soluble in water)}}{2[Ag(CN)_2]^-_{(aq)}} \xrightarrow[\text{as a reducing agent}]{Zn} [Zn(CN)_4]^{2-}_{(aq)} + 2\,Ag\ \text{(pure)}$$

$$\underset{\text{(impure)}}{2\,Au_{(s)}} \xrightarrow[\text{in presence of } O_2]{CN^-_{(aq)}} \underset{\text{(Soluble in water)}}{2[Au(CN)_2]^-_{(aq)}} \xrightarrow{Zn} [Zn(CN)_4]^{2-}_{(aq)} + 2\,Au\ \text{(pure)}$$

- **Hydro-metallurgy of Ag_2S**

$$Ag_2S_{(s)} \xrightarrow{2\,NaCN_{(aq)}} Na_2S + 2[Ag(CN)_2]^-_{(aq)} \xrightarrow{Zn} [Zn(CN)_4]^{2-} + 2\,Ag\ \text{(pure)}$$

$$Na_2S \xrightarrow[H_2O]{O_2} Na_2SO_4 + S + NaOH$$

6. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are **(IIT JEE 2012)**

(a) O_2 and CO respectively
(b) O_2 and Zn dust respectively
(c) HNO_3 and Zn dust respectively
(d) HNO_3 and CO respectively

Sol. (b) 1. The Argentite ore (Ag_2S) contains silver.
2. The ore dissolves in cyanide solution forming sodium argentocyanide.
3. In this process, O_2 and Zn act as oxidizing and reducing agents respectively.

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7. Silver jewellery items tarnish slowly in the air due to their reaction with __ . (IIT JEE 1977)

Sol. [H_2S] $Ag + H_2S \rightarrow Ag_2S + H_2$.

Due to reaction of silver jewellery with hydrogen sulphide (H_2S) in the air, silver jewellery items tends to tarnish slowly.

8. In magnetic separation, what type of magnetic material is usually separated from a mixture?

(a) Ferromagnetic (b) Diamagnetic (c) Paramagnetic (d) Non-magnetic

Sol. (a) Ferromagnetic materials (such as iron) are strongly attracted to magnets and are thus effectively separated from mixtures using magnetic separation. Paramagnetic materials have a weaker attraction, while diamagnetic and non-magnetic materials do not respond to magnetic fields.

9. In the Froth Flotation Process, which of the following substances is typically used as a frothing agent?

(a) Sulphuric acid (b) Pine oil (c) Sodium chloride (d) Ethanol

Sol. (b)

10. In the cyanide leaching process, which metal is primarily extracted?

(a) Sodium (b) Gold (c) Aluminium (d) Copper

Sol. (b)

EXTRACTION OF METAL

Concentrated Ore $\xrightarrow[\text{or Roasting}]{\text{Calcination}}$ Metal Oxide $\xrightarrow{\text{Reduction}}$ Metal

Calcination

- Calcinaton involves heating without air when the volatile matter escapes leaving behind the metal oxide.

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2 \uparrow$

$Fe_2O_3.3H_2O \xrightarrow{\Delta} Fe_2O_3 + 3H_2O \uparrow$

$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2 \uparrow$

$FeCO_3 \xrightarrow{\Delta} FeO + CO_2 \uparrow$

$CaCO_3.MgCO_3 \xrightarrow{\Delta} CaO + MgO + 2CO_2 \uparrow$ (Dolomite)

$CuCO_3.Cu(OH)_2 \rightarrow 2CuO + CO_2 + H_2O$

Roasting

- In roasting, sulphide ore is heated in a regular supply of air in a furnace at a temperature below the melting point of the metal.

$PbS \xrightarrow[\Delta]{O_2} PbO + SO_2 \uparrow$

$ZnS \xrightarrow[\Delta]{O_2} ZnO + SO_2 \uparrow$

- Some times roasting may not bring about complete oxidation.

$CuFeS_2 \xrightarrow[\Delta]{O_2} Cu_2S + FeO + SO_2$

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- Volatile impurity (organic matter) can be removed.
- It removes easily oxidisable volatile impurities.

$$As \xrightarrow[\Delta]{O_2} As_2O_3$$
$$Sb \xrightarrow[\Delta]{O_2} Sb_2O_3$$
$$S \xrightarrow[\Delta]{O_2} SO_2$$
$$P_4 \xrightarrow[\Delta]{O_2} P_4O_{10}$$

- In both processes
  - We use reverberatory furnace.
  - Ore becomes porous.
  - Temperature should be below melting point.

REDUCTION PROCESS

Self Reduction in Roasting

- The reduction in which species reacts with itself to form new species.

$$Cu_2S \xrightarrow{O_2} SO_2 + Cu_2O \xrightarrow{Cu_2S} Cu + SO_2\uparrow$$

Copper having blisters of SO_2 gas on its surface is called as blister copper.

- $$HgS \xrightarrow{O_2} SO_2 + HgO \xrightarrow{HgS} Hg + SO_2\uparrow$$
  $$PbS \xrightarrow{O_2} SO_2 + PbO \xrightarrow{PbS} Pb + SO_2\uparrow$$

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Prabhu Sulphide
Cu Hogya self reduce

Note

- Large volume of SO_2 obtained is used to prepare H_2SO_4 in industry.

$$SO_2 \xrightarrow[V_2O_5]{O_2} SO_3 \xrightarrow{H_2O} H_2SO_4$$

Reduction

Li K Ba Sr Ca Na Mg Al	Mn Zn Cr Fe Cd Co Ni Sn Pb Cu Hg	Ag Au Pt
• Highly reactive metals • **Electro-metallurgy** (reduction by using electrolysis)	• Moderate Reactive metals • **Pyro-metallurgy [Smelting]** (reduction by using reducing agent like C, CO, Al with heat)	• Highly Non-reactive Metals • **Hydrometallurgy** 1. Dissolution of impure metal or low grade ore in a solvent 2. Reduction of solution by Zn to get pure metal

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Smelting

The process of converting **metal oxide into metal** using a suitable reducing agent.

- **Reducing agents :** C, CO, Al
- Carbon can be used to reduce a number of oxides and because of the **low cost** and **availability of coke** this method is widely used.

Reduction with carbon ($C \longrightarrow CO$)	Reduction with CO ($CO \longrightarrow CO_2$)
$PbO + C \longrightarrow Pb + CO\uparrow$	$Fe_2O_3 + CO \longrightarrow Fe_3O_4 + CO_2\uparrow$
$Fe_2O_3 + C \longrightarrow Fe + CO\uparrow$	$Fe_3O_4 + CO \longrightarrow FeO + CO_2\uparrow$
$ZnO + C \longrightarrow Zn + CO\uparrow$	$FeO + CO \longrightarrow Fe + CO_2\uparrow$
$SnO_2 + C \longrightarrow Sn + CO\uparrow$	$PbO + CO \longrightarrow Pb + CO_2\uparrow$
	$CuO + CO \longrightarrow Cu + CO_2\uparrow$

11. What is the primary purpose of calcination in metallurgy?

(a) To dissolve ores in acid (b) To separate metals by electrolysis

(c) To decompose metal carbonates into oxides and carbon dioxide

(d) To mix ores with fluxes

Sol. (c)

12. Calcination is often followed by which of the following processes in metal extraction?

(a) Smelting (b) Electroless plating (c) Filtration (d) Precipitation

Sol. (a)

13. What is the main difference between roasting and calcination?

(a) Roasting involves heating in the presence of oxygen, while calcination occurs in the absence of air

(b) Roasting is used for carbonates, while calcination is used for sulphides

(c) Roasting produces oxides, while calcination produces sulphides

(d) Roasting is done at lower temperatures than calcination

Sol. (a)

(IIT JEE 2004)

14. The methods chiefly used for the extraction of lead and tin from their ores are respectively

(a) Self-reduction and carbon reduction (b) Self-reduction and electrolytic reduction

(c) Carbon reduction and self-reduction (d) Cyanide process and carbon reduction

Sol. (a)

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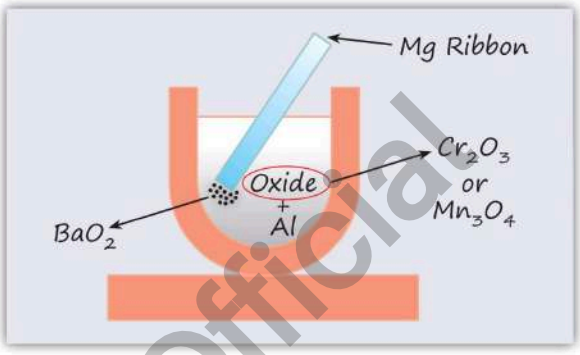
ALUMINOTHERMIC PROCESS

Goldschmidt Process

Thermite mixture: Oxide + Al-powder

Ignition mixture: BaO_2 + Mg (This mixture produce enough heat to ignite the thermite mixture)



Thermite reactions–

$$Cr_2O_3 \xrightarrow[\text{as a reducing agent}]{Al} Cr + Al_2O_3$$

$$Mn_3O_4 \xrightarrow{Al} Mn + Al_2O_3$$

- Metallic oxides (Cr and Mn) can be reduced by a highly electropositive metal such as aluminium that liberates a large amount of energy (1675 kJ/mol) on oxidation to Al_2O_3. The process is known as Goldschmidt or aluminothermic process and the reaction is known as thermite reaction.

Na/Mg Reduction

$$TiCl_4 + Mg \xrightarrow[\text{High Temperature}]{\text{Kroll process}} Ti + 2MgCl_2$$

$$TiCl_4 + Na \xrightarrow{\text{IMI process}} Ti + 4NaCl$$

IMI ≡ Imperial metal industries

- Na and Mg has high reducing power and high solubility of their halides in water so that the reduced metals can be washed free from impurities.

(IIT JEE 1983)

15. In the alumino-thermite process, aluminium acts as

(a) An oxidising agent (b) A flux

(c) A reducing agent (d) A solder

Sol. (c)

ELLINGHAM DIAGRAM

This diagram is a graph between Gibb's energy and temperature and has slope $-\Delta S$. $\Delta G = \Delta H - T\Delta S$

- W H.J.T. Ellingham was the first to use a graphical representation of Gibb's energy. This graph, known as the Ellingham Diagram, serves as a solid foundation for selecting the appropriate reducing agent for oxide reduction. These diagrams are useful for predicting whether the thermal reduction of an ore is feasible. At a given temperature, the criterion for determining feasibility is that, Gibbs energy of the reaction must be negative.

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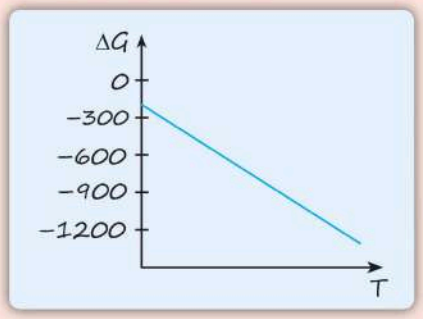
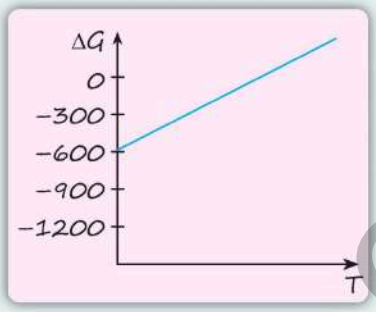
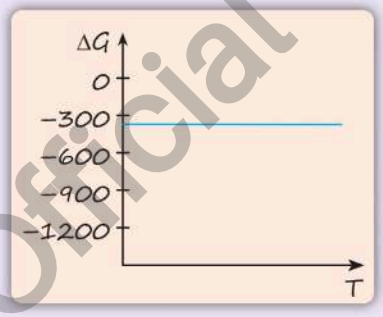
(i) $2C_{(s)} + O_{2(g)} \longrightarrow 2CO$	(ii) $2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)}$	(iii) $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$
❑ Gaseous amount (molecular randomness) is increasing from left to right, which leads to +ve value of ΔS.	❑ Gaseous amount is decreasing from left to right, which leads to −ve value of ΔS.	❑ Gaseous amount is same from left to right, which leads to zero value of ΔS.
❑ Slope = $-\Delta S$ = ⊖ve 	❑ Slope = $-\Delta S$ = −(−) = ⊕ve 	❑ Slope = $-\Delta S$ = zero 

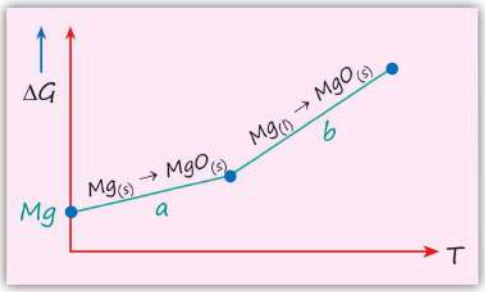
❑ Ellingham diagram normally consists of plots of Δ_f. G° vs T for formation of oxides of elements.

$$2xM_{(s)} + O_{2(g)} \longrightarrow 2M_xO_{(s)}$$

- ♦ Gaseous amount is decreasing from left to right, which leads to −ve value of ΔS.
- ♦ Slope = $-\Delta S$ = −(−) = ⊕ve
- ♦ Most of the reactions for the formation of metal oxide (M_xO) shows +ve slope in the curve.

❑ The entropy is higher if the metal is in liquid state than when it is in solid state.

❑ Phase change (solid → liquid, liquid → gas) is indicated by an abrupt change in the slope on +ve side. For example – Mg / MgO curve



(a) $2Mg_{(s)} + O_{2(g)} \longrightarrow 2MgO_{(s)}$ ΔS = less negative

(b) $2Mg_{(l)} + O_{2(g)} \longrightarrow 2MgO_{(s)}$ ΔS = more negative

- ♦ ΔS depends much on the physical state of the compound.

Since entropy depends on disorder or randomness in the system, it will increase if a compound melts (s → l) or vapourises (l → g) [since molecular randomness increases on changing the phase from solid to liquid or from liquid to gas].

Note

Q **Why is the reduction of metal oxide easier if the metal formed is in liquid state at temperature of reduction?**

A The value of entropy change (ΔS) of the reduction process [Metal oxide (s) → Metal (l)] is more on positive side when the metal formed is in liquid state and the metal oxide being reduced is in solid state. Thus the value of ΔG becomes more on negative side and the reduction become easier.

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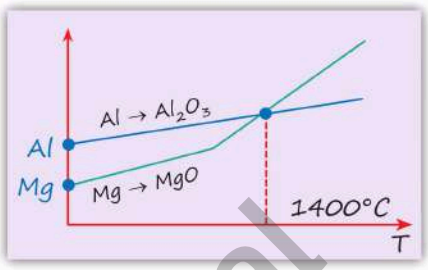
❑ Below the point of intersection of Al / Al_2O_3 and Mg / MgO curve, Mg can reduce Al_2O_3.

$2Mg + O_2 \longrightarrow 2MgO$ $\quad \Delta G^\circ = -x$ KJ/mol of O_2

$\frac{4}{3}Al + O_2 \longrightarrow \frac{2}{3}Al_2O_3$ $\quad \Delta G^\circ = -y$ KJ/mol of O_2

$\frac{2}{3}Al_2O_3 + 2Mg \longrightarrow 2MgO + \frac{4}{3}Al \quad \Delta G^\circ = y - x$

$= \ominus$ve (because $x > y$)



♦ Mg can reduce alumina. But the process will be uneconomical. So in practice, Mg metal is not used for the reduction of alumina in the metallurgy of aluminium.

♦ At the point of intersection of the Al_2O_3 and MgO curves, the ΔG° becomes ZERO for the reaction.

$\frac{2}{3}Al_2O_3 + 2Mg \longrightarrow 2MgO + \frac{4}{3}Al \quad \Delta G^\circ = y - x$

$=$ Zero (because at point of intersection $y = x$)

♦ Above the point of intersection of Al / Al_2O_3 and Mg / MgO curve, Al can reduce MgO.

$\frac{2}{3}Al_2O_3 + 2Mg \longrightarrow 2MgO + \frac{4}{3}Al \quad \Delta G^\circ = y - x$

$= \oplus$ve (because $y > x$)

Reverse of above reaction is feasible (ΔG = –ve)

$2MgO + \frac{4}{3}Al \longrightarrow \frac{2}{3}Al_2O_3 + 2Mg$

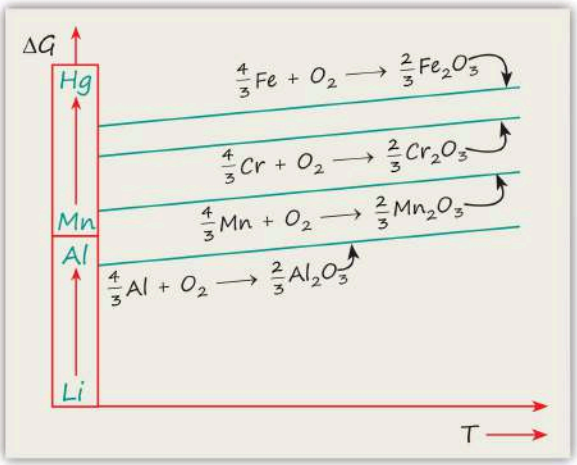
Li K Ba Sr Ca Na Mg Al	Mn Zn Cr Fe Cd Co Ni Sn Pb Cu Hg	Ag Au Pt
• Highly reactive metals (M)	• Moderate Reactive metals (M′)	• Highly Non-reactive Metals
• Metal $\xrightarrow{O_2}$ oxide (more $\ominus$ve ΔG)	• Metal $\xrightarrow{O_2}$ oxide (less $\ominus$ve ΔG)	

♦ Reductions of the oxide of the upper line is feasible by the element represented by the lower line. If the difference is large, the reduction is easier.

$Mn_2O_3 \xrightarrow{Al} Mn + Al_2O_3$

$Fe_2O_3 \xrightarrow{Al} Fe + Al_2O_3$

$Cr_2O_3 \xrightarrow{Al} Cr + Al_2O_3$



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16. The reaction, $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$ ($\Delta_rG^\circ = -421$ kJ) is thermodynamically feasible as is apparent from the Gibbs energy value. Why does it not take place at room temperature?

Sol. The rate of reaction at room temperature is extremely slow due to the high activation energy, so the reaction appears not to proceed.

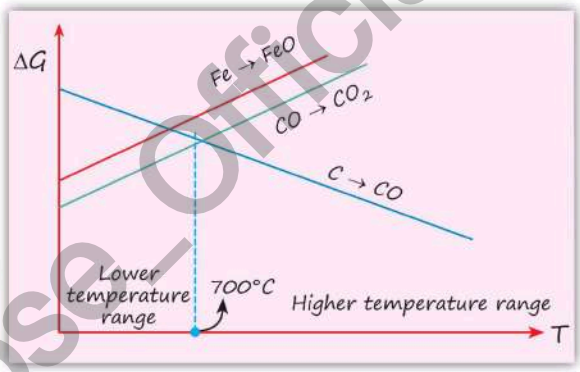
In practice, this reaction is carried out at high temperatures (like in a thermite reaction), where enough energy is provided to overcome the activation barrier and allow the reaction to proceed rapidly.

❑

$2Fe + O_2 \longrightarrow 2FeO$ $\quad \Delta G = -z$ KJ/mol of O_2

$2C + O_2 \longrightarrow 2CO$ $\quad \Delta G = -a$ KJ/mol of O_2

$2CO + O_2 \longrightarrow 2CO_2$ $\quad \Delta G = -b$ KJ/mol of O_2



- At higher temperature, coke will be reducing the FeO to Fe and will itself be oxidised to CO.

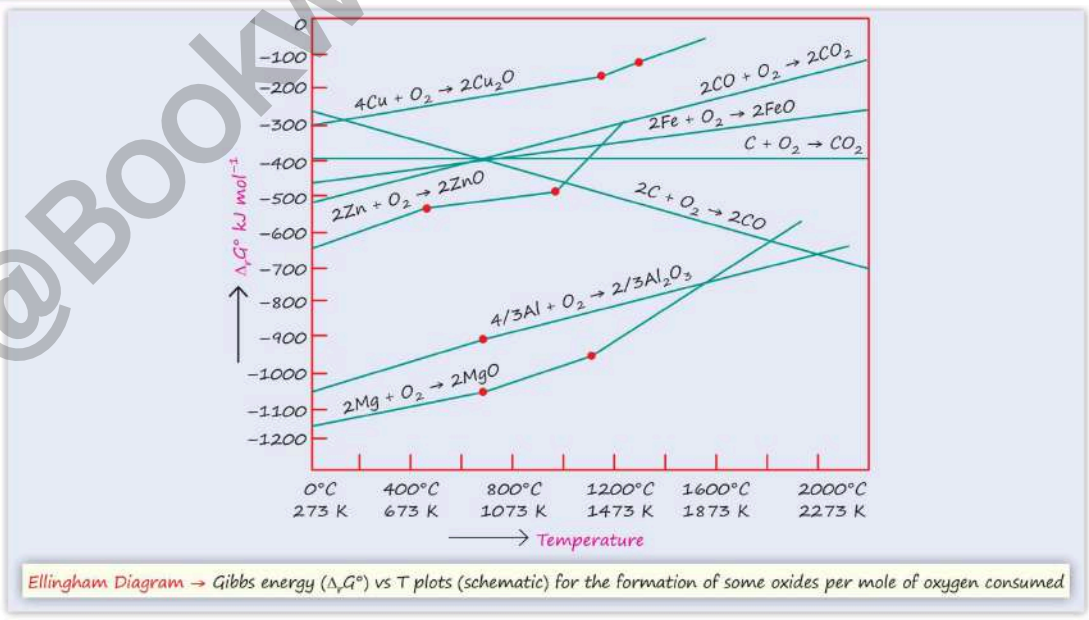
$FeO + C \longrightarrow CO + Fe$ (At higher temperature range: a > z)

- At lower temperature, CO gas will be reducing FeO to Fe and will itself be oxidised to CO_2.

$FeO + CO \longrightarrow CO_2 + Fe$ (At lower temperature range: b > z)

❑ The Cu / Cu_2O line is almost at the top in graph . So it is quite easy to reduce oxide ores of copper directly to the metal by heating with coke because both the lines of C, CO and C, CO_2 are at much lower positions in the graph particularly after 500 – 600K.

$Cu_2O + C \longrightarrow Cu + CO\uparrow$



Ellingham Diagram → Gibbs energy (Δ_rG°) vs T plots (schematic) for the formation of some oxides per mole of oxygen consumed

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17. **Given below are two statements:**

Statement I: The choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature.

Statement II: The value of ΔS increases from left to right in Ellingham diagram.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both Statement I and Statement II are true
(b) Statement I is false but Statement II is true
(c) Both Statement I and Statement II are false
(d) Statement I is true but Statement II is false

Sol. (d) In an Ellingham diagram, the value of entropy change (ΔS) decreases from left to right.

EXTRACTION OF IRON FROM ITS OXIDES

- In a blast furnace, the reduction of iron oxides occurs across various temperature ranges.
- Hot air is introduced from the bottom of the furnace.
- Coke is burned, reaching temperatures of up to about 2200K in the lower section.
- The combustion of coke supplies the majority of the heat required for the process.
- Generated CO and heat rise to the upper part of the furnace.
- In the upper region, the temperature is lower.
- Iron oxides (Fe_2O_3 and Fe_3O_4) descending from the top are gradually reduced to FeO.
- Reduction reactions occurring at both lower and higher temperature ranges depend on the corresponding intersection points in the $\Delta_r G^\circ$ vs T plots.

Blast Furnace

Lower Temperature Range (500 – 800 K)

$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$

$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$

$CaCO_3 \longrightarrow CaO + CO_2$

$CaO + SiO_2 \longrightarrow CaSiO_3$ (slag)

$FeO + CO \longrightarrow Fe + CO_2$

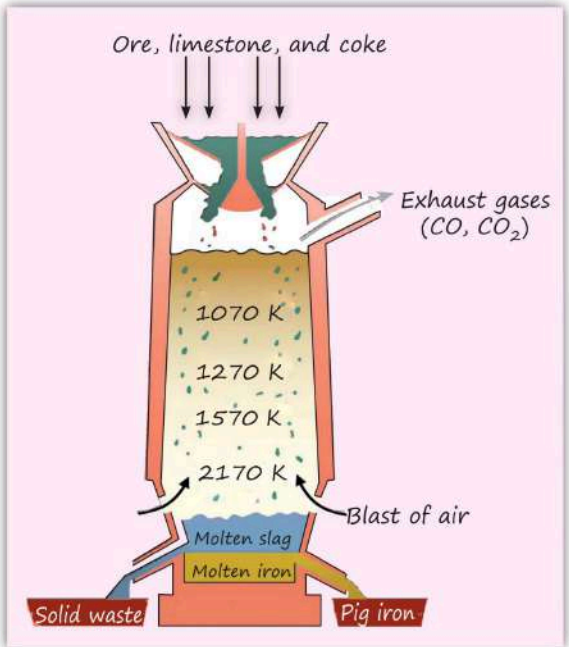
$C + CO_2 \longrightarrow 2CO$

$C + O_2 \longrightarrow CO_2$

$FeO + C \longrightarrow Fe + CO$

Higher Temperature Range (900 – 1500 K)

- Limestone (flux) is decomposed to CaO which removes silicate impurity of the ore as slag. The slag is in molten state and separates out from iron.



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- Sulphur, silicon and phosphorus are oxidised and passed into the slag. The metal is removed and freed from the slag by passing through rollers.
- The iron obtained from Blast furnace contains about 4% carbon known as pig iron and cast into variety of shapes.
- Cast iron is different from pig iron and is made by melting pig iron with scrap iron and coke using hot air blast. It has slightly lower carbon content (about 3%) and is extremely hard and brittle.
- Wrought iron or malleable iron is the purest form of commercial iron and is prepared from cast iron by oxidising impurities in a reverberatory furnace lined with haematite. This haematite oxidises carbon to carbon monoxide.

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO\uparrow$$

- Order of M.P. → Wrought Iron > Steel > Cast Iron or Pig Iron

Different form of iron		% of C
(1)	Wrought iron	0.1 to 0.25 (purest form)
(2)	Steel	0.25 to 2.0
(3)	Cast iron	2.6 to 4.3 (about 3)
(4)	Pig iron	2.3 to 4.6 (about 4)

18. In the commercial electrochemical process for aluminium extraction, the electrolyte used is (IIT JEE 1999)

(a) $Al(OH)_3$ in NaOH solution (b) An aqueous solution of $Al_2(SO_4)_3$

(c) A molten mixture of Al_2O_3 and Na_3AlF_6 (d) A molten mixture of AlO(OH) and $Al(OH)_3$

Sol. (c)

19. The correct option(s) related to the extraction of iron from its ore in the blast furnace operating in the temperature range 900–1500 K is (are): (JEE Adv. 2022)

(a) Limestone is used to remove silicate impurity.

(b) Pig iron obtained from blast furnace contains about 4% carbon.

(c) Coke (C) converts CO_2 to CO.

(d) Exhaust gases consist of NO_2 and CO.

Sol. (a, b, c)

EXTRACTION OF ALUMINIUM FROM RED BAUXITE ($Al_2O_3.2H_2O$)

Bayer's Process

Leaching Agent: $NaOH_{(aq)}$ Impurity: Ferric oxide + Silicates

$$Al_2O_3.2H_2O + 2NaOH \longrightarrow 2NaAlO_2 + 3H_2O$$

$$NaAlO_2 + 2H_2O \longrightarrow NaOH + Al(OH)_3\downarrow$$

$$2Al(OH)_3 \xrightarrow{1200^\circ C} Al_2O_3 + 3H_2O$$

(pure)

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Hall's Process

Leaching Agent: $Na_2CO_{3(aq)}$ Impurity: Ferric oxide + Silicates

$$Al_2O_3.2H_2O + Na_2CO_3 \longrightarrow 2NaAlO_2 + CO_2\uparrow + 2H_2O$$

$$2NaAlO_2 + CO_2 + 3H_2O \longrightarrow 2Al(OH)_3\downarrow + Na_2CO_3$$

$$2Al(OH)_3 \xrightarrow{1200^\circ C} Al_2O_3 + 3H_2O$$

(pure)

Serpeck's Process

Ore: White Bauxite ($Al_2O_3.2H_2O$) Impurity: SiO_2

$$Al_2O_3.2H_2O + 3C + N_2 \xrightarrow{1800^\circ C} 2\,AlN + 3CO\uparrow + 2H_2O\uparrow$$

$$2AlN + 3H_2O \longrightarrow 2Al(OH)_3\downarrow + NH_3\uparrow$$

(ppt)

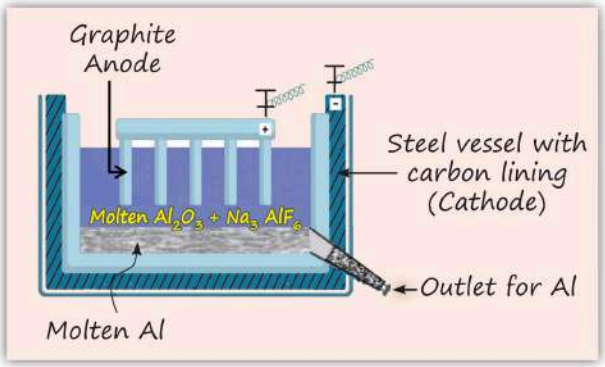
$$SiO_2 + 2CO \longrightarrow 2CO_2\uparrow + Si\uparrow$$ (Silicon volatilizes at this temperature)

$$2Al(OH)_3 \xrightarrow{1200^\circ C} Al_2O_3 + 3H_2O$$

(pure)

Electrolytic Reduction of Al (Hall-Heroult Process)

- Fused Matrix (electrolyte): Purified Al_2O_3 + Na_3AlF_6 (cryolite) or CaF_2 (fluorspar)
- Addition of CaF_2 or Na_3AlF_6 in Al_2O_3 lowers the melting point of the mixture and brings conductivity.



- Cathode

$$Al^{3+}_{(melt)} + 3e^- \longrightarrow Al_{(l)}$$

- Anode

$$2O^{2-}_{(melt)} \longrightarrow O_2 + 4e^-$$

$$O_2 \xrightarrow{C(s)} CO_2 \text{ or } CO$$

- The oxygen liberated at anode reacts with the carbon of anode producing CO and CO_2. This way for each kg of aluminium produced, about 0.5 kg of carbon anode is burnt away.

Overall reaction: $2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$

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20. Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out (IIT JEE 2000)

(a) In the presence of NaCl

(b) In the presence of fluorite

(c) In the presence of cryolite which forms a melt with lower melting temperature

(d) In the presence of cryolite which forms a melt with higher melting temperature

Sol. (c)

21. The electrochemical extraction of aluminium from bauxite ore involves: (JEE Adv. 2022)

(a) The reaction of Al_2O_3 with coke (C) at a temperature > 2500 °C.

(b) The neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($Al_2O_3 \cdot 3H_2O$).

(c) The dissolution of Al_2O_3 in hot aqueous NaOH.

(d) The electrolysis of Al_2O_3 when mixed with Na_3AlF_6 gives Al and CO_2.

Sol. (b, c, d)

22. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Aluminium is extracted from bauxite by the electrolysis of molten mixture of Al_2O_3 with cryolite.

Reason (R): The oxidation state of Al in cryolite is +3.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) (A) is true but (R) is false

(b) (A) is false but (R) is true

(c) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(d) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Sol. (d) Aluminium is a reactive metal, so it is extracted by the electrolysis of alumina using a molten mixture of cryolite.

Cryolite is Na_3AlF_6, and in this compound, the oxidation state of aluminium is +3.

23. Match List-I with List-II:

List-I		List-II	
(A)	Haematite	(i)	$Al_2O_3.xH_2O$
(B)	Bauxite	(ii)	Fe_2O_3
(C)	Magnetite	(iii)	$CuCO_3.Cu(OH)_2$
(D)	Malachite	(iv)	Fe_3O_4

Choose the most appropriate answer from the options given below :

(a) (A)-(i), (B)-(iii), (C)-(ii), (D)-(iv)

(b) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

(c) (A)-(ii), (B)-(iii), (C)-(i), (D)-(iv)

(d) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

Sol. (b)

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24. **The point of intersection and sudden increase in the slope, in the Ellingham diagram respectively, indicates:**

(a) $\Delta G > 0$ and decomposition of the metal oxide

(b) $\Delta G < 0$ and decomposition of the metal oxide

(c) $\Delta G = 0$ and reduction of the metal oxide

(d) $\Delta G = 0$ and melting or boiling point of the metal oxide

Sol. (d) From the Ellingham diagram given, the point of intersection represents $\Delta G = 0$ and the temperature at which sudden increase in the slope occurs is indicated by melting or boiling.

25. **In the leaching of alumina from bauxite, the ore expected to leach out in the process by reacting with NaOH is :**

(a) TiO_2 (b) Fe_2O_3 (c) ZnO (d) SiO_2

Sol. (d) In bauxite, the impurities present include Fe_2O_3, TiO_2, and SiO_2. Fe_2O_3 and TiO_2 are basic oxides and do not react or dissolve in NaOH. On the other hand, SiO_2 is an acidic oxide, and it dissolves in NaOH. The reaction is as follows:

$SiO_2 + 2NaOH \rightarrow Na_2SiO_3$ (aqueous) + H_2O

26. **Calamine, malachite, magnetite and cryolite, respectively are:** **(JEE Adv. 2019)**

(a) $ZnSO_4$, $CuCO_3$, Fe_2O_3, AlF_3 (b) $ZnCO_3$, $CuCO_3 . Cu(OH)_2$, Fe_3O_4, Na_3AlF_6

(c) $ZnSO_4$, $Cu(OH)_2$, Fe_3O_4, Na_3AlF_6 (d) $ZnCO_3$, $CuCO_3$, Fe_2O_3, Na_3AlF_6

Sol. (b)

27. **The carbon-based reduction method is not used for the extraction of** **(JEE Adv. 2013)**

(a) Tin from SnO_2 (b) Iron from Fe_2O_3

(c) Aluminium from Al_2O_3 (d) Magnesium from $MgCO_3.CaCO_3$

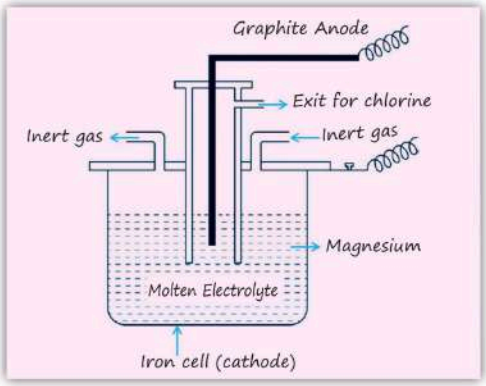
Sol. (c, d) Carbon can be used to reduce the oxide of metals (Fe and Sn) below aluminium in the reactivity series.

ELECTROLYSIS OF FUSED Anhyd. $MgCl_2$

- Pure $MgCl_2$ is fused and mixed with NaCl and $CaCl_2$ to lower the melting point of $MgCl_2$.
- The molten mixture is electrolysed.
- Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

Cathode: $Mg^{2+} + 2e^- \longrightarrow Mg$

Anode: $2Cl^- \longrightarrow Cl_2\uparrow + 2e^-$



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(IIT JEE 2002)

28. Which of the following process is used in extractive metallurgy of magnesium?

(a) Fused salt electrolysis (b) Self-reduction

(c) Aqueous solution electrolysis (d) Thermite reduction

Sol. (a)

EXTRACTION OF Cu

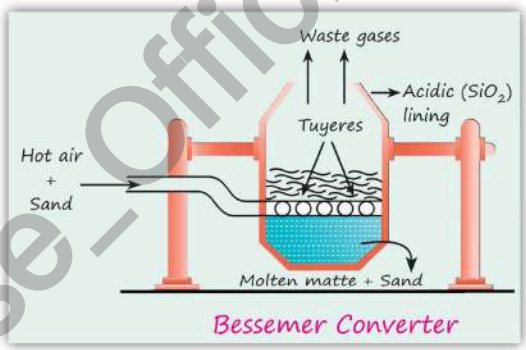
Ore: Copper pyrite ($CuFeS_2$)

(i) $CuFeS_2 + O_2 \longrightarrow Cu_2S + FeO + SO_2 \uparrow$

(ii) $Cu_2S + FeO + SiO_2 \longrightarrow FeSiO_3$ (slag) + Cu_2S (matte)
(Impurity) (Flux) (upper layer) (lower layer)

(iii) $FeS + O_2 \longrightarrow FeO + SO_2$
$FeO + SiO_2 \longrightarrow FeSiO_3$(slag)

(iv) $Cu_2S + O_2 \longrightarrow Cu_2O + SO_2$
$Cu_2O + Cu_2S \longrightarrow Cu + SO_2$



Bessemer Converter

- Ore is heated in a reverberatory furnace after mixing with silica. In the furnace, iron oxide 'slags of' as iron silicate and copper is produced in the form of copper matte (it contains Cu_2S and FeS).
- Copper matte is then charged into silica lined convertor. Some silica is also added and hot air blast is blown to convert the remaining FeS, FeO, Cu_2S and Cu_2O to metallic copper.
- SO_2 gas escaping from the melt gets trapped in the cooler parts of the molten copper surface gives a blistery appearance for copper and hence it is called blister copper.

Copper from Low Grade Ores and Scraps

- Copper is extracted by hydrometallurgy from low grade ores. It is leached out using acid or bacteria. The solution containing Cu^{2+} is treated with scrap iron / H_2.

$$Cu^{2+}_{(aq)} + H_{2(g)} \longrightarrow Cu_{(s)} + 2H^+_{(aq)}$$

- Less reactive metal will come out of the solution and the more reactive metal will go to the solution.

EXTRACTION OF ZINC FROM ORE (ZnS)

$$\text{Zinc blende (ZnS)} \xrightarrow[\text{Froth floatation process}]{\text{Concentration by}} \underset{\text{(Concentrated)}}{ZnS} \xrightarrow{\text{Roasting}} ZnO \xrightarrow{\text{Coke}} Zn + CO \uparrow$$

The reduction of zinc oxide is done using coke at high temperature.

$$ZnO + C \xrightarrow{1400^\circ C} Zn + CO \uparrow$$

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Note

Q **At a site, there are low-grade copper ores as well as zinc and iron metal scraps. Which of the two metals-zinc or iron-would be better for reducing the copper from the leached ore, and why?**

A Since zinc is higher than iron in the electrochemical series, it is more reactive and can reduce copper from its solution more quickly. However, zinc is more expensive than iron, so using iron scraps would be more practical and cost-effective.

29. Upon heating with Cu_2S, the reagent(s) that give copper metal is/are (JEE Adv. 2014)

(a) $CuFeS_2$ (b) CuO (c) Cu_2O (d) $CuSO_4$

Sol. (b, c, d)

(b) $4CuO \xrightarrow{1100^\circ C} 2Cu_2O + O_2$

$2Cu_2O + Cu_2S \xrightarrow{\Delta} 6Cu + SO_2$

(c) $Cu_2S + 2Cu_2O \xrightarrow{\Delta} 6Cu + SO_2$

(d) $CuSO_4 \xrightarrow{720^\circ C} CuO + SO_2 + \frac{1}{2}O_2$

$4CuO \xrightarrow{1100^\circ C} 2Cu_2O + O_2$

$2Cu_2O + Cu_2S \xrightarrow{\Delta} 6Cu + SO_2$

So, the correct answer is CuO, Cu_2O and $CuSO_4$.

30. The chemical composition of 'slag' formed during the smelting process in the extraction of copper is (IIT JEE 2001)

(a) $Cu_2O + FeS$ (b) $FeSiO_3$ (c) $CuFeS_2$ (d) $Cu_2S + FeO$

Sol. (b)

31. Extraction of copper from copper pyrite ($CuFeS_2$) involves (JEE Adv. 2016)

(a) Crushing followed by concentration of the ore by froth-floatation

(b) Removal of iron as slag

(c) Self reduction step to produce 'blister copper' following evolution of SO_2

(d) Refining of 'blister copper' by carbon reduction

Sol. (a, b, c)

32. Passage:- Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4 \cdot 5H_2O$), atacamite ($Cu_2Cl(OH)_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($Cu_2(OH)_2CO_3$).
However, 80% of the world copper production comes from the ore chalcopyrite ($CuFeS_2$).

The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. Partial roasting of chalcopyrite produces (IIT JEE 2010)

(a) Cu_2S and FeO (b) Cu_2O and FeO (c) CuS and Fe_2O_3 (d) Cu_2O and Fe_2O_3

Sol. (b) Partial roasting of Chalcopyrite follows given reaction:

(i) $2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2 \uparrow$ (ii) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2 \uparrow$

(iii) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2 \uparrow$

Thus option (b) is correct.

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33. Extraction of zinc from zinc blende is achieved by (IIT JEE 2007)

(a) Electrolytic reduction

(b) Roasting followed by reduction with carbon

(c) Roasting followed by reduction with another metal

(d) Roasting followed by self-reduction

Sol. (b)

34. Match the conversions in Column-I with the type(s) of reaction(s) given in Column-II. (IIT JEE 2008)

Column-I		Column-II	
A.	$PbS \rightarrow PbO$	p.	Roasting
B.	$CaCO_3 \rightarrow CaO$	q.	Calcination
C.	$ZnS \rightarrow Zn$	r.	Carbon reduction
D.	$Cu_2S \rightarrow Cu$	s.	Self-reduction

Sol. A → p; B → q; C → p, r; D → p, s.

35. The correct statement(s) related to processes involved in the extraction of metals is(are) (JEE Adv. 2023)

(a) Roasting of Malachite produces Cuprite.

(b) Calcination of Calamine produces Zincite.

(c) Copper pyrites is heated with silica in a reverberatory furnace to remove iron.

(d) Impure silver is treated with aqueous KCN in the presence of oxygen followed by reduction with zinc metal.

Sol. (b, c, d)

36. The addition of silica during the extraction of copper from its sulphide ore

(a) converts copper sulphide into copper silicate

(b) converts iron oxide into iron silicate

(c) reduces copper sulphide into metallic copper

(d) reduces the melting point of the reaction mixture

Sol. (b) The addition of silica during the extraction of copper from its sulphide ore serves to form a slag that helps in separating the impurities. Specifically, it reacts with iron oxide to form iron silicate, which is removed as slag.

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37. **Match List-I with List-II: (Both having metallurgical terms)**

List-I		List-II	
(A)	Concentration of Ag ore	(i)	Reverberatory furnace
(B)	Blast furnace	(ii)	Pig iron
(C)	Blister copper	(iii)	Leaching with dilute NaCN solution
(D)	Froth floatation method	(iv)	Sulphide ores

Choose the most appropriate answer from the options given below:

(a) A-(iii), B-(ii), C-(i), D-(iv)
(b) A-(iii), B-(iv), C-(i), D-(ii)
(c) A-(iv), B-(i), C-(iii), D-(ii)
(d) A-(iv), B-(iii), C-(iv), D-(i)

Sol. (a)

(A) Concentration of Ag is performed by leaching with dilute NaCN solution

(B) Pig iron is formed in blast furnace

(C) Blister copper is obtained during the reverberatory furnace process

(D) Froth floatation method is used for sulphide ores.

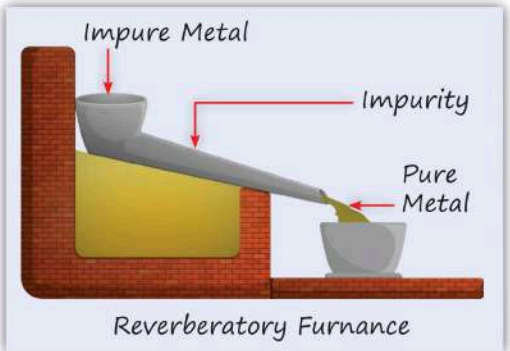
REFINING

Removal of unwanted substance from impure metal

1. Liquation
2. Distillation
3. Electrolytic refining
4. Zone refining
5. Vapour phase refining
6. Chromatography
7. Polling

Liquation

- This method is used when impurities have high melting point than metals.
- Liquation is used for metals having low melting point (Sn, Pb, Bi).
- Crude metal is heated on sloping floor of a furnace.
- Fused metal flows down leaving behind infusible impurities.



Reverberatory Furnance

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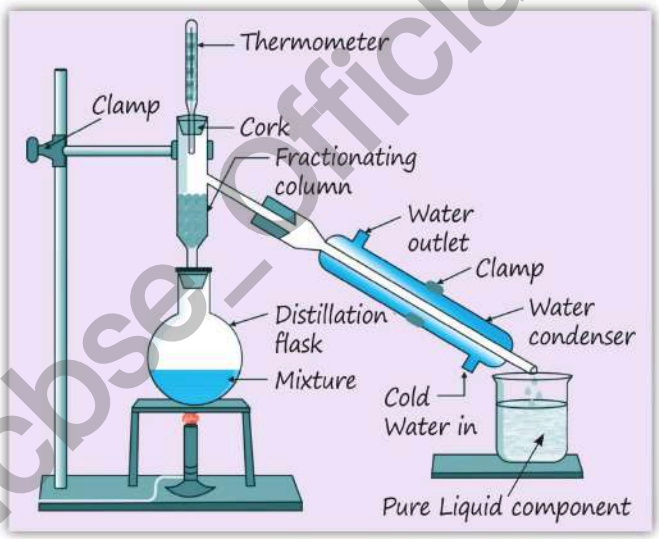
Son → Sn Bhi → Bi Pub → Pb में liquid → liquation पीने गया

Distillation

- This method is useful for low boiling point metal like (Zn, Hg, Cd).
- Distillation is used for the elements that are liquids at room temperature or can be converted to liquid at room temperature easily.



38. The metal that can be purified economically by fractional distillation method is:

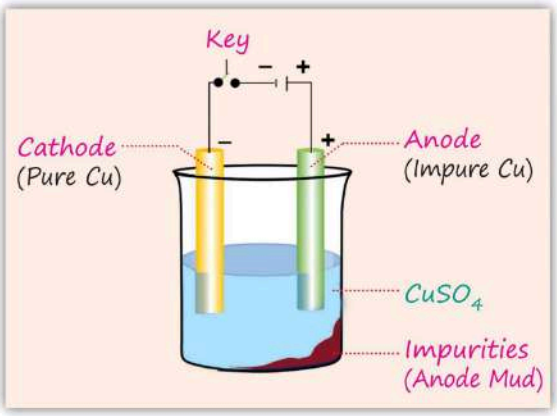
(a) Fe (b) Zn

(c) Cu (d) Ni

Sol. (b) The metal that can be purified economically by fractional distillation is Zn.

Electrolytic Refining

- In this method, the impure metal is made to act as anode.
- A strip of the same metal in pure form is used as cathode.
- Anode and cathode are put in a suitable electrolytic bath containing soluble salt of the same metal.
- Impurities go to the anode mud from impure metal anode.
- By this method Cu and Zn can be refined.
- For copper,



At anode: $Cu \longrightarrow Cu^{2+}_{(aq)} + 2e^-$ (Impure)

At cathode: $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu$ (Pure)

- Impurities from the blister copper deposit as anode mud which contains antimony, selenium, tellurium, silver, gold and platinum; recovery of these elements may meet the cost of refining.

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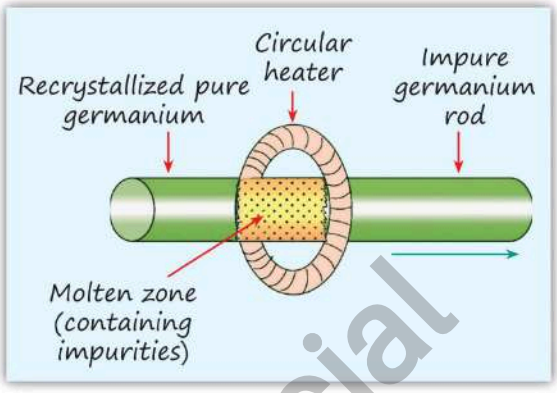
Zone Refining

- Highly pure semiconductors (Ge, Si, B, Ga, In) are generally refined by zone refining.
- This process is used when metals are required in very high purity, for specific application.
- An impure molten metal on gradual cooling will deposit crystals of pure metal.
- Impurities will be left in the remaining part of molten metal.



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Indian → In
Zone → Zone refining
me
Sita → Si
Ge → Ge
B → B
Gana → Ga
gati hain.

Vapour Phase Refining

Two major requirements for this process are:

(i) The metal must form a volatile compound with an available reagent

(ii) the volatile compound must be easily decomposable to facilitate easy recovery.

$$\text{Impure Metal} \longrightarrow \text{Volatile Compound} \xrightarrow{\Delta} \text{Pure Metal}$$

1. Mond Process

- Any impurity in the nickel sample remains in the solid state and the gas is heated to 230 °C, when it decomposes, giving pure metal and CO, which is recycled. $Ni(CO)_4$ is gaseous and may be produced by warming nickel with CO at 50 °C.

$$\underset{\text{Impure}}{Ni_{(s)}} + 4CO_{(g)} \xrightarrow{50^\circ C} [Ni(CO)_4]_{(g)} \xrightarrow{230^\circ C} \underset{\text{Pure}}{Ni_{(s)}} + 4CO\uparrow$$

2. Van Arkel Method

- Small amounts of very pure metals (Ti, Zr) can be produced by this method.
- This process is based on the fact that iodides are the least stable of the halides.
- The impure element is heated with iodine, producing a volatile iodide, TiI_4, ZrI_4.

$$\underset{\text{Impure}}{Ti} + 2I_2 \xrightarrow{100^\circ C} TiI_{4(g)} \xrightarrow{1400^\circ C} \underset{\text{Pure}}{Ti} + 2I_2$$

$$\underset{\text{Impure}}{Zr} + 2I_2 \xrightarrow{100^\circ C} ZrI_{4(g)} \xrightarrow{1500^\circ C} \underset{\text{Pure}}{Zr} + 2I_2$$

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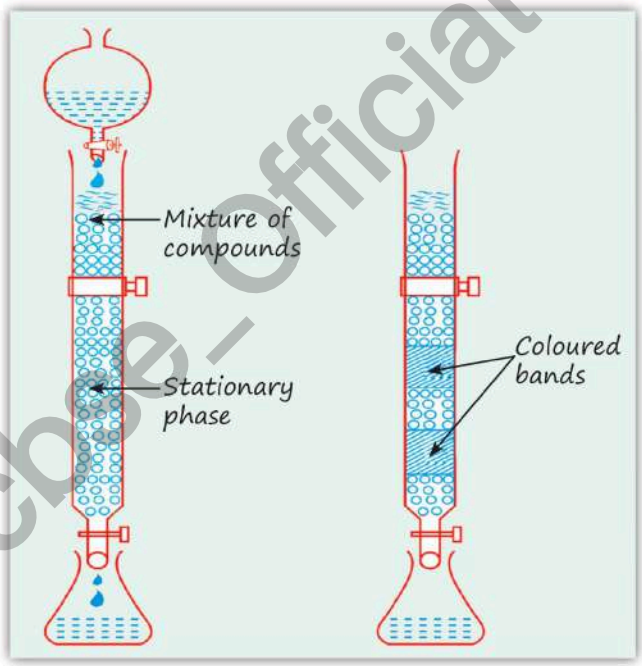
Mo ↓ Mond Ni ↓ Ni baba ne VAMpire ↓ Van Arkel method का Ti ↓ Ti Zar ↓ Zr देखा

Chromatographic Methods



- **Basic Principle:** Different components of a mixture are differently adsorbed on an adsorbent.
- Useful for purification of the elements which are available in minute quantities and the impurities are not very different in chemical properties from the element to be purified.
- The mixture is placed in a liquid or gaseous medium that is passed through the adsorbent.
- Different components are adsorbed at various levels on the column.
- The adsorbed components are later removed (eluted) using suitable solvents (eluant).
- The chromatographic method is named based on the physical state of the moving medium, the adsorbent material, and the process of passage of the moving medium.
- In column chromatography, a column of Al_2O_3 is prepared in a glass tube, and the moving medium containing a solution of the components is in liquid form.

Poling Process

Green wood, at high temperature, liberates hydrocarbon gases, which are converted into methane (CH_4). Methane thus obtained reduces metal oxide to metal.

1. Purification of impure copper

$$4CuO + CH_4 \longrightarrow \underset{\text{pure metal}}{Cu} + CO_2 + H_2O$$

2. Purification of impure tin

$$2SnO_2 + CH_4 \longrightarrow \underset{\text{pure metal}}{2Sn} + CO_2 + H_2O$$

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39. Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is/are **(JEE Adv. 2015)**

(a) Impure Cu strip is used as cathode
(b) Acidified aqueous $CuSO_4$ is used as electrolyte
(c) Pure Cu deposits at cathode
(d) Impurities settle as anode-mud

Sol. (b, c, d)

40. Match the extraction processes listed in Column-I with metals listed in Column-II. **(IIT JEE 2006)**

	Column-I		Column-II
A.	Self-reduction	p.	Lead
B.	Carbon reduction	q.	Silver
C.	Complex formation and displacement by metal	r.	Copper
D.	Decomposition of iodide	s.	Boron

Sol. A → p, r ; B → p,r ; C → q ; D → s.

41. The electrochemical extraction of aluminium from bauxite ore involves: **(JEE Adv. 2022)**

(a) The reaction of Al_2O_3 with coke (C) at a temperature > 2500 °C.
(b) The neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($Al_2O_3 \cdot 3H_2O$).
(c) The dissolution of Al_2O_3 in hot aqueous NaOH.
(d) The electrolysis of Al_2O_3 when mixed with Na_3AlF_6 gives Al and CO_2.

Sol. (b, c, d)

42. The correct statement(s) related to the metal extraction processes is (are): **(JEE Adv. 2021)**

(a) A mixture of PbS and PbO undergoes self reduction to produce Pb and SO_2.
(b) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.
(c) Partial oxidation of sulphide ore of copper by roasting, followed by self reduction produces blister copper.
(d) In cyanide process, zinc powder is utilized to precipitate gold from $Na[Au(CN)_2]$.

Sol. (a, c, d)

43. Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite? **(JEE Adv. 2020)**

(a) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.
(b) Addition of Na_3AlF_6 lowers the melting point of alumina.
(c) CO_2 is evolved at the anode during electrolysis.
(d) The cathode is a steel vessel with a lining of carbon.

Sol. (a, b, c, d)

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44. Which refining process is generally used in the purification of low melting metals?

(a) Chromatographic method (b) Liquation

(c) Electrolysis (d) Zone refining

Sol. (b) The refining process generally used in the purification of low melting metals is Liquation.

45. Match List-I with List-II:

List-I		List-II	
(A)	Mercury	(i)	Vapour phase refining
(B)	Copper	(ii)	Distillation refining
(C)	Silicon	(iii)	Electrolytic refining
(D)	Nickel	(iv)	Zone refining

Choose the most appropriate answer from the options given below:

(a) A-(ii), B-(iii), C-(iv), D-(i) (b) A-(iii), B)-(iv), C-(iii), D-(i)

(c) A-(ii), B-(iii), C-(i), D-(iv) (d) A-(i), B)-(iv), C-(ii), D-(iii)

Sol. (a)

(A) Mercury → Distillation refining (B) Copper → Electrolytic refining

(C) Silicon → Zone refining (D) Nickel → Vapour phase refining

46. The method used for the purification of Indium is:

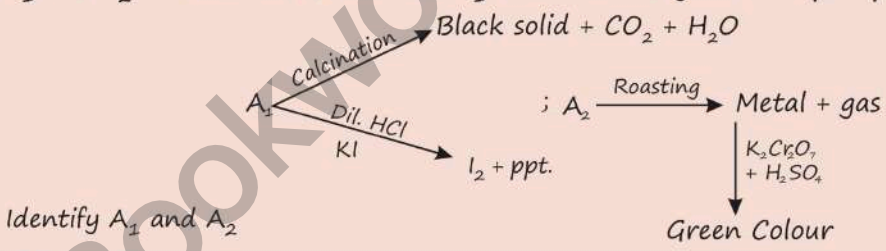
(a) Zone refining (b) Vapour phase refining

(c) Liquation (d) Van Arkel method

Sol. (a) The method used for the purification of Indium is zone refining.

(IIT JEE 2004)

47. A_1 and A_2 are two ores of metal M. A_1 on calcination gives black precipitate, CO_2 and water.



Identify A_1 and A_2

Sol. (a) A_1 is malachite: $CuCO_3 \cdot Cu(OH)_2$

- $CuCO_3 \cdot Cu(OH)_2 \rightarrow 2CuO + CO_2 \uparrow + H_2O$ (On heating)

  CuO is the black solid.
- $CuCO_3 \cdot Cu(OH)_2 \rightarrow CuCl_2 + CO_2 + H_2O$ (in presence of Dil. HCl)

  $CuCl_2 + KI \rightarrow Cu_2I_2 + KCl + I_2$

(b) A_2 is copper glance Cu_2S

- $Cu_2S \rightarrow Cu_2O + SO_2$ (in presence of oxygen gas)
- $Cu_2S + Cu_2O \rightarrow Cu + SO_2$

The gas SO_2 gives green colour with acidified $K_2Cr_2O_7$

- $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O$

  $Cr_2(SO_4)_3$ is green in colour.

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EXTRACTION OF Tin FROM SnO_2 [Cassiterite]

- Tin is obtained by reducing cassiterite (SnO_2) with coal in a reverberatory furnace.

$$SnO_2 + 2C \rightarrow Sn + 2CO$$

- Limestone is added to produce a slag with the impurities, which can get separated.
- Crude tin so obtained is contaminated with iron, copper, lead and other metals.
- It is, remelted on an inclined furnace.
- The easily fusible tin melts away and the less fusible impurities are left behind.
- Molten tin is finally stirred with green poles of wood in contact with air.

Purification of impure tin

$$2SnO_2 + CH_4 \rightarrow \underset{\text{Pure metal}}{2Sn} + CO_2 + H_2O$$

48. Extraction of metal from the ore cassiterite involves (IIT JEE 2011)

(a) Carbon reduction of an oxide ore
(b) Self-reduction of a sulphide ore
(c) Removal of copper impurity
(d) Removal of iron impurity

Sol. (a, c, d)

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